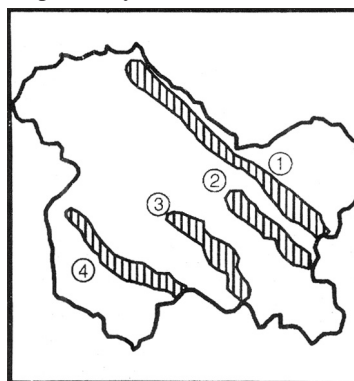


63. Which among the following states of India have common borders with Pakistan?
 (a) Jammu and Kashmir, Himachal Pradesh, Punjab and Rajasthan
 (b) Punjab, Jammu and Kashmir, Rajasthan and Gujarat
 (c) Jammu and Kashmir, Punjab, Haryana and Rajasthan
 (d) Punjab, Himachal Pradesh, Gujarat and Rajasthan
64. Which of the following three rivers of the peninsula India have the Amarkantak region as their source?
 (a) Narmada, Krishna Godavari
 (b) Son, Mahanadi, Narmada
 (c) Godavari, Krishna, Cauvery
 (d) Chambal, Betwa, Luni
65. Which one among the following rivers does not flow into the Bay of Bengal?
 (a) Mahanadi (b) Cauveri
 (c) Tapti (d) Godavari
66. The country that shares longest border with India is
 (a) China (b) Bangladesh
 (c) Nepal (d) Pakistan
67. Amarkantak plateau in the Maikal hills marks the origin of the river
 (a) Gandak (b) Chambal
 (c) Narmada (d) Ghaggar
68. Which one among the following states does not form part of the Narmada basin ?
 (a) Madhya Pradesh (b) Gujrat
 (c) Rajasthan (d) Maharashtra
69. Match the following

List I (Hydroelectric power station)	List II (Location in map)
A. Nagarjuna Sagar	
B. Mettur	
C. Hirakund	
D. Sileru	

Codes :

- | | | | |
|-------|---|---|---|
| A | B | C | D |
| (a) 3 | 4 | 1 | 2 |
| (b) 3 | 1 | 4 | 2 |
| (c) 2 | 1 | 4 | 3 |
| (d) 2 | 4 | 1 | 3 |
70. Which one among the following is nearest to the Tropic of Cancer ?
 (a) Patna (b) Ranchi
 (c) Rourkela (d) Varanasi
71. In which one of the following states is Zojila pass located?
 (a) Arunachal Pradesh (b) Jammu and Kashmir
 (c) Himachal Pradesh (d) Sikkim
72. Which one of the following pairs is not correctly matched?
 (a) Indus : Shyok
 (b) Ganga : Gandak
 (c) Godavari : Musi
 (d) Krishna : Tungabhadra
73. Where is the Nanda Devi peak located ?
 (a) Himachal Pradesh (b) Uttarakhand
 (c) Sikkim (d) Nepal
74. The river Sone is a tributary of which one of the following rivers ?
 (a) Ganga (b) Yamuna
 (c) Narmada (d) Mahanadi
75. Which one of the following is the correct sequence of passes when one travels along the Himalayas from Kashmir to Sikkim?
 (a) Zozila-Nathula-Shipkila
 (b) Nathula-Shipkila-Zozila
 (c) Nathula-Zozila-Shipkila
 (d) Zozila-Shipkila-Nathula
76. Which one among the following peaks is the highest?
 (a) Dhaul Giri (b) Namcha Barwa
 (c) Nanda Devi (d) Nanga Parvat
77. From West to East, which one of the following is the correct sequence of Hills?
 (a) Mahadeo Hills—Maikala Hills—Garhjat Hills
 (b) Mahadeo Hills—Garhjat Hills—Maikala Hills
 (c) Maikala Hills—Mahadeo Hills—Garhjat Hills
 (d) Maikala Hills—Garhjat Hills—Mahadeo Hills
78. The Tropic of Cancer passes through which one of the following?
 (a) Assam (b) Manipur
 (c) Mizoram (d) Nagaland
79. Mahatma Gandhi Hydroelectric project is on which river ?
 (a) Godavari (b) Sharavati
 (c) Cauvery (d) Krishna
80. Indian Standard Time refers to the local time of which one of the following places in India?
 (a) Allahabad (b) Bhopal
 (c) Delhi (d) Lucknow
81. Which one of the following rivers flow between Vindhyan and Satpura ranges?
 (a) Narmada (b) Son
 (c) Mahe (d) Netravati
82. Amravati, Bhavani, Hemavati and Kabini are tributaries of which one of the following rivers?
 (a) Mahanadi (b) Godavari
 (c) Kaveri (d) Krishna
83. The Tropic of Cancer does not pass through
 (a) Orissa (b) Tripura
 (c) Chhattisgarh (d) Rajasthan
84. Which one of the following mountain ranges is spread over only one state in India?
 (a) Aravalli (b) Satpura
 (c) Ajanta (d) Sahyadri
85. Examine the map of Jammu and Kashmir given below:
 The mountains ranges marked 1, 2, 3 and 4 are respectively:



- (a) Ladakh, Zaskar, Karakoram and Pir Panjal
 (b) Karakoram, Ladakh, Zaskar and Pir Panjal
 (c) Karakoram, Zaskar, Pir Panjal and Ladakh
 (d) Ladakh, Pir Panjal, Karakoram and Zaskar

86. Which one of the following east flowing rivers of India has rift valley due to down warping?
 (a) Damodar (b) Mahanadi
 (c) Sone (d) Yamuna
87. If it is 10.00 am. I.S.T., then what would be the local time at Shillong on 92° E longitude?
 (a) 9.38 a.m. (b) 10.38 a.m.
 (c) 10.22 a.m. (d) 9.22 a.m.
88. The correct sequence of the eastward flowing rivers of the peninsular India from north to south is :
 (a) Subarnarekha, Mahanadi, Godavari, Krishna, Pennar, Cauvery and Vagai
 (b) Subarnarekha, Mahanadi, Krishna, Godavari, Cauvery and Vagai
 (c) Mahanadi, Subarnarekha, Godavari, Krishna, Cauvery, Pennar and Vagai
 (d) Mahanadi, Subarnarekha, Krishna, Godavari, Cauvery, Vagai and Pennar
89. Among the following cities, which one is nearest to the Tropic of Cancer?
 (a) Delhi (b) Kolkata
 (c) Jodhpur (d) Nagpur
90. Gandhi Sagar Dam is a part of which one of the following?
 (a) Chambal Project
 (b) Kosi Project
 (c) Damodar Valley Project
 (d) Bhakra Nangal Project
91. Where are Shevaroy hills located?
 (a) Andhra Pradesh (b) Karnataka
 (c) Kerala (d) Tamil Nadu
92. Out of the four southern States: Andhra Pradesh, Karnataka, Kerala and Tamil Nadu, which shares boundaries with the maximum number of Indian States?
 (a) Andhra Pradesh only
 (b) Karnataka only
 (c) Each of Andhra Pradesh and Karnataka
 (d) Each of Tamil Nadu and Kerala
93. Which of the following hills are found where the Eastern Ghats and the Western Ghats meet?
 (a) Anamalai Hills (b) Cardamom Hills
 (c) Nilgiri Hills (d) Shevaroy Hills
94. Which one of the following statements is not true?
 (a) Ghaggar's water is utilised in the Indira Gandhi canal
 (b) Narmada rises from Amarkantak region
 (c) Nizam Sagar is situated on the Manjra river
 (d) Penganga is a tributary of the Godavari
95. Which one of the following statements is not correct?
 (a) The Western Ghats are relatively lower in their northern region.
 (b) The Anai Mudi is the highest peak in the Western Ghats.
 (c) Tapi river lies to the south of Satpura.
 (d) The Narmada and Tapi river valleys are said to be old rift valleys.

Exercise -2

Statement Based MCQ

1. If there were no Himalayan ranges, what would have been the most likely geographical impact on India?
 1. Much of the country would experience the cold waves from Siberia.
 2. Indo-gangetic plain would be devoid of such extensive alluvial soils.
 3. The pattern of monsoon would be different from what it is at present.

Which of the statements given above is/are correct?

(a) 1 only (b) 1 and 3
(c) 2 and 3 (d) 1, 2 and 3
2. Consider the following statements:
 1. The Brahmaputra plain extends in Assam for about 640 Kms from Dhubri to Sadiya.
 2. The Peninsular plateau is made up of highly denuded rocks.

Which of the above statements is/are not correct?

(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2
3. Consider the following statements with regard to Himalaya.
 1. Mt Everest, Kanchenjunga fall in the inner Himalaya.
 2. The inner Himalaya has an average altitude of 6000 metres.
 3. Kothri Dun and Patli Dun are longitudinal valley.

Which of the above statements is/are correct?

(a) 1, 2 and 3 (b) 1 and 2
(c) 1 and 3 (d) 3 only
4. Consider the following statements about Telengana
 1. It is situated to the south-east of the Deccan lava plateau.
 2. It is high plateau highly denuded and dissected.
 3. Monad rocks are also found in the plateau.
 4. Its northern parts have scanty vegetation.

Choose the incorrect statements:

(a) 1, 2 and 4 (b) 2 and 3
(c) 1 and 4 (d) 2 only
5. Which of the following are true with respect to the Indian Peninsular Plateau?
 1. The southern plateau block is formed mainly of granites and gneiss.
 2. The Deccan lava plateau is an elevated tableland consisting of horizontally arranged lavasheets.
 3. The Malwa plateau dominates the Vindhyan scarps forming the northern flank of the plateau.
 4. The troughs of the Narmada and the Tapi are interposed between the Vindhyan and the Satpura ranges

(a) 1, 2 and 3 (b) 1 and 2
(c) 1, 2, 3 and 4 (d) 1, 3 and 4
6. Which of the following is true with regard to the characteristics of the Himalayan rivers ?
 1. Many of them have their sources in the inner Himalays.
 2. They have a perennial flow.
 3. They have a great capacity for erosion.
 4. They do not form gorges.

(a) 1 and 2 (b) 1, 2 and 3
(c) 3 and 4 (d) 2 and 4
7. When you travel in Himalayas, you will see the following
 1. Deep gorges
 2. U-turn river courses
 3. Parallel mountain ranges
 4. Steep gradients causing land-sliding

Which of the above can be said to be the evidences for Himalayas being young fold mountains?

(a) 1 and 2 (b) 1, 2 and 4
(c) 3 and 4 (d) 1, 2, 3 and 4
8. Consider the following statements:
 1. A valley glacier consists of not only ice but rocks also.
 2. The largest valley glaciers in India are in the Karakoram range.
 3. Mountain glaciers can dig new valleys.

Which of the statements given above is/are correct?

(a) 1 only (b) 1 and 2
(c) 2 and 3 (d) 1, 2 and 3
9. Consider the following statements:
 1. Ranjit Sagar Dam is on the river Beas.
 2. Hirakund Dam is on the river Damodar.
 3. The reservoir Gandhi Sagar is on the river Chambal.

Which of the statements given above is/are correct?

(a) 1 and 2 (b) 3 only
(c) 2 and 3 (d) 1 and 3
10. Consider the following statements :
 1. Alamatti dam is on the Cauvery river.
 2. Mettur dam is on the Krishna river.
 3. Gandhi Sagar Reservoir is on the Chambal river.

Which of the statements given above is/are correct?

(a) 1 and 2 (b) 2 only
(c) 1 and 3 (d) 3 only
11. Identify from the following states of India through which the Tropic of Cancer passes and arrange them from East to West.

1. Gujarat	2. West Bengal
3. Uttar Pradesh	4. Jharkhand
5. Madhya Pradesh	6. Bihar
7. Chhattisgarh	

Select the correct answer using the codes given below.

(a) 2-5-7-4-1 (b) 2-4-7-5-1
(c) 3-2-6-7-5 (d) 3-7-4-6-2

12. Which of the following statements is/are true?
 (I) Longitudes of Jabalpur's location is between those of Indore and Bhopal.
 (II) Latitude of Aurangabad's location is between those of Vadodara and Pune
 (III) Bangalore is situated more south ward than Chennai.
 (a) I and III (b) only II
 (c) II and III (d) I, II and III
13. Tank irrigation is practised mainly in Peninsular India because
 1. undulating relief and hard rocks make it difficult to dig canals and wells
 2. rivers are rainfed
 3. of compact nature of population and agricultural fields
 Select the correct answer using the codes given below
 Codes
 (a) 1 and 2 (b) 2 and 3
 (c) 1 and 3 (d) All of these
14. Which of the following statements about Nathu la Pass are correct?
 1. It links Sikkim with Tibet.
 2. It was the main artery of the ancient Silk Route.
 3. It was reopened in the year 2006.
 Select the correct answer using the codes given below
 (a) 1, 2 and 3 (b) 1 and 2
 (c) 2 and 3 (d) 1 and 3
15. Which of the following is/are West flowing river(s) of India?
 1. Mahanadi 2. Krishna
 3. Narmada 4. Cauvery
 Select the correct answer using the codes given below
 (a) 1, 2 and 4 (b) 2 and 3
 (c) Only 3 (d) 1 and 3
16. Arrange the following features formed by rivers in its course starting from upstream:
 1. Meanders 2. Falls
 3. Delta 4. Oxbow Lake
 Select the correct answer using the code given below:
 (a) 2 - 1 - 3 - 4 (b) 2 - 1 - 4 - 3
 (c) 1 - 2 - 3 - 4 (d) 1 - 4 - 2 - 3
17. Arrange the following tributaries of river Indus from North to South :
 1. Chenab 2. Jhelum
 3. Ravi 4. Sutlej
 Select the correct answer using the code given below :
 (a) 4-3-1-2 (b) 2-3-1-4
 (c) 1-2-3-4 (d) 2-1-3-4
18. Consider the following statements:
 1. Ranjit Sagar Dam is on the river Beas.
 2. Hirakund Dam is on the river Damodar.
 3. The reservoir Gandhi Sagar is on the river Chambal.
 Which of the statements given above is/are correct?
 (a) 1 and 2 (b) 2 only
 (c) 2 and 3 (d) 1 and 3
19. Which one of the following pairs is correctly matched?

Rivers	Major Tributaries
(a) Godavari:	Pairi, Hasdo, Tel
(b) Maha:	Puma, Penganga, Wain ganga
(c) Krishna:	Manjra, Dudhana, Indravati
(d) Cauvery:	Kabani, Hemavati, Amaravati
20. Consider the following statements :
 1. Alamatti dam is on the Cauvery river.
 2. Mettur dam in on the Krishna river.
 3. Gandhi Sagar Reservoir is on the Chambal river.
 Which of the statements given above is/are correct?
 (a) 1 and 2 only (b) 2 only
 (c) 1 and 3 only (d) 3 only
21. Consider the following rivers:
 1. Betwa
 2. Kosi
 3. Gandak
 Which of the above join(s) Yamuna river?
 (a) 1 only (b) 1 and 2
 (c) 1 and 3 (d) 2 and 3
22. Identify from the following states of India through which the Tropic of Cancer passes and arrange them from East to West.
 1. Gujarat 2. West Bengal
 3. Uttar Pradesh 4. Jharkhand
 5. Madhya Pradesh 6. Bihar
 7. Chhattisgarh
 Select the correct answer using the codes given below.
 (a) 2-5-7-4-1 (b) 2-4-7-5-1
 (c) 3-2-6-7-5 (d) 3-7-4-6-2
23. Consider the following statements:
 1. Longitude of Jabalpur's location is between those of Indore and Bhopal.
 2. Latitude of Aurangabad's location is between those of Vadodara and Pune.
 3. Bangalore is situated more southward than Chennai
 Which of these statements is/are correct?
 (a) 1 and 3 (b) Only 2
 (c) 2 and 3 (d) 1, 2 and 3
24. Consider the following pairs:

Tributary River	Main River
1. Chambal	Narmada
2. Sone	Yamuna
3. Manas	Brahmaputra

 Which of the pairs given above is/are correctly matched?
 (a) 1, 2 and 3 (b) 1 and 2 only
 (c) 2 and 3 only (d) 3 only
25. The Narmada river flows to the west, while most other large peninsular rivers flow to the east. Why?
 1. It occupies a linear rift valley.
 2. It flows between the Vindhyas and the Satpuras.
 3. The land slopes to the west from Central India.
 Select the correct answer using the codes given below.
 (a) 1 only (b) 2 and 3
 (c) 1 and 3 (d) None

Matching Based MCQ

DIRECTIONS (Qs. 26 to 37) : Match List-I with List-II and select the correct answer using the codes given below the lists.

26. **List-I** **List-II**
 (A) Mahanadi (1) Manjra
 (B) Godavari (2) Orisan
 (C) Narmada (3) Aner
 (D) Tapi (4) Ib
 (a) A - 4 ; B - 1 ; C - 3 ; D - 2
 (b) A - 2 ; B - 1 ; C - 4 ; D - 3
 (c) A - 4 ; B - 1 ; C - 2 ; D - 3
 (d) A - 2 ; B - 1 ; C - 3 ; D - 4
27. **List-I** **List-II**
 (A) Krishna (1) Chambal
 (B) Brahmaputra (2) Indravati
 (C) Godavari (3) Tisha
 (D) Yamuna (4) Bhima
 (a) A - 4 ; B - 3 ; C - 2 ; D - 1
 (b) A - 3 ; B - 4 ; C - 1 ; D - 2
 (c) A - 4 ; B - 3 ; C - 1 ; D - 2
 (d) A - 3 ; B - 4 ; C - 2 ; D - 1
28. Which one of the following pairs is not correctly matched?

River	City
A. Gomati	Lucknow
B. Sarya	Ayodhya
C. Alakananda	Badrinath
D. Narmada	Satna

29. Match the following

List I River	List II Tributary
A. Brahmaputra	1. Musi
B. Krishna	2. Tawa
C. Narmada	3. Bhavani
D. Cauveri	4. Dikhow

Codes :

- A B C D
 (a) 4 2 1 3
 (b) 4 1 2 3
 (c) 3 2 1 4
 (d) 3 1 2 4

30. Match the following

List I (Pass)	List II (State)
A. Zoji La Pass	1. Sikkim
B. Bara Lacha Pass	2. Uttarakhand
C. Jelep La Pass	3. Himachal Pradesh
D. Niti Pass	4. Jammu and Kashmir

Codes :

- A B C D
 (a) 4 1 3 2
 (b) 2 3 1 4
 (c) 4 3 1 2
 (d) 2 1 3 4

31. Match the following.

List I	List II
(River)	(Source)
A. Ganga	1. Amarkantak
B. Sone	2. Gaumukh
C. Godavari	3. Mahabaleshwar
D. Krishna	4. Trimbakeshwar

Codes :

- A B C D
 (a) 1 2 4 5
 (b) 2 1 3 4
 (c) 4 3 1 2
 (d) 2 1 4 3

32. Match the following

List I	List II
(River)	(Tributary)
A. Ganga	1. Chambal and Ken
B. Indus	2. Wainganga and Indravati
C. Yamuna	3. Chenab and Sutlej
D. Godavari	4. Gomti and Kosi

Codes :

- A B C D
 (a) 4 1 3 2
 (b) 4 3 1 2
 (c) 2 1 3 4
 (d) 2 3 1 4

33. Match list I with List II and select the correct answer using the codes given below the lists:

	List I (Peak)		List II (State)
A.	Dodabetta	1.	Maharashtra
B.	Guru Shikhar	2.	Madhya Pradesh
C.	Kalsubai	3.	Tamil Nadu
D.	Dhupgarh	4.	Rajasthan

Codes:

- A B C D
 (a) 3 1 4 2
 (b) 2 1 4 3
 (c) 3 4 1 2
 (d) 2 4 1 3

34. Match List I with List II and select the correct answer using the codes given below the lists:

	List I (Himalayan Peaks)		List II (States)
A.	Nanga Parbat	1.	Jammu & Kashmir
B.	Nanda Devi	2.	Sikkim
C.	Kanchanjunga	3.	Uttarakhand

Codes :

- A B C
 (a) 1 2 3
 (b) 1 3 2
 (c) 2 3 1
 (d) 3 1 2

35. Match List-I (Valley) with List-II (State) and select the correct answer using the codes given below the lists:

List-I (Valley)	List-II (State)
A. Markha Valley	1. Sikkim
B. Dzukou Valley	2. Himachal Pradesh
C. Sangla Valley	3. Jammu & Kashmir
D. Yumthang Valley	4. Nagaland

Codes :

- (a) A-2; B-4; C-3; D-1
 (b) A-3; B-1; C-2; D-4
 (c) A-2; B-1; C-3; D-4
 (d) A-3; B-4; C-2; D-1
36. Match List I with List II and select the correct answer using the code given below the Lists :

List I (Himalayan Peak)	List II (State)
A. K2	1. Uttarakhand
B. Nanda Devi	2. Jammu & Kashmir
C. Tara Pahar	3. Sikkim
D. Kanchenjunga	4. Himachal Pradesh

Code:

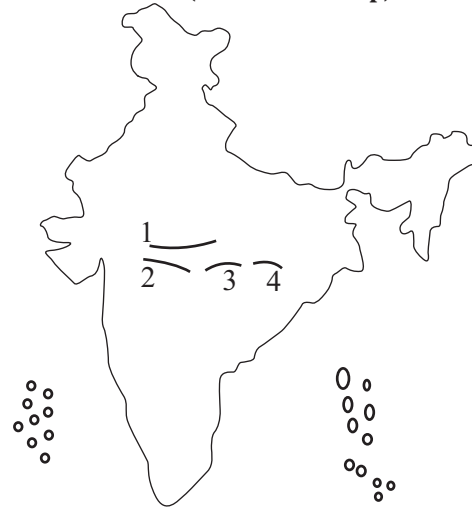
	A	B	C	D
(a)	2	4	1	3
(b)	2	1	4	3
(c)	3	1	4	2
(d)	3	4	1	2

37. Match List I with List II and select the correct answer using the code given below the Lists :

List I
(Hill range of Central India)

- A. Satpura
 B. Mahadeo
 C. Vindhya
 D. Maikala

List II
(Location in map)



Code :

	A	B	C	D
(a)	4	3	1	2
(b)	4	1	3	2
(c)	2	1	3	4
(d)	2	3	1	4

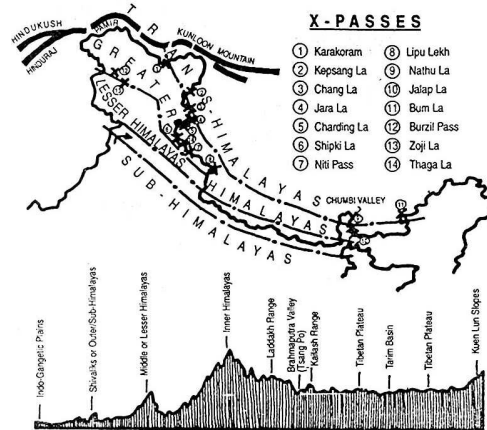
Hints and Explanations

EXERCISE-1

1. (d) Luni river originates from Aravalli range and flows in south west direction through the hills and finally ends up in the marshy land of Rann of Kutch.
2. (b) Brahmaputra originates near Mt. Kailash and is known to take a U turn near Mount Namcha Barwa. This U turn is also known as Great Bend. The U Turn is because of the 180° bend of the Himalayan structural trends.
3. (b) 4. (d) 5. (c) 6. (b) 7. (d)
8. (a) 9. (d) 10. (a) 11. (c) 12. (a)
13. (c) 14. (b) 15. (b) 16. (d) 17. (c)
18. (b) 19. (a)
20. (d) Rivers like Beas, Chenab, Ravi, Sutlej and Yamuna passes through Himachal Pradesh.
21. (a) Chilka, Periyar and Pulicate lake are lagoons. They are formed due to separation of sea water by deposition and clay, but Ashtamudi lake is not a lagoon.
22. (b) 23. (c)
24. (c) Both Andhra Pradesh and Karnataka shares boundaries with maximum number of Indian states.
25. (d) 26. (c) 27. (b)
28. (c) Wardha river is a tributary of River Godavari.
29. (a)
30. (c) Musi is the tributary of Krishna, Panganga, Wanganga, Wardha, Indiravati, Pravada etc. are the tributaries of Godavari.
31. (b) Nanda Devi Peak (7817 m) is situated in Uttarakhand.
32. (a) Ramganga, Gomti, Ghaghra, Gundak, Kosi, Son, Mahananda are the tributaries of Ganga.
33. (d)
34. (a) From West to East the sequence of Hills : Nanda Devi Hills — Maikala Hills — Grahjat hills
35. (c) Tropic of cancer passes through Mizoram.
36. (d) Port Blair is the largest town and a municipal council in Andaman district in the Andaman Islands and the capital of the Andaman and Nicobar Islands, a Union Territory of India. It lies on the east coast of South Andaman Island and is the main entry point to the islands.
37. (a) 38. (d)
39. (d) The Colorado river is the primary river of the American Southwest. The original name of this river is Rio Colorado or Red River by the Spanish.
40. (d) The Narmada river (Rewa) is the fifth largest river in the Indian subcontinent. It is third largest river which is completely flows in India after Ganga and Godavari.
41. (d) Govind Ballabh Pant Sagar is on the Rihand River which is the tributary of the Son River.
42. (a) The Luni is a river of western Rajasthan state. It originates in the Pushkar valley of the Aravalli Range near Ajmer and ends in Rann of Kutch in Gujarat.
43. (d) The Gandhi Sagar Dam is one of the four major dams built on India's Chambal River. The dam is located in the Mandsaur district of Madhya Pradesh.
44. (b) With a view to restoring the past glory for Kolleru lake, government had taken up 'Operation Kolleru' on the Krishna and West Godavari district borders. Kolleru Lake is one of the largest freshwater lakes in India located in state of Andhra Pradesh
45. (c) Cauvery river dispute is among Kerala, Karnataka, Tamil Nadu and Puducherry.
46. (b) The Palk Strait is situated between the Tamil Nadu and the Mannar district of the Northern Province of Sri Lanka. It connects the Bay of Bengal in the northeast with the Palk Bay and thence with the Gulf of Mannar in the southwest.
47. (c) Godwin Austen (K^2) is the highest peak in the given options. It is the world's second tallest peak. It is located in the Karakoram range of the Himalaya in Northern Kashmir. Its height is 8611m.
48. (a) Bangladesh shares longest border with India. Both countries share a 4,096-kilometer) long international border.
49. (d) The Son flows north-northwest through Madhya Pradesh.
50. (c) Nasik is located on the banks of the Godavari.
51. (d) The Rihand Dam was constructed across the Rihand River near Pipri in Sonbhadra district of Mirzapur division in 1962 for hydropower generation.
52. (a)
53. (b) The Latitude & Longitude for New Delhi are 28.38° N and 77.12°.
54. (d) Narmada flows into the Gulf of Khambhat (Arabian Sea).
55. (c)
56. (c) 1. Rihand - Pipri
2. Gandak- Balmikinagar
3. Chambal- Kota
4. Mahanadi- Hirakund
57. (b) The Aravalli range are the oldest fold mountains in India.
58. (d) Indian Standard Time is calculated on the basis of 82.5° E longitude in Shankargarh Fort Mirzapur (Allahabad).
59. (c)
60. (b) The Ten Degree Channel is a channel that separates the South Andaman and Car Nicobar in the Bay of Bengal.
61. (a) Correct sequence is Godavari- Penner- Cauveri- Periyar

The Godavari is the second longest river in India after the river Ganga. It originates from Western Ghats of central India near Nasik in Maharashtra. The Penner originates from Nandi Hills in Chikballapur District of Karnataka. This river flows from north as well as east corners across Andhra Pradesh. The Cauvery originates from Western Ghats in Karnataka and flows generally south and east through Karnataka and Tamil Nadu. The Periyar originates from Western Ghats range near the border with Tamil Nadu and flows in Kerala.

62. (c) Uttarkashi is located on a confluence of two rivers varuna and
63. (b) Punjab, Jammu and Kashmir, Rajasthan and Gujarat have common borders with Pakistan.
64. (b) Son, Mahanadi and Narmada rivers originate from Amarkantak region.
65. (c) Tapi does not flow in to the Bay of Bengal. The river rises in the eastern Satpura Range of southern Madhya Pradesh and flows into the Gulf of Cambay of the Arabian Sea.
66. (b) Bangladesh shares longest border with India. India and Bangladesh share a 4,096 kilometres long international border. It is the fifth-longest land border in the world.
67. (c) The origin of the river is a tiny reservoir named as Narmada Kund which is situated on the Amarkantak Hill in Anuppur District of East Madhya Pradesh. Amarkantak region is a unique natural heritage area and is the meeting point of the Vindhyas and the Satpuras, with the Maikal Hills being the fulcrum.
68. (c) The Narmada River travels a distance of 1,312 km before it falls into Gulf of Cambay in the Arabian Sea near Bharuch in Gujarat. The first 1,079 km of its run is in Madhya Pradesh. In the next length of 35 km, the river forms the boundary between the States of Madhya Pradesh and Maharashtra. Again, in the next length of 39 km, it forms the boundary between Maharashtra and Gujarat.
69. (b)
70. (b) Tropic of Cancer comes across Jharkhand and Ranchi is nearest of it.
71. (b) Zojila Pass is situated in Jasker range in Jammu and Kashmir state. It passes through Srinagar-Leh National Highway.
72. (c) Musi Rives is a tributary of the krishna River in the Deccan ptaeleau flowing through telangana stata in India.
73. (c)
74. (b) Nanda Devi Peak (7817 m) is situated in Uttarakhand.
75. (a) Ramganga, Gomati, Ghaghra, Gundak, Kosi, Son, Mahananda are the tributaries of Ganga.
76. (d) The correct sequence of passes given in the question is shown in the following diagram.



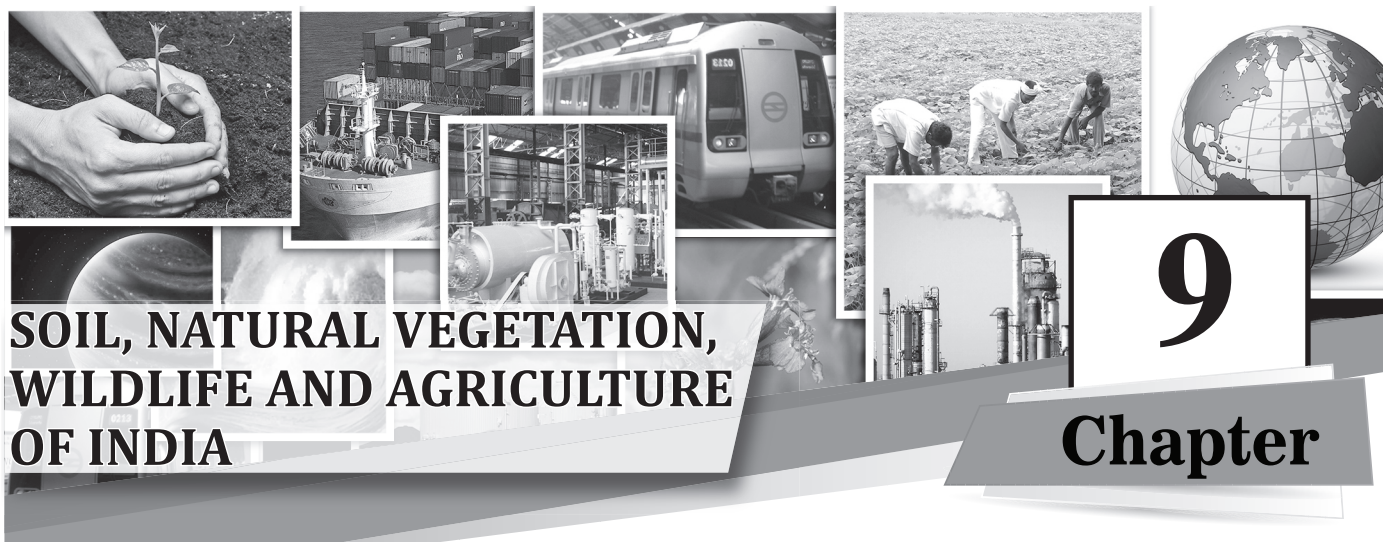
77. (a) Dhaulagiri is some of highest peak in great Himalaya range. Its height is approximately 8172 m. Height of Nanda devi is 7816 m.
78. (a) From West to East The sequence of Hills : Nanda Devi Hills — Maikala Hills — Grahjat hills
79. (b)
80. (a) The standard time in India is the local time of a place at 82.5° E longitude near Mirzapur. This meridian also divides India approximately into half.
81. (a)
82. (c) The Kaveri rises from the Brahmagiri hills in the Coorg district at 1,341 m above sea level and drain the Maysore plateau before flowing into the plain.
83. (a) The tropic of cancer passes through tripura, Chhattisgarh and Rajasthan states in India.
84. (c) Ajanta mountain range a short range, which spreads within Maharashtra.
 - Aravali ranges covers Rajasthan, Haryana and Delhi
 - Satpura ranges are found in Gujrat and Madhya Pradesh.
 - Sahyadris ranges starts from Gujrat, Maharastra border and crossing Goa, Karnataka. It reaches Kerala tip upto Cape Comrin.
 - Ajanta is found in the Aurangabad district of Maharastra.
85. (b) According to the physiography map of India the mountains ranges found in Jammu and Kashmir is Karakoram which is marked as '1' in figure. Ladakh range which is marked as '2', Zanskar range which is marked as '3', and Pir Panjal which is marked as '4'.
86. (a) Damodar is a east flowing river with rift valley due to down warping. Down warp denotes a segment of the earth's crust that is broadly bent downward.
87. (b)
88. (a) The correct sequence of eastward flowing river of the peninsular India from north to south is Subarnarekha, Mahanadi, Godavari, Krishna, Pennar, Cauvery and Vagai.
89. (b) The Tropic of Cancer located at 23.5° north. Location of Kolkata is 22°33' N. So, Kolkata is nearer to Tropic of Cancer. Delhi is 28.38°N.

90. (a) Gandhi Sagar Dam is situated on the river Chambal in 1960, near Bhanpura of Madhya Pradesh.
91. (d) Shevaroy hill is situated near Salem of Tamil Nadu. This hill range covers an area of fifty square kilometers.
92. (c) Both Andhra Pradesh and Karnataka shares boundaries with maximum number of Indian states.
93. (c) Nilgiri hills are at the junction of the eastern and western ghats of the Sahayadri hills. The heights of the hills range varies between 2,280 and 2,290 metres.
94. (a) Indira Gandhi Canal originated from Harike barrage at Sultanpur on Sutlej but Ghaggar is a tributary of river Saraswati, which ends in the Thar desert.
95. (a) Western Ghats are relationship higher in the Southern region.
15. (c) West flowing rivers:
Narmada River(1057 km)- Rises in Amarkantak Plateau and flows into gulf of khambat.
Tapti(724 km) - Rises from Betul district in Maharashtra. Luni and Payaswani are also west flowing rivers.
16. (b) Falls are possible only when a river is losing height suddenly. Next comes meanders. When a river reaches flatter land,flow is slowed down and meanders are formed through erosion of the river banks and deposition on the inside of bends.
Oxbow lake is formed when a wide meander is cut off from the main river, creating a free-standing body of water.
Delta is formed at the mouth of a river, where the river flows into an ocean, sea, estuary, lake, or reservoir.

EXERCISE-2

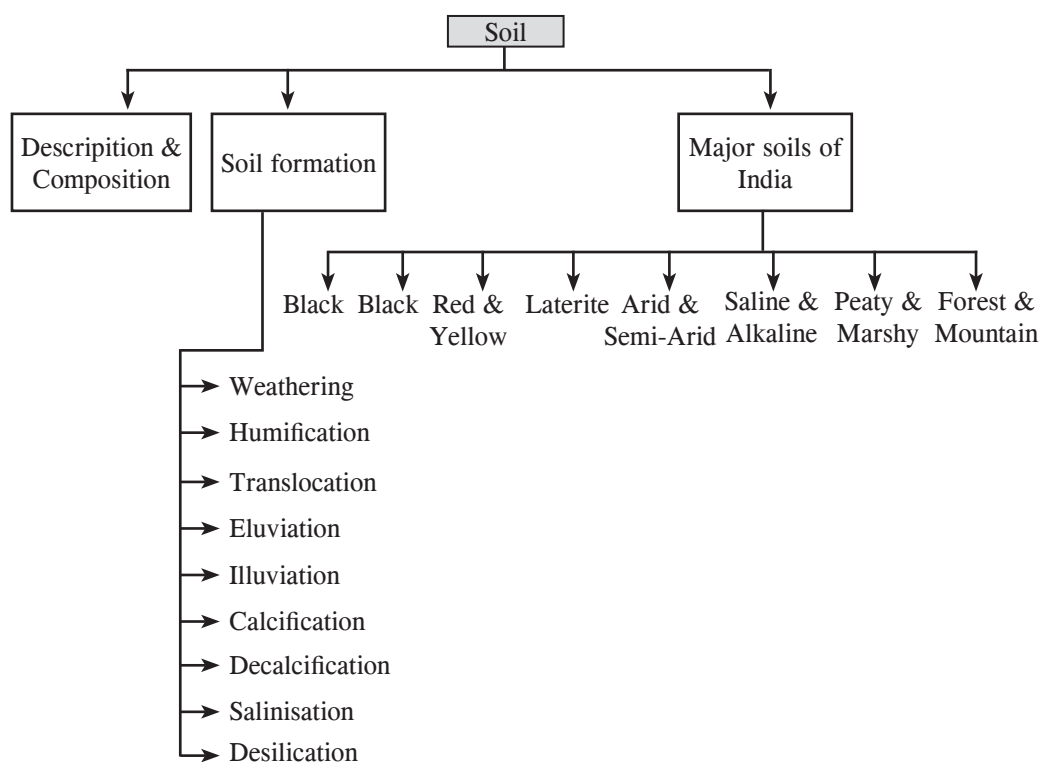
1. (d) All the statements given in the question are correct.
2. (d) 3. (a) 4. (d) 5. (c) 6. (b)
7. (d) All are correct.
8. (d)
9. (d) Hirakund Dam is situated on the river Mahanadi.
10. (d) Gandhi Sagar Reservoir on Chambal river, jointly executed by Madhya Pradesh and Rajasthan.
11. (b)
12. (c)
13. (a) The tank irrigation is practised mainly in peninsular India due to the following reasons:
 1. The undulating relief and hard rocks make it difficult to dig canals and wells.
 2. There is little percolation of rain water due to hard rock structure and ground water is not available in large quantity.
 3. Most of the rivers of this region are seasonal and dry up in summer season. Therefore, they cannot supply water to canals throughout the year.
 4. There are several streams which become torrential during rainy season. The only way to make best use of this water is to impound it by constructing bunds and building tanks. Otherwise this water would go waste to the sea.
 5. The scattered nature of population and agricultural fields also favour tank irrigation.
14. (a) Nathu La is a mountain pass in the Himalayas. It connects the Indian state of Sikkim with China's Tibet Autonomous Region. Nathu La is located on the 563 km Old Silk Route, an offshoot of the historic Silk Road. It was sealed by India after the 1962 Sino-Indian War and was re-opened in 2006 following numerous bilateral trade agreements.
17. (d) The Indus River originates near the Mansarovar Lake in the Tibetan plateau, on the northern slopes of the Kailash Mountain Range. Given below are the main tributaries of the Indus River from north to south:
 - Jhelum
 - Chenab
 - Ravi
 - Sutlej
18. (b) Hirakund Dam is situated on the river Mahanandi.
19. (d) Godavari is the largest in the peninsular region. It has formed an extensive delta in Andhra Pradesh. It rises near Nasik in Maharashtra.
Cauvery is known as the Ganga of the South. It rises in the Coorg district of Karnataka.
Krishna rises in the Western Ghat near Mahabaleshwar and flows through Andhra Pradesh.
20. (d) Gandhi Sagar Reservoir on Chambal river, jointly executed by Madhya Pradesh and Rajasthan.
21. (a) 22. (b)
23. (c) The longitude of Jabalpur's location is to the east of Bhopal, Bangalore is north of Chennai.
24. (d) Tributaries of Brahmaputra in India are the Manas, Pagladiya, Puthimari, Dhanisri, Jia Bhariti and Subansiri.
Manas is a tributary of Brahmaputra Chambal is the chief tributary of Yamuna and sone is a tributary of Ganga.
25. (a) 26. (c) 27. (a)
28. (d) The source of the Narmada is Amarkantak in the Anuppur District (eastern Madhya Pradesh).
29. (b)
 1. Brahmaputra- Dihow
 2. Krishna- Musi
 3. Narmada- Tawa
 4. Cauveri- Bhavani

30. (c)
1. Zoji La Pass- Jammu and Kashmir
 2. Bara Lacha Pass- Himachal Pradesh
 3. Jalep La Pass- Sikkim
 4. Niti pass- Uttarakhand.
31. (d)
1. Ganga - Gaumukh
 2. Son- Amarkantak
 3. Godavari- Trimbakeshwar
 4. Krishna- Mahabaleshwar
32. (b) **River Tributary**
- | | |
|----------|-------------------------|
| Ganga | Gomti and Kosi |
| Indus | Chenab and Sutlej |
| Yamuna | Chambal and Ken |
| Godavari | Wainganga and Indravati |
33. (c) Dodabetta (Tamil Nadu), Guru Shikhar (Rajasthan), Kalsubai (Maharashtra) and Dhupgarh (Madhya Pradesh).
34. (b) Nanga parbat (8126 m)-Jammu & Kashmir, Nanda Devi (7817 m) – Uttarakhand and Kanchenjunga (8598 m) Sikkim.
35. (d)
36. (b) K2-Jammu and Kashmir, Nanda Devi-Uttarakhand, Tara Pahar-Himachal Pradesh, Kanchenjunga-Sikkim.
37. (a)



Introduction

Soil is the mixture of rock debris and organic materials which develop on the earth's surface. Soil forms different layers of particles of different sizes. A vertical section that shows different layers of soil is called a Soil Profile. Each layer is called a Horizon.



Basic Facts

- Soil formation process is called **pedogenesis** and the scientific study of soils is known as **pedology**.
- Under the soil layer are gravels of parent rock and concretions formed by the accumulation of leached materials collectively known as **sub-soil**.
- There are two classes of minerals abundant in soils. **Primary minerals** and **Secondary minerals**.
- **Primary minerals** are mostly silicate minerals-compounds of silicon and oxygen, with varying proportions of aluminium, calcium, sodium, iron and magnesium. But they play no important role in sustaining plant or animal life.
- **Secondary minerals** - clay minerals, mineral oxides, etc. are essential for soil development and for soil fertility.
- The nature of the clay minerals in a soil determines its base status. If the clay minerals can hold abundant base ions, the soil is of high base status and generally will be highly fertile, and vice versa.

Soil Forming Processes – The transformation of rocks into soil is called soil formation.

The fundamental & specific process of soil formation are as follows –

1. **Weathering** - the process that results in the break down and chemical changes of the parent rocks in situ.
2. **Humification** - It is the process of transformation of raw organic matter into humus. It helps in formation of surface layer.
3. **Translocation** - It refers to the material movement within the solid body. Two process of translocation are eluviation and illuviation.
4. **Eluviation** - the downward transport of fine particles, particularly the clays and colloids, from the uppermost part of the soil.
5. **Illuviation** - the accumulation of materials that are brought downward, in the underlying zone.
6. **Calcification** – occurs in dry regions where due to lack of excessive moisture, the soil accumulates considerable amount of soluble materials of calcium carbonate and magnesium in some part of soil profile.
7. **Decalcification** - the leaching of calcium carbonate from the entire soil. It generally happens in moist climates.
8. **Salinisation or alkalinisation** takes place in the case of extreme evaporation where soluble salts or sodium salts accumulate on the soil surface as a result of the capillary action of water from a water table.
9. **Desilication** - In this process, silica, together with many bases, is removed from a soil profile by intense weathering and leaching. It leads to development of ferralsol soils. Desilicified soils are known as **feralsols**.

Major Soil Groups of India

(i) Alluvial Soil

- It covers about 15 lakh sq km (45.6%) of the total land area of the country.
- It contributes largest share of our agricultural wealth.
- It is yet immature and has weak profiles.
- It is rich in potash, phosphoric acid, lime and organic matter but deficient in nitrogen and humus contents. It is one of the most fertile in the world.
- It is found in river plains and also along with the coastal regions of India. Its extent is from Punjab in west to west Bengal & Assam in the east.
- It is divided mainly into younger **Khadar** and older **Bhangar** soils.
- Khadar soils are found in the low areas of valley bottom which are flooded almost every year.
- The Bhangar is found on the higher areas of the flood plains. **A few metres below the surface of the bhangar are beds of lime nodules known as Kankar.**
- Along the Shiwalik foothills, there are alluvial fans having coarse, often pebbly soils. This zone is called **Bhabar**.
- To the south of the bhabar is a long narrow strip of swampy lowland with silty soils known as **Terai**. The terai soils are rich in nitrogen and organic matter but are deficient in phosphate.

Difference between Khadar and Bhangar

	Khadar	Bhangar
(i)	khadar is the newer and younger deposits of the flood plains	Bhangar is the older alluvium. It forms the largest part of the Northern plains
(ii)	This type of alluvial soil found in the lower levels in the plains near the rivers.	Bhangar found in higher up in the plains at river terraces away from rivers
(iii)	It is loamy and porous soil.	It is clayey and non-porous soil
(iv)	It is more fertile than Bhangar as new layers are deposited year after year during monsoonal floods	It is less fertile than khadar as it is not renewed.
(v)	It is pale brown, sandy clays & loams, more dry & leached less calcareous & carbonaceous	It is generally dark coloured & of a more clayey composition.

(ii) Black soils

- They are also known as '**Regur**' and '**Black cotton**' soils.
- They are spread over 5.46 Lakh sq km (16.6%) of the total geographical area of the country.
- The black colour of these soils has been attributed to the presence of a small proportion of titaniferous magnetite or even to iron and black constituents of the parent rock.
- The black soil is very retentive of moisture. It swells greatly and become sticky when wet in rainy season, and shrink when dried in dry season and develop wide cracks. Thus, there occurs a kind of 'self ploughing'.
- They contain lime, iron, magnesium, alumina and potash but lack phosphorous, nitrogen, organic matter and humus.
- Generally, the black soils of uplands are of low fertility but they are darker, deeper and richer in the valleys.

(iii) Red and Yellow Soils

- Most of the red soils have come into existence due to weathering of ancient crystalline and metamorphic rocks.
- The soil develops a reddish colour due to a wide diffusion of iron in crystalline and metamorphic rocks. It looks yellow when it occurs in a hydrated form.
- It covers about 3.5 lakh sq km (10.6%) of the total geographical area of the country.
- The red soils are poor in lime, magnesia, phosphates nitrogen and humus, but are fairly rich in potash.
- They are lying on the periphery of the peninsular plateau.
- They are not retentive of moisture.

(iv) Laterite soils

- Derives from Latin word 'Laterite' meaning 'brick'.
- The laterite soils are formed in the areas of high temperature and high rainfall with alternate wet and dry periods.
- These are the result of intense leaching. With rain, lime and silica are leached away and soils rich in iron oxide and aluminium compound are left behind.
- They are poor in organic matter, nitrogen, phosphate and calcium, while iron oxide & potash are in excess

- They cover an area of 2.48 lakh sq km.
 - The laterite soils on the higher areas are generally more acidic than those in the low-lying areas.
 - They are widely used in formation of bricks for construction of house.
- (v) **Arid and semi-arid soils**
- They cover an area of 1.42 lakh sq km (4.32%) of the total area of India.
 - The desert soils consist of sand (90 to 95%) and clay (5 to 10%).
 - Nitrogen is insufficient and the phosphate content is normal.
 - Lower horizons of the soil are occupied by 'Kankar' layers because of the increasing calcium content downwards.
 - These soils are poor and contain little humus and organic matter.
- (vi) **Saline and Alkaline soils**
- They are known by different names such as reh, kallar, usar, thur, rakar, karl & chopan.
 - Saline soils contain a larger proportion of sodium, potassium and magnesium and thus, they are infertile.
- They occur in arid and semi-arid regions and in waterlogged and swampy areas.
 - They lack nitrogen and calcium.
- (vii) **Peaty and Marshy soils**
- They are found in the areas of heavy rainfall and high humidity.
 - They are rich in humus and organic content, but deficient in potash and phosphate.
 - Organic matter may go even up to 40-50%.
 - They are normally heavy and black in colour & highly acidic.
- (viii) **Forest and Mountain soils**
- These soils occupy about 2.85 lakh sq km (8.67%) of the total land area of the country.
 - Such soils are mainly found on the hills slopes covered by forests.
 - These soils are heterogeneous in nature and their character changes with parent rocks, ground-configuration and climate.
 - The forest soils are very rich in humus but are deficient in potash, phosphorus and lime.
 - The forest soils are loamy and silty on valley sides and coarse-grained in the upper slopes.

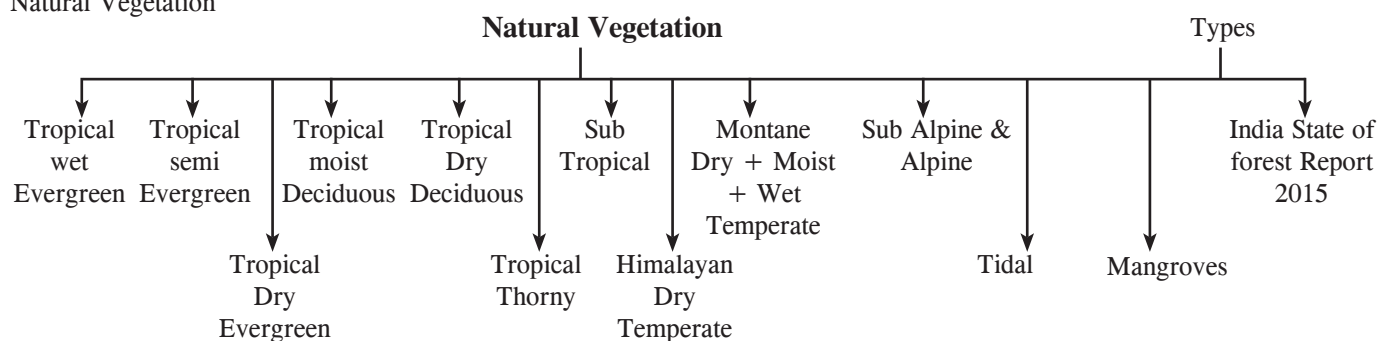
Soils of India: Types, Depth, pH Range and Regional Distribution

Type	pH range	Distribution	Predominant Crops
Alluvial Soil	6.5-8.4	Ganga and Brahmaputra river valleys; deltas of Godavari and Krishna; Plains of Uttar Pradesh, Uttarakhand, Punjab, Haryana, West Bengal and Bihar; Coastal strip of peninsular India	Rice, wheat, sugarcane, oilseeds, Jute, maize, vegetables & fruits.
Desert Soil	7.6-8.4	Rajasthan, northern Gujarat and southern Punjab	Wheat, grams, melon, bajra (with irrigation), barley, cotton, maize, pulses.
Black Soil	6.5-8.4	Maharashtra and Malwa plateaus, Kathiawar peninsula, Telengana and Rayalaseema region of Andhra and northern part of Karnataka, some parts of Tamil Nadu.	Cotton, millets, tobacco, sugarcane (Millets include jowar, bajra and ragi), castor, sunflower.
Red & yellow Soil	Below 5.5-7.5	Scattered in peninsular India, Eastern parts of Deccan plateau, southern states of Kerala, Tamil Nadu and Karnataka, and Chhota Nagpur plateau (Jharkhand), Semi-arid tract of Rajasthan.	Millets, wheat, tobacco, rice, cotton, sugarcane, pulses, groundnut, potatoes, fruits, Oilseeds.
Laterite Soil	Below 5.5	Assam hills, Tamil Nadu, Madhya Pradesh, Kerala, Karnataka, and Eastern Ghat region of Orissa.	Coffee, rubber, cashewnut, tapioca
Mountain Soil (It includes peat, forest and hill soils)	5.0-6.5	Coniferous forest belt of Jammu and Kashmir, Himachal Pradesh, Uttarakhand and Sikkim.	Fruits, tea, coffee, wheat, maize, barley.
Saline Soils		Western Gujarat, deltas of Eastern Coast & in Sunderban areas of West Bengal	
Peaty & Marshy Soils		Northern part of Bihar, Southern part of Uttaranchal & the coastal areas of West Bengal, Orissa & Tamil Nadu	

Rainfall

- The average annual rainfall of India is about 118 cm. There are large spatial and temporal variations.
- Of the country's total rainfall, about 75 per cent is received in the monsoon months from June to September, 13 per cent comes in the post-monsoon season, 10 per cent in the pre-monsoon season and the remaining 2 per cent in the winter season.
- In January and February, north-west India gets rainfall from the western disturbances.
- Coromandel coast receives rainfall by the north-east monsoons in the winter season.
- The highest record, is 103.6 cm in 24 hours at Cherrapunji in Meghalaya.

Natural Vegetation



Natural Vegetation refers to a plant community that has been left undisturbed over a long time, so as to allow its individual species to adjust themselves to climate & soil conditions as fully as possible.

- The geographical factors which influence natural vegetation include climate, soil and topography.
- Areas receiving 200 cm or more rainfall per annum have **evergreen rain forests**.
- Monsoon deciduous** forests dominate in areas which receive rainfall between 100 and 200 cm.
- In areas having 50 to 100 cm rainfall, there are drier deciduous or **tropical savanna** grading into open thorny scrub.
- The areas with less than 50 cm rainfall have only dry thorny scrub and low open bush merging into **semi-desert**.
- As the temperature falls with altitude in the Himalayan region, the vegetal cover changes from tropical to sub-tropical, temperate and finally alpine.
- The forests are very unevenly distributed. They are more scarce in Gangetic area.

India State of Forest Report 2015

The word forest is derived from Latin 'fores' meaning outside, the reference being to a fence & it must have included all uncultivated & unhabited land.

- Total forest and tree cover is 79.42 million hectare, which is 24.16% of the total geographical area.

- India's forest and tree cover has increased by 5, 081 sq km. While the total forest cover of the country has increased by 3, 775 sq km, the tree cover has gone up by 1, 306 sq km.
- Open Forest area has increased by 4, 744 sq km, which is 9.14% of the geographical area.
- Very Dense Forest area has increased by 2, 404 sq kms, which is 2.61% of the geographical area.
- About 40% forest cover is in 9 big patches of 10, 000 sq km and more.
- Increase in total forest cover also includes an increase in Mangrove cover.
- Maximum increase in forest cover has been observed in – Tamil Nadu (2, 501 sq km), followed by Kerala (1, 317 sq km) and Jammu & Kashmir (450 sq km).
- Madhya Pradesh has the largest forest cover of 77, 462 sq km in the country, followed by Arunachal Pradesh, with a forest cover of 67, 248 sq km & Chhattisgarh with 55, 586 sq km.
- Mizoram (88.93 %) has the highest forest cover in percentage terms, followed by Lakshadweep with 84.56 %.
- 7 States/UTs – Mizoram, Lakshadweep, Andaman & Nicobar Island, Arunachal Pradesh, Nagaland, Meghalaya and Manipur – have more than 75% forest cover.
- 8 states – Tripura, Goa, Sikkim, Kerala, Uttarakhand, Dadra & Nagar Haveli, Chhattisgarh and Assam – have forest cover between 33% to 75%.
- Total carbon stock in country's forest is 7, 044 million tones, an increase of 103 million tonnes.

India's Natural Vegetation

No.	Vegetation type	Distribution	Characteristic vegetation
1.	Tropical Wet Evergreen Vegetation	In areas where the annual rainfall is over 250 cm and the average annual humidity exceeds 77 per cent (North-East states, western portions of the Western Ghats and Andaman and Nicobar).	Lofty, very dense, multilayered forest with mesosphytic evergreens, e.g., bamboos, ebony, rosewood, champa, toon, jamun, mesa, white cedar, mahogany.
2.	Tropical Semi-Evergreen Vegetation	Found in regions where the annual rainfall is between 200-250 cm and the humidity approaches 75% (Upper Assam, lower eastern Himalayas, Orissa, the Andaman and Nicobar islands & Western Coast).	Evergreen trees mixed with deciduous ones, less dense but more gregarious e.g., aini, semul, gutel, kadam, irul, thorny bamboo, rosewood, Kusum, hollock bonsum, white cedar, Indian chestnut, champa, mango, bamboos etc.
3.	Tropical Dry Evergreen Vegetation	In areas where the mean rainfall is about 100 cm, mostly from north-east monsoon, the mean annual humidity is 75%, and the mean annual temperature is 28°C (Tamil Nadu coast).	Short statured trees, with complete canopy, coriaceous leaved trees of short boles, no canopy layer differentiation. e.g., khirmi, jamun, kokko, toddy palm, tarrina, ritha, neem etc.

4.	Tropical Moist Deciduous Vegetation (Also known as monsoon forest)	In areas having moderate rainfall of 100-200 cm, a mean annual temperature of about 27°C, and an average relative humidity of 60 to 75% (Western Ghats, Odisha, eastern coastal plains, and Himalayan foothills).	Trees which shed their leaves during spring & early summer season, very useful forests because they yield valuable timber & several other forest products, heavily buttressed trees, shrubby undergrowth with patches of bamboos, climbers & canes. e.g. sal, teak, sandalwood, siris, palas, mahua, sisam, amla etc.
5.	Tropical Dry Deciduous Vegetation	In areas where rainfall is less than 150 cm and dry period is relatively long (eastern Rajasthan, Kathiawar, rain shadow area of the Deccan plateau, Central India).	Trees which grow relatively shorter than the tropical moist deciduous trees closed & uneven canopy. Examples include teak, sal, bijasal, palas, khair, tendu, rosewood, anjar etc.
6.	Tropical Thorny Vegetation	Mostly prevalent in areas having very low rainfall, i.e., 50 cm to 75 cm, the annual mean temperature between 25°C and 30°C, and the annual humidity less than 50% (Kutchh, Saurashtra, Punjab, Haryana, Rajasthan, Upper Ganga plains and the Deccan plateau).	Open stunted forest breaking down into xerophytic bush, e.g., babul, accasia, senegal, ber, khair, ak, neem, Cactii etc.
7.	Sub-Tropical Vegetation	Found at 1,000-2000 m altitude in eastern and western Himalayas, and drier areas of Kashmir.	Luxurious forests of evergreen species. Eg. – Oak, chestnut, ash, birch, pine, sal, chir pine, oak, wild olives.
8.	Himalayan Dry Temperate Vegetation	In the inner dry ranges of the Himalayas where precipitation is below 10 cm. (Ladakh Lahul, Chamba, Kinnaur, Garhwal, Sikkim).	Predominantly coniferous forests with xerophytic shrubs, e.g., chilgoza, deodar, oak, maple, ash, celtis, etc.
9.	Himalayan or Montane/ Mountain Moist Temperate Vegetation	In the temperate eastern and western Himalayas, between 1,500 m and 3,300 m (Jammu and Kashmir, Himachal Pradesh, Uttarakhand, Sikkim and Darjeeling).	Broad leaved evergreens mixed with dominant coniferous species, generally 30 to 50 m high, open forest with shrubby undergrowth. e.g., deodar, spruce, maple, ash, pine, fir, beach, etc.
10.	Himalayan or Montane/ Mountain Wet Temperate Vegetation	Found between 1,800 m and 3,000 m of altitude (eastern Himalayas, Nilgiris, the Annamalai, Palni hills of South India, Sikkim, Nagaland, hills of West Bengal, Assam, Arunachal Pradesh) where average rainfall is between 150 cm and 300 cm. & humidity is over 80%.	Evergreen forests, short-boleds branchy trees, dense and rounded leaves. Examples include oak, magnolia, chilauni, birch, plum, deodar, laurel and maple.
11.	Sub-Alpine and Alpine Vegetation	Above 2,900 m of altitude in the eastern Himalayas and above 3,500 m of altitude in the western Himalayas and extends up to the snowline.	Dwarf shrubs which degenerate into a low evergreen scrubs and into xerophytic vegetation; examples include fir, juniper, pine birch and <i>Rhododendron</i> , <i>spruce</i> .
12.	Littoral or Tidal or Delta or Swamp Vegetation	In and around the tidal creeks and along the deltas of the rivers Ganga, Mahanadi, Krishna and Godavari.	Evergreen trees having profuse growth and stiltlike roots, e.g., sundari in the great Sunderban delta, <i>Rhizophora</i> , nipa fruiticans (a type of palm), palms, keora, amur etc.

Mangroves

- Mangroves are very specialized forest ecosystem of tropical and subtropical region of the world bordering sheltered sea-coasts.

Mangroves of India	
State	Mangrove
East coast	
West Bengal	Sunderbans
Orissa	Mahanadi, Bhitarkanika
Andhra Pradesh	Godavari, Krishna
Tamil Nadu	Pichavaram, Muthupet, Point Calimere
West coast	

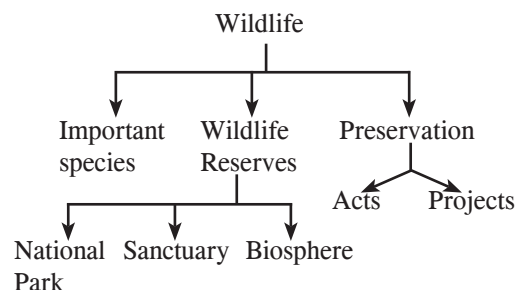
Gujarat	Gulf of Kutch, Gulf of Khambat
Goa	Goa
Karnataka	Coondapur
Maharashtra	Achra/Ratnagiri
Kerala	Vembanad
Other mangroves	
Andaman & Nicobar Islands	Andaman islands & Nicobar islands

- They occur all along the Indian coastline in the sheltered estuaries, tidal creeks, backwaters, salt marshes and mudflats, covering a total area of 0.67 m ha.
- Mangroves are dominated by salt tolerant halophytic plants of diverse structure, and are invaluable marine nurseries for a large variety of fish and other marine fauna.

- Mangroves have a dense network of aerial roots which help to aerate the root system and anchor the tree.
- In world's total mangrove vegetation, India's share stands at 3%.

State/UT	Forest cover as Percentage of geographical area (%)
Andhra Pradesh	16.86
Arunachal Pradesh	80.50
Assam	35.28
Bihar	37.27
Chhattisgarh	41.18
Delhi	11.88
Goa	59.94
Gujarat	7.46
Haryana	3.64
Himachal Pradesh	26.37
Jammu & Kashmir	10.14
Jharkhand	28.82
Karnatka	18.87
Kerala	44.52
Madhya Pradesh	25.21
Lakshadeep	84.96
Puduchery	10.43
State/UT	Forest Cover as Percentage of Geographical Area (%)
Maharashtra	16.46
Manipur	76.54
Meghalya	77.02
Mizoram	90.68
Nagaland	80.33
Orissa	31.41
Punjab	3.50
Rajasthan	4.70
Sikkim	47.34
Tamil Nadu	18.16
Tripura	76.04
Uttar Pradesh	5.95
West Bengal	14.64
Andman & Nicobar	81.51
Chandigarh	14.72
Dadra & Nagar Haveli	42.97
Daman & Diu	5.49
Uttarakhand	45.80

WILDLIFE



- Wildlife comprises animals, birds, and insects living in forests.
- With large regional variations in physiography, climate and edaphic types. Indian forests offer a wide range of habitat types, which is responsible for a large variety of wild life in India.
- **Elephant is the largest Indian mammal**, which only a few centuries ago, was found in large numbers in vast forest tracts of India.
- **The one-horned rhinoceros, India's second largest mammal** was once found throughout the Indo-Gangetic Plain as far west as Rajasthan. The number of this mammal has drastically decreased and now there are less than 1,500 rhinoceros in India, confined to the restricted locations in Assam and West Bengal.
- Rhinoceros are protected in Kaziranga and Manas sanctuaries of Assam and the Jaldapara sanctuary of West Bengal.
- The wild buffalo is found in Assam and in Bastar district of Chhattisgarh.
- The gaur or the Indian bison is one of the largest existing bovine and is found in the forests of Central India.
- There are about 3,000 tigers in India mainly found in the forests of eastern Himalayan foothills and in parts of the peninsular India.
- The number of Cheetahs had fallen to less than two hundred until successful breeding programme in the Gir sanctuary in Gujarat resulted in some recovery.
- The arboreal clouded leopard is found in northern Assam while the Black Panther is widely distributed predator.
- Brown, Black and Sloth Bear are found at high altitudes in the northwestern and central Himalayas.
- Yak, the ox of snows is largely found in Ladakh and is tamed to be used as a draught animal.
- Stag or barasingha is found in Assam and Madhya Pradesh.
- The Mungjac or barking deer are found extensively in the lower wooded slopes of the Himalayas and in the forests of southern India.
- The kastura or the musk deer, much sought after for its musk pod, live in the birch woods in the higher forests of the Himalayas.
- India's first National Park is Jim Corbett Park in Uttarakhand, established in 1936

- Thamin is a pretty deer found in Manipur.
- India is extremely rich in bird life. There are about 2,000 species of birds in India.
- Although most of the bird has their origin in India, a number of them have their source in other areas. Some birds such as ducks, cranes, swallows, and flycatchers migrate from central Asia to the wetlands of Bharatpur every winter. Recently, some migratory birds have been seen near Mathura.

National Park: A reserved area meant for preserving its natural vegetation, wildlife and natural beauty.

Sanctuary: A reserved area meant for preservation and development of endangered species.

Biosphere: Multipurpose protected areas to preserve genetic diversity in representative ecosystems.

As of July 2015, there were 105 National Parks.

Number of wildlife Sanctuaries – 531

Number of Biosphere Reserves – 18.

Preservation of Wildlife

- The fast dwindling forest cover in India has adversely affected wildlife in the country.
- The number of several species has been drastically reduced, some are endangered species, and the others are on the verge of extinction while some of them have already disappeared.
- **Indian Board for Wildlife was constituted in 1952.** The main purpose of the board was to advise the Government on the means of conservation and protection of wildlife, construction of national parks, sanctuaries and zoological gardens as well as promoting public awareness regarding conservation of wildlife.
- The Wildlife (protection) Act, 1972 is a comprehensive law which gives firm status to the national parks and sanctuaries and other

Distinction between National Park, Sanctuary and Biosphere Reserve

National Park

- Habitat for particular wild animal species
- The general size range is 0.04 to 3162 sq km.
- Boundaries fixed by legislation
- Except the buffer zone, no biotic interference
- Tourism permissible
- Research and Scientific management lacking
- So far no gene pool and conservation

Sanctuary

- Generally species-oriented such as citrus, pitcher plant, etc.
- The general size range is 0.61 to 7818 sq km.
- Boundaries are not sacrosanct
- Limited biotic interference
- Tourism permissible
- Research and Scientific management lacking
- So far no gene pool and conservation given

Biosphere Reserve

- Ecosystem-oriented i.e. All forms of life
- The general size range is over 5670 sq km
- Boundaries fixed by legislation
- Except the buffer zone, no biotic interference
- Tourism normally not permissible
- Managed attention

- Endangered species of plants and animals have been brought under the purview of this act. **Project Tiger**, one of the premier conservation efforts in the country was **launched in April, 1972**. It is a centrally financed scheme under which 40 tiger reserves have been set up.
- As a result of the tiger project the tiger population is now well over 3000. A tiger crisis cell has also been formed in the Ministry of Environment and Forest.
- The National Wildlife Action Plan adopted in 1983 provides the framework and programme for conservation of wildlife.
- A Central Zoo Authority has been set up for the proper management of zoological parks in the country. It coordinates the activities of over 200 zoos and also supervises the exchange of animals.

Endangered Species Projects

Project Tiger: The sort of India has taken a pioneering initiative for conserving tiger by launching the 'Project Tiger' in 1973. India is home to 70% of tigers in the world. **In 2014, there were 2,226 tigers. Statewise, Karnataka has the highest**

number of tigers (406) followed by 340 in Uttarakhand, 308 in Madhya Pradesh, 229 in Tamil Nadu. The project tiger aims to foster an exclusive tiger agenda in the core area of tiger reserves, which is an inclusive people oriented agenda in the buffer. The largest tiger reserve is the Nagarjunsagar-Srisailem tiger reserve of Andhra Pradesh which covers the area of 3538 km².

Project Elephant: Project elephant, a centrally sponsored scheme was launched in February 1992 to provide surgical and technical support to major elephant bearing states in the country for protection of elephants, their habitats and corridor. The project is being implemented in 13 states/UT's viz. Andhra Pradesh, Arunachal Pradesh, Assam, Jharkhand, Karnataka, Kerala, Meghalaya, Nagaland, Orissa, Tamil Nadu, Uttar Pradesh and West Bengal. There are 28 notified elephant reserves in India covering approximately 60,000 sq km area.

Project Snow Leopard: This project was launched to safeguard and conserve India's unique natural habitats of high altitude wildlife population and their habitats by promoting conservation through participatory policies and actions. This project was drafted by Ministry of Environment and Forests,

Govt of India. It was launched in January. There are nearly 750 snow leopards in the country.

Memorandum of Understanding (MOU) or Siberian Crane: This memorandum came into effect 1st July, 1993 and was amended in January, 1999. This memorandum focuses on conserving the Siberian crane as one of these rarest crane species. India had signed the MOU on 13th Dec. 1998. Siberian crane are migratory visitors to India in winter season.

MOU for Marine Turtle: Major threats to marine turtle include unexceptionable exploitation, destruction of resting and feeding habitats and incidental mortality is fishing operations.

The objectives of this memorandum are conservation and management of Marine turtles and their habitats. India had signed this memorandum on 20 February, 2007.

MOU for Dugong: The dugong is a seagrass dependent marine's mammal of tropical and subtropical coastal water. The dugong are vulnerable to human related influences due to their life history. Dugong is commonly known as sea cow. In India, these are found in Indian waters. From Gujarat to Andaman and Nicobar islands India had signed MOU for Dugong on 28 May, 2008.

Major Wildlife Reserves

Jammu and Kashmir

National Parks

Dachigam

Hemis

Kishtwar

Sanctuaries

Baltal

Changthang

Gulmarg

Hirpora

Hokarsar

Kanji

Karakoram

Lachipora

Limber

Nandini

Overa

Overa-Aru

Ramnagar

Surinsar-Mansar

Tongri

Himachal Pradesh

National Parks

Great Himalayan

Pin Valley

Sanctuaries

Bandli

Chail

Churdhar

Daranghati

Darlaghat

Gamgul Siahbehi

Gobind Sagar and Naina Devi

Kais

Kalatop & Khajjiar

Kanawar

Khokhan

Kugti

Lippa Asrang

Majathal

Manali

Naina Devi

Nargu

Pong Dam Sanctuary

Raksham Chitkul

Renuka

Rupi Bhabha

Sechu Tuan Nala

Shikari Devi

Shilli

Simbalbara

Talra

Tirthan

Tundah

Punjab

Sanctuaries

Abohar

Hariker Lake

Bir Gurdialpura

Bir Bunnerheri

Bir Motibagh

Haryana

Sultanpur

Sanctuaries

Bir Shikargarh

Chautala

Delhi

Indira Priyadarshini

Chandigarh

Sukhna

Rajasthan

National Parks

Desert

Keoladeo

Ranthambore

Sariska

Sanctuaries

Bandh Bartha

Bassi

Bhensrodgarh

Darah

Jaisamand

Jamwa Ramgarh

Jawahar Sagar

Keladevi

Kumbhalgarh

Mount Abu

Nahargarh

National Chambal

National Garhial

Phulwari

Ramgarh Bundi

Sawai Mansingh

Shergarh

Sita Mata

Sunda Mata

Todgarh Raoli

TalChapper

Van Vihar

Uttar Pradesh

Dudwa

Sanctuaries

Chandra Prabha

Chilla

Hastinapur

Kaimur

Kateraniaghat

Kishanpur

Maahavir Swamy

National Chambal

Nawabganj

Ranipur

Samaspur

Sohagabarwa

Uttaranchal

National Parks

Corbett

Gangotri

Govind

Nanda Devi

Rajaji

Valley of flowers

Sanctuaries

Askot

Binsar

Govind Pashu

Kedarnath

Sonanadi

Madhya Pradesh

National Parks

Bandhavgarh

Dinosaur Fossils

Ghati	<u>National Parks</u>	<u>National Parks</u>	Kutch Desert
Kanha	Bhitarkanika	Gugamal	Marine (Gulf of Kutch)
Madhav	Simlipal	Nawegaon	Nalsarovar
Panna	<u>Sanctuaries</u>	Sanjay Gandhi	Narayan (Chenkars)
Pench	Badrama	Tadoba	Sarovar
Sanjay	Baisipalli	<u>Sanctuaries</u>	Paniya
Satpura	Balukhand Konark	Andhari	Purna
Van Vihar	Bhitarkanika	Aner Dam	Rampara
<u>Sanctuaries</u>	Chandaka Dampara	Bhimashankar	Ratanmahal
Achanakmar	Chilka Lake (Nalaban)	Bor	Shoolpaneshwar
Badankhoh	Debrigarh	Chandoli	Wild Ass
Bagdara	Hadgarh	Chaprala	
Barnawapara	Karlapat	Dhakna Kolkaz	Goa, Daman and Diu
Bhairamgarh	Khalasuni	Gandhari	Bhagwan Mahavir
Bori	Kotgarh	Gautala Autramghat	<u>Sanctuaries</u>
Gandhi Sagar	Kuldiha	Great Indian Bustard	Bhagwan Mahavir
Ghatigaon	Lakhari Valley	Jaikwadi	Cotigao
Gomardah	Satkosia Gorge	Kalsubai Harishchandra	
Karera	Simlipal	Katepurna	West Bengal
Ken Gharial	Sunabeda	Koyna	<u>National Parks</u>
Kuno-Palpur	Ushakothi	Malvan Marine	Neora Valley
Kheoni		Melghat	Singalila
Narsingarh	Bihar	Nagzira	Sundarban
National Chambal	Valmiki	Nandur Madmeshwar	<u>Sanctuaries</u>
Noradehi	<u>Sanctuaries</u>	Painganga	Buxa
Pachmarhi	Bhimbandh	Phansad	Bibnutibhushan
Palpur	Chandra Prabha	Radhangiri	Betuadahary
Pamed Wild Buffalo	Kabar	Sagareswar	Ballavpur
Panpatha	Kaimur	Tansa	Chapramari
Pench	Nakti Dam	Wainganga	Gorumara
Fensatallite	Rajgir	Yawal	Halliday Island
Ratapani	Valmikinagar		Jaldapara
Sailana Florican	Udaipur	Gujarat	Jorpokhri
Sanjay (Dubri)		Vansda	Lothian Island
Sardarpur Florican	Jharkhand	Gir	Mahananda
Samarsot	Dalma	Marine (Gulf of Kutch)	Narendrapur
Singhori (Sindhari)	Gautam Buddha	Velavadar/Blackbuck	Pamadhan
Sitanadi	Hazaribagh		Ramnabagan
Sone Gharial	Koderma	<u>Sanctuaries</u>	Raiganj
Tamor Pingla	Lawalong	Balaram-Ambaji	Sajnakhali
Udanti Wild Buffalo	Mahuadanr	Barda	Senchal
Chhattisgarh	Palamau	Dhumkhal	
Indravati	Parasnath	Gir	Sikkim
Kanger Ghati	Topchanchi	Jambughoda	<u>National Parks</u>
Orissa	Maharashtra	Jessore	Khangchendzonga
		Khijadiya	

Sanctuaries

Fambong Lho
Kyongasia Alpine
Maenam
Shingba Rhododendron

Manipur

Keibul Lamjao
Siroi

Sanctuary

Yagoupokpi Lokchao

Meghalaya

Balphakram
Nokrek

Sanctuary

Bhagmara
Nongkhyllem
Siju

Arunachal Pradesh

Mouling
Namdapha

Sanctuaries

D'Ering Memorial
Dibang Valley
Eagle nest
Itanagar
Kamlang
Kane
Mehao
Pakhui
Sessa Orchid

Mizoram

Murlen
Phawngpui

Sanctuaries

Dampa
Khawnglung
Ngengpui

Nagaland

Nitangki
Pulebatze
Fakim

Tripura

Charilam
Sepahijala
Rishna
Bison
Clouded Leopard

Assam

Kaziranga
Nambiar
Dibru-Saikhowa

Sanctuaries

Dipor Beel
Garampani
Laokhowa

Manas

Nameri

Orang

Pabha

Pobitara

Sonai Rupai

Andhra Pradesh**National Parks**

Shri Venkataswara

Sanctuaries

Coringa
Eturnagaram
Gundlabrahmeswaram
Koundinya
Kawai

Kinnerasani

Kolleru

Krishna

Lanjamadugu

Manjira

Nagarjunasagar-Srisailem

Neelapattu

Pakhal

Papikonda

Pocharam

Pranahita

Pulicat

Rollapadu

Srilanka Malleswara

Siwaram

Karnataka**National Parks**

Anshi
Bandipur
Bannerghatta
Kudremukh
Nagarhole

Sanctuaries

Arabithittu
Bhadra
Rangaswamy- Temple
Biligiri
Bramhagiri
Cauvery
Dandeli

Ghataprabha

Melkote Temple

Mukambika

Nugu

Pushpagiri

Ranganthittu

Ranebennur

Sharavathi Valley

Shettihally

Someswara

Talakaveri

Kerala

Eravikulam
Periyar
Silent Valley

Sanctuaries

Aralam
Chimony
Chinnar
Idukki
Neyyar
Parambikulam
Peechi Vazhani
Peppara
Periyar
Shendurney
Thattekkad Bird
Wynad

Tamil Nadu

Guindy
Indira Gandhi
or Anamalai
Sanctuaries
Kalakad
Karikili
Mudumalai
Mukurthi
Mundanthurai
Point Cali mere
Pulicat
Srivilliputhur
Vedantangal
Vettangudi

The Andaman Islands**National Parks**

Campbell Bay
Galathea
Wandur
Mount Harriet Island
Saddle Peak
Sanctuaries
Barren Island
Battimalve Island
Benett Island
Bluff Island
Bondoville Island
Buchaan Island
Cinque Island
Crocodile (Lohabrack)
Defence Island

East (Inglis) Island
East Island
Flat Island
Interview Island
James Island
Kyd Island
Landfall Island
Narcondum Island
North Reef Island
Paget Island
Pitman Island
Point Island
Ranger Island

Fleef Island	Sir Hugh Rose Island	Swamp Island	Tillonchang Island
Roper Island	South Brother Island	Table (Delgarno) Island	West Island
Ross Island	South Reef Island	Table (Excelsior) Island	
Sandy Island	South Sentinel Island	Talabaicha Island	
Shearme Island	Spike Island	Temple Island	

Biosphere Reserves of India

S. No.	Name	Type	Key Funna	Area (In Km ²)
1.	Nilgiri Biosphere Δ Reserve, Tamilnadu, Kerala Karnataka	Western Ghat	Nilgiri Tahr, Lion talked macaque	5520
2.	Nanda Devi, Utrakhand Δ	Western Himalyas Coast	—	5860
3.	Gulf of Mannar, Tamilnadu Δ	East Himalayas	Dugong	10500
4.	Nokrek, Meghalya Δ	Gangatic Delta	Red Panda	820
5.	Sunderbans, West Bengal Δ	East Himalayas	Royal Bengal Tiger	9630
6.	Manas, Assam	Deccan Peniusula	Red Panda	2837
7.	Simlipal, Odissa Δ		Royal Bengal Tiger, Ganv, Wild elephant	4374
8.	Dihang - dibang, Arunachal Pradesh	Eastern Himalayas Semi-Arid Maikals Hills	—	5112
9.	Panchmarhi Biosphere Δ	Desert	Flying squirrel	4981.72
10.	Amerkantak, Madhya Padesh Chhatisgarh Δ	Western Himalayas	Giant Squirrel	3835
11.	Rann of Kutch, Gujarat		Indian wild ass	12454
12.	Pin valley national park (Cold Desert), Himachal Pradesh	East Himalayas Western Glats	Show Leopased	7770
13.	Khangchendzonga, Sikkim		Red Panda	2620
14.	Agasthyamalai Biosphere resurve, Kerala, Tamilnadu	Islands	Nilgiri Tahr Elephants	1828
15.	Nicobar Biosphere Reserve, A & Nicobar, Islands Δ	East Himalayas Eastern Ghats	Saltwater Crocodile	885
16.	Dibru-Saikhowa, Assam	Catchment area of Ken River	Golden Langur	765
17.	Seshachalam Hills, Andhra Pradesh			
18.	Panna, Madhya Pradesh		Tiger, Chital, Chinkara, Sembhar and Sloth bear	543

Δ Biosphere Reserves listed in UNESCO World Network of Biosphere Reserves.

Specialities of National Park and Sanctuary

- Gir National Park in Gujarat is the only existent habitat for the nearly extinct Asiatic Lions in India.
- The Kaziranga Sanctuary in Assam is a major habitat of the endangered Rhinoceros.
- Periyar in Kerala is famous for the the wild Elephants
- Dachigam National Park is natural habitat of Kashmiri Stag.
- Nandankanan national park in Orissa serves as the natural habitat of white tigers.
- The Padmaja Naidu Himalayan Zoological Park in Darjeeling (West Bengal) has undertaken Project Red Panda with a view to conserving this species.
- The entire Little Rann of Kutch in Gujarat is known as the “Indian Wild Ass Sanctuary”.
- Bhitarkanika in Orissa is a hot-spot of biodiversity. It’s home to largest population of giant salt water crocodile in India.
- Gahirmatha, the only marine wildlife sanctuary of Orissa, is the most suitable natural place for Olive Ridley sea turtles’ for mating and nesting.
- The National Chambal Sanctuary is famous for the rare gangetic dolphin and also for magar (crocodile) and gharial (alligator).
- Ranganathittu Bird Sanctuary in Karnataka is a favourite abode for birds

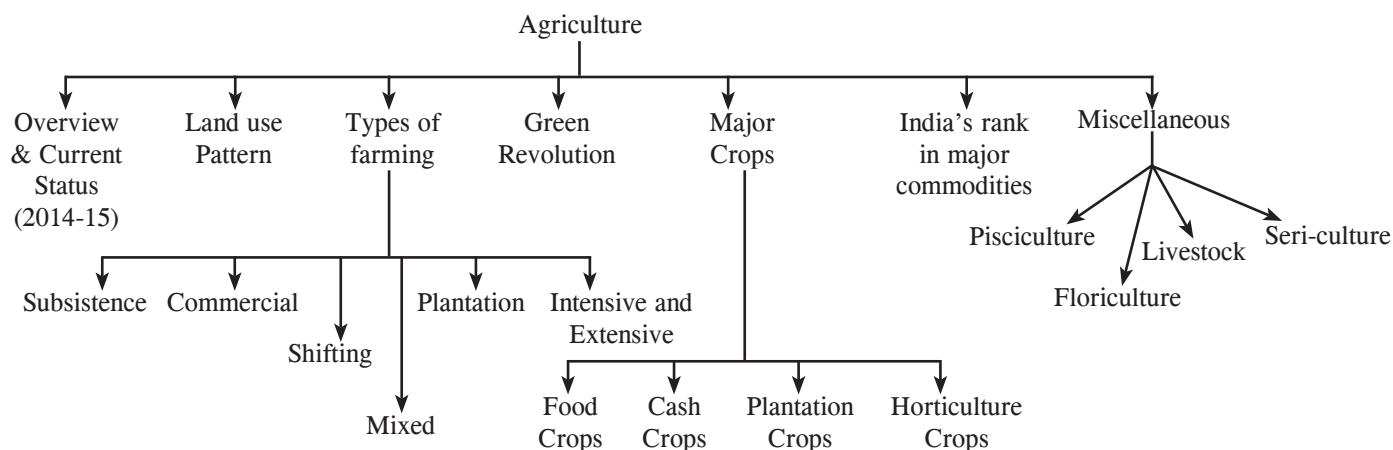
National Parks by the largest area			Wildlife Sanctuaries by the largest area		
National Parks	State	Area (sq km)	Wildlife Sanctuary	State	Area (sq km)
1. Hemis	Jammu & Kashmir	4,100.00	1. Great Indian Bustard	Maharashtra	8,496.44
2. Desert	Rajasthan	3,162.00	2. Kachchh Desert	Gujarat	7,506.22
3. Namdapha	Arunachal Pradesh	1,807.82	3. Karakoram	Jammu & Kashmir	5,000.00
4. Khangchendzonga	Sikkim	1,784.00	4. Wild Ass	Gujarat	4,953.71
5. Gangotri	Uttarakhand	1,552.00	5. Dibang	Arunachal Pradesh	4,149.00

Western Ghats

The Western Ghats (or Sahyadri) is a mountain range that runs parallel to the western coast of the Indian peninsula. It is a UNESCO World Heritage Site and is one of the 8 “hottest hotspots” of biological diversity in the world. The Western Ghats stretches into 6 states – Tamil Nadu, Kerala, Karnataka, Gujarat, Goa & Maharashtra. It hosts a large list of exceptional endemic species, the International Union for Conservation (IUCN) lists 5,000 vascular plant species, 228 freshwater fish species, 179 amphibians, 157 reptiles, 508 birds & 139 mammal species, some of the very rare species include the Great Indian Hornbill, Lion-Tailed Macaque, Travancore Turtles & Nilgiri Martens.

In recent times 2 panels have been set up, under Madhav Gadgil & Kasturirangan, for protection of the Western Ghats.

Agriculture



Agriculture includes raising of Crops from the land, animal husbandry, agroforestry & pisciculture. India is an agriculturally important country. Two-thirds of its population is engaged in agricultural activities. Agriculture is a primary activity, which produces most of the food that we consume and raw material for various industries.

Overview of Agricultural Sector: Highlights

(Source: - Economic Survey 2014-15)

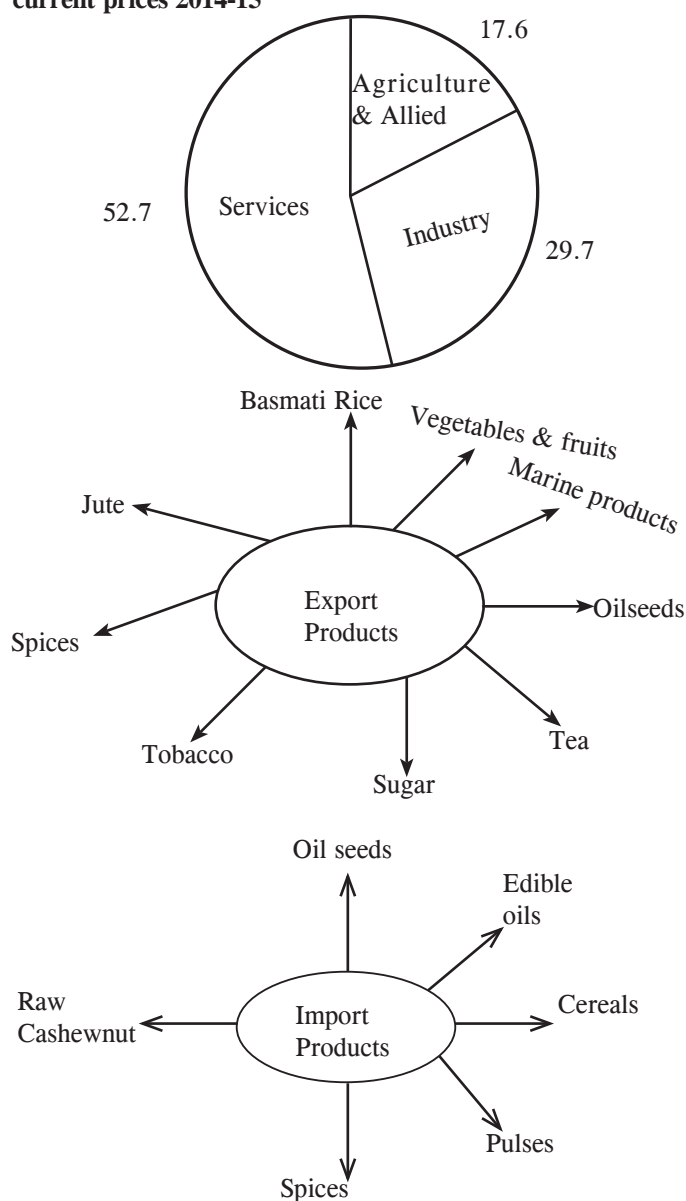
- Share of agriculture in total GDP is 18%.
- Target of 4% for agriculture & allied sectors in the 12th plan.
- Growth registered in 2014-15 is 1.1%.
- During 2014-15, banks have disbursed ₹. 8,40,643 crore (provisional) credit to the agriculture sector, against a target of ₹. 8,00,000 crore.
- India's agricultural exports were valued at US \$ 47 billion in 2013.
- India ranks first in milk production, accounting for 17% of world production.
- The National livestock Mission has been launched in 2014-15 with an approved outlay of ₹. 2,800 crore during the 12th plan.
- The revamped National Food Security Mission (NFSM) is being implemented from 2014-15, in 619 districts of 28 states. In addition to rice, wheat & Pulses, crops like coarse cereals & commercial crops (sugarcane, cotton, & jute) have been included since 2014-15.
- With effect from 2014-15, the mission for Integrated Development of Horticulture has been operationalized by bringing all ongoing schemes on Horticulture under a single umbrella.
- The following initiatives announced in Budget 2014-15 have brought the issue of sustainability & climate adaptation to the forefront:
 - (i) The Pradhan Mantri Krishi Sinchayee Yojana.
 - (ii) 'Neeranchal' a new programme to give additional impetus to watershed development in the country.
 - (iii) The National Adaptation Fund for climate change, with an initial sum of ₹. 100 crore.

(iv) A scheme to provide a “Soil Health Card” to every farmer.

- 4th Advanced Production Estimates of major crops during 2014-15 (released by the Departments of Agriculture & Cooperation) is as under: –

(A.) Foodgrains	–	252.68 MT
• Rice	–	104.80 MT
• Wheat	–	88.94 MT
• Coarse Cereals	–	41.75 MT
• Maize	–	23.67 MT
• Pulses	–	17.20 MT
• Tur	–	2.78 MT
• Gram	–	7.17 MT
(B.) Oilseeds	–	26.68 MT
• Soyabean	–	10.53 MT
• Groundnut	–	6.56 MT
• Rapeseed & Mustard	–	6.31 MT
(C.) Cotton	–	35.48 million bales
(D.) Sugarcane	–	359.33 MT

Diagram – Sectoral Share (%) in GVA at factor cost at current prices 2014-15



Land use Pattern

- Cropped area in the year under consideration is called Net Sown Area.
- The net sown area occupies the highest category (above 55% of the reporting area) in Punjab, Haryana, West Bengal, Maharashtra, Uttar Pradesh, Bihar and Kerala.
- Medium category (30-55%) in Karnataka, Gujarat, Tamil Nadu, Rajasthan, Madhya Pradesh, Andhra Pradesh, Chhattisgarh, Orissa, Goa and Assam.
- Low category (below 30%) in Himachal Pradesh, Jammu and Kashmir, Meghalaya, Nagaland, Manipur, Jharkhand, Uttarakhand, Mizoram and Arunachal Pradesh.

Area sown more than once: This area is used to grow more than one crop in a year. This accounts for 34.3% of the net sown area and 16.6% of the total reporting area of the country. This type of area comprises land with rich fertile soil and regulars water supply.

Forests: It is the area which the govt. has identified & demarcated for forest growth.

Land not available for cultivation : This class consists of two types of land (i) Land put to non-agricultural uses (ii) barren and unculturable waste.

Permanent pastures and other grazing land: It amounts to about 3.45% (i.e. 11.8 mha) of the total reporting area. The area presently under pastures is not sufficient keeping in view the large population of livestock in the country.

Land under miscellaneous tree crops and groves: It includes all cultivable land which is not included under net area sown, but is put to some agricultural use.

Culturable Wasteland: It includes all lands available for cultivation, but not cultivated for one reason or the other.

Types of Farming

India is a vast country and had various climatic patters and geographical condition, so these are different types of farming.

1. **Subsistence Farming:** In this type of farming farmer produce for his own consumption. These is no surplus left for sale. This involves cultivation of food crops like rice, wheat, pulses etc.
2. **Commercial Farming:** In this farming, food crops produced specifically for sale in the market by using improved variety of seeds and machinery. Normally it is characterised by large farms and only one crop is grown. Advance machinery, chemical fertilizers, hybrid seeds and pesticides are used. Cotton, sugarcane, tobacco, oil seeds, chillies etc. are commercial crops.
3. **Shifting Cultivation:** Shifting cultivation means the migratory subsistence farming. Under this system, a plot of land is cultivated for few years and when the crop yield declines the plot of land is changed. Dry paddy, buck wheat, maize, small millets, tobacco & sugarcane are the main crops grown under this type of agriculture. It is known by different names in different parts of the country. It is “Jhumming” in –north eastern states;

‘podu’ in Andhra Pradesh, ‘Bewar’ in M.P., ‘Kumari’ in Western Ghats.

4. **Mixed Farming:** Mixed farming is raising of crops and rearing of cattle, poultry, bee keeping, seri culture etc. on the same cattle or poultry do not need extra expenditure as they thrive on the farm wastes. Livestock animals provide substitute income when crops are not ready. This type of farming is done in densely populated areas.
5. **Plantation Farming:** Predominance of a single crop (only for sale) farming in tropical regions is called plantation farming. Important crops grown under this type of farming are cotton, tea, rubber, spices, coconuts etc. This farming outlay. Latest knowledge and modern methods of agriculture are used in this farming.
6. **Intensive Agriculture:** System of cultivation using large amount of labour and capital with application of fertilizers and insecticides is called intensive agriculture. Use of high efficiency machinery for planting, cultivating and harvesting as well as latest irrigation equipment.
7. **Extensive Agriculture:** System of crop cultivation using small amounts of labour and capital in relation to area of land being farmed. The crop yield in extensive agriculture depends primarily on the natural fertility of the soil, terrain climate and the availability of water.

Main crops season in India

Season	Ground is Prepared	Harvest	Major crops of the season
Kharif	In April, May and the seeds are sown in June on arrival of rain.	Beginning of November	Rice, maize, Jowar, bajara, cotton, jute, groundnut, pulses etc.
Rabi	By end of October or beginning of November.	April to June	Wheat, barley, peas, gram, oilseeds, tobacco etc.
Zaid	It is summer season crops sown at beginning of the season in February and March	April and May	Urad, moong, melons, cucumber

Wheat is cultivated with high yielding varieties of seeds like Lerma, Rajo, Sonora 63 and 64 (Mexican varieties), Sona 227, Kalyan Sona, Sonalika, Chhoti Lerma, Sharbati Sonora, Shera, Heera, Safed Lerma, UP 302, Saran, Champaran and C-1-7 etc.

Green Revolution

The term “Green Revolution” is applied to the period from 1967 to 1978. The green revolution started by Dr. Norman Borlaug in Mexico. Between 1947 and 1967, efforts at achieving food self sufficiency were not entirely successful. Population was growing at a much faster rate than good production. This called for drastic action to increase yield. The action came in the form of the green revolution. The term

green revolution is a general one that is applied to successful agricultural experiments in many countries. But it was most successful in India.

There were three basic elements in India regarding Green revolution

- Continued expansion of farming areas.
- Double cropping of existing farm and
- Using new and scientific treated seeds with improved genetics.

12 Components of the Green Revolution

High yield varieties (seeds), irrigation, use of fertilizers, use of insecticide and pesticide, command area development, consolidation of holding, land reform, supply of agricultural credit, rural electrification, rural roads and marketing, farm mechanisation, agricultural universities.

Impacts of Green Revolution

Impacts of Green Revolution are as follows

Positive Impact Increase in agricultural production, reduction of the import of foodgrains, capitalistic farming, industrial growth and rural employment.

Negative Impact Inter-crop imbalance, increase in regional imbalance, unemployment due to mechanisation. Increase in inter-regional migration, ecological problems and social conflict between large and small farmers.

Various Revolutions in Agriculture

Revolutions	Area
Green Revolution	Agriculture (Food Production)
Yellow Revolution	Oil seeds production (Edible oil)
White Revolution	Milk
Blue Revolution	Fish
Pink Revolution	Shrimp, food processing
Brown Revolution	Coffee/Cocoa
Red Revolution	Meat/Tomato
Golden Revolution	Fruits/Apple/Honey/Horticulture
Grey Revolution	Fertilizers
Silver Revolution	Eggs/Poultry
Golden Fibre	Jute
Silver Fibre	Cotton

MAJOR CROPS

Indian crops can be divided into following categories:

- **Food crops :** Rice, wheat, maize, millets-jowar, bajra etc.
- **Cash crops :** Cotton, jute, sugarcane, tobacco, groundnut etc.
- **Plantation crops :** Tea, Coffee, spices, coconut, rubber etc.
- **Horticulture crops :** Apple, mango, banana, citrus etc.

Crops	Temp. (0°C)	Rainfall (cm)	Soil	Distribution
Cash Crops				
Cotton (Gossypium)	21-30	50-75	Black soil	Gujarat, M.P., Karnataka, Maharashtra, Punjab
Jute (Corchorus Capsularis)	24-35	125-200	Sandy or clayed loams, deep rich	West Bengal Odisha Bihar Assam Meghalaya
Sugarcane (Saccharum officinarum)	20-26	75-150	Loamy soil	Uttar Pradesh Maharashtra, Tamil nadu, Karnataka
Tobacco (Nicotiana)	15-38	50-100	Friable sandy soil	Uttar Pradesh, Andhra Pradesh, Gujarat, Karnataka
Food Crops				
Rice (oryza sativa)	24-27	150	Clayed and loamy soil	West Bengal, Karnataka, Andhra Pradesh, Assam Odisha, Uttar Punjab, Pradesh Telangana, Chhattisgarh
Wheat (triticum)	10-15	75	Light, sandy, clayed loamy soil	Uttar Pradesh Punjab, Haryana, Rajasthan
Jowar (Sorghum)	27-32	30-65	Black clayed loamy soil	Maharashtra, Karnataka, Madhya Pradesh, Andhra Pradesh
Bajra (Penisetu Typhoidum)	25-35	40-50	Loamy soil	Rajasthan, Uttar Pradesh, Haryana, Maharashtra, Gujarat
Plantation Crops				
Tea (Camellia Thea)	24-30	150-250	Loamy forest soil	Kerala, Tamil Nadu, West Bengal, Assam
Coffee (coffea)	16-28	150-250	Friable forest loamy soils	Karnataka, Kerala, Tamil Nadu

Rubber (Hevea Brasiliensis)	25-35	300	Loamy soils	Kerala, Karnataka, Tamil Nadu
-----------------------------	-------	-----	-------------	-------------------------------

Spices (plantation crop)

- India is the second largest producer of pepper.
- India is the largest producer of ginger, cardamom and arecanut.
- India is the third largest producer of coconut next to Philippines and Indonesia.
- Kerala leads in the production of pepper, cardamom, ginger, arecanut and coconut among Indian states.
- Andhra Pradesh is the largest producer of chillies and turmeric.
- India is the largest producer, processor, consumer and exporter of cashew nut in the world.

India's Ranks in Major Commodity

Rice	–	2
Wheat	–	2
Millet	–	1
Tea	–	2
Sugarcane	–	2
Jute	–	1
Silk	–	2
Banana	–	1
Mangoes	–	1
Apple	–	3
Cotton	–	2
Buffalo Milk	–	1
Pulses	–	1

Fisheries/pisciculture

- Fish catch in India is of two types - marine fisheries and inland fisheries.
- India is the third largest producer of fish and second largest producer of inland fishing in the world.
- It accounts for about one per cent of the total agricultural production in India.
- About 75 % of marine fish landings are on the west coast and only 25 % in the east coast.
- Important fish caught along the coast are shark, sardine, herring, Mumbai duck, fly fish ribbon fish and Mackerel.
- West Bengal is the largest producer of fish in India and is the largest producer of inland fish (31 %) also.
- Kerala has about 85 % of India's total processing facilities and processes the largest amount of fish in the country.
- India exports about 8 % of the total fish production. Sri Lanka alone purchases 80 % of our fish and fish products.
- Sasson Dock in Mumbai is a major fishing harbour.
- There are six major fishing harbours and 38 minor fishing harbours. The major harbours are –Cochin, Chennai, Vishakhapatnam, Roy Chowk, Paradip and Sasson dock.
- The Central Institute of Fisheries, Nautical and Engineering Training is at Kochi.
- The Central Institute of Coastal Engineering for Fisheries is in the Bengaluru.

Livestock and Dairy Farming

Livestock includes domestic animals such as cattle, buffaloes, sheep, goats, horses, ponies, donkeys, camels, pigs etc. India's animal wealth is both large and varied. India has about 20% of the world's livestock population.

Dairy Farming includes a class of agricultural enterprise for long-term production of milk which is processed for eventual sale of a dairy product. India is endowed with largest livestock population in the world. It accounts for about 57.3% of the world's buffalo population and 14.7% of the cattle population.

Different Breeds of Animals

Milch Breeds of Cattle Gir, Sindhi, Red Sindhi, Sahiwal, Tharparkar and Deoni

Draught Breeds of Cattle Nagori, Bauchaur, Malvi Hallikar, Ponwar, Siri, Bargur.

Dual Purpose Breeds of Cattle Tharparkar, Haryana, Mewati, Kankrej, Rath, Nimari, Dangi, Ongole.

Goats Breeds Angora, Pashmina, Barabari, Marwari, Mehsana Beetal, Kathiawari and Zalwadi.

Buffaloes Breeds Murrah, Jafarabadi, Shruti, Mehsana, Nagpuri, Nili Ravi, Bhadawari.

Horses and Ponies Breeds Marwari, Kathiawari, Manipuri, Bhutani, Spiti and Chummarti.

Sericulture

Sericulture refers to the rearing of the silk worms for the raw silk production. Silk is a protein produced from the salivary gland of silk worms.

- Important features of Indian sericulture are as follows.
- It is a agro-based labour intensive, export oriented and cottage industry.
- Silk is exported to more than 80 countries like USA, UK, Italy, UAE, Saudi Arabia etc.
- India enjoys the unique distinction of being the only country in the world to produce all the four varieties of silk such as Mulberry, Tasar, Eri and Muga.
- Muga is the monopoly of India. India ranks second in the world after China in Silk production.

Horticulture

Horticulture is a comprehensive term and includes fruits, vegetables, spices, floriculture and coconut. Some of the most important crops grown in India as a part of the horticulture sector are: mango, cashewnut, apple, banana, orange, grape, peach, pear, apricot, strawberry and vegetables. Some important information regarding these fruits is given in the table.

Fruits Crops and their Favourable Climate, Distribution and Relevant Information

Fruit Crops	Favorable Climate	Distribution	Relevant Information
Apple	Temperate fruit crop- It requires average temperature from 21 °C to 4 °C during the active growing season, 100-125 cm rainfall well distributed throughout the growing season. These conditions are found on the hill slopes at altitudes ranging from 1500-2700 m above sea level.	Kullu and Shimla in Himachal Pradesh, Kashmir valley and hilly areas of Uttarakhand.	Loamy soil, rich in organic matter, free from water logging are suitable for apple cultivation.
Banana	Primarily a tropical and sub-tropical crop requiring average temperature of 20°C to 30°C throughout the growing period and rainfall fairly above 150 cm.	Tamil Nadu and Maharashtra are the two main producers	India is the largest producer of banana in the world.
Mango	It is native of monsoon land and is grown in areas with temperature 20°C to 30°C and rainfall 75 cm to 250 cm.	Uttar Pradesh, Bihar, Andhra Pradesh, West Bengal, Odisha, Kerala, Tamil Nadu are the major producers.	India is the largest producer of mango and contributes 54% of the world production of mango.
Grapes	It requires long summer, short winter and moderately fertile well drained soil.	The major producing States are Uttarakhand, Himachal Pradesh, Jammu and Kashmir.	In Northern India, the plant gives only one crop during summer, but in South India, the plant grows throughout the year, one in March, April and the other in August and September.
Strawberry	It requires above 16°C temperature during its growing season and lots of water because its fields are sub-merged under 10 cm of fresh and slowly moving water for at least three months.	The main producers are the hilly areas of Jammu and Kashmir, Himachal Pradesh, Uttarakhand.	Water retaining fertile soil is most suitable.

Floriculture

Government of India has identified floriculture as a sunrise industry and accorded it 100% export oriented status. Owing to steady increase in demand of flower, floriculture has become one of the important commercial trades in agriculture. Floriculture products mainly consist of cut flowers, pot plants, cut foliage, seeds, bulbs, tubers, rooted cuttings and dried flowers or leaves. Maharashtra, Karnataka, Andhra Pradesh, Haryana, Tamil Nadu, Rajasthan, West Bengal have emerged as major floriculture centers.

Exercise -1

1. A geographic area with an altitude of 400 metres has following characteristics.

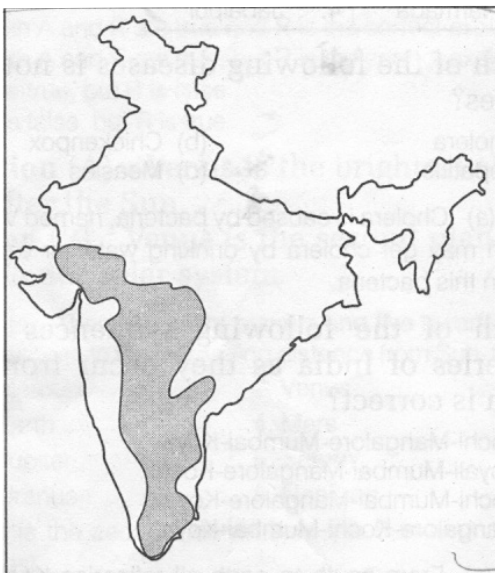
Month	J	F	M	A	M	J	J	A	S	O	N	D
Average maximum temp °C	31	31	31	31	30	30	29	28	29	29	30	31
Average minimum temp °C	21	21	21	21	21	21	20	20	20	20	20	20
Rainfall (mm)	51	85	188	158	139	121	134	168	185	221	198	86

If this geographic area were to have a natural forest, which one of the following would it most likely be?

- Moist temperate coniferous forest
 - Montane subtropical forest
 - Temperate forest
 - Tropical rain forest
2. The lower Gangetic plain is characterised by humid climate with high temperature throughout the year. Which one among the following pairs of crops is most suitable for this region ?
- Paddy and Cotton
 - Wheat and Jute
 - Paddy and Jute
 - Wheat and Cotton
3. Salinization occurs when the irrigation water accumulated in the soil evaporates, leaving behind salts and minerals. What are the effects of salinization on the irrigated land ?
- It greatly increases the crop production
 - It makes some soils impermeable
 - It raises the water table
 - It fills the air spaces in the soil with water
4. Which one of the following statements is correct?
- Red soil lie on the periphery of the Peninsular plateau
 - The laterite soils are more acidic on the low-lying areas than on the higher areas
 - Alluvial soils are poor in potash and phosphorus
 - Black soils are rich in phosphorous, nitrogen and organic matter
5. What do you understand by the word regur ?
- Black soil found in the Deccan
 - Grey-brown soil found in Russia
 - Red colour sticky soil found in the Amazon area
 - Black soil found in northern Argentina
6. Which of the following is not a tropical type of soil ?
- Red soil
 - Desert soil
 - Black soil
 - Brown soil
7. The retreating monsoon withdraws from the
- west coast to the east
 - north to the south
 - north-east India to the west coast
 - north-west India to Bengal
8. Orographic rainfall occurs along the
- Eastern Ghats
 - Himalayas
 - Aravallis
 - Jaintia hills
9. Black soil requires least tilling because
- it is fertile
 - it develops cracks on drying
 - it is a lava origin
 - it has plenty of iron and aluminium compounds
10. Why is it that India has one of the lowest agricultural yields in the world ?
- Summers are very hot
 - Lack of fertile soil
 - Farmers being poor and illiterate do not use modern farming techniques
 - Occurrence of frost at night
11. The state with the largest area under waste land is
- Gujarat
 - Madhya Pradesh
 - Jammu and Kashmir
 - Rajasthan
12. Which of the following would be most suited for an area which has only two months of scanty rainfall in a year ?
- Sugarcane
 - Tea
 - Cotton
 - Pulses
13. Sucrose content in sugarcane decreases :
- if high rainfall occurs during the period of growth of the plant
 - if frost occurs during the period of ripening
 - if there is fluctuation in temperature during the period of growth of the plant
 - if there is high temperature during the time of ripening
14. Doon valley is able to grow rice because
- it has warm summers and snow-melt waters for irrigation.
 - people in the valley are rice eaters
 - other crops cannot be grown
 - there is a huge export demand
15. India has the highest productive potential of fisheries in
- inland water bodies
 - shallow continental shelf
 - deep sea areas
 - brackish water lagoons
16. During winter, the northern half of India is warmer than areas at similar latitudes outside the country by 3° to 8°C. This is due to
- India experiencing a tropical monsoon climate.
 - the warm currents
 - the Tropic of Cancer passing through the middle of the country
 - the presence of the Himalayas with their east and west extent

17. The extreme of temperature between summer and winter is quite low in southern part of Peninsular India mainly because
 - (a) the sun's rays are almost vertical throughout the year
 - (b) adjoining oceans moderate the temperature
 - (c) sky is generally cloudy
 - (d) strong winds blow throughout the year
18. The irregularity in the amount of rain in different parts of the north Indian plains, during different years, is mainly due to the
 - (a) irregular intensity of low pressure in the northwestern part of India
 - (b) difference in frequency of cyclones
 - (c) variations in the location of the axis of the low pressure trough
 - (d) the amount of moisture carried by the winds not being the same every year
19. How do the 'western disturbances' affect the crops in north India?
 - (a) They cause heavy damage to the standing crops
 - (b) They bring in locusts which destroy the crops
 - (c) They are beneficial to the crops by causing winter rain
 - (d) They help in keeping the plants warm to some extent in winter
20. Most of the precipitation in India is _____ in nature.
 - (a) cyclonic
 - (b) convectional
 - (c) orographic
 - (d) frontal
21. Which soil requires the least tilling ?
 - (a) Red
 - (b) Black
 - (c) Laterite
 - (d) Alluvial
22. What kind of soil predominates in the Sunderbans area ?
 - (a) Red
 - (b) Laterite
 - (c) Black
 - (d) Alluvial
23. Which one of the following does not belong to biosphere reserves set-up so far?
 - (a) Great Nicobar
 - (b) Sunderbans
 - (c) Nanda Devi
 - (d) Gulf of Kachchh
24. Which one of the following regions of India is now regarded as an 'ecological hot spot'?
 - (a) Western Himalayas
 - (b) Eastern Himalayas
 - (c) Western Ghats
 - (d) Eastern Ghats
25. Out of all the Biosphere Reserves in India, four have been recognized on the World Network by UNESCO. Which one of the following is not one of them?
 - (a) Gulf of Mannar
 - (b) Kanchenjunga
 - (c) Nanda Devi
 - (d) Sunderbans
26. Which of the following forest species is not characteristics of Deciduous forests?
 - (a) Teak
 - (b) Sal
 - (c) Sandalwood
 - (d) Deodar
27. In India, in which one of the following areas are the tropical evergreen forests found ?
 - (a) The Western Ghats
 - (b) The Eastern Ghats
 - (c) The Western Himalaya
 - (d) The Central Himalaya
28. On the basis of the process of their formation, which of the following soils is formed differently from the other three?
 - (a) Khadar
 - (b) Bangar
 - (c) Bhabar
 - (d) Regur
29. Which one among the following is the best reason for the marked increase in the agricultural production in India in the past decades?
 - (a) Increases in the area under cultivation
 - (b) Conversion of barren land into agricultural land
 - (c) Use of improved agricultural methods and technologies
 - (d) Priority status given by the successive governments to agricultural sector over the industry sector
30. Which one among the following is not a reason for practising tank irrigation in Peninsular India?
 - (a) The undulating relief and hard rocks
 - (b) Little percolation of rain water due to impervious rock structure
 - (c) Most of the rivers of Peninsular India are perennial
 - (d) There are many streams which become torrential during rainy season
31. Contour bunding is a method of soil conservation used in
 - (a) desert margins, liable to strong wind action
 - (b) low flat plains, close to stream courses, liable to flooding
 - (c) scrublands, liable to spread of weed growth
 - (d) None of the above
32. In India, cluster bean (Guar) is traditionally used as a vegetable or animal feed, but recently the cultivation of this has assumed significance. Which one of the following statements is correct in this context?
 - (a) The oil extracted from seeds is used in the manufacture of biodegradable plastics
 - (b) The gum made from its seeds is used in the extraction of shale gas
 - (c) The leaf extract of this plant has the properties of antihistamines
 - (d) It is a source of high quality biodiesel
33. Among the following forest types in India, which one occupies the largest area?
 - (a) Tropical wet evergreen forest
 - (b) Tropical moist deciduous forest
 - (c) Sub-tropical dry evergreen forest
 - (d) Montane wet temperate forest
34. The National Forestry Action Plan aims to bring 33% of land area in India under tree cover by which year?
 - (a) 2008
 - (b) 2012
 - (c) 2016
 - (d) 2020
35. What is mixed farming?
 - (a) Growing of several crops in a planned way
 - (b) Growing rabi as well as kharif crops
 - (c) Growing several crops and also rearing animals
 - (d) Growing of fruits as well as vegetables
36. Which one of the following states has the largest forest area to its total land area?
 - (a) Mizoram
 - (b) Arunachal Pradesh
 - (c) Sikkim
 - (d) Jammu and Kashmir

37. In which one of the following states is Ranganathittu Bird Sanctuary located?
 (a) Tamil Nadu (b) Kerala
 (c) Karnataka (d) Andhra Pradesh
38. In which one of the following places is the Forest Survey of India (FSI), a national organisation engaged in forest cover mapping, forest inventory and training in the field of remote sensing and GIS, located?
 (a) Dehradun (b) Itanagar
 (c) Ahmedabad (d) Aizawl
39. Which one of the following states is the largest producer of black pepper in India?
 (a) Tamil Nadu (b) Kerala
 (c) Karnataka (d) Andhra Pradesh
40. The shaded area in the map given below is the major producer of which one of the following?



- (a) Cotton (b) Groundnut
 (c) Wheat (d) Mustard
41. Which one of the following is categorised as millet?
 (a) Wheat (b) Rice
 (c) Sorghum (d) Maize
42. Leaching is the maximum in the soil type of
 (a) Laterite (b) Red
 (c) Regur (d) Desert
43. Sal trees are the typical species of
 (a) Tropical rain forest (b) Tropical monsoon forest
 (c) Taiga forest (d) Tundra forest
44. Which one of the following is the correct sequence of the given tiger reserves of India from North to South?
 (a) Dudwa-Kanha-Indravati-Bandipur
 (b) Kanha-Bandipur-Dudwa-Indravati
 (c) Indravati-Kanha-Dudwa-Bandipur
 (d) Dudwa-Kanha-Bandipur-Indravati
45. In wildlife conservation which one among the following best defines an 'endemic species'?
 (a) When the critical number of a species declines in a forest due to parasitic attack
 (b) A species which is cosmopolitan and can be commonly found in biosphere
 (c) An endangered species which is found in a few restricted areas on the Earth
 (d) A species confined to a particular region and not found anywhere else
46. Red soil colour is caused by
 (a) aluminium compounds (b) mercury compounds
 (c) iron compound (d) clay
47. Which one of the following is the example of subsistence farming?
 (a) Shifting cultivation
 (b) Commercial farming
 (c) Extensive and intensive farming
 (d) Organic farming
48. In India, in which one of the following areas are the tropical evergreen forests found ?
 (a) The Western Ghats (b) The Eastern Ghats
 (c) The Western Himalaya (d) The Central Himalaya
49. In which of the following Hills is tea cultivated?
 (a) Maikala Hills (b) Nallamalai Hills
 (c) Nilgiri Hills (d) Shivalik Hills
50. Which one of the following types of forest covers the maximum area in India?
 (a) Tropical rain forest
 (b) Tropical moist deciduous forest
 (c) Tropical dry deciduous forest
 (d) Tropical dry evergreen forest
51. The minimum land area recommended for forest cover to maintain proper ecological balance in India is :
 (a) 25% (b) 33%
 (c) 43% (d) 53%
52. The first marine sanctuary in India, within its bounds coral reefs, mollusca, dolphins, tortoises and various kinds of sea birds, has been established in:
 (a) Sundarbans (b) Chilka Lake
 (c) Gulf of Kutch (d) Lakshadweep
53. Amongst the following Indian States which one has the minimum total forest cover?
 (a) Sikkim (b) Goa
 (c) Haryana (d) Kerala
54. Which one of the following is not a Biosphere reserve?
 (a) Agasthyamali (b) Nallamalai
 (c) Nilgiri (d) Panchmarhi
55. Which one of the following is also known as Top Slip?
 (a) Ismlipal National Park
 (b) Periyar Wildlife Sanctuary
 (c) Manjira Wildlife Sanctuary
 (d) Indira Gandhi Wildlife Sanctuary and National Park
56. Which one of the following is located in the Bastar region?
 (a) Bandhavgarh National Park
 (b) Dandeli Sanctuary
 (c) Rajaji National Park
 (d) Indravati National Park
57. Which one of the following is not essentially a species of the Himalayan vegetation?
 (a) Juniper (b) Mahogany
 (c) Silver fir (d) Spruce

58. Which one among the following has the maximum number of National Parks?
 (a) Andaman and Nicobar Islands
 (b) Arunachal Pradesh
 (c) Assam
 (d) Meghalaya
59. When you travel in certain parts of India, you will notice red soil. What is the main reason for this colour?
 (a) Abundance of magnesium
 (b) Accumulated humus
 (c) Presence of ferric oxides
 (d) Abundance of phosphates
60. Contour bunding is a method of soil conservation used in
 (a) desert margins, liable to strong wind action
 (b) low flat plains, close to stream courses, liable to flooding
 (c) scrublands, liable to spread of weed growth
 (d) None of the above
61. Which one of the following sets of conditions is necessary for a good crop of wheat?
 (a) Moderate temperature and moderate rainfall
 (b) High temperature and heavy rainfall
 (c) High temperature and moderate rainfall
 (d) Low temperature and low rainfall
62. Which one of the following agricultural practices is eco-friendly?
 (a) Organic farming
 (b) Shifting cultivation
 (c) Cultivation of high yielding varieties
 (d) Growing plants in glass-houses
63. The Genetic Engineering Approval Committee, whose permission is required for cultivation of any genetically modified crop such as Bt Cotton in India, is under the Union Ministry of :
 (a) Agriculture
 (b) Environment and Forests
 (c) Commerce and Industry
 (d) Rural-Development
64. An objective of the National Food Security Mission is to increase the production of certain crops through area expansion and productivity enhancement in a sustainable manner in the identified districts of the country. What are those crops?
 (a) Rice and wheat only
 (b) Rice, wheat and pulses only
 (c) Rice, wheat, pulses and oil seeds only
 (d) Rice, wheat, pulses, oil seeds and vegetables
65. Given below are the names of four energy crops. Which one of them can be cultivated for ethanol ?
 (a) Jatropha (b) Maize
 (c) Pongamia (d) Sunflower
66. Which of the following is the chief characteristic of 'mixed farming'?
 (a) Cultivation of both cash crops and food crops
 (b) Cultivation of two or more crops in the same field
 (c) Rearing of animals and cultivation of crops together
 (d) None of the above
67. Which one of the following regions of India has a combination of mangrove forest, evergreen forest and deciduous forest? [CSAT 2015-I]
 (a) North Coastal Andhra Pradesh
 (b) South-West Bengal
 (c) Southern Saurashtra
 (d) Andaman and Nicobar Islands
68. Which one of the following best describes the main objective of 'Seed Village Concept'? [CSAT 2015-I]
 (a) Encouraging the farmers to use their own farm seeds and discouraging them to buy the seeds from others
 (b) Involving the farmers for training in quality seed production and thereby to make available quality seeds to others at appropriate time and affordable cost
 (c) Earmarking some villages exclusively for the production of certified seeds
 (d) Identifying the entrepreneurs in village and providing them technology and finance to set up seed companies
69. The agricultural production in different parts of India is very much affected by varying intensities of floods and droughts. Which one of the following measures would not be sustainable in this respect?
 (a) Provision for extensive irrigation facilities
 (b) Change in the crop calendar
 (c) Avoidance of flood and drought prone areas for agriculture
 (d) Emphasis on selection of crops best suited to flood and drought conditions
70. Kanha National Park belongs to which one among the following biogeographical areas in the world?
 (a) Tropical Sub-humid Forests
 (b) Tropical Humid Forests
 (c) Tropical Dry Forests
 (d) Tropical Moist Forests
 (e) Middle and South Andaman
71. What is the sequential order of vegetation types observed while moving from Assam Valley to Rajasthan Plains?
 (a) Tropical Wet Evergreen
 Tropical Moist Deciduous
 Tropical Dry Deciduous
 Tropical Thorn Forest
 (b) Tropical Thorn Forest
 Tropical Dry Deciduous
 Tropical Moist Deciduous
 Tropical Wet Evergreen
 (c) Tropical Moist Deciduous
 Tropical Wet Evergreen
 Tropical Dry Deciduous
 Tropical Thorn Forest
 (d) Tropical Dry Deciduous
 Tropical Thorn Forest
 Tropical Moist Deciduous
 Tropical Wet Evergreen

72. Which one of the following is the correct sequence in the decreasing order of production (in million tonnes) of the given foodgrains in India?
- (a) Wheat - Rice - Pulses - Coarse Cereals
 - (b) Rice - Wheat - Pulses - Coarse Cereals
 - (c) Wheat - Rice - Coarse Cereals - Pulses
 - (d) Rice - Wheat - Coarse Cereals - Pulses
73. In India, cluster bean (Guar) is traditionally used as a vegetable or animal feed, but recently the cultivation of this has assumed significance. Which one of the following statements is correct in this context?
- [CSAT 2014 - I]
- (a) The oil extracted from seeds is used in the manufacture of biodegradable plastics
 - (b) The gum made from its seeds is used in the extraction of shale gas
 - (c) The leaf extract of this plant has the properties of antihistamines
 - (d) It is a source of high quality biodiesel
74. In India, markets in agricultural products are regulated under the _____ [CSAT 2015 - I]
- (a) Essential Commodities Act, 1955
 - (b) Agricultural Produce Market Committee Act enacted by States
 - (c) Agricultural Produce (Grading and Marking) Act, 1937
 - (d) Food Products Order, 1956 and Meat and Food Products Order, 1973
75. The terms 'Agreement on Agriculture', 'Agreement on the Application of Sanitary and Phytosanitary Measures' and 'Peace Clause' appear in the news frequently in the context of the affairs of the _____ [CSAT 2015 - I]
- (a) Food and Agriculture Organization
 - (b) United Nations Framework Conference on Climate Change
 - (c) World Trade Organisation
 - (d) United Nations Environment Programme

Exercise -2

Statement Based MCQ

1. Consider the following statements :

1. Biodiversity hotspots are located only in tropical regions.
2. India has four biodiversity hotspots i.e., Eastern Himalayas, Western Himalayas, Western Ghats and Andaman and Nicobar Islands.

Which of the statements given above is/are correct ?

- (a) 1 (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

2. With reference to soil conservation, consider the following practices:

1. Crop rotation
2. Sand fences
3. Terracing
4. Wind breaks

Which of the above are considered appropriate methods for soil conservation in India?

- (a) 1, 2 and 3 (b) 2 and 4
(c) 1, 3 and 4 (d) 1, 2, 3 and 4

3. Consider the following statements:

1. Silent Valley National Park in the Nallamalai range.
2. Pathrakkadavu Hydroelectric project is proposed to be built near the Silent Valley National Park.
3. The Kunthi river originates in Silent Valley's rainforests.

Which of the statements given above is/are correct?

- (a) 1 and 3 (b) 2 only
(c) 2 and 3 (d) 1, 2 and 3

4. A state in India has the following characteristics :

1. Its northern part is arid and semi-arid.
2. Its central part produces cotton.
3. Cultivation of cash crops is predominant over food crops.

Which one of the following states has all of the above characteristics ?

- (a) Andhra Pradesh (b) Gujarat
(c) Karnataka (d) Tamil Nadu

5. Consider the following crops of India

1. Cowpea
2. Green gram
3. Pigeon pea

Which of the above is/are used as pulse, fodder and green manure?

- (a) 1 and 2 (b) 2 only
(c) 1 and 3 (d) 1, 2 and 3

6. Consider the following statements

1. India is the only country in the world producing all the five known commercial varieties of silk.
2. India is the largest producer of sugar in the world.

Which of the statements given above is/are correct?

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

7. Though coffee and tea both are cultivated on hill slopes, there is some difference between them regarding their cultivation. In this context, consider the following statements:

1. Coffee plant requires a hot and humid climate of tropical areas whereas tea can be cultivated in both tropical and subtropical areas.
2. Coffee is propagated by seeds but tea is propagated by stem cuttings only.

Which of the statements given above is/ are correct?

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

8. The active agents of soil forming process are

1. Climate
2. Parent material
3. Biota
4. Topography
5. Time

Select the correct answer from the codes given below:

- (a) 1 and 2 (b) 1 and 3
(c) 4 and 5 (d) 3, 4 and 5

9. Which of the following factors have caused water logging in the black lava soil of Deccan

1. Introduction of irrigation
2. Sugarcane cultivation
3. High-rainfall

Select the correct answer

- (a) 1, 2 and 3 (b) 1 and 2
(c) 1 only (d) 1 and 3

10. Which of the following are the rivers at the delta mouths of which mangroves occur in abundance?

1. Narmada
2. Subarnarekha
3. Krishna
4. Ganga

Select the correct answer:

- (a) 1 and 3 (b) 3 and 4
(c) 2 and 4 (d) 1, 3 and 4

11. Consider the following statements:

Andhra Pradesh is one of the leading producers of rice because

1. It has fertile alluvial soil in the coastal plain.
2. It receives about 125 cm of rainfall per annum in the coastal tract.
3. It has a lot of tank irrigation in the coastal plain.

Of these statements:

- (a) 1, 2 and 3 (b) 1 and 2
(c) 2 and 3 (d) 1 and 3

12. Consider the following statements about Indian forestry:

1. About 40 per cent of the Indian forests are in the inaccessible mountainous regions which impede the speedy transfer of felled trees.
2. Although according to India's forest policy, a high proportion of the land area shall be covered by forest except in some states, the percentage of forest cover is precariously low.
3. Exploitation of forests is less time consuming and less costly, as almost in every forest a single type of tree dominates.
4. The annual productivity of the forests is 3.5 cubic metres per hectare.

Of these statements

- (a) 1 and 2 (b) 1, 2 and 3
(c) 3 and 4 (d) 1, 2, 3 and 4

13. What are the major causes of droughts in India?
1. Most of the rainfall occurs within a few months.
 2. The absence of forests results in rapid run-off and thus droughts.
 3. Rainfall is highly variable.
 4. The amount of rainfall is inadequate.
- (a) 1, 3 and 4 (b) 2, 3 and 4
(c) 1, 2 and 3 (d) 1, 2 and 4
14. Consider the following programmes:
1. Afforestation and development of wastelands.
 2. Reforestation and replantation in existing forests.
 3. Encouraging the wood substitutes and supplying other types of fuel.
 4. Promotion of wide use of insecticides and pesticides to restrict the loss of forest area from degradation caused by pests and insects.
- The National Forest Policy of 1988 includes:
- (a) 1, 2, 3 and 4 (b) 2 and 4
(c) 1, 3 and 4 (d) 1, 2 and 3
15. Consider the following statements regarding environment issues of India:
1. Gulf of Mannar is one of the biosphere reserves.
 2. The Ganga Action Plan, phase II has been merged with the National River Conservation Plan.
 3. The National Museum of Natural History at New Delhi imparts non-formal education in environment and conservation.
 4. Environmental Information System (ENVIS) acts as a decentralized information network for environment information.
- Which of these statements are correct?
- (a) 1, 2 and 4 (b) 1, 2, 3 and 4
(c) 2 and 3 (d) 1, 3 and 4
16. Consider the following statements:
- The black cotton soil of India
1. is mainly distributed over the river valleys of the lava region of the Deccan.
 2. has a great capacity of retaining moisture
 3. is extremely fertile at surface
 4. occupies about 10 per cent of the total area of the country of these statements
- (a) 1, 2 and 3 (b) 1, 3 and 4
(c) 1 and 2 (d) 3 and 4
17. Consider the following statements :
1. The equatorial forests are evergreen.
 2. The Mediterranean region has deciduous forests of broad leaved trees.
 3. Taiga forests are found in Kashmir.
- Which of the statements given above are correct ?
- (a) 1 and 2 (b) 2 and 3
(c) 1 and 3 (d) 1, 2 and 3
18. Tamil Nadu is a leading producer of mill-made cotton yarn in the country. What could be the reason?
1. Black cotton soil is the predominant type of soil in the State.
 2. Rich pool of skilled labour is available.
- Which of the above is/are the correct reasons?
- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2
19. Which of the following is/are unique characteristic/ characteristics of equatorial forests?
1. Presence of tall, closely set trees with crowns forming a continuous canopy
 2. Coexistence of a large number of species
 3. Presence of numerous varieties of epiphytes
- Select the correct answer using the codes given below.
- (a) 1 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3
20. Consider the following pairs:
- | National Park | River flowing through the Park |
|-----------------------------------|---------------------------------------|
| 1. Corbett National Park: | Ganga |
| 2. Kaziranga National Park: | Manas |
| 3. Silent Valley: National Park : | Kaveri |
- Which of the above pairs is/are correctly matched?
- (a) 1 and 2 (b) 3 only
(c) 1 and 3 (d) None of these
21. The Narmada river flows to the west, while most other large peninsular rivers flow to the east. Why?
1. It occupies a linear rift valley.
 2. It flows between the Vindhyas and the Satpuras.
 3. The land slopes to the west from Central India.
- Select the correct answer using the codes given below.
- (a) 1 only (b) 2 and 3
(c) 1 and 3 (d) None
22. Which of the following statements regarding laterite soils of India are correct?
1. They are generally red in colour.
 2. They are rich in nitrogen and potash.
 3. They are well-developed in Rajasthan and UP.
 4. Tapioca and cashew nuts grow well on these soils.
- Select the correct answer using the codes given below.
- (a) 1, 2 and 3 (b) 2, 3 and 4
(c) 1 and 4 (d) 2 and 3 only
23. If you travel through the Himalayas, you are likely to see which of the following plants naturally growing there?
1. Oak
 2. Rhododendron
 3. Sandalwood
- Select the correct answer using the code given below.
- (a) 1 and 2 only (b) 3 only
(c) 1 and 3 only (d) 1, 2 and 3
24. Consider the following crops:
1. Cotton
 2. Groundnut
 3. Rice
 4. Wheat
- Which of these are Kharif crops?
- (a) 1 and 4 (b) 2 and 3 only
(c) 1, 2 and 3 (d) 2, 3 and 4
25. With reference to Neem tree, consider the following statements :
1. Neem oil can be used as a pesticide to control the proliferation of some species of insects and mites.
 2. Neem seeds are used in the manufacture of biofuels and hospital detergents.
 3. Neem oil has applications in pharmaceutical industry.
- Which of the statements given above is/are correct?
- (a) 1 and 2 only (b) 3 only
(c) 1 and 3 only (d) 1, 2 and 3

26. Consider the following statements :
1. Maize can be used for the production of starch.
 2. Oil extracted from maize can be a feedstock for biodiesel.
 3. Alcoholic beverages can be produced by using maize.
- Which of the statements given above is/are correct?
- (a) 1 only (b) 1 and 2 only
(c) 2 and 3 only (d) 1, 2 and 3
27. What are the significances of a practical approach to sugarcane production known as 'Sustainable Sugarcane Initiative'?
1. Seed cost is very low in this compared to the conventional method of cultivation.
 2. Drip irrigation can be practiced very effectively in this.
 3. There is no application of chemical/inorganic fertilizers at all in this.
 4. The scope for intercropping is more in this compared to the conventional method of cultivation.
- Select the correct answer using the code given below.
- (a) 1 and 3 only (b) 1, 2 and 4 only
(c) 2, 3 and 4 only (d) 1, 2, 3 and 4
28. What are the benefits of implementing the 'Integrated Watershed Development Programme'?
1. Prevention of soil runoff
 2. Linking the country's perennial rivers with seasonal rivers
 3. Rainwater harvesting and recharge of groundwater table
 4. Regeneration of natural vegetation
- Select the correct answer using the code given below.
- (a) 1 and 2 only (b) 2, 3 and 4 only
(c) 1, 3 and 4 only (d) 1, 2, 3 and 4
- Matching Based MCQ**
29. Consider the following statements
1. National parks are a special category of protected areas of land and sea coasts where people are an integral part of the system.
 2. Sanctuaries are concerned with conservation of particular species.
 3. Biosphere reserves are connected with the habitat of a particular wild animal.
- Which of the statements given above is/are correct?
- (a) 1, 2 and 3 (b) Only 2
(c) 1 and 2 (d) 1 and 3
30. Which one of the following statements on biosphere reserves is not correct?
- (a) In 1973, UNESCO launched a worldwide programme on man and biosphere
(b) Biosphere reserves promote research on ecological conservation
(c) Nanda Devi Biosphere Reserve is located in Madhya Pradesh
(d) Biosphere reserves are multipurpose protected areas to preserve the genetic diversity in ecosystems
31. Consider the following statements about black soil of India
1. Black soil becomes sticky when it is wet.
 2. Black soil contains adequate nitrogen as well as phosphorus required for the growth of plants.
- Which of the statements given above is/are correct?
- (a) Only 1 (b) Only 2
(c) Both 1 and 2 (d) Neither 1 nor 2
32. Which of the following statements regarding red soils of India is/are correct?
1. The colour of the soil is red due to ferric oxide content.
 2. Red soils are rich in lime, humous and potash.
 3. They are porous and have friable structure.
- Select the correct answer using the code given below
- (a) Only 1 (b) 1 and 3
(c) 2 and 3 (d) 1, 2 and 3
33. Which of the following statements regarding jhum cultivation in India are correct ?
1. It is largely practised in North-Eastern Indian states.
 2. It is referred to as 'slash and burn' technique.
 3. In it, the fertility of soil is exhausted in a few years.
- Select the correct answer using the code given below :
- (a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3
34. Which of the following statements relating to Indian agriculture is/are correct ?
1. India has the World's largest cropped area.
 2. Cropping pattern is dominated by cereal crop.
 3. The average size of an Indian farm holding is too small for several agricultural operations.
- Select the correct answer using the code given below :
- (a) 1 only (b) 1 and 2 only
(c) 2 and 3 only (d) 1, 2 and 3
35. Consider the following statements:
1. The forest cover in India constitutes around 20% of its geographical area. Out of the total forest cover, dense forest constitutes around 40%.
 2. The National Forestry Action Programme aims at bringing one third of the area of India under tree forest cover.
- Which of the statements given above is/are correct?
- (a) 1 only (b) 2 only
(c) both 1 and 2 (d) Neither 1 nor 2
36. Consider the following statements:
1. The boundaries of a National Park are defined by legislation.
 2. A Biosphere Reserve is declared to conserve a few specific species of flora and fauna.
 3. In a Wildlife Sanctuary, limited biotic interference is permitted.
- Which of the statements given above is/are correct?
- (a) 1 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3
37. Following are the characteristics of an area in India:
1. Hot and humid climate
 2. Annual rainfall 200 cm
 3. Hill slopes up to an altitude of 1100 metres
 4. Annual range of temperature 15°C to 30°C.

- Which one among the following crops are you most likely to find in the area described above?
- (a) Mustard (b) Cotton
(c) Pepper (d) Virginia tobacco
38. Which of the following statements regarding laterite soils of India are correct?
- They are generally red in colour.
 - They are rich in nitrogen and potash.
 - They are well-developed in Rajasthan and UP.
 - Tapioca and cashew nuts grow well on these soils.
- Select the correct answer using the codes given below.
- (a) 1, 2 and 3 (b) 2, 3 and 4
(c) 1 and 4 (d) 2 and 3 only
39. The following are the major oilseeds produced in India:
- Sesamum
 - Mustard
 - Groundnut
 - Soyabean
- Which one of the following is the correct sequence of the descending order of the quantity of their production?
- (a) 1, 2, 3, 4 (b) 3, 2, 4, 1
(c) 2, 4, 3, 1 (d) 3, 4, 2, 1
40. Consider the following high yielding varieties of crops in India:
- Arjun
 - Jaya
 - Padma
 - Sonalika
- Which of these are of wheat?
- (a) 1 and 2 (b) 2 and 3
(c) 1 and 4 (d) 3 and 4
41. Consider the following statements:
- India is the original home of the cotton plant
 - India is the first country in the world to develop hybrid cotton variety leading to increased production
- Which of these statements is/are correct?
- (a) Only 1 (b) Only 2
(c) Both 1 and 2 (d) Neither 1 nor 2
42. Consider the following statements:
- India ranks first in the world in fruit production
 - India ranks second in the world in export of tobacco
- Which of these statements is/are correct?
- (a) Only 1 (b) Only 2
(c) Both 1 and 2 (d) Neither 1 nor 2
43. Consider the following crops:
- Cotton
 - Groundnut
 - Maize
 - Mustard
- Which of the above are kharif crops?
- (a) 1 and 2 (b) 1, 2 and 3
(c) 3 and 4 (d) 1, 2, 3 and 4
44. Consider the following statements:
- India is the only country in the world producing all the five known commercial varieties of silk.
 - India is the largest producer of sugar in the world.
- Which of the statements given above is/are correct?
- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2
45. Due to their extensive rice cultivation, some regions may be contributing to global warming. To what possible reason/reasons is this is attributable?
- The anaerobic conditions associated with rice cultivation cause the emission of methane.
 - When nitrogen based fertilizers are used, nitrous oxide is emitted from the cultivated soil.
- Which of the statements given above is/are correct?
- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2
46. Consider the following crops of India :
- Cowpea
 - Green gram
 - Pigeon pea
- Which of the above is/are used as pulse, fodder and green manure?
- (a) 1 and 2 only (b) 2 only
(c) 1 and 3 only (d) 1, 2 and 3
47. Consider the following crops of India :
- Groundnut
 - Sesamum
 - Pearl millet
- Which of the above is/are predominantly rained crop/crops?
- (a) 1 and 2 only (b) 2 and 3 only
(c) 3 only (d) 1, 2 and 3
48. Consider the following crops:
- Cotton
 - Groundnut
 - Rice
 - Wheat
- Which of these are Kharif crops?
- (a) 1 and 4 (b) 2 and 3 only
(c) 1, 2 and 3 (d) 2, 3 and 4
49. With reference to Neem tree, consider the following statements :
- [CSAT 2014-I]
- Neem oil can be used as a pesticide to control the proliferation of some species of insects and mites.
 - Neem seeds are used in the manufacture of biofuels and hospital detergents.
 - Neem oil has applications in pharmaceutical industry.
- Which of the statements given above is/are correct?
- (a) 1 and 2 only (b) 3 only
(c) 1 and 3 only (d) 1, 2 and 3
50. Consider the following pairs : [CSAT 2014-I]
- | Region | Well-known for the production of |
|---------------|----------------------------------|
| 1. Kinnaur | : Areca nut |
| 2. Mewat | : Mango |
| 3. Coromandel | : Soya bean |
- Which of the above pairs is/ are correctly matched?
- (a) 1 and 2 only (b) 3 only
(c) 1, 2 and 3 (d) None
51. Consider the following statements : [CSAT 2014 - I]
- Maize can be used for the production of starch.
 - Oil extracted from maize can be a feedstock for biodiesel.
 - Alcoholic beverages can be produced by using maize.
- Which of the statements given above is/are correct?
- (a) 1 only (b) 1 and 2 only
(c) 2 and 3 only (d) 1, 2 and 3
52. What are the significances of a practical approach to sugarcane production known as 'Sustainable Sugarcane Initiative'?
- [CSAT 2014-I]
- Seed cost is very low in this compared to the conventional method of cultivation.
 - Drip irrigation can be practiced very effectively in this.

3. There is no application of chemical/inorganic fertilizers at all in this.
4. The scope for intercropping is more in this compared to the conventional method of cultivation.

Select the correct answer using the code given below.

- (a) 1 and 3 only (b) 1, 2 and 4 only
(c) 2, 3 and 4 only (d) 1, 2, 3 and 4

53. What can be the impact of excessive / inappropriate use of nitrogenous fertilizers in agriculture? [CSAT 2015 - I]

1. Proliferation of nitrogen-fixing microorganisms in soil can occur.
2. Increase in the acidity of soil can take place.
3. Leaching of nitrate to the ground-water can occur.

Select the correct answer using the code given below.

- (a) 1 and 3 only (b) 2 only
(c) 2 and 3 only (d) 1, 2 and 3

54. Consider the following States [CSAT 2015-I]

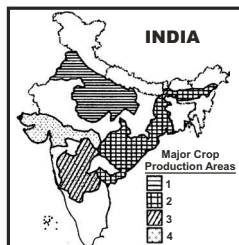
1. Arunachal Pradesh
2. Himachal Pradesh
3. Mizoram

In which of the above States do 'Tropical Wet Evergreen Forests' occur?

- (a) 1 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

55. In the map the given four areas are differently shaded, three of which indicate cereal crops production areas and one indicates non-cereal crop production area.

As per the index given, the non-cereal crop production area is :



- (a) 1 (b) 2
(c) 3 (d) 4

56. Consider the following pairs :

Region	Well-known for the production of
1. Kinnaur	: Areca nut
2. Mewat	: Mango
3. Coromandel	: Soya bean

Which of the above pairs is/ are correctly matched?

- (a) 1 and 2 only (b) 3 only
(c) 1, 2 and 3 (d) None

57. Consider the following pairs :

Programme/Project	Ministry
1. Drought-Prone Area : Ministry of Agriculture Programme	
2. Desert Development : Ministry of Environment and Forests Programme	
3. National Watershed : Ministry of Rural Development Project for Rainfed Areas	

Which of the above pairs is/ are correctly matched?

- (a) 1 and 2 only (b) 3 only
(c) 1, 2 and 3 (d) None

58. Match the following

List I (Natural Vegetation of India)	List II (Annual Rainfall Received)
A. Tropical evergreen forests	1. 100-200 cm
B. Tropical deciduous forests	2. Above 200 cm
C. Tropical dry forests	3. Less than 50 cm
D. Arid forests	4. Above 300 cm
	5. 50-100 cm

Codes:

- A B C D A B C D
(a) 1 2 5 3 (b) 4 3 1 5
(c) 2 1 5 3 (d) 2 1 3 4

59. Match the following

List I (Biosphere Reserve)	List II (Places)
A. Manas	1. Meghalaya
B. Pachmarhi	2. Asom
C. Nokrek	3. Madhya Pradesh
D. Achanakmar Amarkantak	4. Chhattisgarh

Codes :

- A B C D A B C D
(a) 4 3 1 2 (b) 2 1 3 4
(c) 4 1 3 2 (d) 2 3 1 4

60. Match the following

List I (Tiger Reserve)	List II (State)
A. Indravati	1. Karnataka
B. Periyar	2. Odisha
C. Simlipal	3. Kerala
D. Bandipur	4. Chhattisgarh

Codes :

- A B C D A B C D
(a) 1 2 3 4 (b) 1 3 2 4
(c) 4 3 2 1 (d) 4 2 3 1

61. Arrange the following tiger reserves of India from North to South :

1. Indravati 2. Dudhwa
3. Bandipur 4. Simlipal

Select the correct answer using the code given below :

- (a) 3 - 4 - 1 - 2 (b) 4 - 2 - 3 - 1
(c) 2 - 4 - 1 - 3 (d) 2 - 1 - 4 - 3

62. Match List-I and List-II and select the correct answer using the codes given below :

List-I (Product)	List-II (Major Producer)
A. Tea	1. Andhra Pradesh
B. Jute	2. Kerala
C. Rubber	3. Orissa
D. Tobacco	4. Tamil Nadu

Codes :

	A	B	C	D
(a)	2	3	4	1
(b)	4	1	2	3
(c)	2	1	4	3
(d)	4	3	2	1

63. Match List-I with List-II and select the correct answer using the code given below the Lists :

	List-I		List-II
	(Type of vegetation)		(State)
A.	Mangrove	1	Madhya Pradesh
B.	Scrub	2	Karnataka
C.	Teak	3	Rajasthan
D.	Coniferous	4	Arunachal Pradesh

Code:

	A	B	C	D
(a)	4	1	3	2
(b)	2	1	3	4
(c)	4	3	1	2
(d)	2	3	1	4

64. Which one of the following pairs is correctly matched?

(a) Teak	:	Jammu and Kashmir
(b) Deodar	:	Madhya Pradesh
(c) Sandalwood	:	Kerala
(d) Sundari	:	West Bengal

65. Match List-I (Mangrove) with List-II (State) and select the correct answer using the codes given below the lists:

List-I (Mangrove)	List-II (State)
A. Achra Ratnagiri	1. Karnataka
B. Coondapur	2. Kerala
C. Pichavaram	3. Andhra Pradesh
D. Vembanad	4. Maharashtra
	5. Tamil Nadu

Codes:

- (a) A-2; B-1; C-5; D-4 (b) A-4; B-5; C-3; D-2
(c) A-2; B-5; C-3; D-4 (d) A-4; B-1; C-5; D-2

66. Match List I (National Park/Sanctuary) with List II (State) and select the correct answer using the codes given below:

List-I	List-II
A. Kanger Ghati National Park	1. Chhattisgarh
B. Nagerhole National Park	2. Haryana
C. Kugti Wildlife Sanctuary	3. Himachal Pradesh
D. Sultanpur Bird Sanctuary	4. Karnataka

Codes :

- (a) A-3; B-2; C-1; D-4
(b) A-1; B-4; C-3; D-2
(c) A-3; B-4; C-1; D-2
(d) A-1; B-2; C-3; D-4

67. Match List-I (Biosphere Reserve) with List-II (States) and select the correct answer using the codes given below:

List-I

- A. Similipal
B. Dehong Deband
C. Nokrek
D. Kanchenjunga

List-II

1. Sikkim
2. Uttaranchal
3. Arunachal Pradesh
4. Orissa
5. Meghalaya

Codes:

- (a) A-1; B-3; C-5; D-4
(b) A-4; B-5; C -2; D-1
(c) A-1; B-5; C-2; D-4
(d) A-4; B-3; C-5; D-1

68. Match List-I with List-II and select the correct answer using the codes given below the lists:

List-I (National Park/Wildlife Sanctuary)

- A. Bondla Wildlife Sanctuary
B. Kangerghat National Park
C. Orang Sanctuary
D. Ushkothi Wildlife Sanctuary

List-II (State)

1. Orissa
2. Assam
3. Chhattisgarh
4. Goa
5. Tripura

Codes :

- (a) A-2; B-1; C-5; D-3 (b) A-4; B-3; C-2; D-1
(c) A-2; B-3; C-5; D-1 (d) A-4; B-1; C-2; D-3

69. Match List-I (National Park/Wildlife Sanctuary) with List-I (Nearby Town) and select the correct answer using the codes given below the lists:

List-I**(National Park/Wildlife Sanctuary)**

- A. Chandra Prabha
B. Karera
C. Jaisamand
D. Nahargarh

List-II**(Nearby Town)**

1. Jaipur
2. Jhansi
3. Agra
4. Varanasi
5. Udaipur

Codes :

- (a) A-4; B-4; C-1; D-1 (b) A-5; B-2; C-3; D-1
(c) A-4; B-2; C-5; D-1 (d) A-5; B-1; C-3; D-2

70. Which of the following pairs of states and their important crops are correctly matched?

1. Kerala-Tapioca 2. Maharashtra-Cotton
3. West Bengal-Jute 4. Gujarat-Groundnut
(a) 1, 2 and 3 (b) 1, 2 and 4
(c) 1, 3 and 4 (d) 2, 3 and 4

71. Match List-I with List-II and select the correct answer:

**List-I
(Crops)**

- A. Barley
B. Rice
C. Millets
D. Tea

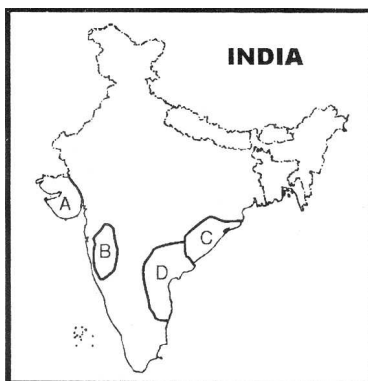
List-II**(Geographical conditions)**

1. Hot and dry climate with poor soil
2. Cool climate with poorer soil
3. Warm and moist climate with high attitude
4. Hot and moist climate with rich soil

Codes:

- (a) A - 2; B - 4; C - 1; D - 3
(b) A - 3; B - 4; C - 1; D - 2
(c) A - 2; B - 1; C - 4; D - 3
(d) A - 3; B - 2; C - 4; D - 1

72. Consider the map given below:



The place marked A, B, C and D in the map are respectively known for the cultivation of :

- groundnut, ragi, tobacco and sugarcane
- groundnut, sugarcane, ragi and tobacco
- ragi, sugarcane, groundnut and tobacco
- ragi, groundnut, sugarcane and tobacco

73. Match List-I with List-II and select the correct answer using the codes given below the lists:

List-I (Agricultural products)	List-II (Foremost producer)
A. Cotton	1. Madhya Pradesh
B. Gram	2. Gujarat
C. Black pepper	3. West Bengal
D. Pineapple	4. Kerala

Codes:

- A-2; B-1; C- 4; D-3
- A-2; B-1; C- 3; D-4
- A-1; B-2; C- 4; D-3
- A-1; B-2; C- 3; D-4

74. Match List I with List II and select the correct answer using the codes given below the lists:

List-I	List-II
A. Cotton	1. Rainfall 1000–1500 mm; Temperature 40°–60°C
B. Flax	2. Rainfall 1500–2000 mm; Temperature 25°–35°C
C. Sugar beet	3. Rainfall 600–800 mm; Temperature 5°–18° C
D. Jute	4. Rainfall 500–1000 mm; Temperature 18°–22°C
	5. Rainfall 500–600 mm; Temperature 18°–22°C

Codes:

- A-1; B-3; C-4; D-2
- A-2; B-3; C-5; D-4
- A-4; B-5; C-2; D-1
- A-4; B-3; C-5; D-2

75. Consider the following pairs:

National Park	River flowing through the Park
---------------	--------------------------------

- Corbett National Park : Ganga
- Kaziranga National Park : Manas
- Silent Valley: National Park : Kaveri

Which of the above pairs is/are correctly matched?

- 1 and 2
- 3 only
- 1 and 3
- None of these

76. Consider the following pairs : [CSAT 2015 - I]

Programme/Project	Ministry
1. Drought-Prone Area Programme	: Ministry of Agriculture
2. Desert Development	: Ministry of Environment and Forests
3. National Watershed Development Project for Rainfed Areas	: Ministry of Rural Development

Which of the above pairs is/ are correctly matched?

- 1 and 2 only
- 3 only
- 1, 2 and 3
- None

77. Consider the following pairs :

Wetlands	Confluence of rivers
1. Harike Wetlands	: Confluence of Beas and Satluj/Sutlej
2. Keoladeo Ghana National Park	: Confluence of Banas and Chambal
3. Kolleru Lake	: Confluence of Musi and Krishna

Which of the above pairs is/ are correctly matched?

- 1 only
- 2 and 3 only
- 1 and 3 only
- 1, 2 and 3

78. Consider the following pairs :

Hills	Region
1. Cardamom Hills	: Coromandel Coast
2. Kaimur Hills	: Konkan Coast
3. Mahadeo Hills	: Central India
4. Mikir Hills	: North-East India

Which of the above pairs are correctly matched?

- 1 and 2
- 2 and 3
- 3 and 4
- 2 and 4

Hints and Explanations

EXERCISE-1

1. (d) Tropical rain forest temperature gets higher than 34°C and drops below 20°C and average rain fall 50–260 inches yearly.
2. (c) The low and deltaic plains of the Ganges and Brahmaputra rivers that is characterised by swamps and Sundarbans. So, the people out there grow Paddy and Jute.
3. (b) Unless the salts and minerals are washed down into the groundwater, the sodium and other ions are absorbed by the colloidal clay particles. It leads to the deflocculating of the particles and the soil becomes structure less and impermeable to water.
4. (a) 5. (a) 6. (d) 7. (d) 8. (c)
9. (b) When it rains, the water carries the soil into cracks and thus to the lower layers; it has the same effect as tilling.
10. (c) 11. (c) 12. (d) 13. (a) 14. (a)
15. (c) 16. (d) 17. (a) 18. (a) 19. (c)
20. (c)
21. (b) Black soil becomes sticky when wet and develops cracks on drying. When it rains, top soil is carried into the cracks with the rain water. Thus, the soil of the upper layers is continuously taken to lower horizons, with the same effect as tilling.
22. (d)
23. (d) Great Nicobar, Sundarbans, Nanda Devi are biosphere reservoir declared by Govt. of India, but Gulf of Kachchh is not a biosphere reservoir, it is dry sandy plain.
24. (c) Western Ghat is known as ecological hot spots in India by the Wild Life Act of 1972 of Indian Constitution.
25. (b) 26. (d) 27. (a)
28. (d) First three soils are formed by the silt brought up by the river.
29. (c) The Deccan Traps are a large igneous province, one of the largest volcanic features on Earth. It is located on the Deccan Plateau of west-central India.
30. (c)
31. (b) Contour Bunding is one of the simple method of soil and water conservation. This technique is used at places where the land is sloppy. Due to slope, soil and nutrients erode fast which makes agriculture on this land very uneconomical. To adopt this technique the agriculture fields contours are marked and then the bunds are taken along the contours.
32. (b) Guar gum is used in hydraulic fracturing technology during shale gas extraction.
33. (b) The Eastern Highlands moist deciduous forests is a tropical moist broadleaf forest of east-central India. It covers an area of 341,100 square kilometers extending across portions of Andhra Pradesh, Chhattisgarh, Jharkhand, Madhya Pradesh, Maharashtra, Orissa, and Telangana states.
34. (b) The National Forestry Action Plan aims to bring 33% of land area in India under tree cover by 2012.
35. (c) Mixed farming is an agrarian system that mixes arable farming with the raising of livestock. When on a farm along-with crop production, some other agriculture based practice like poultry, dairy farming or bee keeping etc. is adopted then this is known as mixed farming.
36. (a) According to 2011 Forest Survey of India. Mizoram has third highest total forest cover with 1,594,000 hectares and highest percentage area (90.68%) covered by forests, among the states of India.
37. (c) Ranganathittu Bird Sanctuary is located in Karnataka.
38. (a) Forest Survey of India (FSI) is an organisation under the Ministry of Environment & Forests. Government of India conducts survey and assessment of forest resources in the country. The organization came into being in June, 1981. The headquarters of FSI are at Dehradun.
39. (b) Kerala is the largest producer of pepper in India. Kerala accounts for 97.4 per cent of the total area under the crop in the country.
40. (a) The shaded area in the map is major cotton producer states in India. Cotton producing states in India are Gujarat, Maharashtra, Andhra Pradesh, Karnataka, Tamil Nadu etc.
41. (c) Sorghum is a genus of plants in the grass family. One species is grown for grain and many of which are used as fodder plants. Millets are a group of highly variable small-seeded grasses, widely grown around the world as cereal crops or grains for fodder and human food. Thus, Sorghum is categorized as millet.
42. (a) In agriculture, leaching refers to the loss of water-soluble plant nutrients from the soil, due to rain and irrigation. The rate of leaching increases with the amount of rainfall, high temperatures, and the removal of protective vegetation. In areas of extensive leaching, many plant nutrients are lost, leaving quartz and hydroxides of iron, manganese, and aluminium. This remainder forms a distinctive type of soil, called laterite.
43. (b) The trees in Monsoon Forests shed their leaves for about six to eight weeks in summer on account of a long dry spell. The rainfall suited for such forests is between 75 to 200 cm. Sal is of the dry deciduous variety of tropical monsoon forest.

44. (a) 1. Dudwa National park - Uttar Pradesh
2. Kanha National Park- Madhya Pradesh
3. Indravati National Park - Chattisgarh
4. Bandipur National Park- Karnataka
45. (d) An endemic species is one whose habitat is restricted to a particular area. The term could refer to an animal, a plant, a fungus, or even a microorganism. Some of the endemic species in India are Grey-headed Bulbul, Malabar Lark, Nilgiri Flycatcher and Grey Jungle fowl etc.
46. (c) Red soils develop in a warm, temperate, wet climate under deciduous forests and have thin organic and mineral layers overlying a yellowish-brown leached layer resting on red layer made of iron oxide(ferric oxide). Red soils generally form from iron-rich sediments. They are usually poor growing soils, low in nutrients and humus and difficult to cultivate.
47. (a) Subsistence Farming is a type of farming in which most of the produce (subsistence crop) is consumed by the farmer and his family, leaving little or nothing to be marketed.
48. (a)
49. (c) In India, tea growing areas are Darjeeling in West Bengal, the Nilgiri in the South Wynaad, Cochin and Malabar in Kerala.
50. (c) Tropical dry deciduous forests run from the Himalayan foot hills to Kanyakumari and comprise important trees like bijasal, teak, tendu, amaltas, khair, palas, rosewood and axlewood.
51. (b) For proper ecological balance 33% of forest land is recommended, but in India we have only 20.14% of forest coverage.
52. (c) Gulf of Kutch in 1980, 270 km from Obha to Sadiya.
53. (c) Haryana has 6.83% of area is forest cover area of the total land coverage, whereas Sikkim has 36%, 38.5% in Goa and 28.9% of Kerala has forest area of their total land mass.
54. (b) Nallamalai is not a biosphere reserve. It is hill of Eastern ghats which stretches over Kurnool, Mahabubnagar, Guntur and Kadapa districts of the state of Andhra Pradesh.
55. (d) Indira Gandhi Wildlife Sanctuary and National Park is known as top slip. It is a part of Western Ghats is located above 300 metres from the sea level on the Anamalai mountain ranges.
56. (b) Indravati National Park is located in the Bastar region. The park is situated at the distance of 97.4 km Bastar.
57. (b) Mahogany is a species of tree found throughout the Caribbean, central and south America but not in Himalayas.
58. (a) Andaman and Nicobar has 9 national parks; Assam has 6, Arunachal Pradesh and Maghalaya both have 2 each.
59. (c) Red soil in India is largely found in Deccan plateau. Red soil is less clayey and sandier in nature and has a rich content of iron and small amount of humus. Red soil is also known as yellow soil. The redness of the soil is due to Ferric oxide, is 2nd largest soil found in India from south of Bundelkhand to North of Raj mahal hill.
60. (d) Contour Bunding is one of the simple method of soil and water conservation. This technique is used at places where the land is sloppy. Due to slope, soil and nutrients erode fast which makes agriculture on this land very uneconomical. To adopt this technique the agriculture fields contours are marked and then the bunds are taken along the contours.
61. (a) Wheat is a temperate region crop thus require moderate temperature and rainfall both. Temperature should not exceed above 20°C and rainfall should be between 50-100 cm alongwith a first free growing season.
62. (a) Organic farming is a production system of crops which avoids the use of synthetic and chemical inputs like fertilizers, pesticides, growth regulators and livestock feed additives.
63. (a) The genetic engineering approval committee comes under Ministry of Environment, Forest and Climate change.
64. (b) National Food Security Mission launched by government of India to increase the food production of rice, wheat and pulses and commercial crops like sugarcane jute. The target is to increase the production of rice by 10 million tones, wheat by 8 million tones and pulses, by 2 million tones by the end of Eleventh Plan.
65. (b)
66. (c) Mixed farming refer to rearing of animals and cultivation of crops together.
67. (d) Andaman and Nicobar Islands is rich in biodiversity. Different types of forests as Tropical Evergreen forests, Moist Deciduous forests, Mangrove forests, Littoral forests are found here.
68. (b) Seed village concept is to promote the quality seed production of foundation and certified seed classes. The area which is suitable for raising a particular crop will be selected, and raised with single variety of a kind. Suitable area for seed production will be identified by the Scientists. The foundation/ certified seeds or University labelled seeds will be supplied by the University through Krishi Vigyan Kendras (KVKs) and Research Stations at 50% subsidy cost to the identified farmers in the area. The farmers will use these quality seeds and take up their own seed production in a small area (1 acre) for their own use. The crops are Rice, Pulses and Oilseeds.

69. (c) Avoidance of flood and drought prone areas for agriculture would not be sustainable in this respect.
70. (c) Kanha National Park belongs to tropical moist dry deciduous forest. It is a tiger reserve of India and the largest national park of Madhya Pradesh.
71. (a) The sequential order of vegetation types observed while moving from Assam valley to Rajasthan is : Tropical wet evergreen ® Tropical moist deciduous Teropical dry deciduous ® Tropical thorn forest.
72. (d) Production of rice is 96.43 MT, wheat is 78.4 MT, pulses is 15.11 MT and coarse cereals are 40.73 MT in India. So, the correct sequence in decreasing order is Rice – Wheat – Coarse cereals – Pulses.
73. (b) Guar gum is used in hydraulic fracturing technology during shale gas extraction.
74. (b) Agricultural Produce Market Committee Acts of respective states are responsible for markets in agricultural products.
75. (c) 'Agreement on Agriculture', 'Agreement on the Application of Sanitary and Phytosanitary Measures' and 'Peace Clause' are related to World Trade Organisation.
18. (b) The predominant soil in Tamil Nadu is red which supports cotton cultivation and a rich pool of skilled labour is available in Tamil Nadu.
19. (d) The canopy is the primary layer of the forest forming a roof over the two remaining layers. The densest of the biodiversity is found here along with a large variety of epiphytes.
20. (d) Through Corbett National Park Ramganga flows (not Ganga) which is a tributary of Ganges. Through Silent Valley National Park river Bhavani flows which is a tributary of Kaveri. Kaziranga and Manas are both national parks.
21. (a) The land slopes west ward from central India due to deposition of alluvial fan. The rift valley which is occupied by the Narmada is one of the reasons of Narmada flowing to the west. The rift is formed due to complex natural processes leading to either subsidence or upliftment of the earth's surface. The area is seismically active and has seen major upheavals. Such movements reshape the landscape. There is readjustment of slopes and rivers either start bringing more material or change course.
22. (c) Laterite soil is rusty red in colour due to iron oxide present in it. In the lateritic soil cashews and tapiocas can be grown.

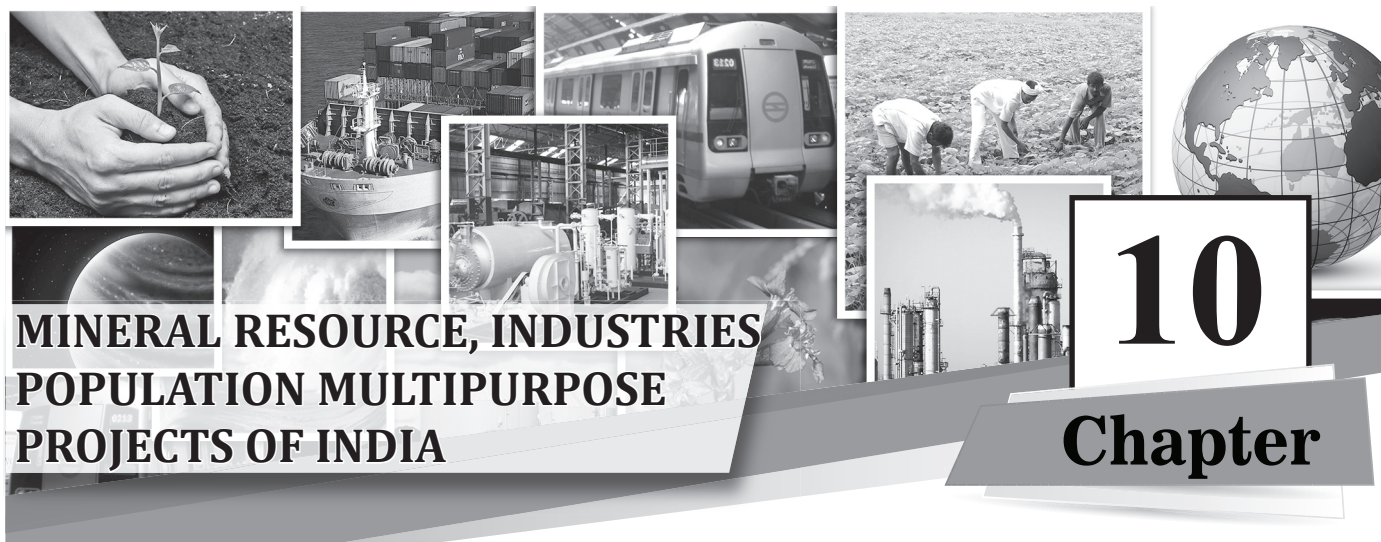
EXERCISE-2

1. (d) Biodiversity hot spots are found in temperature regions only. Eastern Himalayas, Western Ghats and Andaman Islands are such hot spots.
2. (c)
3. (d) Silent Valley National Park is situated in Nilgiri Hills of Western Ghats. The park is bound by Attappadi reserved forest to the east and vested forest of Palaghat division and Nilamber division to the south-west respectively.
4. (b)
5. (d) All are used as pulse, fodder and green manure.
6. (a) Five varieties of silkworms are reared in India for producing five varieties of silk. Mulberry silk, Muga silk, Tasar silk, Oak tasar silk and Eri silk. India is the 2nd largest producer of sugar in the world.
7. (a) Both tea and coffee can be propagated by seeds and stem cutting, so (b) is wrong.
8. (b) 9. (c) 10. (b) 11. (b) 12. (b)
13. (c)
14. (d) National Forest Policy 1988 describes the protection of forest and development of forest, it does not describe the use of insecticide and pesticide in forest areas.
15. (b) All the statements given in the options are correct.
16. (a)
17. (c) The Mediterranean region has deciduous forests of thick bark and thick leaves. Because it get high sun rays and much rainfall.
23. (a) Oak and Rhododendron is found in Himalayan region while sandalwood is found in South India.
24. (c) Rice is the main kharif crop and groundnut and cotton are also the kharif crops.
25. (d) All 1,2 and 3 are true.
26. (d) All three statements are correct as Maize can be used for the production of starch. Oil extracted from maize can be a feedstock for biodiesel. Alcoholic beverages can be produced by using maize.
27. (b) NPK (fertilizers) can be applied at the rate of 112 kg, 25 kg and 48 kg per acre, respectively through inorganic or organic methods. Hence statement #3 is wrong, this eliminates (a), (c) and (d). therefore, answer (b) 1,2,4 only.
28. (c) Integrated Watershed Management Programme (IWMP) was launched during 2009-10. It aims at Prevention of soil runoff, Rainwater harvesting and recharge of groundwater table and Regeneration of natural vegetation.
29. (c) National Park an area of countryside, or occasionally sea or fresh water, protected by the state for the enjoyment of the general public or the preservation of wildlife.
An animal sanctuary is a facility where animals are brought to live and be protected for the rest of their lives.
A biosphere reserve is an area of land or water that is protected by law in order to support the conservation of ecosystems, as well as the sustainability of mankind's impact on the environme

30. (c) Nanda Devi Biosphere Reserve is located in Uttarakhand. It was inscribed a World Heritage Site by UNESCO in 1988.
31. (a) Black soils become sticky when it is wet. It is rich in iron, lime, calcium, Magnesium, carbonates, and alumina and poor in Phosphorous, Nitrogen and Organic matter. The soil is black in colour because it is volcanic in origin. It is created from igneous rocks, and is also called 'regur soil'.
32. (b) Red soils are very poor in lime, humous and potash.
33. (c) Jhum is a traditional form of shifting cultivation common in the North-East. Slash-and-burn is a mere land clearing method used by many people around the globe to open up forest land and use it for permanent agriculture. On the contrary, shifting cultivation is an integrated farming system involving forestry, agriculture and strong social organisation on the part of the communities. The village or group of villages practice jhum on one particular tract of land until the soil is depleted of nutrients and then move on to another allowing the former tract of land to regenerate.
34. (d) India has the world's largest cropped and irrigated area. The cropping pattern of foodgrains in India is dominated by cereal crop. The average size of an Indian farm holding is below 2 hectares which constitute almost 80% of all Indian farmers.
35. (b) Out of the total forest cover, dense forest constitute around 2.54%; very dense forest and 8.77% are moderately dense forest. National Forestry Action Programme aims a long term plan to achieve the target of 33% forest cover.
36. (c) The fix boundary of a National Park is described in the Wildlife Protection Act, 1972. The legislation states the actual area of the National Park which is notified by the state government. A biosphere reserve conserves an ecosystem and not just few specific species of plants and animals.
37. (c)
38. (c) Laterite soil is rusty red in colour due to iron oxide present in it. In the lateritic soil cashews and tapiocas can be grown.
39. (c)
40. (c) Arjun and Sonalika are hybrid variety of wheat produced by Indian Council of Agricultural Research.
41. (c) India is the original home of cotton plant, and India ranks number one in the world accounting for 20% of the total area planted under cotton. The cotton hybrid variety H-4 developed first and Raj-16, Dhanalaxmi, and Fateh are the variety which were developed in India.
42. (d) India ranked 2nd in the fruit production in the world and ranked 6th in production of tobacco.
43. (b) Cotton, Groundnut and Maize are kharif crop whereas Mustard is a rabi crop.
44. (a) Five varieties of silkworms are reared in India for producing five varieties of silk. Mulberry silk, Muga silk, Tasar silk, Oak tasar silk and Eri silk. India is the 2nd largest producer of sugar in the world.
45. (c)
46. (d) All are used as pulse, fodder and green manure ?
47. (d) All of them are predominantly rain fed crops
48. (c) Rice is the main kharif crop and groundnut and cotton are also the kharif crops wheat is a rabi.
49. (d) All 1,2 and 3 are true.
50. (d) Kinnaur (Himachal) is famous for apples. Areca nut mostly confined to Karnataka, Kerala and Assam. Mewat in haryana is not famous for mango. UP is famous for mango. Similarly, coromandel coast is not famous for soya beans. Top two producers of soya are Madhya Pradesh and Maharashtra which are outside coromandel coast. Therefore, (d) is the correct option.
51. (d) All three statements are correct as Maize can be used for the production of starch. Oil extracted from maize can be a feedstock for biodiesel. Alcoholic beverages can be produced by using maize.
52. (b) NPK (fertilizers) can be applied at the rate of 112 kg, 25 kg and 48 kg per acre, respectively through inorganic or organic methods. Hence statement #3 is wrong, this eliminates (a), (c) and (d). therefore, answer (b) 1,2,4 only.
53. (c) Excessive/ inappropriate use of nitrogenous fertilizers increase the acidity of soil and Leaching of nitrate to the ground water.
54. (c) Tropical Wet Evergreen Forests are found in Assam, Arunachal Pradesh, Meghalaya, Nagaland, Tripura, West Bengal and Andaman and Nicobar Island and on the eastern and western slopes of the Western Ghats in such states as Tamil Nadu, Karnataka, Kerala and Maharashtra.
55. (d) Cotton grows there
56. (d) Kinnaur (Himachal) is famous for apples. Areca nut mostly confined to Karnataka, Kerala and Assam. Mewat in haryana is not famous for mango. UP is famous for mango. Similarly, coromandel coast is not famous for soya beans. Top two producers of soya are Madhya Pradesh and Maharashtra which are outside coromandel coast. Therefore, (d) is the correct option.
57. (d) The draught prone area programme is by ministry of Rural development to minimise the adverse effects of drought on production of crops and livestock and productivity of land, water and human resources. The

- desert development programme is also by ministry of Rural development to minimise the adverse effect of drought and control desertification. National Watershed Development Project for Rainfed Areas is by ministry of agriculture.
58. (c) A. Tropical evergreen forests- Above 200 cm
B. Tropical deciduous forests - 100-200 cm
C. Tropical Dry forest- 50-100 cm
D. Arid forest- Less than 50 cm
59. (d) 1. Manas- Asom
2. Pachmarhi - Madhya Pradesh
3. Nokrek - Meghalaya
4. Achanakmar-Amarkantak- Chhattisgarh,
60. (c) Tiger Reserves States
1. Indravati Tiger Reserve - Chhattisgarh
2. Periyar Tiger Reserve - Kerala,
3. Simlipal Tiger Reserve - Odisha
4. Bandipur Tiger Reserve - Karnataka
61. (c) The correct arrangement from north to south is as follows:
Dudhwa- Uttar Pradesh
Simlipal - Odisha
Indravati - Chattisgarh
Bandipur- Karnataka
62. (d) Chief Producer States
Tea — Assam, West Bengal, Tamil Nadu
Jute — West Bengal, Bihar, Assam, Orissa
Rubber — Kerala, Tamil Nadu, Karnataka
Tobacco — Andhra Pradesh, Gujarat, Karnataka, Tamil Nadu
63. (c) Mangrove, Scrub, teak and coniferous are the various types of vegetations found in Karnataka, Rajasthan, Madhya Pradesh and Arunachal Pradesh respectively.
64. (d) Sundari tree is found in Sundarban area of West Bengal. Sundarban is a largest wetland area in the world. Sundarban got its name itself due to abundance of 'Sundari' tree in this area.
65. (d) Achra Ratnagiri Mangrove is in Maharashtra, Coondapur Mangrove is in Karnataka, Pichavaram Mangrove is in Tamil Nadu and Vembanad Mangrove is in Kerala.
66. (b) National Park/ Sanctuary
A. Kanger Ghati National Park - Chhattisgarh
B. Nagerhole National Park - Karnataka
C. Kugti Wildlife Sanctuary - Himachal Pradesh
D. Sultanpur Bird Sanctuary - Haryana
67. (d) Similipal biosphere reserve is in Orissa. Dehong Deband biosphere reserve is in Arunachal Pradesh, Nokrek biosphere reserve is in Meghalaya and Kanchenjunga biosphere reserve is situated in Sikkim.
68. (b) National Park/ Wildlife Sanctuary
Bondla Wildlife Sanctuary - Goa
Kangerghat National Park - Chhattisgarh
Orang Sanctuary - Assam
Ushkothi Wildlife Sanctuary - Orissa.
69. (c)
70. (d) Option 1 is not correct because, Tapioca is not an important crop of Kerala.
71. (a) Barley requires cool climate with poor soil quality. Barley can be grown in high latitude even beyond the Arctic circle. Whereas rice needs hot and moist climate with rich soil. Millets are grown in hot and dry climate with poor soil, whereas tea needs warm and moist climate with high altitude.
72. (b) The place marked in the given map of India. 'A' refers to Gujarat where groundnut is a major crop. 'B' refers to western part of Maharashtra which is known for sugarcane. 'C' refers to southern part of Orissa and northern part of Andhra Pradesh, where 'Ragi' is cultivated abundantly. 'D' refers Andhra Pradesh and tobacco is cultivated in this area.
73. (a) Gujarat is the foremost producer of cotton. Gram is produced in Madhya Pradesh, Black pepper is produced in Kerala and Pineapple is produced highest in West Bengal.
74. (b) A. Cotton cultivation needs 1500–2000 mm and the suitable temperature for cotton growth is 25° – 35°C.
B. Flax cultivation needs 600–800 mm and temperature varies from 5° – 18° C.
C. The suitable climatic condition for sugar beet requires 500–600 mm and temperature 18°–22° C.
D. Jute cultivation needs 500–1000 mm rainfall and temperature between 18°–22°C
75. (d) Through Corbett National Park Ramganga flows (not Ganga) which is a tributary of Ganges. Through Silent Valley National Park river Bhavani flows which is a tributary of Kaveri. Kaziranga and Manas are both national parks.
76. (d) The draught prone area programme is by ministry of Rural development to minimise the adverse effects of drought on production of crops and livestock and productivity of land, water and human resources. The desert development programme is also by ministry of Rural development to minimise the adverse effect of drought and control desertification. National Watershed Development Project for Rainfed Areas is by ministry of agriculture.

77. (a) Harike Wetlands is at Confluence of Beas and Satluj/Sutlej. The Keoladeo National Park formerly known as the Bharatpur Bird Sanctuary in Bharatpur is at the confluence of two rivers, the Gambhir and Banganga. Kolleru Lake is one of the largest freshwater lakes in India located in state of Andhra Pradesh. Kolleru is located between Krishna and Godavari delta.
78. (c) The Cardamom Hills are southern hills of India and part of the southern Western Ghats located in southeast Kerala and southwest Tamil Nadu. They are not in coromandel coast. Kaimur Range is the eastern portion of the Vindhya Range extending from Madhya Pradesh to Bihar. They are not in konkan coast. The Mahadeo Hills are in Madhya Pradesh, state of central India. Mikir hills are in assam i.e. in North East India.



MINERAL RESOURCE, INDUSTRIES POPULATION MULTIPURPOSE PROJECTS OF INDIA

10 Chapter

MINERAL RESOURCE OF INDIA

Minerals are the natural resources which are used in many industries as raw materials. Iron ore, manganese, bauxite, copper, etc. are such minerals.

Minerals are of two types: **metallic** and **non-metallic**. Iron ore and copper are metallic minerals while limestone and dolomite are non-metallic minerals.

Metallic minerals are further sub-divided into ferrous and **non-ferrous minerals**. Those metallic minerals which have iron content belong to ferrous group. The metallic minerals belonging to non-ferrous group do not have iron content.

India is rich in iron, mica, manganese, bauxite; self sufficient

in antimony, building materials, cement materials, clay, chromite, lime, dolomite, and gold, but deficient in copper, lead, mercury, zinc, tin, nickel, petroleum products, rock phosphate, sulphur, and tungsten.

Mineral resources like potassium are totally absent and have to be imported. Minerals like crude petroleum (which accounts for about 80 per cent of the total value of Indian imports) diamonds (uncut), sulphur, and rock phosphorus are imported. The state with the highest mineral output is Jharkhand. India is rich in ferrous metals but its reserves of non-ferrous metals are poor.

Mineral	Ore	Found in	Features
Iron	Magnetite —the best quality of iron ore and contains 72% pure iron. Haematite —contains 60 to 70% pure iron. Limonite —contains 40 to 60% pure iron. Siderite —contains many impurities and has just 40 to 50% pure iron.	Odisha (Sonai, Mayubhanj, Keonjhar), Jharkhand and Bihar (Singhbhum Hazaribagh, Palamau, Shahbad), Chhattisgarh and Madhya Pradesh (Raipur, Durg, Bastar, Raigarh, Bilaspur, Jabalpur, Balaghat), Andhra Pradesh (Krishna, Kurnool, Chittor, Cuddapha, Warangal, Guntur), Tamil Nadu (Salem, Tiruchirapalli), Karnataka (Ballary, Chitradurg, Chikmagalur), Maharashtra (Ratnagiri, Chanda), Goa	India has the world's largest reserves, approximately one-fourth of world's known reserves; Jharkhand has the largest reserves accounting for about 25% of the total reserves of iron ore in India.
Coal (Black Gold)	Anthracite Coal —the best quality of coal and contains 80 to 95% carbon. It is found only in Jammu and Kashmir in small quantity. Bituminous coal —The most widely used coal and contains 40 to 80% carbon. It is found in Jharkhand, Orissa, West Bengal, Chhattisgarh and Madhya Pradesh. Lignite — Also known as brown coal. It is a lower grade coal and contains about 40 to	Bihar-Jharkhand-Bengal belt (Raniganj, Jharia, Giridih, Bokaro, Karanpur), Madhya Pradesh and Chhattisgarh belt (Singrauli, Korba, Raigarh, Sonhat, Sohagpur, Umaria), Odisha (Desgarh, Talcher), Maharashtra (Chand), Andhra Pradesh (Singreni), Assam (Makum, Lakhimpur); in small quantities in Arunachal Pradesh, Meghalaya, Jammu and Kashmir, and Nagaland	“About one-fourth of India's coal reserves lie in the Damodar Valley, across Bihar, Jharkhand, and West Bengal. India is the fourth largest coal producing country in the world according to 1992 coal production in the country.”

	50% carbon. It is found in Palna of Rajasthan, Neyveli of Tamil Nadu, Lakhimpur of Assam and Karewa of Jammu and Kashmir. Peat —It is the first stage of transformation of wood into coal and contains less than 40% carbon.		
Manganese	India has the second largest manganese ore reserves in the world after Zimbabwe. India is the fifth largest producer in the world after Brazil, Gabon, South Africa and Australia.	Odisha (Keonjhar, Kalahandi, Mayurbhaj, Talcher) Madhya Pradesh (Balaghat, Seoni, Chhindwara, Jabalpur), Maharashtra (Nagpur, Bhandara, Ratnagiri), Gujarat (Panchmahal), Karnataka (Chitradurg, Tumkur, Shimoga, Chikmagalur, Belgaum, North Canara, Dharwar), Jharkhand (Singbhum), Andhra Pradesh (Visakhapatnam), Rajasthan (Udaipur, Bansawara)	Orissa is the leading producer of manganese in the country. India ranks third in world in manganese production.
Mica	The three major types of mica found in India are – Muscovite, Phlogopite and Biotite.	Bihar (Gaya Monghyr), Jharkhand (Hazaribagh), Rajasthan (Ajmer, Shahpur, Tonk, Bhilwara, Jaipur), Andhra Pradesh (Nellore)	India has largest deposits of mica in world India alone contributes about two-thirds of the world's production
Bauxite (aluminium ore)		Jharkhand (Palamu), Gujarat (Kaira), Madhya Pradesh (Katni, Jabalpur, Balaghat, Bilaspur, Bastar), Tamil Nadu (Salem), Karnataka (Chitradurg, Belgaum), Maharashtra (Kolhapur), Jammu and Kashmir (Kotli)	Third largest producer in the world.
Copper	India contributes to about 3.5 to 4% of the world's total production of copper.	Jharkhand (Singbhum, Hazaribagh), Rajasthan (Khetri, Alwar, Bhilwara, Jhunjhunu, Sirohi), Andhra Pradesh (Guntur, Khamman, Agnigundala), Karnataka (Chitradurg, Hassan, Chikmagalur, Raichur), Madhya Pradesh (Balaghat), Gujarat (Banaskantha); some quantities also found in Sikkim, Punjab, Uttar Pradesh, and Tamil Nadu.	Very meager reserves; almost all copper comes from Singbhum and Hazaribagh in Jharkhand and Khetri in Rajasthan.
Crude oil	51.08 crore tones	Assam, Tripura, Manipur, West Bengal, Ganga Valley, Himachal Pradesh, Kutch of West Bengal coast, Orissa, Andhra Pradesh, Maharashtra, and Gujarat.	
Lignite	429 crore tones	Tamil Nadu (Neyveli fields) Some deposits also found in Gujarat, Punducherry, Rajasthan (Palana fields), Jammu and Kashmir (Riasi fields).	Maximum deposits of about 383 crore tones, are found in Tamil Nadu.
Gold	India's contribution to gold production across the world is less than one percent (0.75%).	Karnataka (Kolar gold fields, Hutti Mines), in small quantities in Andhra Pradesh (Ramgiri gold fields and Anantpur).	Karnataka was the leading producer of gold accounting for 99% of the total production. The remaining production came from Jharkhand.
Magnesite	23.91 crore tones	Tamil Nadu (Salem), Uttranchal (Almora, Chamoli, Pithoragarh), Karnataka (Mysore, Hassan)	

INDUSTRY

Agro-Based Industry

This group of industries depend on the raw material produced by agricultural sector. The products comprise mostly of the consumer goods.

Cotton Textiles Industry

Cotton accounts for 70% of the total fabric produced. The first successful cotton textile mill was set up at Mumbai by a Parsi entrepreneur.

Geographical distribution

Mumbai, Ahmedabad, Surat, Solapur, Pune, Nagpur (Maharashtra and Gujarat). Coimbatore, Madurai and Chennai (Tamil Nadu), Ludhiana (Punjab), Bengaluru (Karnataka), Kolkata (West Bengal), Kanpur (Uttar Pradesh).

Woolen Textile Industry

- The first woolen textiles mill was set up in 1876 at Kanpur, because Kanpur was the principal depot for the British Indian Army.
- The woolen textiles industry in India is partly a cottage industry and partly, a factory industry.

Geographical spread

Kanpur (Uttar Pradesh), Dhariwal and Ludhiana (Punjab), Mumbai (Maharashtra), Bengaluru (Karnataka), Jamnagar (Gujarat), Srinagar (Jammu and Kashmir).

Jute Textiles Industry

- This jute mill was set up at Rishra near Calcutta in 1855.
- After independence, this sector made rapid progress as an export-oriented industry.

Geographical location

- Nearly 90% of the manufacturing capacity is located in a narrow belt about 100 km long and 3 km wide along river Hooghly.
- Recently slight dispersal of industries have been marked as the use of gunny bag has increased to many folds in sugar and cement industries. These industries are producing gunny bags using local fibres like mesta and Bimlipatlan as the raw material for production.
- West Bengal accounts for 84.1 per cent of the total jute manufactures of the country.
- Andhra Pradesh contributes another 10 percent of the production.
- India ranks number two in the export of jute goods in the world.

Sugar Industry

Indian sugar industry is the second largest agro-based industry in India.

Geographical distribution

Uttar Pradesh and Bihar alone account for 70% of the productive capacity and 75% of the total employment of 30 lakh.

- **Uttar Pradesh:** There are two belts one in western Uttar Pradesh and the other in eastern Uttar Pradesh. The western belt includes Meerut, Saharanpur, Muzaffarnagar, Bijnor and Moradabad, and the eastern belt includes Gorakhpur, Deoria, Basti and Gonda.
- **Bihar:** This is an extension of the eastern Uttar Pradesh belt, which includes Darbhanga, Saran, Champaran and Muzaffarpur.
- The reasons for concentration of sugar industry in Uttar Pradesh and Bihar are–
 - (i) fertile alluvial soil, rich in lime and potash;
 - (ii) level topography-suitable for irrigation;
 - (iii) abundant water for washing and processing;
 - (iv) sugar industry is relatively independent of coal and electricity, because bagasse is enough to run steam;
 - (v) densely populated market with excellent transport links;
 - (vi) availability of cheap labour;
 - (vii) cultivation is done in compact block, which ensures ready availability of fresh cane to factories.
- In Maharashtra, Nasik, Pune, Satara, Sangli, Kolhapur, Sholapur are the centres well integrated in the cooperative sector in terms of cultivation and sugar factories.
- In Punjab, Centres exist mainly in the eastern side, in Phagwara, Dhuri.
- In Karnataka, Munirabad, Shimoga and Mandya are the main centres.
- In Tamil Nadu, Nalikipuram, Pugulur, Coimbatore and Pandyarajpuram are famous for producing sugar.
- Andhra Pradesh. Nizamabad, Medak, west and east Godavari, Visakhapatnam and Chittoor produce sugar.
- In Odisha Bargarh and Rayagada in Odisha produce sugar.
- In Madhya Pradesh, Sehore is the sugar producing centre.

Differences in Sugar production in North India and Peninsular India.

1. Yields are higher in south India
2. The southern sugarcane, being of the tropical variety, has more sucrose content.
3. The crushing season is longer in the south, where it lasts from October to May-June. In the north, it lasts from November to February.

Silk Industry

- India is the only country producing all the five known commercial varieties of silk, viz. Mulberry, Tasar (Tropical), Oak Tasar, Eri and Muga.
- India is the second largest producer of raw silk, accounting for 20% of the world production, next to China.
- Karnataka is the foremost silk producing state in India, which accounts for 50% of Mulberry silk of the country.

- About 50% of India's silk cloth is also manufactured in Karnataka, followed by West Bengal (13% of country's total silk).

Metallurgical Industries

- These industries form the economic backbone of a developing country.

Iron and Steel Industry

- The first iron and steel unit on modern lines was established in 1830 at Porto Novo in Tamil Nadu.
- But the real beginning of modern iron and steel industry was made in 1907 when TISCO was set up at Sakchi, Jamshedpur.
- IISCO was set up in 1919 at Burnpur on the bank river Barakar.
- Mysore steel works at Bhadravati came into existence in 1923, produces alloy and pig iron.
- SAIL was established in 1973.
- India is eighth largest producer of steel in the world.
- The first on-shore steel plant in India was set up at Vishakhapatnam (Andhra Pradesh)
- India is the leading country in the production of sponge iron.
- Steel Authority of India Ltd (SAIL) has the highest sales followed by Tata Steel Ltd (2004-05).

Different Steel Plants and their location

TISCO (Jharkhand)

IISCO (West Bengal)

Visweswaraya Steel Plant (Karnataka)

Bhilai Steel Plant (Chhattisgarh)

Bokaro Steel Plant (Jharkhand)

Rourkela Steel Plant (Orissa)

Durgapur Steel Plant (West Bengal)

Aluminium Smelting Industry

- Aluminium is extracted from bauxite at Muri in Jharkhand.
- The reduction plant of alumina is located at Alupuram in Kerala and the fabrication plant at Belur manufactures aluminium sheets, rods, aluminium paste, electric conductors, domestic utensils etc.
- In order to achieve the economics of scale, a second smelter was set up at Hirakud.
- The Hindustan Aluminium Corporation Limited (HINDALCO) is located at Renukoot near Mirzapur in Uttar Pradesh and the Bharat Aluminium Company (BALCO) has set up two units at Korba and Ratnagiri, to utilize bauxite ores of Amarkantak in Madhya Pradesh and of Udaigiri Dhangarvadi region in Maharashtra, respectively.
- Integrated Aluminium plant of NALCO at Damanjodi near Jeypur in Koraput district of Orissa is the largest complex in India.
- NALCO, the lowest cost producer of aluminium in the world, at present ranked among the top 10 global companies and is the second to HINDALCO in India.

- Aluminium companies with the highest sales in descending order: HINDALCO, NALCO, INDAL, MALCO
- A huge public sector aluminium company, the National Aluminium Company (NALCO) was set up with assistance from a French company at Damanjodi near Jeypore (Koraput district in Odisha)

Engineering Industries

- These industries contribute about 10% of the total exports of the country.
- Heavy Engineering Corporation Ltd was set up at Ranchi (Jharkhand) in 1958.
- Drills for drilling holes in rocky areas are manufactured at Naroda (Ahmedabad).
- Kirloskar Brothers Ltd is the pioneer company in the manufacturing of machine tools.
- HMT is the largest manufacture of machine tools in the country.
- HMT's plants: Bengaluru, Pinjore, Kalamassery (Kerala), Hyderabad, Sri Nagar and Ajmer.
- Locomotives: Chittaranjan Locomotive Works, Diesel Locomotive Works (Varanasi), Tata Engineering and Locomotive Works (Jamshedpur) in 1964.
- BHEL, Bhopal has been developed to manufacture electric locomotive for the Indian railways.
- Wheel and Axle Plant was set up at Bengaluru in 1984.
- The Integral Coach Factory at Perampur near Chennai was set up in 1955 with Swiss collaboration.
- Rail Coach Factory at Kapurthala (Punjab) was set up in 1988.
- Bharat Movers Limited at Bengaluru also produces railways coaches.
- Most of wagons are produced in private sector. About 60% of wagons are produced in West Bengal.

Automobile Industry

- Automobile Industry contributes about 5% to GDP.
- Top four automobile companies with the highest sales:
 - Tata Motors
 - Maruti Udyog Ltd.
 - Mahindra & Mahindra Ltd.
 - Ashok Leyland Ltd.
- In medium and heavy commercial vehicles, TELCO produces over 70% of such vehicles in India.
- In passenger cars, Maruti Udyog Ltd is at the top position.
- Almost the entire production of jeeps comes from Mahindra, Mumbai.
- Indian has second largest market for two - wheelers in the world after China.

Fertilizer Industry

- The first super-phosphate factory was set up at Ranipet in Tamil Nadu in 1906.
- India is now the third largest producer of nitrogenous fertilizers in the world.
- Gujarat has more than 14% of the country's total installed capacity in India followed by Tamil Nadu (11%) and UP (9%).

Aircraft Industry

- The first aircraft industry was set up at Bengaluru in 1940 under the name of Hindustan Aircraft Ltd.
- Later, Hindustan Aircraft Ltd was merged into Aeronautics India Ltd in 1964 to form Hindustan Aeronautics Ltd.
- Different Divisions of HAL and production:
 - Nashik division – MIG airframe
 - Koraput division – Engine of MIG aircraft
 - Hyderabad division – Electronic equipment of MIG.
 - Transport aircrafts are manufactured at Kanpur.
- Recently, a factory was set up at Lucknow for producing equipment for aircraft.
- Fertilizer companies with the highest sales are – National Fertilizer Ltd, Tata Chemicals Ltd, Rashtriya Chemicals & Fertilizers Ltd (in descending order).
- India is the third largest producer and consumer of fertilizers in the world.

Cement Industry

- The Indian Cement industry is the second largest in the world after that of China.
- States with the largest cement capacity are Andhra Pradesh, Rajasthan, Madhya Pradesh and Gujarat (in descending order).

Glass Industry

- The main raw material for the industry is silica sand.
- Uttar Pradesh is the leading producer of glass in India followed by West Bengal and Maharashtra.
- Ferozabad in Agra district is the largest producer of glass, followed by Bahjoi, Naini, Hathras, Sasni and Allahabad in Uttar Pradesh.

Indian Towns Associated with Industries

Town	State	Industries
Ahmedabad	Gujarat	Cotton Textiles
Agra	U.P.	Leather, Marble, Carpet, glass
Aligarh	U.P.	Locks, Cultery
Ankleshwar	Gujarat	Oil Fields
Ambernath	Maharashtra	Machine Tools
Anand	Gujarat	Milk and its Products
Ambala	Haryana	Scientific Instruments
Bangalore	Karnataka	Telephones Aircrafts, Motors, Cotton, Textiles, Toys
Batanagar	West Bengal	Shoes
Barielly	Uttar Pradesh	Resin Industries, Match Factory
Bhilai	Chhattisgarh	Steel Plant
Bhandara	Maharashtra	Explosives
Bhadravati	Karnataka	Iron & Steel
Bhadohi	Uttar Pradesh	Carpets

Churk	Madhya Pradesh	Cement
Cyberabad	Andhra Pradesh	Electronics, Computers, Information Technology
Chitranjan	West Bengal	Locomotive
Kolkata	West Bengal	Jute, Leather, Electric goods
Cochin	Kerala	Coffee, Coconut
Digboi	Assam	Petroleum
Darjeeling	West Bengal	Tea
Frazabad	Madhya Pradesh	Bangal Works
Hardwar	Uttarkhand	Heavy electricals
Jamshedpur	Jharkhand	Iron & Steel, Locomotives, Railway Coaches
Jabalpur	Madhya Pradesh	Bidi Industry
Jainakot	Jammu & Kashmir	H.M.T. Watch
Kanpur	Uttar Pradesh	Cotton and Woollen mills, Leather, sugar
Katni	Madhya Pradesh	Cement
Koyna	Maharashtra	Aluminium factory
Kanchipuram	Tamil Nadu	Silk Clothes
Karnal	Haryana	Dairy product
Kandla	Gujarat	Chemical fertiliser, famous port
Khetri	Rajasthan	Copper Industries
Ludhiana	Punjab	Hosiery
Lucknow	Uttar Pradesh	Embroidery work, chicken work
Mathura	Uttar Pradesh	Oil refinery
Meerut	Uttar Pradesh	Publication work, sports goods, scissors making
Modinagar	Uttar Pradesh	Nylon thread
Nagpur	Maharashtra	Cotton mills
Nepanagar	Madhya Pradesh	Newsprint
Panna	Madhya Pradesh	Diamond Mining
Pinjore	Haryana	Hindustan Machines Tools
Raniganj	West Bengal	Coal mining
Renukoote	Uttar Pradesh	Aluminium plant
Rishikesh	Uttarkhand	Antibiotic plant
Saharanpur	Uttar Pradesh	Cigarette factory, News print
Srinagar	Jammu & Kashmir	Woolen shawls, silk, woodwork
Surat	Gujarat	Cotton textiles, Diamond Cutting
Vishakhapatnam	Andhra Pradesh	Ship building, Iron, and steel, oil refinery
Varanasi	Uttar Pradesh	Rail Engines and Saari Industries

ENERGY

Conventional Energy

- Power development in India commenced with the commissioning of electricity supply in Darjeeling during 1897, followed by a hydropower station at Sivasamudram in Karnataka during 1902.
- India is the fifth largest electricity producing in the world.
- The installed power generation capacity in the country has been increased to 144564.97 MW (31 October 2007) comprising 88216MW thermal, 34391MW hydro, 6190.86MW wind and 3360 MW nuclear.

Major Power Plants

Thermal Power Plant

Power Station	Operator	Year of Establishment	Location	District	State
Talcher Super Thermal Power Station	NTPC	1995	Kaniha	Angul	Odisha
Sipat Thermal Power Plant	NTPC	2008	Sipat	Bilaspur	Chhattisgarh
Vindhyachal Super Thermal Power Station	NTPC	2013	Singrauli	Vindhya Nagar	Madhya Pradesh
Mundra Ultra Mega Power Project	Tata Power	2009	Mundra	Kutch	Gujarat
Korba Super Thermal Power Plant	NTPC	1983	Jamani Palli	Korba	Chhattisgarh
Bhusawal Thermal Power Station	MAHAG ENCO	1968	Deepnagar Jalgaon	Jalgao	Maharashtra
Satpura Thermal Power Station	MPPGCL	1967	Sarni	Betul	Madhya Pradesh
Sterlite Jharsuguda Power Station	Vedanta	2006	Jharsuguda	Jharsuguda	Odisha
Durgapur Thermal Power Station	DVC	1996	Durgapur	Bardhaman	West Bengal

Nuclear Power Plant

Power Station	Operator	Year of Establishment	Location	District	State
Tarapur Atomic Power Station	NPCIL	1969	Tarapur	Thane	Maharashtra
Rajasthan Atomic Power Station	NPCIL	1973	Rawatbhata	Chittorgarh	Rajasthan
Kakrapar Atomic Power Station	NPCIL	1993	Kakrapar	Surat	Gujarat
Kudankulam Nuclear Power Plant	NPCIL	2013	Kudankulam	Tirunelveli	Tamil Nadu
Kaiga Nuclear Power Plant	NPCIL	2000	Kaiga	Uttara Kannada	Karnataka
Madras Atomic Power Station	NPCIL	1984	Kalpakkam	Kancheepuram	Tamil Nadu
Narora Atomic Power Station	NPCIL	1991	Narora	Bulandshahar	Uttar Pradesh
Gorakhpur Atomic Power Station	NPCIL		Fatehabad	Fatehabad	Haryana

Hydro Power Plant

Power Station	Operator	Year of Establishment	Location	State
Bhakra Dam	BBMB	1963	Bilaspur	Himachal Pradesh,
Tehri Dam	THDC India Limited	2006	Tehri	Uttarakhand
Machkund	APGENCO	1955	Jeypore	Andhra Pradesh
Hirakund – I	OHPC	1957	Burla	Odisha
Saradar Sarovar-RBPH	SSNNL	2006	Navagam	Gujrat

Wind Power Plant

Power Station	Operator	Year of Establishment	Location	State
Muppandal Wind Farm	Muppandal Winds	1985	Kanyakumri	Tamil Nadu
Jaisalmer Wind Farm	Suzlon Energy	2001	Jaisalmer	Rajasthan
Brahmanvel Wind Farm	Parakh Agro Industry	2006	Dhule	Maharashtra
Dhalgaon	Grade Mission Export	2008	Sangli	Maharashtra
Damanjodi wind Plant	Suzlon Energy	2014	Koraput	Odisha

India's Major Photovoltaic (PV) Solar Power Plants

Power Station	Operator	Year of Establishment	Location	State
Charanka Solar Power Plant	GMR group	2012	Patan	Gujarat
Neemuch Solar Power Plant	Welspun Solar.	2014	Neemuch	Madhya Pradesh
Sakri Power Plant	Maharashtra state power generation company	2013	Dhule	Maharashtra
GEDCOL Solar Power Plant	Green Energy Development Corporation Ltd.	2014	Boudh district	Odisha
Dhirubhai Ambani Solar Plant	Reliance Industries	2012	Jaisalmer	Rajasthan

OTHER ENERGY SOURCES

- A five KW, geothermal pilot power plant has been commissioned at Manikaran in Kullu district of Himachal Pradesh. Puga valley in Ladakh and Tattapani in Chhattisgarh have been finalised for geothermal plant.
- In tidal energy, India has the potential of 8000-9000 MW. The Gulf of Khambhat (7000 MW), followed by Gulf of Kachchh (1000 MW) and Sunderbans (100 MW) are significant sites for tidal power. The feasibility of a tidal power project at Durgaduani creek in Sunderbans area of West Bengal is also being examined.
- One wave energy power plant has been installed at Vizhinjam near Thiruvananthapuram. Another plant is being set up in the Andaman and Nicobar Islands.
- The first plant of ocean thermal energy conversion of 100 MW is proposed to be set up off the coast of Tamil Nadu.

MAJOR PORTS

On the West Coast

1. **Cochin Port, Kerala:** It is major natural port. It deals in fertilizers, petroleum and general cargo.
2. **Jawaharlal Nehru Port, Maharashtra:** Major port in Mumbai. It is ranked among world's top 30 ports. Earlier it was called as Nhava Sheva.
3. **Kandla Port, Gujarat:** It is a tidal port and a free trade zone has been set up here. Important traffic handles are crude oil, petroleum products, edible oil, foodgrains, salt, cotton etc.
4. **Marmugao Port, Goa:** One of the oldest natural harbour on west coast. It was declared a major port. In 1963. It occupies the fifth position in terms of total quantum of traffic handled.
5. **New Mangalore Port, Karnataka:** It is an all-weather port. Iron ore of Kudremukh is being exported now here. Other items are petroleum products, fertilizers, molasses, etc. It is an Artificial Lagoon port.
6. **Port of Mumbai, Maharashtra:** Port of Mumbai has long been the principal gateway of India. Mumbai handles the maximum traffic. It is a natural harbour with wet dock. It handles over one fifth of the total traffic of ports, mostly petroleum products and dry cargo.

On the East Coast

7. **Kolkata Port, West Bengal:** Kolkata is the oldest major port in the country. Kolkata Port is India's only riverine port. It has the most sophisticated port facilities with extensive storage facility for diverse cargo.
8. **Paradip Port, Odisha:** One of India's major ports, located in Paradip. Government of India declared Paradip as the Eighth Major Port of India on 18 April 1966 making it the first major port in the east coast commissioned in Independent India. It handles iron ore and some quantities of coal and dry cargo. It is a wet dock.
9. **Port of Chennai, Tamil Nadu:** It is the second largest port in the volume of traffic handled. Important items are petroleum products, crude oil, fertilizers, iron ore and dry cargo and oldest artificial port of India. It has artificial harbour and a wet dock.
10. **Port of Visakhapatnam, Andhra Pradesh:** It is described as the Brightest Jewel of all Indian major ports for its outstanding performance and productivity. It is the deepest port. An outer harbour has been developed for exporting iron ore and a berth for crude oil is located here. It has both artificial and natural harbour.
11. **Tuticorin Port, Tamil Nadu :** Tuticorin Port is an artificial deep-sea harbour. It is an all-weather port. It handles mainly coal, salt, edible oil, dry cargo and petroleum products.
12. **Ennore Port Limited, Tamil Nadu:** It is the 12th major port and first corporatised major port in India; a Government of India undertaking. It handles coal, iron ore, LNG, chemical & other liquids, & crude since 2001. Artificial harbour is found here.
13. **Port Blair-Andman Nicobar:** It is the latest addition to the major port on June 2010, the thirteen port in the country.

Minor and Intermediate Ports

There are 140 such ports which include Rediport (Maharashtra), Kakinada (Andhra Pradesh) and Calicut (Kerala). Other ports proposed to be developed as minor ports are Andaman and

Nicobar, Lakshadweep and Puducherry. These ports can relieve the overloaded major ports and can be used as bases for deep-sea fishing. These ports mainly serve coastal trade and support passenger traffic where there is no proper rail or road link.

Tribal Groups of India			
Tribal Group	Found in	Tribal Group	Found in
Abhor	Arunachal Pradesh	Khond	Jharkhand
Adivasi	AP, Bihar, Odisha, Jharkhand, Madhya Pradesh, Maharashtra, Rajasthan, Tamilnadu, Some Northeastern States, West Bangal, Andaman and Nicobar	Khasi	Meghalaya
Ahgani	Manipur	Kharia	Jharkhand, Odisha
Apatani	Arunachal Pradesh	Kol	Madhya Pradesh
Baiga	Madhya Pradesh	Kolam	Maharashtra, Andhra Pradesh, Telengana, MP
Bakarwal	Jammu and Kashmir	Kota	Karnataka
Bhil	M.P and Rajasthan	Kuki	Mizoram
Birhor	M.P and Bihar	Lahaula	Himachal Pradesh
Chang	Nagaland	Lepcha	Sikkim
Chenchuas	Telengana, Karnataka	Lushai	Mizoram, Manipur
Sutiya	Assam	Muria	Chhattisgarh
Gaddis	Himachal Pradesh	Miha	Rajasthan
Gallong	Arunachal Pradesh	Moplah	Malabar
Garo	Meghalaya	Munda	West Bengal, Jharkhand, Odisha, Chhattishgarh
Gond	M.P and Bihar	Nishi	Assam
Gujjar	Rajasthan	Naga	Nagaland
Irula	Tamil Nadu	Oraon	MP, Bihar and Odisha, Chhotanagpur, WB,
Jaintia	Meghalaya	Onges	Andaman & Nicobar
Jarawa	Andamans	Singpho	Assam, Arunachal Pradesh
Kanikar	Tamil Nadu and Kerala	Santhal	WB, Odisha & Bihar, Jharkhand, Assam
Kalkari	Maharashtra	Sangtam	Nagaland
Kharia	Maharashtra	Sema	Nagaland
		Sentinelese	Andaman & Nicobar
		Shompen	Andaman & Nicobar
		Toda	Tamil Nadu
		Uralis	Kerala
		Wancho	Arunachal Pradesh
		Warli	Maharashtra, Daman and Diu, Bihar, Madhaya Pradesh, West Bengal Dadra, Nagar Haveli

1. Highest Population (State)

- (i) U.P. – 19.981 crore
- (ii) Maharastra – 11.237 crore
- (iii) Bihar – 10.409 crore

Highest Population (Union Territory)

- (i) Delhi – 16787941

- (ii) Puducherry – 1247953

- (iii) Chandigarh – 1055450

2. Lowest Population (State)

- (i) Sikkim – 610,577

- (ii) Mizoram – 1,097,206

Lowest population (U.T.)

- (i) Lakshadweep – 64,473
- (ii) Daman & Diu – 243,247

3. Highest Literacy (State)

- (i) Kerala – 94%
- (ii) Mizoram – 91.3%
- (iii) Goa – 88.7%
- (iv) Tripura – 87.2%

4. Lowest Literacy (State)

- (i) Bihar – 61.8%
- (ii) Arunachal Pradesh – 65.4%
- (iii) Rajasthan – 66.1%

Highest Literacy (U.T.)

- (i) Lakshadweep – 91.8%
- (ii) Daman & Diu – 87.1%
- (iii) Andaman & Nicobar Island – 86.6%

Lowest Literacy (U.T.)

- (i) Dadra & Nagar Haveli – 76.2%
- (ii) Puducherry – 85.8%

5. Highest Sex Ratio (State)

- (i) Kerala – 1084
- (ii) Tamil nadu – 996
- (iii) Andhra pradesh – 993

Lowest Sex Ratio (State)

- (i) Haryana – 879
- (ii) Jammu & Kashmir – 889
- (iii) Sikkim – 890

Highest Sex Ratio (U.T.)

- (i) Puducherry – 1087
- (ii) Lakshadweep – 947
- (iii) Andaman & Nicobar Island – 876

Lowest Sex Ratio (U.T.)

- (i) Daman & Diu – 618
- (ii) Dadra & Nagar Haveli – 774

6. Highest Density (State)

- (i) Bihar – 1106
- (ii) W. Bengal – 1028

- (iii) Kerala – 860

Highest Density (U.T.)

- (i) Delhi – 11320
- (ii) Chandigarh – 9258
- (iii) Puducherry – 2547

Lowest Density (State)

- (i) Arunachal Pradesh – 17
- (ii) Mizoram – 52
- (iii) Sikkim – 86

Lowest Density (U.T.)

- (i) Andaman & Nicobar – 46
- (ii) Dadra and Nagar Haveli – 700

7. Highest population growth (State)

- (i) Meghalaya – 27.9
- (ii) Arunachal Pradesh – 26.0
- (iii) Bihar – 25.4

Lowest population growth (State)

- (i) Nagaland – (–) 0.6
- (ii) Kerala – 4.9
- (iii) Lakshadweep – 6.3

Highest population growth (U.T.)

- (i) Dadra & Nagar Haveli – 55.90
- (ii) Daman & Diu – 53.8

Lowest population growth (U.T.)

- (i) Lakshadweep – 6.3
- (ii) Andaman & Nicobar Island – 6.9

Ethnic Groups		
Tribes	Race	Islands
Onges	Negroid	Little Nicobar
Sentinelese	Negroid	Sentinel Island
Jarawa	Negroid	Middle & South Andaman
Andamanese	Negroid	Strait Island
Shompen	Mongloid	Great Nicobar
Nicobarese	Mongloid	Great Nicobar

Multipurpose Projects

Sr. No.	Name of the Project	River	Purpose	Beneficiary States
1.	Bhakra-Nangal Project The project consists of: (i) Bhakra dam (second highest in the world) - 518 m long, 226 high (ii) Nangal dam (iii) Nangal hydel channels (iv) 4 power houses (Biggest in Asia)	Sutlej (Hoshiarpur district in Punjab)	Power and irrigation	Punjab, Himachal Pradesh, Haryana, and Rajasthan
2.	Damodar Valley Corporation Project The project consists of: (i) Tilaiya dam (ii) Konar dam (iii) Maithon dam (iv) Panchet Hill dam (v) Power houses at Bokaro, Durgapur, and Chandrapur	Damodar	Power, irrigation, flood control	Bihar and West Bengal, shared by Madhya Pradesh

3.	Hirakund This is a dam project. The main dam is 4,800 m long, 28.9 m high (World's largest mainstream dam)	Mahanadi	Power and irrigation	Odisha
4.	Tungabhadra Project	Tungabhadra (tributary of river Krishna)	Power and irrigation	Andhra Pradesh and Karnataka
5.	Mayurakshi Project	Murali	Power and irrigation	West Bengal
6.	Nagarjunasagar Project	Krishna	Power and irrigation	Andhra Pradesh
7.	Gandak River Project	Gandak (tributary of Ganga)	Power and irrigation	Bihar, Uttar Pradesh, Nepal (Joint venture of India and Nepal)
8.	Kosi Project	Kosi	Flood control, power and irrigation	Bihar
9.	Farakka Project	Ganga, Bhagirathi	Power, irrigation, avoid accumulation of silt to improve navigation	West Bengal
10.	Beas Project Units : Beas-Sutlej link Beas dam at Pong Beas transmission system	Beas	Irrigation and power	Rajasthan, Haryana, Punjab, and Himachal Pradesh
11.	Rajasthan Canal Project	Sutlej in Punjab, Beas, and Ravi	Irrigation	Rajasthan, Punjab and Haryana
12.	Chambal Project Units (a) Gandhisagar dam (b) Rana Pratap Sagar dam (c) Jawahar Sagar dam	Chambal	Power and irrigation	Madhya Pradesh and Rajasthan
13.	Kakrapara Project	Tapti	Irrigation	Gujarat
14.	Nagpur Power Station		Koradi	Thermal power Maharashtra
15.	Ukai Project	Tapti	Power and irrigation	Gujarat
16.	Tawa Project	Tawa (Narmada)	Irrigation	Madhya Pradesh
17.	Poochampad Project	Godavari	Irrigation	Andhra Pradesh
18.	Malaprabha Project	Malaprabha	Irrigation	Karnataka
19.	Durgapur Barrage	Damodar	Irrigation, Navigation, between Kolkata and Raniganj	West Bengal and Bihar
20.	Mahi	Mahi	Irrigation	Gujarat
21.	Mahanadi Delta Project	Mahanadi	Irrigation	Orissa
22.	Idukki Project	Periyar	Hydroelectricity	Kerala
23.	Koyna Project	Koyna	Hydroelectricity	Maharashtra
24.	Upper Krishna Project	Krishna	Irrigation	Karnataka
25.	Ramaganga Multipurpose Project	Chisot stream near kala	Power and irrigation	Uttar Pradesh
26.	Matatilla project	Betwa	Multipurpose power and irrigation	Uttar Pradesh and Madhya Pradesh
27.	Tehri Dam Project	Bhilangana, Bhagirath	Hydroelectricity	Uttar Pradesh
28.	Narmada Sagar Valley Project	Narmada	–	Madhya Pradesh, Gujarat, Rajasthan, and Maharashtra
29.	Obra Power Station	Obra	Thermal power	Uttar Pradesh
30.	Rihand Scheme	Rihand	Hydroelectricity	Uttar Pradesh
31.	Kundah Project	Kundah	Hydroelectricity and irrigation	Tamil Nadu

Exercise -1

- The iron ore mined at Bailadila is mostly
 - Haematite
 - Siderite
 - Limonite
 - Magnetite
- Why is Gujarat, the largest producer of salt in India?
 - Gujarat possesses the largest dry coastal area
 - The sea water here is more saline
 - Gujarat farmers are highly skilled in making salt
 - Gujarat has large areas of shallow sea
- The largest industry in India is
 - Textiles industry
 - Steel industry
 - Cement industry
 - Automobile industry
- Among sources of power, India has largest reserves of
 - coal
 - oil
 - natural gas
 - atomic power
- Rangap in Sikkim is famous for
 - bauxite
 - mica
 - copper
 - aluminium
- The correct sequence in decreasing order in terms of total production of the given non-ferrous metals in India is
 - lead, zinc, copper, aluminium
 - zinc, lead, aluminium, copper
 - aluminium, zinc, copper, lead
 - aluminium, copper, zinc, lead
- The refining capacity of crude oil is highest in the
 - Haldia oil refinery
 - Mathura oil refinery
 - Gujarat oil refinery
 - Vishakhapatnam oil refinery
- Contact metamorphism is
 - small scale heating and alteration of rock by localized igneous intrusion
 - structural alteration of rock through pressure
 - large scale heating and modification of rocks at subduction zones
 - None of these
- In the Indian context the term 'De-notified tribes' refers to:
 - tribes which are aboriginals
 - nomadic tribes
 - tribes practising shifting cultivation
 - tribes which were earlier classified as criminal tribes
- Which one of the following major sea ports of India does not have a natural harbour?
 - Bombay
 - Cochin
 - Marmagao
 - Paradeep
- The correct sequence in the ascending order of the given cities in terms of altitude above mean sea level is:
 - Marmugao, Mumbai, Kolkata, Chennai
 - Mumbai, Kolkata, Chennai, Marmugao
 - Chennai, Marmugao, Mumbai, Kolkata
 - Kolkata, Mumbai, Chennai, Marmugao
- Which of the following is not associated with oilfields in India?
 - Bombay High
 - Digboi
 - Ankleshwar
 - Bijapur
- In terms of the mineral resources/deposits, the leading state in India is :
 - Bihar
 - Madhya Pradesh
 - Karnataka
 - Rajasthan
- The Central Building Research Institute is located at :
 - Bareilly
 - Lucknow
 - Roorkee
 - Madurai
- Which one of the following National Highways passes through Maharashtra, Chhattisgarh and Odisha?
 - NH 4
 - NH 5
 - NH 6
 - NH 7
- Which one of the following is a 'foot loose' industry?
 - Paper
 - Cement
 - Electronics
 - Iron & Steel
- Which one of the following pairs is not correctly matched?

Important Location (Industry/Mining)	State
(a) Himgiri	— Uttar Pradesh
(b) Koraput	— Orissa
(c) Palana	— Rajasthan
(d) Nellore	— Andhra Pradesh
- Which one of the following cities is NOT connected by NH -3 ?
 - Agra
 - Bhopal
 - Dhule
 - Gwalior
- Among the following Union Territories, which one has the highest density of population?
 - Dadra and Nagar Haveli
 - Daman and Diu
 - Lakshadweep
 - Pondicherry
- The leading states in the production of cotton in India are:
 - Maharashtra and Gujarat
 - Gujarat and Andhra Pradesh
 - Maharashtra and Punjab
 - Gujarat and Punjab
- Which one of the following Indian States is the largest producer of natural rubber ?
 - Tamil Nadu
 - Kerala
 - Assam
 - Andhra Pradesh
- Which one of the following National Highways links Jabalpur, Nagpur, Hyderabad, Bangalore and Madurai ?
 - NH 5
 - NH 7
 - NH 9
 - NH 11
- Haridwar is well-known for which one of the following industries?
 - Fertilizers
 - Cement
 - Heavy Electricals
 - Silk Textiles
- The Nuclear Power Station Rawatbhata is in which State?
 - Maharashtra
 - Uttar Pradesh
 - Rajasthan
 - Tamil Nadu
- Mahatma Gandhi Hydroelectric project is on which river?
 - Godavari
 - Sharavati
 - Cauvery
 - Krishna

26. The National Highway No. 7 connects which of the following?
 - (a) Mumbai with Varanasi
 - (b) Mumbai with Bhubaneswar
 - (c) Delhi with Kanyakumari
 - (d) Varanasi with Kanyakumari
27. Among the following, which state has highest level of literacy?
 - (a) Mizoram
 - (b) West Bengal
 - (c) Kerala
 - (d) Punjab
28. Which among the following is the major item of export from Paradeep Port?
 - (a) Rice
 - (b) Tea
 - (c) Fish
 - (d) Iron ore
29. Which one of the following types of coal has highest amount of carbon and burns without smoke ?
 - (a) Anthracite
 - (b) Bituminous coal
 - (c) Lignite
 - (d) Peat
30. Which one among the following is a correct sequence of production of coal in the Indian States in descending order?
 - (a) Jharkhand – Madhya Pradesh – West Bengal – Meghalaya
 - (b) West Bengal – Madhya Pradesh – Jharkhand – Meghalaya
 - (c) Jharkhand – West Bengal – Meghalaya – Madhya Pradesh
 - (d) Madhya Pradesh – Jharkhand – West Bengal – Meghalaya
31. The Ruhr basin is the famous industrial region of
 - (a) China
 - (b) Japan
 - (c) Germany
 - (d) United Kingdom
32. Monoculture is a distinct characteristic of
 - (a) Commercial grain farming
 - (b) Shifting cultivation
 - (c) Subsistence farming
 - (d) Organic farming
33. Kimberley is famous for
 - (a) Gold Mining
 - (b) Diamond mining
 - (c) Steel industry
 - (d) Automobile industry
34. Which one of the following is a fossil source of energy?
 - (a) Wood
 - (b) Solar radiation
 - (c) Tidal waves
 - (d) Petroleum
35. The largest producer of tungsten in the world is
 - (a) Australia
 - (b) China
 - (c) Russia
 - (d) U.S.A.
36. Which one of the following deserts is famous for its nitrate deposits?
 - (a) Atacama
 - (b) Gobi
 - (c) Kalahari
 - (d) Sahara
37. The per capita CO₂ emission is the highest in
 - (a) China
 - (b) India
 - (c) Japan
 - (d) U.S.A.
38. Which of the following countries is the greatest producer of bauxite in the world?
 - (a) Argentina
 - (b) India
 - (c) South Africa
 - (d) Brazil
39. The lines of equal transport cost in the industrial location model of Alfred Weber are known as
 - (a) Isoline
 - (b) Isobar
 - (c) Isodapen
 - (d) Isotim
40. Krivoirog is known for
 - (a) Textile industry
 - (b) Chemical industry
 - (c) Iron-ore mining
 - (d) Bauxite mining
41. The largest iron-ore producing region in the U.S.A. is
 - (a) North East Appalachian region
 - (b) Albama region
 - (c) Lake superior region
 - (d) Sierra Nevada region
42. The major agricultural regions of the world were first delineated by
 - (a) L.D. Stamp
 - (b) H. Bobek
 - (c) D. Whittlesey
 - (d) J.E. Spencer
43. Plantation agriculture is practical mainly in the
 - (a) Arid region
 - (b) Mediterranean region
 - (c) Temperate region
 - (d) Tropical region
44. Which region is called the bread basket of the world?
 - (a) Temperate grassland
 - (b) Savanna grassland
 - (c) Mediterranean region
 - (d) Mid latitude forest
45. The Broken Hill of Australia is famous for the mining of
 - (a) Zinc
 - (b) Silver
 - (c) Lead
 - (d) All of these
46. Which one of the following countries is the largest producer of Geo-thermal energy in the world?
 - (a) Australia
 - (b) New Zealand
 - (c) U.S.A.
 - (d) U.K.
47. Katanga province of Zaire is famous for
 - (a) Copper
 - (b) Coal
 - (c) Gold
 - (d) Diamond
48. Which of the following crops are grown mostly under subsistence farming?
 - (a) Millets and Rice
 - (b) Cotton and Tobacco
 - (c) Tea and Coffee
 - (d) Vegetables and Fruits
49. Teak and Sal are products of
 - (a) Tropical dry deciduous Forest
 - (b) Tropical Evergreen Forests
 - (c) Tropical Thorn Forests
 - (d) Alpine Forests
50. Dapog method of rice nursery was developed in
 - (a) China
 - (b) Indonesia
 - (c) Japan
 - (d) Philippines
51. Which one of the following districts does not have a gold field?
 - (a) Anantpur
 - (b) Kolar
 - (c) Raichur
 - (d) Vishakhapatnam
52. Which one of the following iron and steel plants was established with the British collaboration?
 - (a) Bhilai
 - (b) Rourkela
 - (c) Bokaro
 - (d) Durgapur
53. Which one of the following is the correct sequence of the

- nuclear power plants of India in the increasing order of their installed power generation capacity?
- Rawatbhata-Narora-Kaiga-Tarapur
 - Narora-Kaiga-Rawatbhata-Tarapur
 - Kaiga-Tarapur-Narora-Rawatbhata
 - Tarapur-Narora-Kaiga-Rawatbhata
54. Rihand Valley Project is located in which one of the following states?
- Odisha
 - Gujarat
 - Himachal Pradesh
 - Uttar Pradesh
55. In which of the following State is Kakrapar Nuclear Power Station located? [2008-II]
- Karnataka
 - Tamil Nadu
 - Maharashtra
 - Gujarat
56. Which of the following sequences of the oil refineries of India as they occur from South to North is correct?
- Kochi-Mangalore-Mumbai-Koyali
 - Koyali-Mumbai-Mangalore-Kochi
 - Kochi-Mumbai-Mangalore-Koyali
 - Mangalore-Kochi-Mumbai-Koyali
57. Which water body separates Australia from New Zealand?
- Cook Straits
 - Megallan
 - Tasman Sea
 - Great Barrier Reef
58. Which one of the following does not have a heavy water plant?
- Narora
 - Sriharikota
 - Kakrapar
 - Kota

Exercise -2

Statement Based MCQ

- Which of the following statements are true with respect to iron ore ?
 - Limestone and coal are important for smelting iron ore.
 - More than 20 per cent iron ore deposits of the world are in India.
 - Odisha and Bihar top in the production of iron ore.
 - Iran is the largest buyer of Indian iron ore.

Select the correct answer using the code given below:

 - 1, 2 and 3
 - 2 and 3
 - 3 and 4
 - 2, 3 and 4
- Which of the following minerals found in a natural way in the state of Chhattisgarh?
 - Bauxite
 - Dolomite
 - Iron ore
 - Tin

Select the correct answer using the code given below:

 - 1, 2 and 3
 - 1 and 3
 - 2 and 4
 - 1, 2, 3 and 4
- Which of the following are true with regard to coal in India ?
 - Coal is found in sedimentary rocks.
 - The best quality of coal is lignite.
 - The largest and oldest coalfield in India is Raniganj.
 - The Damodar river valley is popularly known as the 'Ruhr of India'.

Which of the statements are correct?

 - 1 and 4
 - 1, 2 and 3
 - 1, 3 and 4
 - 2, 3 and 4
- Which of the following statement(s) is/are correct?
 - Broad gauge railway engines-Varanasi.
 - Electric locomotives-Bhopal.
 - Integral Coach Factory-Calcutta.

Which of the statement(s) is/are correct?

 - 1 only
 - 2 only
 - 3 only
 - 1, 2 and 3
- Which of the following is/are matched correctly?
 - Garnet gem - Kanyakumari (Tamil Nadu)
 - Corundum - Tonk (Rajasthan)
 - Gold - Chittor (Andhra Pradesh)
 - Magnesite - Salem (Tamil Nadu)

Select the correct answer using the code given below:

 - 1 and 2
 - 2 and 3
 - 1, 2 and 3
 - 2, 3 and 4
- Which of the following is true with regard to petroleum?
 - It is found in the anticlines
 - It is found over underground water
 - It is found in wide, horizontal rocks
 - It is found over impermeable rocks

Select the correct answer using the code given below:

 - 1 and 2
 - 1, 2 and 4
 - 2 and 4
 - 2 and 3
- Consider the following sites /monuments :
 - Champaner-Pavagadh Archaeological Park
 - Chhatrapati Shivaji Railway Station, Mumbai
 - Mahallapuram
 - Sun Temple (Konark Temple)

Which of the above are included in the World Heritage List of UNESCO?

 - 1, 2 and 3
 - 1, 3 and 4
 - 2 and 4
 - 1, 2, 3 and 4
- Between India and East Asia, the navigation-time and distance can be greatly reduced by which of the following?
 - Deepening the Malacca straits between Malaysia and Indonesia.
 - Opening a new canal across the Kra isthmus between the Gulf of Siam and Andaman Sea.

Which of the statements given above is/are correct ?

 - 1 only
 - 2 only
 - Both 1 and 2
 - Neither 1 nor 2
- Arrange the following states of north-east in descending order in accordance with the population density
 - Nagaland
 - Manipur
 - Tripura
 - Meghalaya

Select the correct answer:

 - 1, 2, 3 and 4
 - 3, 1, 2 and 4
 - 1, 2, 4 and 3
 - 3, 1, 4 and 2

10. Consider the following statements:
1. Nagaland has no Scheduled Caste population.
 2. Literacy among Scheduled Caste is better than that among the Scheduled Tribes.
 3. Haryana has no Scheduled Tribe population.
- Select the correct statements:
- (a) 1, 2 and 3 (b) 1 and 3
(c) 2 only (d) 1 only

11. Consider the following areas:
1. Himalayas
 2. Chhota Nagpur
 3. Chhattisgarh
 4. Sone Valley
- What is the correct sequence of these areas in descending order of their coal reserves ?
- (a) 1, 4, 3 and 2 (b) 2, 4, 3 and 1
(c) 1, 3, 4 and 2 (d) 2, 3, 4 and 1

12. Consider the following statements :
1. The first newsprint plant set up in India was in Napanagar.
 2. The first **modern** steel plant in India was set up at Bhadravati.

Which of these statements is/are correct ?

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

13. Consider the following statements
1. In India, the largest concentration of roads is found in the northern plains.
 2. The ratio of surfaced road to the total road length is lower in the northern plains .

Which of the statements given above is/are correct?

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

14. Which of the following are not the examples of shifting cultivation?

Select the correct answer from the codes given below:

1. Ladang
 2. Hacienda
 3. Fazenda
 4. Pondu
- (a) 1 and 2 (b) 1 and 3
(c) 2 and 4 (d) 3 and 4

15. Which of the following is/are the characteristic/ characteristics of Indian coal?

1. High ash content
 2. Low sulphur content
 3. Low ash fusion temperature
- Select the correct answer using the codes given below.

- (a) 1 and 2 only (b) 2 only
(c) 1 and 3 only (d) 1, 2 and 3

16. Consider the following statements:
1. Natural gas occurs in the Gondwana beds.
 2. Mica occurs in abundance in Kodarma.
 3. Dharwars are famous for petroleum.
- Which of the statements given above is/are correct?

- (a) 1 and 2 (b) 2 only
(c) 2 and 3 (d) None

17. Consider the following towns of India:
1. Bhadrachalam
 2. Chanderi
 3. Kancheepuram
 4. Karnal
- Which of the above are famous for the production of traditional sarees/ fabric?

- (a) 1 and 2 only (b) 2 and 3 only
(c) 1, 2 and 3 (d) 1,3 and 4.

18. Arrange the following countries in the descending order of their wheat production and select the correct answer from the codes given below:

1. China
 2. India
 3. Russia
 4. U.S.A.
- (a) 1, 2, 3, 4, (b) 1, 2, 4, 3
(c) 2, 3, 4, 1 (d) 4, 1, 2, 3

19. Which of the following is accredited with the Geographical Indication (GI) mark?

1. Handwoven Pashmina shawls of Kashmir.
2. Bhagalpur Silk, Bihar.
3. Madurai Idly, Tamil Nadu.
4. Darjeeling Tea, West Bengal.

Select the answer from the codes given below:

- (a) 1, 2, and 3
(b) 2, 3, and 4
(c) 1, 3, and 4
(d) All of the above

20. Match the following

List I (Place)	List II (Industry)
A. Muzaffarnagar	1. Cotton textiles
B. Adoni	2. Sugar
C. Ranchi	3. Sports goods
D. Jalandhar	4. Engineering goods

Codes

- | | | | | |
|-----|---|---|---|---|
| | A | B | C | D |
| (a) | 2 | 1 | 4 | 3 |
| (b) | 1 | 3 | 4 | 2 |
| (c) | 2 | 3 | 4 | 1 |
| (d) | 1 | 4 | 3 | 2 |

Matching Based MCQ

DIRECTIONS (Qs. 18 to 43) : Match List-I with List-II and select the correct answer using the codes given below the lists.

- 21.
- | List-I
(Aluminium Company) | List II
(Location) |
|-------------------------------|-----------------------|
| (A) BALCO | (1) Hirakund |
| (B) HINDALCO | (2) Korba |
| (C) Indian Aluminium Company | (3) Koraput |
| (D) NALCO | (4) Renukoot |
- (a) A - 3 ; B - 1 ; C - 4 ; D - 2
(b) A - 2 ; B - 4 ; C - 1 ; D - 3
(c) A - 3 ; B - 4 ; C - 1 ; D - 2
(d) A - 2 ; B - 1 ; C - 4 ; D - 3

- 22.
- | List-I | List-II |
|-----------------------|----------------|
| (A) Paper | (1) Vijaynagar |
| (B) Heavy Engineering | (2) Varanasi |
| (C) Locomotives | (3) Hatia |
| (D) Steel production | (4) Napanagar |
- (a) A - 3 ; B - 4 ; C - 1 ; D - 2
(b) A - 4 ; B - 3 ; C - 2 ; D - 1
(c) A - 3 ; B - 1 ; C - 2 ; D - 4
(d) A - 4 ; B - 2 ; C - 1 ; D - 3

23. **List-I (Minerals)** **List-II (States)**

- (A) Uranium (1) Rajasthan
 (B) Copper (2) Bihar
 (C) Zinc (3) Karnataka
 (D) Lignite (4) Tamil Nadu
 (E) Gold (5) Odisha

- (a) A - 1 ; B - 2 ; C - 3 ; D - 4 ; E - 5
 (b) A - 2 ; B - 5 ; C - 1 ; D - 4 ; E - 3
 (c) A - 4 ; B - 5 ; C - 2 ; D - 1 ; E - 3
 (d) A - 5 ; B - 1 ; C - 3 ; D - 1 ; E - 2

24. **List-I (Minerals)** **List-II (Areas)**

- (A) Gold (1) Khetri
 (B) Coal (2) Kolar
 (C) Copper (3) Kudremukh
 (D) Iron (4) Jharia

- (a) A - 2 ; B - 4 ; C - 1 ; D - 3
 (b) A - 1 ; B - 2 ; C - 3 ; D - 4
 (c) A - 4 ; B - 3 ; C - 2 ; D - 1
 (d) A - 3 ; B - 4 ; C - 1 ; D - 2

25. **List-I** **List-II**
Atomic power stations **States**

- (A) Kakrapar (1) Rajasthan
 (B) Narora (2) Gujarat
 (C) Kalpakkam (3) Tamil Nadu
 (D) Tarapur (4) Maharashtra
 (E) Kota (5) Uttar Pradesh

- (a) A - 1 ; B - 2 ; C - 3 ; D - 4 ; E - 5
 (b) A - 5 ; B - 4 ; C - 3 ; D - 2 ; E - 1
 (c) A - 2 ; B - 5 ; C - 3 ; D - 4 ; E - 1
 (d) A - 1 ; B - 4 ; C - 2 ; D - 3 ; E - 5

26. **List-I** **List-II**

- (A) BHEL (1) Atomic
 (B) HAL (2) Electronic
 (C) SAIL (3) Aeronautics
 (D) BPCL (4) Chemicals
 (E) NPTC (5) Iron and steel

- (a) A - 2 ; B - 3 ; C - 5 ; D - 4 ; E - 1
 (b) A - 5 ; B - 4 ; C - 3 ; D - 2 ; E - 1
 (c) A - 1 ; B - 2 ; C - 3 ; D - 4 ; E - 5
 (d) A - 4 ; B - 2 ; C - 3 ; D - 1 ; E - 5

27. **List-I** **List-II**
Public Sector **Location Undertaking**

- (A) Fertilizer Corporation of India (1) Pimpri
 (B) Hindustan Antibiotics Ltd. (2) Sindri
 (C) Indian Rare Earths Ltd. (3) Nepanagar
 (D) National Newsprint Ltd. (4) Alwaye

- (a) A - 2 ; B - 1 ; C - 4 ; D - 3
 (b) A - 1 ; B - 2 ; C - 3 ; D - 4
 (c) A - 4 ; B - 3 ; C - 2 ; D - 1
 (d) A - 3 ; B - 1 ; C - 2 ; D - 4

28. **List-I** **List-II**

- (A) Central Building Research Institute (1) Madras

- (B) Central Road Research Institute (2) Roorkee
 (C) Central Drug Research Institute (3) Lucknow
 (D) Central Leather Research Institute (4) New Delhi

(5) Dhanbad

- (a) A - 1 ; B - 2 ; C - 3 ; D - 5
 (b) A - 2 ; B - 4 ; C - 3 ; D - 1
 (c) A - 4 ; B - 3 ; C - 2 ; D - 1
 (d) A - 5 ; B - 4 ; C - 1 ; D - 3

29. **List-I** **List-II**

- (A) Highest decadal growth in population (1) Kerala
 (B) Highest rural population growth rate (2) Nagaland
 (C) Lowest rural population growth rate (3) Tamil Nadu
 (D) Lowest decadal growth in population (4) Odisha

(5) Karnataka

- (a) A - 2 ; B - 3 ; C - 4 ; D - 1
 (b) A - 5 ; B - 2 ; C - 3 ; D - 1
 (c) A - 2 ; B - 4 ; C - 3 ; D - 5
 (d) A - 3 ; B - 4 ; C - 1 ; D - 5

30. **List-I** **List-II**
(Industry) **(Industrial Centre)**

- (A) Aluminium (1) Coimbatore
 (B) Heavy Electricals (2) Renukoot
 (C) Petrochemicals (3) Vadodara
 (D) Cotton Textiles (4) Jagdishpur

- (a) A - 2 ; B - 4 ; C - 3 ; D - 1
 (b) A - 1 ; B - 3 ; C - 4 ; D - 2
 (c) A - 1 ; B - 4 ; C - 3 ; D - 2
 (d) A - 2 ; B - 3 ; C - 4 ; D - 1

31. **List-I** **List-II**
(National Highway No.) **(Cities connected)**

- (A) NH 5 (1) Bhopal-Jaipur
 (B) NH 7 (2) Bhubaneswar- Chennai
 (C) NH 9 (3) Nagpur-Varanasi
 (D) NH 12 (4) Pune-Hyderabad

- (a) A - 2 ; B - 1 ; C - 4 ; D - 3
 (b) A - 4 ; B - 3 ; C - 2 ; D - 1
 (c) A - 2 ; B - 3 ; C - 4 ; D - 1
 (d) A - 4 ; B - 1 ; C - 2 ; D - 3

32. **List-I** **List-II**
(Country) **(Coal fields)**

- A. China 1. Pennsylvania
 B. Germany 2. Saar
 C. Ukraine 3. Shensi
 D. U.S.A. 4. Donetsk Basin

- | | A | B | C | D |
|-----|---|---|---|---|
| (a) | 4 | 2 | 1 | 3 |
| (b) | 3 | 2 | 4 | 1 |
| (c) | 3 | 1 | 4 | 2 |
| (d) | 4 | 3 | 2 | 1 |

33. **List-I** (Iron and steel centre) **List-II** (Country)
- | | |
|--------------|------------|
| A. Cleveland | 1. Canada |
| B. Essen | 2. Russia |
| C. Hamilton | 3. U.S.A. |
| D. Tula | 4. Germany |
- | | | | | |
|-----|---|---|---|---|
| | A | B | C | D |
| (a) | 1 | 4 | 2 | 3 |
| (b) | 3 | 4 | 1 | 2 |
| (c) | 3 | 1 | 2 | 3 |
| (d) | 4 | 3 | 1 | 2 |
34. **List-I** (Mineral) **List-II** (Mine)
- | | |
|------------|--------------|
| A. Diamond | 1. Butte |
| B. Coal | 2. Kimberley |
| C. Cobalt | 3. Katanga |
| D. Silver | 4. Saar |
- | | | | | |
|-----|---|---|---|---|
| | A | B | C | D |
| (a) | 4 | 3 | 2 | 1 |
| (b) | 2 | 4 | 3 | 1 |
| (c) | 3 | 4 | 1 | 2 |
| (d) | 2 | 1 | 3 | 4 |
35. **List-I** **List-II**
- | | |
|---------------|-------------------|
| A. Detroit | 1. Cutlery |
| B. Pittsburgh | 2. Ship building |
| C. Plymouth | 3. Iron and steel |
| D. Sheffield | 4. Automobile |
- | | | | | |
|-----|---|---|---|---|
| | A | B | C | D |
| (a) | 4 | 3 | 1 | 2 |
| (b) | 2 | 3 | 1 | 4 |
| (c) | 3 | 1 | 4 | 2 |
| (d) | 4 | 3 | 2 | 1 |
36. **List-I** (Crops) **List-II** (Largest producer)
- | | |
|--------------|-----------|
| A. Wheat | 1. Brazil |
| B. Cotton | 2. China |
| C. Sugarcane | 3. U.S.A. |
| D. Tea | 4. India |
- | | | | | |
|-----|---|---|---|---|
| | A | B | C | D |
| (a) | 1 | 2 | 3 | 4 |
| (b) | 2 | 3 | 1 | 4 |
| (c) | 2 | 4 | 3 | 1 |
| (d) | 4 | 1 | 2 | 3 |
37. **List-I** **List-II**
- | | |
|-------------------------|------------------|
| A. Shifting cultivation | 1. Mongolia |
| B. Nomadic herding | 2. Australia |
| C. Livestock ranching | 3. Tundra Region |
| D. Fishing and hunting | 4. Amazon Basin |
- | | | | | |
|-----|---|---|---|---|
| | A | B | C | D |
| (a) | 1 | 3 | 2 | 4 |
| (b) | 4 | 1 | 2 | 3 |
| (c) | 3 | 2 | 1 | 4 |
| (d) | 4 | 1 | 3 | 2 |
38. **List-I** **List-II**
- | | |
|---|-----------|
| A. Largest producer of wheat in the world | 1. U.S.A. |
| B. Largest producer of milk in the world | 2. China |
- C. Largest producer of sugarcane in the world 3. India
- D. Largest producer of maize in the world 4. Brazil
- | | | | | |
|-----|---|---|---|---|
| | A | B | C | D |
| (a) | 1 | 2 | 4 | 3 |
| (b) | 2 | 3 | 4 | 1 |
| (c) | 3 | 4 | 2 | 1 |
| (d) | 1 | 3 | 2 | 4 |
39. **List-I** **List-II**
- | | |
|--------------------|--------------|
| A. Iron and steel | 1. Atlanta |
| B. Ship building | 2. Bradford |
| C. Automobile | 3. Cleveland |
| D. Woollen textile | 4. Yakohama |
- | | | | | |
|-----|---|---|---|---|
| | A | B | C | D |
| (a) | 3 | 4 | 1 | 2 |
| (b) | 1 | 2 | 4 | 3 |
| (c) | 2 | 4 | 1 | 3 |
| (d) | 3 | 2 | 1 | 4 |
40. **List-I** **List-II**
- | | |
|------------------|------------------------------|
| A. Hydropower | 1. France |
| B. Coal | 2. Congo Democratic Republic |
| C. Petroleum | 3. Poland |
| D. Nuclear power | 4. Iraq |
- | | | | | |
|-----|---|---|---|---|
| | A | B | C | D |
| (a) | 2 | 3 | 4 | 1 |
| (b) | 3 | 4 | 1 | 2 |
| (c) | 1 | 3 | 2 | 4 |
| (d) | 4 | 1 | 3 | 2 |
41. **List-I** (Industrial region) **List-II** (Name of country)
- | | |
|--------------------|------------|
| A. Ruhr | 1. U.S.A. |
| B. New England | 2. Brazil |
| C. Kinki | 3. Germany |
| D. Belo Horizontal | 4. Japan |
- | | | | | |
|-----|---|---|---|---|
| | A | B | C | D |
| (a) | 3 | 1 | 4 | 2 |
| (b) | 2 | 3 | 1 | 4 |
| (c) | 4 | 2 | 3 | 1 |
| (d) | 1 | 4 | 2 | 3 |
42. **List-I** (Minerals) **List-II** (Important centres)
- | | |
|------------|------------------|
| A. Copper | 1. Butte |
| B. Diamond | 2. Katanga |
| C. Gold | 3. Kimberley |
| D. Silver | 4. Witwatersrand |
- | | | | | |
|-----|---|---|---|---|
| | A | B | C | D |
| (a) | 2 | 4 | 3 | 1 |
| (b) | 2 | 3 | 4 | 1 |
| (c) | 1 | 3 | 2 | 4 |
| (d) | 3 | 1 | 4 | 2 |
43. **List-I** (Industrial production) **List-II** (Place of production)
- | | |
|------------------------|-----------------|
| (A) Brassware | (1) Kanchipuram |
| (B) Silk Sarees | (2) Lucknow |
| (C) Chikkan Embroidery | (3) Moradabad |
| (D) Sports Goods | (4) Jalandhar |
- | | | | | |
|-----|---|---|---|---|
| | A | B | C | D |
| (a) | 3 | 1 | 2 | 4 |
| (b) | 3 | 2 | 1 | 4 |
| (c) | 4 | 2 | 1 | 3 |
| (d) | 4 | 1 | 2 | 3 |

44. **List-I (Industrial activities)** **List-II (Explanations)**
- (A) Processing activities (1) Inputs are processed goods
- (B) Fabricating activities (2) Major inputs are raw material
- (C) Integrative activities (3) Involve neither inputs nor outputs
- (D) Administrative activities (4) Inputs are processed goods undergoing little change
- A B C D
- (a) 1 2 3 4
- (b) 2 1 4 3
- (c) 1 4 3 2
- (d) 3 2 4 1

45. Match the following

List I (Multi purpose River Projects)	List II (Hydel Power Station)
A. Rihand	1. Hirakund
B. Gandak	2. Balmikinagar
C. Chambal	3. Pipri
D. Mahanadi	4. Kota

- Codes
- A B C D
- (a) 3 4 2 1
- (b) 1 2 4 3
- (c) 3 2 4 1
- (d) 1 4 2 3

46. Match the following

List I (Place)	List II (Famous for)
A. Balaghat	Oil field
B. Katni	Iron ore
C. Sigraili	Manganese
D. Kalol	Bauxite
	Coal

- Codes
- A B C D
- (a) 1 2 4 3
- (b) 3 4 5 1
- (c) 3 5 4 1
- (d) 1 2 5 3

Hints and Explanations

EXERCISE-1

1. (a) 2. (d) 3. (a) 4. (a) 5. (c)

6. (d) 7. (c) 8. (a)

9. (a) De-notified tribes refers to tribes which are aboriginals. These tribes are extremely backward and are struggling to come out of stone age of existence. Such tribes live in forests.

10. (c) 11. (d) 12. (d) 13. (a) 14. (c)

15. (c) National Highways passes through Gujarat, Maharashtra, Chhatisgarh, Orissa, Jharkhand and West Bengal. This road is 1,949 km long.

16. (c) 17. (a)

18. (a) Agra is connected by N.H.- 2.

19. (d)

20. (a) For production of cotton, Black soil is most suitable and in the area of Maharashtra and Gujarat black soil is found.

21. (b)

22. (b) National Highway (NH 7) deals with Varansi and Kanyakumari. It is the longest Highway of India.

23. (c) Heavy electric equipments plants was set up with Soviet assistance at Ranipur (Haridwar) with the capacity of producing 2.7 lakh KW of steam and water turbines and generators and 5.2 lakh KW of industrial electric motors.

24. (c) 25. (b) 26. (d) 27. (c) 28. (d)

29. (a) Anthracite coal has highest amount of carbon (90%) and burns without smoke.

30. (a)

31. (c) The Ruhr basin is the famous Industrial region of Germany. Formerly Germany's coal-mining region, the Ruhrgebiet forms one of the largest conurbations in Europe with 5 million residents and is now known for its diverse and vibrant cultural scene.

32. (a) Monoculture is a distinct characteristic of commercial grain farming. Monoculture is the agricultural practice of producing or growing a single crop or plant species over a wide area and for a large number of consecutive years.

33. (b) Kimberley is famous for diamond mining. The Big Hole, Open Mine or Kimberley Mine is an open-pit and underground mine in Kimberley, South Africa, and claimed to be the largest hole excavated by hand.

34. (d) 35. (b)

36. (a) Atacama deserts is famous for its nitrate deposits. The desert is littered with approximately 170 abandoned nitrate (or "saltpetre") mining towns, almost all of which were shut down decades after the invention of synthetic nitrate in Germany at the turn of the 20th century. The towns include Chacabuco, Humberstone, Santa Laura, Pedro de Valdivia, Puelma, María Elena, and Oficina Anita.

37. (a)

38. (d) Brazil is the greatest producer of bauxite in the world. Para state dominates bauxite production in Brazil accounting for 85% of overall output, while Minas Gerais produces the country's remainder.

39. (c) 40. (c) 41. (c)

42. (c) The major agricultural region of the world were first delineated by D. Whittlesey. One of the most satisfactory classifications of agricultural regions in the world was proposed by D. Whittlesey in 1936. Whittlesey employed five criteria to classify agricultural regions in the world: (a) crop and livestock combination; (b) intensity of land use; (c) processing and marketing of farm produce; (d) degree of mechanisation; and (e) types and associations of buildings and other structures associated with agriculture.

43. (d)

44. (a) Temperate grassland is called the 'bread basket' of the world. The roots of perennial grasses usually penetrate far into the soil, and grassland soil tends to be deep and fertile. In North America, the prairies were once inhabited by huge herds of bison and pronghorns, which were hunted by wolves, bears, and other predators. These herds are almost gone now, and most of the prairies have been converted into the richest agricultural region on earth.

45. (d) The Broken Hill of Australia is famous for the mining of zinc, lead and silver. Broken Hill is an isolated mining city in the far west of outback New South Wales, Australia. The world's largest mining company, BHP Billiton, has roots in the town.

46. (c) USA is the largest producer of Geo-thermal energy in the world.

47. (a) Katanga province of Zaire is famous for copper. Copper mining is an important part of the economy of Katanga province.

48. (a) Millets and Rice are grown mostly under subsistence farming. Subsistence agriculture is self-sufficiency farming in which the farmers focus on growing enough food to feed themselves and their families. The typical subsistence farm has a range of crops and animals needed by the family to feed and clothe themselves during the year.

49. (a) Teak and Sal are products of tropical dry deciduous forests. The tropical and subtropical dry broad leaf forest biome, also known as tropical dry forest, is located at tropical and subtropical latitudes. Though these forests occur in climates that are warm year-round, and may receive several hundred centimeters of rain per year, they have long dry seasons which last several months and vary with geographic location. These seasonal droughts have great impact on all living things in the forest.

50. (d) Dapog method of rice nursery was developed in Philippines. The dapog method of raising seedling originated in the Philippines and is now, fairly common in South and Southeast. The dapog nursery is constructed for the raising of seedlings without any soil whatsoever. Rice seeds contain sufficient food in the endosperm to permit the young seedling to grow for up to 14 days without receiving any outside nutrients except air, water, and sunlight. Consequently, it is possible to nurse seedlings without actually sowing them in soil.

51. (d) Vishakhapatnam does not have a gold field.
52. (d) Durgapur Steel Plant is located in Durgapur (West Bengal.) It was Set up in the late 1950s with assistance from United Kingdom. It is one of the integrated steel plants of Steel Authority of India Limited.
53. (b) 1. Narora(Uttar Pradesh)- 2×220 MW
2. Kaiga(Karnataka)- 4×220 MW
3. Rawatbhata(Rajasthan)- 1×100 MWe; 1×200 MWe; 4×220 MWe
4. Tarapur (Maharashtra)- 1400 MW
54. (d) The Rihand Dam was constructed across the Rihand River near Pipri in Sonbhadra district of Mirzapur division in 1962 for hydropower generation.
55. (d) Kakrapar Atomic Power Station is situated in Gujarat. It consists of two 220 MW pressurized water reactors with heavy water as moderator (PHWR). It was commissioned in 1993.
56. (a) The correct sequence is Kochi- Mangalore- Mumbai- Koyli
57. (c) The Tasman Sea separates Australia from New Zealand.
58. (b) Sriharikota does not have a heavy water plant. Satish Dhawan Space Centre is a rocket launch centre located in Sriharikota in Andhra Pradesh.
14. (c) Hacienda and Pondu are not the examples of shifting cultivation. Shifting cultivation is known as ladang cultivation in south east Asia. Fazendas (meaning "farms") were plantations found throughout Brazil; during the colonial period (16th - 18th centuries), they were concentrated primarily in the northeastern region, where sugar was produced.
15. (a) Indian coal has high ash content and low calorific value. It has low sulphur and low phosphorous content but high ash fusion temperature.
16. (a) Dharwar rocks are non fossiliferous rather they are metalliferous. They bear out gold, iron ore, manganese mica, cobalt, chromium, copper, tungsten, lead, nickel, precious stones and budding stones. Kodarma is a store house of mica and Gondwana beds have natural gases.
17. (b) Chanderi in Madhyapradesh and Kancheepuram in Tamil Nadu are famous for Silk sarees.
18. (b) Copper mineral located in Katanga. Diamond mineral located in Kimberley. Gold mineral located in Witwatersrand. The Witwatersrand Gold Rush was a gold rush in 1886 that led to the establishment of Johannesburg, South Africa. It was part of the Mineral Revolution. Silver located in Butte.
19. (b) The countries in the descending order of their wheat production are; China, India, U.S.A. and Russia.
20. (a) 1. Muzaffarnagar- Sugar
2. Adoni- Cotton textile
3. Ranchi- Engineering goods
4. Jalandhar- Sports goods
21. (b) 22. (b) 23. (b) 24. (a) 25. (c)
26. (a) 27. (a) 28. (b) 29. (b)
30. (a) **Industry**
Aluminium : Alwaye (Kerala), Asansol (West Bengal), Renukoot (U.P.), Belur (Karnataka), Hiradud (Orissa).
Petrochemical : Indian Refineries Ltd (Barauni) Bihar Noohamati (Assam), Koyali oil Refinery, Koyali (Gujrat), Chochin oil Refinery Kochi (Kerala).
Cotton Textiles : Ahmedabad (Gujrat) Bangalore, Mumbai, Kolkata, Coimbtore (Tamilnadu), Kanpur (UP), Ludhiana and Amritsar (Punjab), Indore (MP).
Heavy Electricals : Bharat Heavy Electricals Ltd Ranipur Hardwar (Uttarakhand)
Heavy Electricals India Ltd (Bhopal)

EXERCISE-2

1. (a) Japan is the largest importer of Indian iron ore.
2. (d) The minerals deposits in Chhatisgarh is Bauxite (96 million ton), Dolomite (606 million ton), & iron ore (2336 million ton). Tin is not found in Chhatisgarh.
3. (c) 4. (b) 5. (d) 6. (b)
7. (d) All the monuments / sites are included in World Heritage list of UNESCO. Sun Temple of Orissa, group of monuments at Mahabalipuram were included in 1985. Champaner – Pavagadh Archeological Park of Gujrat and Chhatrapati Shivaji Terminus of Maharastra are included in year 2004 in UNESCO list.
8. (b) Opening a new canal in Kra isthums can reduce the navigation time and distance. Malacca strait is the main shipping channel between the Indian Ocean and the Pacific Ocean and links India, China, Japan and South Korea. The issue of deepening of the Malacca strait is "linked" to its economic importance rather than "time of navigation and distance". The issue is that most of the ships can not pass through it and the size of the biggest ships which can enter through it is called **Malaccamax**. Now the deepening of the strait would certainly help in "Increasing the volume of the business" because ships of larger sizes can pass through it, there is no significance of distance and navigation.
9. (b) 10. (a)
11. (d) In India maximum coal reserve found in Chhota Nagpur in Jharkhand.
12. (a) First sentence is correct, second is incorrect. First modern steel plant in India was set up at Bhilai, not in Bhadravati.
13. (c) The Northern Plains of India are expanded mainly in the state of Punjab, Haryana, Rajasthan, UP and Eastern Bihar. Here the total road length is more than total surface roads.
31. (c) **N H No. Cities connected**
NH 1 Delhi and Amritsar
NH 2 Delhi and Kolkata
NH 3 Agra and Mumbai
NH 4 Thane and Chennai
NH 5 Behrabad and Chennai
NH 6 Dhulia and Kolkata
NH 7 Varanasi and Kanyakumari
NH 8 Delhi and Mumbai
NH 9 Pune and Vijaywada
NH 10 Delhi and Fazilka
32. (b) Shensi is the coal field of China. It is a province of the People's Republic of China, officially part of the northwest China region. Saar is known as the coal mining in Germany. The Saarland is one of Germany's sixteen federal states. Its capital is at Saarbrücken. Donetz Basin is the coal field of

- Ukraine. It comprises the Donbas Foldbelt, which is the uplifted and compressionally deformed part of the Pripyat–Dniepr–Donets (PDD) Basin. Pennsylvania is known for coal mining in U.S.A. Pennsylvania, officially the Commonwealth of Pennsylvania, is a U.S. state that is located in the northeastern and mid-Atlantic regions of the United States, and the Great Lakes region.
33. (b) Regarding to the Iron and steel centre, Cleveland is located in U.S.A. Essen is located in Germany. Hamilton is located in Canada. Tula is located in Russia.
 34. (b) Diamond extracts from the Kimberley mines. The first diamond found in South Africa, was discovered less than 30 miles away and within a few years, the mining town of Kimberley. Coal founded in Saar region. In the past, a coal mining was an important branch of industry. Cobalt founded in Katanga region. Katanga is one of the provinces of the Democratic Republic of the Congo. The eastern part of the province is a rich mining region, which supplies cobalt, copper, tin, radium, uranium, and diamonds. The region's capital, Lubumbashi, is the second largest city in the Congo. Silver founded in Butte. Butte is a census-designated place (CDP) in Matanuska-Susitna Borough, Alaska, United States.
 35. (a) Cutlery industry located in Plymouth. Ship building industry located in Sheffield. Iron and steel industry located in Pittsburgh. Automobile industry located in Detroit.
 36. (b) China is the largest producer of wheat. Roughly two-thirds of the total wheat production came from the north China plain and nearly another third from the central provinces. Winter wheat accounts for about 94% of China's total wheat output. U.S.A. is the largest producer of cotton. In its January report, USDA estimated a '13-14 US crop of 13.19 million bales. Upland production was estimated at 12.55 million bales and extra-long staple production at 636,000 bales. Brazil is the largest producer of sugarcane. Brazil's sugarcane industry association UNICA estimates Brazil's sugar cane production in 2012/13 at 531.4 million ton which is 8% up from the 493.2 million ton produced in 2011/12. India is the largest producer of tea. Tea Board of India shows that during January to August, tea production has risen by 6.2% to 705 million kg in 2013.
 37. (b) Shifting cultivation practices in Amazon Basin. Nomadic herding practices in Mangolia. Live stock ranching belongs to Australia. Fishing and hunting belongs to Tundra region.
 38. (b) Largest producer of wheat in the world is China. Largest producer of milk in the world is India. Largest producer of sugarcane in the world is Brazil. Largest producer of maize in the world is U.S.A.
 39. (a) Iron and steel industry belongs to Cleveland. Ship building industry belongs to Yakohama. Automobile industry belongs to Atlanta. Woollen textile industry belongs to Bradford.
 40. (a) Hydropower associated with Congo Democratic Republic. The Democratic Republic of the Congo has reserves of petroleum, natural gas, coal, and a potential hydroelectric power generating capacity of around 100,000 MW. The Inga Dam, alone on the Congo River, has the potential capacity to generate 40,000 to 45,000 MW of electric power, sufficient to supply the electricity needs of the whole southern Africa region. Coal associated with Poland. Today Poland's coal industry is among the most competitive in Europe, providing jobs to over 100,000 people in the country. Poland is now the 9th largest hard coal producer in the world and the largest coal producer in the EU. It is also the 8th largest coking coal exporter in the world. Petroleum associated with Iraq. In 2006, Iraq's oil production averaged 2.0 million barrels per day ($320 \times 103 \text{ m}^3/\text{d}$), down from around 2.6 Mbbl/d ($410 \times 103 \text{ m}^3/\text{d}$) of production prior to the coalition invasion in 2003. Iraq's reserve to production ratio is 158 years. Nuclear power associated with France. France derives over 75% of its electricity from nuclear energy. This is due to a long-standing policy based on energy security.
 41. (a) Ruhr industrial region belongs to Germany. New England industrial region belongs to U.S.A. Kinki industrial region belongs to Japan. Belo Horizonte industrial region belongs to Brazil.
 42. (d) Geographical Indications of Goods are defined as that aspect of industrial property which refer to the geographical indication referring to a country or to a place situated therein as being the country or place of origin of that product. Typically, such a name conveys an assurance of quality and distinctiveness which is essentially attributable to the fact of its origin in that defined geographical locality, region or country. Under Articles 1 (2) and 10 of the Paris Convention for the Protection of Industrial Property, geographical indications are covered as an element of IPRs. They are also covered under Articles 22 to 24 of the Trade Related Aspects of Intellectual Property Rights (TRIPS) Agreement, which was part of the Agreements concluding the Uruguay Round of GATT negotiations. India, as a member of the World Trade Organization (WTO), enacted the Geographical Indications of Goods (Registration & Protection) Act, 1999 has come into force with effect from 15th September 2003.
 43. (a)
 44. (b)
 45. (c)
 1. Rihand - Pipri
 2. Gandak- Balmikinagar
 3. Chambal- Kota
 4. Mahanadi- Hirakund
 46. (b) About 80% of the manganese production in India comes from Balaghat District of Madhya Pradesh. Katni (Madhya Pradesh) is famous for Bauxite mines. Singrauli Coalfield is spread across the districts of Singrauli (Madhya Pradesh). Most of the coal is dispatched to pithead power plants as Singrauli Super Thermal Power Station, Rihand Thermal Power Station and Vindhyachal Thermal Power Station etc. Kalol in Gujaratis famous for its oil fields.

ECOLOGY AND ENVIRONMENT



Introduction

In this chapter we will study about Environment, their biodiversity, i.e. different species of plants and animals in their habitats and their relation and interaction with environments. We will also learn about different environmental issues and their effects on ecosystem, i.e. plants, animals, their habitats and climates all around them.

Ecosystem

An ecosystem is a functional unit of nature consisting of abiotic and biotic factors, where the living organisms interact among themselves and also with their physical environment (abiotic factors).

I. Abiotic Component (Habitat and Environment)

Abiotic components of ecosystem are the nonliving features of ecosystem on which the living organism depends. It is basically referred to the physical environment and its numerous interacting variables, e.g. inorganic substances and climatic factors.

A. Inorganic Substances

Inorganic substances are generally associated with nonliving things.

Bio-geochemical Cycle

It refers to the circulation of chemical nutrients such as carbon, oxygen, nitrogen, phosphorus, calcium, and water etc. through the biological and physical world. In other words the chemical element get recycled while moving through both biotic (biosphere) and abiotic (lithosphere, atmosphere, and hydrosphere) compartments of Earth. It's a circular series of reaction where the chemical element gets back to its original position for joining the cycle again and again. The most well known cycles in the lists are:

- Nitrogen Cycle
- Oxygen Cycle
- Carbon dioxide Cycles
- Water Cycle
- Phosphorous Cycles

(a) Nitrogen & (N₂) Cycle

Nitrogen cycle is a process by which nitrogen is converted between its various chemical forms. 78% of earth atmosphere is nitrogen. Nitrogen cycle is necessary because plants cannot absorb nitrogen directly; they can only absorb in the form of *nitrate*. Nitrogen cycle have 5 important processes, i.e. fixation, ammonification, nitrification, assimilation and denitrification.

1. Nitrogen fixation

It is the process of conversion of nitrogen(N₂) to ammonia (NH₃) because it is the only method by which organisms can attain nitrogen through atmosphere. Bacteria called *Rhizobium* fix nitrogen, be residing in the roots of plants making root nodules

2. Nitrification

It is the process of conversion of ammonia (NH₃) to nitrate (NO₃-)



3. Assimilation

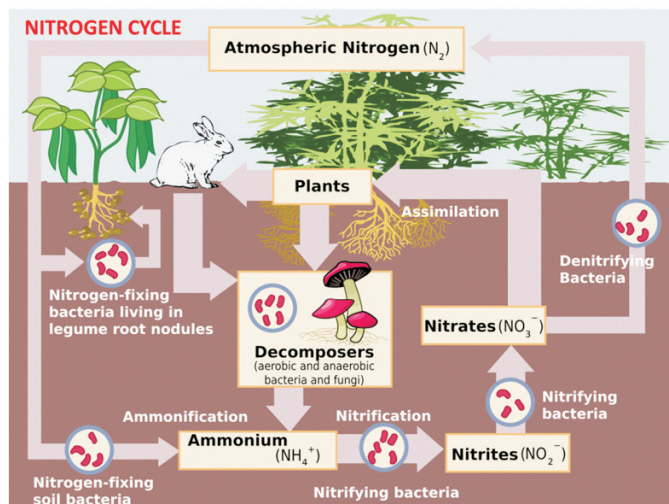
It is done by plants roots. Since nitrogen is present in nitrate form it is absorbed along with water from the soil and then converted into organic matter by plants.

4. Ammonification

It is the process of conversion of organic nitrogen compounds to NH₃. Urea and uric acid are excreted by animals in soil along with nitrogen components, these components are converted into ammonia by bacteria, like- Bacteria→ nitrification→assimilation

5. Denitrification

It is the reduction of NO₃- to gaseous nitrogen N₂. It is done by denitrifying bacteria(*Pseudomonas*) which convert nitrates/nitrites into elemental nitrogen, which escapes to atmosphere completing the cycle.



(b) Oxygen & (O_2) Cycle

- Oxygen is a very important element for the existence of all flora and fauna. Atmosphere contains 21% of oxygen.
- The main source of oxygen is atmosphere. Plants and animals absorb oxygen through respiration either from water or air and leaves through photosynthesis.
- In respiration process some of the oxygen returns to the atmosphere in the form of carbon dioxide and water vapour. During the process of photosynthesis gaseous oxygen is released completing the oxygen cycle.
- The source of ozone is the oxygen in the atmosphere. Ozone layer protects the living being from the UV radiation which reaches the earth.
- By burning fossil fuels man decreases the amount of oxygen in the atmosphere and increases the carbon dioxide content.

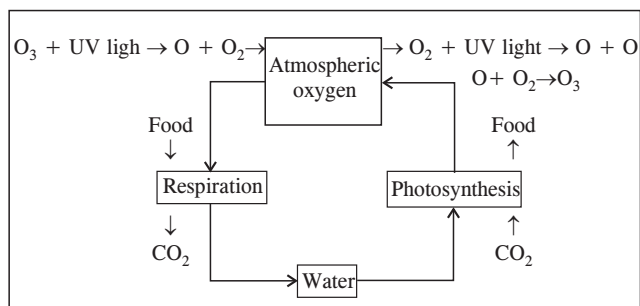


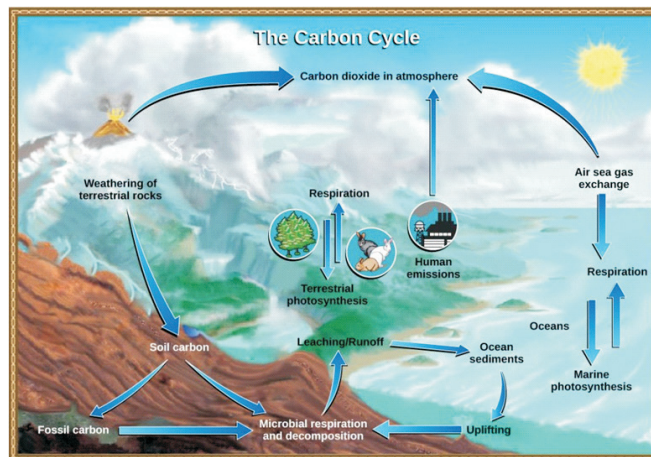
Fig. Oxygen Cycle

As the oxygen concentration in the atmosphere remains constant at 21%, it is likely that natural degradation of ozone must occur to maintain the ozone : oxygen equilibrium. The oxygen cycle is affected most by human activities such as running of automobiles and consumption of fossil fuels, thus releasing more carbon dioxide in the atmosphere.

(c) Carbon dioxide (CO_2) Cycle

Carbon constitutes 49% of the dry weight of an organism, ocean constitutes 71% and atmosphere only constitutes 1% of carbon. Carbon cycle occurs through atmosphere, ocean and through living and dead organisms, decomposers. CO_2 is given out by respiration and fixed into organic matter by the process of photosynthesis. Process can be explained in following steps:

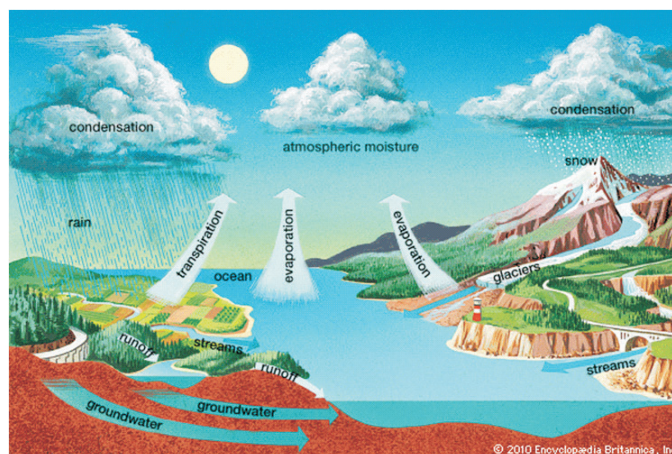
- Through the process of photosynthesis carbon enters into living world in the form of carbon dioxide.
- This organic compound (food) is then passed from the producers to the consumers (herbivores & carnivores).
- By the process of respiration or decomposition of dead bodies of plant and animals by decomposers this carbon returns back to the surrounding medium.
- Recycling of carbon is also done by the burning of fossil fuels.



(d) Water & (H_2O) Cycle

There is a continuous exchange of water between living organisms, air, land and sea. Steps of water cycle can be explained as:

- Evaporation of water takes place from oceans, rivers and lakes which takes water into atmosphere in the form of vapours.
- Clouds and water is formed when these vaporized water is cooled and condensed.
- Ultimately these cooled water vapour return to earth as rain and snow.
- Thus, hydrological cycle is the continuous and balanced process of evaporation, precipitation, transpiration and runoff of water.



Water Cycle

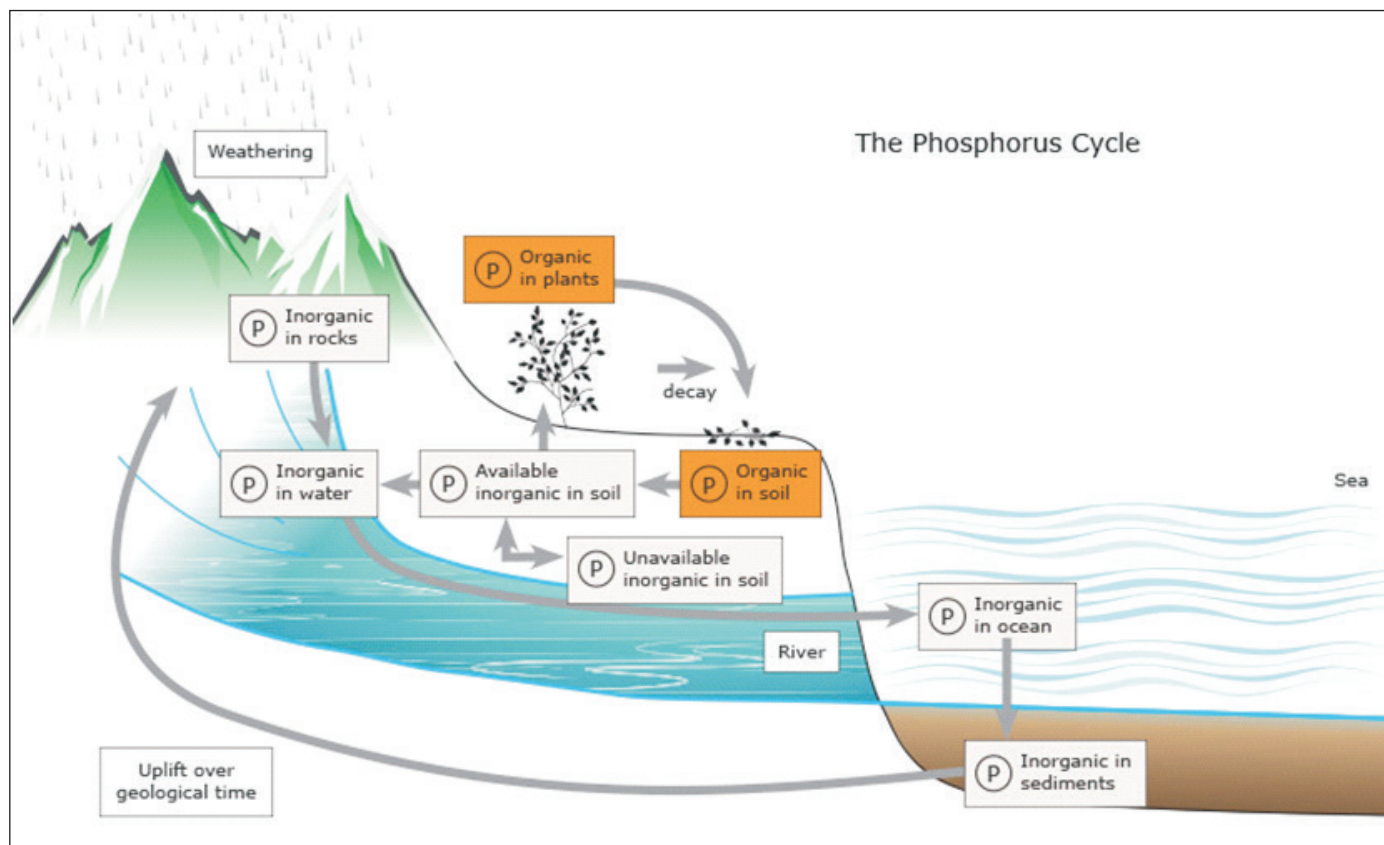
(e) Phosphorus & its cycles

- Phosphorus cycle is also called as sedimentary cycle because the main reservoir is rocks and the earth crust.
- Weathering of these rocks leads to release of Phosphorus and its absorption by plants roots, which in turn gets transferred to higher trophic levels.

- Excretory material of birds on rocks called Guano is also source of Phosphorus. Phosphates are available in rocks and soil in organic form in lithosphere.
- They are converted into organo phosphates by plants. Phosphates are also added into soil by phosphatic fertilizers.
- There is an excess algae growth which leads to

eutrophication when soluble phosphates reach rivers and streams from agricultural land rich in phosphates.

- Return of phosphates to the Earth is by the decay of plant and animal matter and subsequent absorption.
- Phosphorus moves in a cycle through rocks, water, soil and sediments and organisms.



Human activities and their influence on biogeochemical cycles and climate change

As an important biotic element of earth, human being plays a major role in influencing the biogeochemical cycles directly or indirectly. In recent years there has been a growing awareness of the extent to which human activities, such as deforestation and fossil fuel burning has been affecting the environment processes negatively so as the natural cycles. These changes in atmospheric processes are disturbing a variety of cycles in ecosystem services that humans depend upon.

Biogeochemical cycles are always considered as a state of equilibrium that is a balance in the cyclic process of the elements between spheres. However, overall balance may involve elements distributed on a global scale and that is why a disruption in one cycle causes a disruption in all other cycles. The impact can be assessed and analysed through number of such incidents such as:

- **Use of phosphorus fertilizers or commercial synthetic fertilizers** has affected the phosphorus and nitrogen cycles. Sometimes plants unable to utilize all of the phosphate fertilizer applied to them and as a consequence, much of it is lost from the land through the water run-off. The phosphate in the water is eventually precipitated as sediments at the bottom of the water body.

In certain lakes and ponds this may be redissolved and recycled as a problem nutrient. Other important sources of phosphate are in the out flows from municipal sewage treatment plants. Without an expensive tertiary treatment, the phosphate in sewage is not removed during various treatment operations. Again an extra amount of phosphate enters the water.

- **Mining of Fossil fuels** have greatly impacted the carbon cycle where fossil fuels have been mined from the earth crust. Carbon dioxide is number one green house gas contributing to global warming and climate change. Additionally, clearing of vegetation that serve as carbon sinks has increased the concentration of carbondioxide in the atmosphere.
- **Production of Sulphur dioxide** has an impact on the sulfur cycle is primarily in the production of sulfur dioxide (SO_2) from industry (e.g. burning coal) and the internal combustion engine. Sulfur dioxide can precipitate onto surfaces where it can be oxidized to sulphate in the soil (it is also toxic to some plants), reduced to sulphide in the atmosphere, or oxidized to sulphate in the atmosphere as sulphuric acid (a principal component of acid rain). Sulphur compounds play a big role in the climate system because they are important for the formation of clouds.

Additionally, a lot of sulphur is brought into the air by volcanic eruptions. A strong eruption can emit particles up to the stratosphere hence leading to cooling down of the planet.

Cultivation of legumes and use of nitrogen fertilizers:

As a result of extensive cultivation of legumes, creation of chemical fertilizers, and pollution emitted by vehicles and industrial plants, human beings have more than doubled the annual transfer of nitrogen into biologically available forms. Humans have significantly contributed to the transfer of nitrogen gases from Earth to the atmosphere, and from the land to aquatic systems through four main processes:

- The application of nitrogen fertilizers to crops has caused increased rates of denitrification and leaching of nitrate into groundwater. The additional nitrogen entering the groundwater system eventually flows into streams, rivers, lakes, and estuaries. In these systems, the added nitrogen can lead to eutrophication.
- Increased deposition of nitrogen from atmospheric sources because of fossil fuel combustion and forest burning. Both of these processes release a variety of solid forms of nitrogen through combustion.
- Livestock ranching. Livestock release a large amounts of ammonia into the environment from their wastes. This nitrogen enters the soil system and then the hydrologic system through leaching, groundwater flow, and runoff.
- Sewage waste and septic tank leaching.

Much of the nitrogen applied to agricultural and urban areas ultimately enters rivers and near shore coastal systems. In near shore marine systems, increases in nitrogen can often lead to

- (i) *anoxia (no oxygen) or hypoxia (low oxygen)*
- (ii) altered biodiversity
- (iii) changes in food-web structure, and
- (v) general habitat degradation.
- (vi) One common consequence of increased nitrogen is an increase in harmful algal blooms (howarth 2008).
- (vii) Toxic blooms of certain types of dinoflagellates have been associated with high fish and shellfish mortality in some areas. Even without such economically catastrophic effects, the addition of nitrogen can lead to changes in biodiversity and species composition that may lead to changes in overall ecosystem function. Some have even suggested that alterations to the nitrogen cycle may lead to an increased risk of parasitic and infectious diseases among humans and wildlife.

Additionally, increases in nitrogen in aquatic systems can lead to increased *acidification* in freshwater ecosystems.

B. Climatic Factors

Climatic factors are the environmental factors which influence the growth and development of plants, condition in which living organism lives etc. they are defined below:

(a) Light

- The main supply of energy for organisms in ecosystem is sunlight which is an important abiotic factor.

- Through the process of photosynthesis plants with chlorophyll can change light energy to chemical energy. The chemical energy formed is hence stored as complex organic substance (food).
- Besides visible energy electromagnetic spectrum of solar radiation includes UV, infrared radiation (heat) and radio waves.
- Both UV light and infrared radiation are abiotic factors of high importance for ecosystems. To differentiate between flowers insects use UV.
- UV light also works in controlling some biochemical reactions that could be dangerous for the living beings; besides, UV light annihilates pathogens and can cause favorable mutations in all life forms.

(b) Temperature

- Atmospheric temperature of the place depends upon the slope, altitude, latitude, topography, etc. temperature lowers as we go from equator to poles – tropical, subtropical, temperate and arctic.
- Organisms living in tropical regions are called megatherms, in subtropical regions are called mesotherms, in temperate regions are called microtherms, and in arctic regions are called hekisthotherms.
- Following are the examples of temperature change in ecosystems: fruit tree such as peach require cold period so that they can blossom in springs, in winters deciduous trees loss their leaves and enter in dormancy etc.

(c) Pressure

- Atmospheric pressure is the pressure caused by the weight of the atmosphere. Atmospheric pressure decreases with the height and is felt maximum at the surface.
- It is measured by the instrument called barometer.
- All organisms can survive only in a specific range of atmospheric pressure and when the air pressure is low in the higher altitudes it becomes difficult for some species to breathe because of the insufficient oxygen present in the higher altitudes.
- In the depth of the oceans the atmospheric pressure increases with the increasing depth and again this makes only definite types of animals and plants to survive in certain specific ocean regions.

(d) Humidity

- Humidity is the water vapor or water in gaseous form present in the atmosphere. It can also be defined as: molecules of water/unit volume.
- It is measured by Hygrometer. Humidity is of three types:
- Specific humidity: it is defined as the ratio of weight of moist vapour to the weight of moist air.
- Absolute humidity: it is the mass or weight of vapour per unit volume of air.
- Relative humidity: it measures the current absolute humidity relative to the maximum for that temperature. It is measured in percentage.
- It affects water loss from plants (transpiration rate) and evaporation from animals; relative humidity (%) expressed as the amount of water vapour in the air compared to what the air would hold if completely saturated at that temperature.

(e) Miscellaneous

Some other climatic factors include water, soil and chemical components.

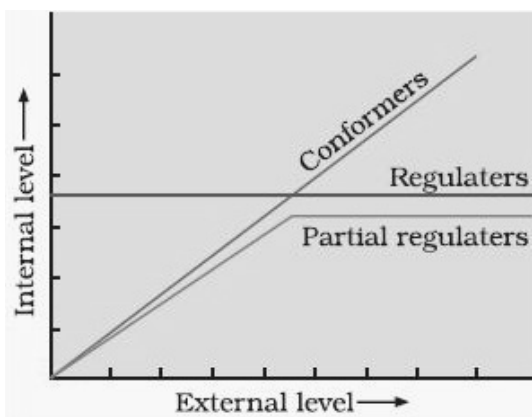
How do organisms Respond to Abiotic Factors:

The fitness of a species in a particular environment depends upon the efficiency of all physiological and biochemical processes within the body.

Some of the animals maintain their body temperature or osmotic concentration of body fluid to the optimum level and maintain a constant internal environment (a process called **Homeostasis**). Their performance is the maximum regardless of the weather conditions outside the body. To maximize the working efficiency, human uses artificial means, like air conditioner, in summer, or heaters / blowers, in winter. The animals, however, have natural means, mainly physiological, to manage such stressful conditions.

There are 3- categories of animals that have specific ways to respond to external environment (or abiotic factors).

1. Conformers
2. Regulators
3. Partial regulators



Response of organisms to abiotic factors

1. Conformers

Majority of the animals (up to 99%) and nearly all plants, cannot maintain a constant internal environment. Their body temperature and osmotic concentration changes according to the surrounding conditions. Such animal and plants are called *Conformers*. Both, the thermal regulation and osmo- regulation are energetically expensive processes, that's why such a vast number of animals have not adopted such ways to maintain a constant internal environment.

2. Regulators

Some animals maintain their homeostasis by physiological or behavioural means. All birds and mammals (Homeothermic or Warm blooded) and a few lower vertebrates and invertebrates can maintain constant body temperature and constant osmotic concentration regardless of the surrounding conditions. Such organisms are called *Regulators*.

Human constantly maintain their body temperature at 37°C. When it is hot outside (in summer) we start

sweating (primarily for cooling the body) and when the surrounding temperature is lower than the body temperature (in winter), we start shivering. The muscle contraction produces heat to raise the body temperature.

3. Partial regulators

If the stressful condition are localized, the animals can '*escape in place*' and migrate to nearby region, having better environmental conditions. If the unfavourable conditions are for a shorter duration then the animals can '*escape in time*' and avoid those conditions suspending their activities. Such organisms are partial regulators.

Based upon the situation, the partial regulators may proceed for migration or for suspension of body activities.

(i) **Migration** : The animal can move away temporally from a stressful environment to a more hospitable area and return when the environmental condition are favourable. Every winter thousands of birds migrate from Siberia (extremely cold) to Keoladeo National Park in Bharatpur (Rajasthan)

(ii) **Suspension of body activities** : During unfavourable conditions the lower plants, fungi and bacteria, produce thick-walled spores which germinate when the environmental condition are favourable. The seeds and the other vegetative reproductive-structures of higher plants similarly undergo '*dormancy*' and reduce their metabolic activities when the conditions are unfavourable, and form new plant when the moisture and the temperature conditions are suitable. The animals that cannot migrate, avoid the stressful condition for the time period (*escape in time*) and undergo either Hibernation (in winter), e.g., Bear; or *Aestivation* (in summer), e.g. snail and few fishes. Frog undergoes hibernation in winter and aestivation in summer.

When the condition are unfavourable, many species of zooplanktons, in lakes and pond, are known to enter '*diapause*', i.e., a stage of suspended development. Such diapauses also occur in eggs, larvae or pupae of insects in adverse conditions and breaking occurs when favourable conditions returns. The diapauses, i.e. arrested development, in insects may depend upon the photoperiod, temperature and the hormonal conditions.

Population

The population is a group of individuals of a particular species, which potentially interbreed and live in a well defined geographical area, and also share or compete for similar resources.

For ecological point of view a group of asexual individuals is also known as population.

Examples of populations are

1. Tigers in a national park
2. Rats in an abandoned building

The population has certain attributes or peculiar features which are not represented by the individuals. The important attributes are *Life expectancy, sex ratio, birth rate, death rate, age distribution* etc.

Birth or Natality rate – It is the number of births per thousand of a population per year. It can also be represented in percentage, or per capita, i.e., per individual.

If there were 50 lotus plants in a pond last year. This year the population has increased to 58 due to reproduction, then birth rate per capita can be calculated as

$$58 - 50 = 8$$

$$8 / 50 = 0.16 \text{ offspring per lotus per year}$$

$$\text{Or per capita birth rate per year} = 0.16$$

Death or Mortality rate – It is the number of deaths occurring in a population of one thousand per year.

This can also be represented in percentage or per capita.

If in a population of 50 house flies, 5 died in a week, then the per capita death rate per week can be calculated as $5 / 50 = 0.1$

- **Population growth** – The gross value of population growth is calculated as the difference of birth rate and death rate.
- **Dispersal** – The movement of the individuals in or out of the population affects the size of population. The movement of individuals into a population is called **Immigration**, and movement, out of the population is called **Emigration**.
- A set of local populations connected by dispersing individuals is called a **Metapopulation**.
- The *accurate population growth* involves migrant individuals also, which means the -
- **Population growth = (Birth rate + Immigrants) – (Death rate + Emigrants)**
- **Zero Population Growth** – When birth rate equals death rate, and the growth of the population is Zero, i.e. the size of the population remains constant, it is called Zero Population Growth or **Demographic Transition**. Considering migration, at zero population growth :
- **Birth rate + immigrants = Death rate + emigrants**
- **Bio-Index number** – It is the ratio of birth rate to the death rate. In case of Zero population growth the bio-index number is one

$$\text{Bio-index number} = \frac{\text{Birth rate}}{\text{Death rate}}$$

- **Population size** : The population size may range from few individuals to millions depending upon species, geographical area, impact of predators, outcome of competition and effect of pesticides etc. Sometimes the population size is either too big to measure or meaningless, then the size is more technically called as population density.
- The **population density** can be defined as the number of individuals per square unit area (in terrestrial organisms) or per cubic unit area (in aquatic or aerial organisms). Sometimes the population density is not required and only Relative density serves the purpose. For example, the fish caught per trap in a lake can be used as population density.

Growth Models

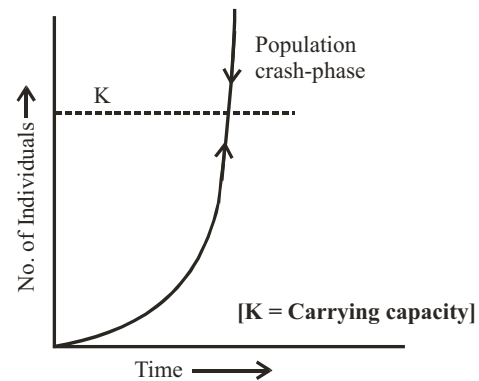
The growth of population with time shows specific and predictable patterns. The 2-common patterns are

1. Exponential or geometric growth pattern
2. Logistic growth pattern

1. Exponential growth

When resources like food and space etc. in a habitat are unlimited, each species realizes its full reproductive or biotic potential and grows in an exponential or geometric fashion. For example if a reindeer population (N) is allowed to grow in a predator free environment, the population in time 't' grows beyond carrying capacity and forms 'J' shaped growth pattern/ curve. Such growth also occurs in algal bloom, insects during rainy season and *Paramecium* (doubling every day)

Thus under unlimited resources and absence of check any species can grow exponentially. Even the slowest breeder, elephant (gestation period –22 months) can grow to enormous number.



Population Growth Curve (J - shaped)

If the population size is 'N' and birth rate and death rate per capita are respectively 'b' and 'd', then the change in population size, i.e., increase or decrease during a unit time period 't', will be

$$\frac{dN}{dt} = (b - d) \times N$$

If $(b - d) = r$, then $\frac{dN}{dt} = rN$

Here 'r' is called '**Intrinsic rate of natural increase**' or **Biotic potential (maximum capacity of reproduction)**, Which indicates the impact of biotic and abiotic factors in population growth. The value of 'r' for human population in 1981 in India was 0.02.

The integral form of exponential growth equation is

$$N_t = N_0 e^{rt}$$

N_t = Population density after time 't'

N_0 = Population density at time zero, '0'

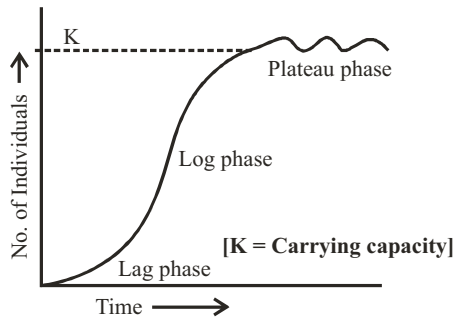
r = Intrinsic rate of natural increase

e = the base of natural logarithms (2.718..)

2. Logistic growth

In nature no species has unlimited resources at its disposal to permit exponential growth. There starts competition between individuals for limited resources. The population growing in a habitat initially shows 'lag phase', followed by phases of acceleration or deceleration (*log phase*) and finally the stage of **asymptote**- when population density reaches carrying capacity (plateau phase or stationary phase).

If the population density 'N' is plotted against time 't', the result is '**S**' shaped growth pattern/ curve (**Sigmoid curve**).



Population Growth Curve (Sigmoid, S-shaped)

This type of growth pattern is also called as '**Verhulst-Pearl logistic growth**' and can be represented as following equation

$$\frac{dN}{dt} = rN \left(\frac{K - N}{K} \right)$$

K = Nature's carrying capacity in that habitat

(K - N)/ K or 1 - N/ K = environmental resistance

Carrying capacity (K) – It can be defined as the maximum number of individuals which the environment can support or sustain. Hence, it is the capacity of environment and not of individuals.

The carrying capacity of the environment depends upon 3-components, i.e.

- Productive system – It includes cropland, orchard etc. to provide food and fibres.
- Protective system – It includes forest, ocean etc. which regulate temperature and humidity.
- Waste assimilative system – It includes the generation of waste material in water, air and soil.

Population Interactions

Thus in nature, plants, animals and microbes do not and cannot live in isolation but interact in various ways to form biological community. The interspecific associations arise from the interaction of populations of different species. Such associations can be beneficial, harmful (detrimental) or neutral (neither beneficial nor harmful).

Following are different interspecific associations/ interactions. The '+' indicates beneficial interaction, '--' detrimental and '0' neutral interaction.

Population Interactions

S.N.	Name of interaction	Species A	Species B
1.	Parasitism	+	-
2.	Commensalism	+	0
3.	Mutualism	+	+
4.	Predation	+	-
5.	Competition	-	-
6.	Amensalism	-	0

Parasitism

In this interaction one species (Parasite) is benefitted and the other (Host) is harmed (+, -). The parasitism ensures free **lodging** and free **meals**. Many parasites are host specific while others can parasitize on various species of host.

The hosts evolve special mechanisms to reject or resist the parasite. The parasite on the other hand evolves the mechanisms to counteract and neutralize their effects.

Majority of the parasites harm the hosts by reducing their growth, reproduction, survival and population density. They may also make the host physically weak and vulnerable to predation. The parasites can be of 3-types

- Ectoparasite
- Endoparasite
- Brood parasite

The **ectoparasites** live or feed on the external surface of the host. The most familiar examples are lice and bed bug on human, tick on dogs and leech on cattle. The mosquitoes cannot be considered as true ectoparasites, since they take only meal from the host-body, and do not make lodging. Amongst plants the common example is **Cuscuta**, which has lost both chlorophyll and leaves during evolution and parasitizes hedge plants. The other parasites are copepods, *Petromyzon* and hagfishes, for which the hosts are marine fishes.



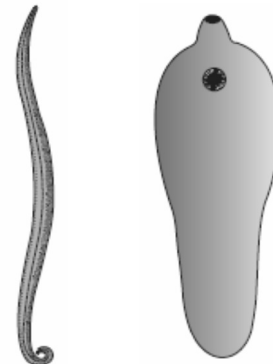
Bed bug

Petromyzon on fish

Tick-Dog

The **endoparasites** live inside the body of the host. Their morphological and anatomical features are greatly simplified, but their life cycles are more complex because of their extreme specialization. Their reproductive potential is very high. They can be **monogenetic** (involving single host) or **digenetic** (involving two hosts). The common monogenetic endoparasites are *Ascaris* (round worm) and *Entamoeba* in small and large intestine respectively. The familiar digenetic endoparasites are *Taenia* (tape worm) and *Fasciola* (liver fluke) in intestine and liver respectively. The *Plasmodium* is also a digenetic endoparasite in human and female *Anopheles* mosquito. The filarial worm is similarly a lymph parasite.

The koel or cuckoo is a **brood parasite** and lays the eggs in the nest of the host, crow, who incubates them. The eggs of the parasitic bird has evolved resemblances with the host's eggs in colour and shape and size.

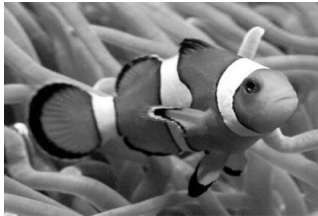


Ascaris Male

Liver fluke

Commensalism

In commensalism one species is benefitted and other is neutral i.e., neither benefitted nor harmed (+, 0). The association of 'egret and the cattle' is the classical example of commensalism. As the cattle move during grazing, they stir up and flush out the insects from the vegetation which otherwise be difficult for the egret to find and catch. Here the cattle gets no benefit from egret. The examples of commensalisms are 'clown fish hiding in the colony of sea anemone', 'orchid growing as an epiphyte on mango tree' and the 'barnacle growing on the back of whale'. Here, sea anemone, mango tree and whale derive no benefit from the association.



Clown fish with sea anemone



Egret and Buffalo

Mutualism

In this interaction both the species are benefitted (+, +). **Lichens** is such relationship between fungus and photosynthetic algae/ cyanobacteria. The **Mycorrhizae** is also a similar association between fungus and the roots of higher plants. The fungi here, helps the plant in absorption of essential nutrients from the soil, while plant provides the food to the fungus. In plant-animal association there has occurred co-evolution of mutualism. In many species of fig trees there is a close one-to-one relationship with the species of wasp, for pollination. The female wasp pollinates the fig inflorescence while searching for suitable site for egg laying (oviposition). She lays the eggs inside the fruit and the seeds of the fruit are also consumed by the developing larvae of the wasp. Thus, both are mutually benefitted.



Fig and Wasp



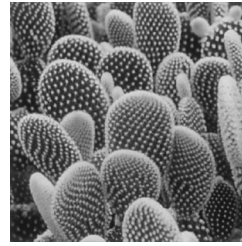
Ophrys (Orchid)

Predation

Like parasitism, in this interspecific interaction also one species is benefitted and the other is harmed (+, -). The common examples of predator and prey are, Tiger and deer, cat and mouse, and lion and zebra. In a broader ecological sense the sparrow feeding on seed, or the herbivorous animals eating plants, are also predators. The predators have following important roles in the community.

1. They (herbivores) act as a conduit for transferring energy, fixed by plants, to higher trophic levels.
2. They keep prey population under check. Otherwise, the

prey may achieve high population density to make the ecosystem unstable.



Prickly pear



Lantana

3. Predation also reduces competition amongst preys species and maintains species diversity. (When *Pisaster* (star fish), a predator, was removed from American pacific coast, about 10 species of invertebrates got extinct because of interspecific competition).

Competition

It is defined as an interspecific process in which the fitness of one species (measured in terms of 'r') is significantly lower in the presence of another species. It is (-, -) relationship. Charles Darwin also considered the interspecific competition a potent force in organic evolution. The competition generally occurs when closely related species compete for the same but limiting or depleting resources.

The competition may ever occur when resources like food and space etc. are abundant, since the feeding efficiency of one species may be reduced due to interfering presence of other species. This competition is called **interference Competition**. Gause's 'competition exclusion principle' states that two closely related species competing for the same resource cannot co-exist indefinitely and the competitively inferior will be eventually eliminated. This is, however, true only when resources are limited.

Amensalism

In this association one species is harmed but the other remains neutral (-, 0). For example the fungus, *Penicillium notatum* growing close to bacteria, like *Staphylococcus*, kills the bacteria due to the secretion of penicillin, whereas the fungus remains unaffected. Similarly the leaves falling from the tree either kill or adversely affect the growth of the seedling population underneath the tree.

Ecological Niche

Ecological Niche : It is the interactions between a species and the its biotic and abiotic environment, which represents a very basic and fundamental ecological concept. Every niche has two sides; one indicates the effects of environment on a species and other indicates effect of species on the environment. If we give more emphasis on the above given aspect within one framework then ecological niche can be defined as an ecological space where species can live and utilize the environmental resources and perform its functions.

It can be defined as a collective set of roles and characteristic that an organism plays in the environment.

Niche as ecological function of the species

Every species has some particular role in an ecosystem and that same role can be fulfilled by different species in different places. Charles Elton, has emphasized the functional role of species. This is the functional niche which can be referred to a species position in food webs and trophic chains.

Fundamental and realized niche

Hutchinson gave the idea of fundamental niche in which he indicated that as, a multidimensional cloud of favorable conditions determined by all abiotic and biotic components where the species can reproduce and flourish, whereas the realized niche is a subset of fundamental niche.

Competitive exclusion principle: This principle was formulated by the biologist G.F. Gause. According to which no species can occupy the same niche at the same time.

Resource partitioning:

It is a concept of niche separation which is ensured by species coexistence. According to Gause's principle no two species can coexist on the same limiting resource. Species avoid competition by partitioning resources and habitat among themselves.

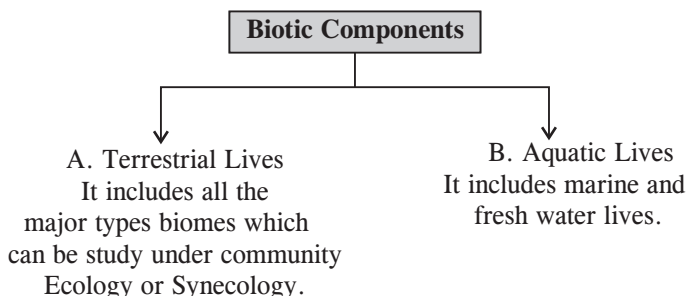
Intra-specific competition and inter-specific

Intra-specific competition can occur between the same species that have nearly identical niche. If natural resources are limited then that can check the population abundance distribution and density due to intra-specific competition.

Inter-specific competition can restrict the size and distribution of competing population as two different species compete for the same natural resource of an ecosystem for example Lion and Leopard shows inter-specific competition as both have the common prey.

II. Biotic Components (Biodiversity)

Biotic components are the living organism which shapes the ecosystem and comes regularly in contact with each other.



A. Terrestrial Lives

Community Ecology or Synecology:

Community ecology is a study of pattern and processes involving at least two species on many spatial and temporal scale which include the distribution, structure, abundance, demography, and interactions between coexisting populations.

Biomes: Biome is a large ecosystem that have common characteristic due to similar climates and can be found over a range of continent. In other words biomes are large natural eco-system wherein we study the total assemblage of plant and animal communities. Though a biome includes both plant and animal communities but a biome is usually identified and named on the basis of its dominant vegetation. Most of the ecologists have recognized at least nine different biomes.

Land biome include Tropical rain forest Coniferous forest, Temperate broad leaf deciduous forest, Mediterranean forest, Tropical deciduous forest, Tropical scrub, Grassland, Tundra and Desert.

1. **Tropical rainforest:** Tropical rainforest are the earth's most complex biome and have great diversity of plants and animals. It is located between the tropic of cancer and tropic of Capricorn. It covers around 5% of the earth's land surface and has 50% of all known species of the earth. These forests have average temperature of 20-25°C which can varies throughout the year. The primary productivity is high but has low net primary productivity. The important characteristics of tropical rainforest that has low nutrient in soil but has tight cycling of nutrients, all the nutrients are present in the buttresses of the trees. Species diversity is highest in the tropical rainforest. Drip tips facilitate drainage of precipitation from the leaves make it rain even when it is not raining. In this forest Strangler are the plants present as an epiphytes in the canopy and send their roots downward to the forest floor. The fig family is well represented among the stranglers.

2. **Coniferous forest:** Coniferous forest is mainly consists of evergreen gymnosperms with needle shaped leaves. It is the largest biome on the Earth. Soil of the forest is thin, nutrient poor and acidic as conifer needles get decomposed here and bacterial activity is slowed down at lower pH. Precipitation is primarily in the form of snow 40-100 cm annually and temperature ranges from 0°C to 5°C. There are two types of Coniferous trees: Northern Boreal Forest and Temperate Coniferous Forest.

Northern Coniferous Forest: South of the tundra and to the north of deciduous forests & grasslands between 50°C to 60°C N latitude in Canada Europe.

Temperate Coniferous Forest: Lower latitude of North America, Europe and Asia in high elevation of mountain.

3. **Temperate Broadleaf Deciduous Forest :** Such type of forest is mostly found in western and central Europe Eastern Asia and eastern North America. The stratum in the forest is trees, understory shrubs, herbaceous substratum and ground layer. Broadleaf forest has moderate to high rainfall and temperature, and the average precipitation being 700mm and average temperature being 10.8°C. Broadleaf trees are dominated with trees such as Oak, Maple and Fagus etc.

4. **Mediterranean Forest :** Located in the west coastal regions between 30° and 40° North and South latitude (mild temperate region) usually surrounded by desert and grassland. Mediterranean forest has hot and dry season in the summer, and cool and moist winter season. Average precipitation is 262 mm and average temperature in the forest is 14.7°C. In such forest plants moves to the dormant condition during dry summer and spiny shrub dominate at that time.

5. **Tropical deciduous Forest or Tropical dry Forest:** Such types of forest are found on the fringes of the tropical rainforest. These forests rank second after tropical rainforest in species richness. Common tree species are Shorea robusta, Tectona grandis, Hopea odorata and Dipterocarpus indicus etc.

6. **Tropical scrub:** Such forest has low tree vegetation type that grow in hot somewhat dry to semiarid lowlands

between desert and Savanna on the one hand and tropical deciduous forest and tropical rain forest on the other hand.

7. **The Grassland :** It is found in all ranges of climate from mesic to xeric and from cold to warm condition. They are generally open and continuous in fairly flat areas dominated by grasses. They experience wide range of temperature from about -20°C to 30°C in extreme winter and warm summer. There are two main division of grassland:

Temperate grassland: Such types of grassland have cold winters and warm summers. Summer temperature ranges to 38°C in summer and as low as -40°C in winter. Various species of grassland which include purple needle grass, blue grass, and buffalo grass are found here. Located between temperate forest at high latitude and desert at subtropical latitude and known by different names in different parts of the world as:

The Prairies of the great plains of North America

The Pampas of South America

The Veldt of South Africa

The Steppes of Central Eurasia

The Savanna in Africa

8. **Tundra:** Tundra is the world coldest and driest biome. This type of biome is totally snow covered vegetation at mountain tops. Soils are rich in organic matter due to slow decomposition rate and it is also one among the Earth's three major carbon dioxide sink. There are three types of tundra- 1. Arctic Tundra 2. Antarctic Tundra 3. The Alpine Tundra

9. **Desert :** Desert biome is the driest of all the biome and is located at 30° North & South and cover about one fifth of the Earth's surface. Most of the desert receive about 250 mm of rain . Due to poor canopy and sparse distribution of plants. The leaf area index is less than one and productivity is also less. Most of the desert receive about 250 mm of rain per year as compared to rainforest which receive over 2,000 mm. The largest desert on Earth is Antarctica. They are classified into:

- **Arid deserts:** these types of deserts are found in North-America, South-America, Africa, and Southern Asia where there is low latitudes. Here generally the temperature is hot and dry with few occurrence of rainfall in winters. In daytime it is very hot as there are no clouds to cover the Earth. The soil here is mostly sand or coarse and rocky. Here vegetation is mostly shrubs and small trees and leaves are adapted to retain water. Here the animals are generally active during night.
- **Semi-arid deserts:** These deserts are found in North-America, Europe, Russia and Northern Asia. Here also there is low rainfall during winters. Here animals can be found in daytime but under the trees or shades.
- **Coastal deserts:** these deserts are found in areas which are warmer to cooler such as in Neotropic and Nearctic area. Here the summers are long and warmer while the winters are short and cold. Here the sand consists of alkaline and the soil is porous so that rain water can enter the ground. Mostly the vegetation here is thick foliage with good water

retention quality. The roots of these plants are near to the surface so that they can get water before water drains to the soil. Animals of these deserts comprise rough skinned amphibians, birds of prey, scavenger mammal's reptiles and insects. Animals here are largely nocturnal during the warmer months.

- **Cold deserts:** they are the deserts which occur mainly in the cold regions. Here the temperature during the warmest month is 10°C . they are mostly covered with snow and ice and due to this they do not support life. Animals in the cold desert are burrowers, even the carnivores and reptiles, which even though cold-blooded, have made their homes in the cold desert. Antarctica is the largest cold desert.

B. Aquatic Lives

Aquatic lives are present in Marine and freshwater ecosystem.

1. Sea Life

Sea regions are broadly divided into coral reefs, estuaries and oceans.

- **Oceans:** they are the biggest and the most varied of the ecosystem. Most of the oxygen in the atmosphere is generated by the algae. Here salt water evaporates and turns to rain which in turn falls on land. Large amount of carbon dioxide is absorbed by the algae in the atmosphere. Inter - tidal zone is the zone which connect ocean to the land. Only few species exist in rocky coastal areas as very few tides reach there.
- **Other oceanic zones:** deep sea which is also called benthic zone is the host to slit, sand and slowly decomposing organisms. Sunlight does not reach these areas so these areas are very cold. There are only few plants here and animals include starfish, anemones, sponges, amongst others, as well as several micro-organisms. Abyssal zone is the deepest part of the ocean. Fishes such as oddities and many species of invertebrates are found here.
- **Coral Reefs:** they are the marine ridges and mounds which are formed due to the decomposition of calcium carbonate of living organisms. Coral consist of animal and algae tissues. It is a living organism. Corals use tentacles to catch microorganisms like animals do and feed by the process of photosynthesis like plants. The coral reef is also host to other species such as starfish, octopi and other mollusks. Coral animals cannot live in water cooler than 65°F (18°C), therefore coral reefs are found mostly in warm, shallow, and tropical seas.
- **Estuaries:** transition area between river and sea is called estuary. They are highly productive and rich in nutrients. There are many different names of estuaries like bays, sounds, inlets, harbors, and sloughs.

2. Fresh water Life

The name freshwater is due to the less salt content in them. They exist in various forms such as lakes, rivers, ponds, swamps or wetland and are host to wide variety of plants and animals.

- **Lakes:** lakes are the water bodies which can exist for centuries other than others like ponds which dry up frequently. The littoral zone, which is close to the shore, is host to a extensive range of species due to its warm and shallow environment. Several species of invertebrates, crustaceans, plants and amphibians bloom in this environment and in turn offer food for predators such as birds, reptiles and other creatures inhabiting the shoreline.
- **Wetlands:** large variety of flora and fauna grows in other still water bodies or wetlands like swamps, glades and marshes. Trees such as Cypress which are highly adaptable to high humidity of this region also grow in wetland. Other plants such as pond lilies and sedges also grow here. Animals found here are different types of reptiles, mammals, amphibians and birds and hundreds of insects. Starting point of rivers and streams are mostly snow and ice melting and spring. At the end they end

up in ocean or other water body. Flora and fauna are different here from the lakes and ponds as the water is continuously flowing. Depending upon water temperature and the exposure of riverbank to the sunlight small fishes such as river trout and crayfish can be found in several areas. Salmon and other vigorous fishes can be found in cold areas while fishes like catfish, carp, and other bottom feeders can be found in warm areas which are rich in sediments and decaying matter. River plants comprise floating weeds and algae, mostly found forming around rocks and submerged tree roots.

The area where freshwater meets saltwater, is called an estuary; this area generally features distinctive features, trees and algae, seaweed, wetland flora, and several species of invertebrates, birds, reptiles and crustaceans congregate into a composite ecosystem, serving as a trade center to the world's aquatic biomes.

Exercise -1

1. What is true of ecosystem?
 - (a) primary consumers are least dependent upon producers
 - (b) primary consumers out-number producers
 - (c) producers are more than primary consumers
 - (d) secondary consumers are the largest and most powerful.
2. Homeostasis is
 - (a) tendency of biological systems to change with change in environment
 - (b) tendency of biological systems to resist change
 - (c) disturbance of self regulatory system and natural controls
 - (d) biotic materials used in homeopathic medicines.
3. Food chain in which microorganisms breakdown the food formed by primary producers is
 - (a) parasitic food chain
 - (b) detritus food chain
 - (c) consumer food chain
 - (d) predator food chain.
4. If we completely remove the decomposers from an ecosystem, its functioning will be adversely affected, because
 - (a) mineral movement will be blocked
 - (b) the rate of decomposition will be very high
 - (c) energy flow will be blocked
 - (d) herbivores will not receive solar energy.
5. The abundance of a species population, within its habitat, is called
 - (a) relative density
 - (b) regional density
 - (c) absolute density
 - (d) niche density.
6. The primary succession refers to the development of communities on a
 - (a) forest clearing after devastating fire
 - (b) newly-exposed habitat with no record of earlier vegetation
 - (c) freshly cleared crop field
 - (d) pond, freshly filled with water after a dry phase.
7. Which one of the following pairs is correctly matched?
 - (a) parasitism – intra-specific relationship
 - (b) uricotelism – aquatic habitat
 - (c) excessive perspiration – xeric adaptation
 - (d) stream lined body – aquatic adaptation.
8. The 'niche' of a species is meant for
 - (a) habitat and specific functions of a species
 - (b) specific place where an organism lives
 - (c) specific species function and its competitive power
 - (d) none of these.
9. Certain characteristic demographic developing countries are
 - (a) high fertility, low or rapidly fall rate, rapid population growth and age distribution
 - (b) high fertility, high density, mortality rate and a very young age
 - (c) high infant mortality, low fertility population growth and a very distribution
 - (d) high mortality, high density, uneven growth and a very old age distribution.
10. What is a keystone species?
 - (a) a species which makes up only a small proportion of the total biomass of a community, yet has a huge impact community's organization and s
 - (b) a common species that has plenty of biomass yet has a fairly low impact on the community organization
 - (c) a rare species that has minimal impact on the biomass and on other species in the community
 - (d) a dominant species that constitutes a large proportion of the biomass and which affects many other species.
11. The population of an insect species shows an explosive increase in numbers during rainy season followed by its disappearance at the end of the season. What does this show?
 - (a) the food plants mature and die at the end of the rainy season
 - (b) its population growth curve is of J-type
 - (c) the population of its predators increases enormously
 - (d) S-shaped or sigmoid growth of this insect.
12. Which one of the following statements is correct?
 - (a) Both Azotobacter and Rhizobium fix atmospheric nitrogen in root nodules of plants.
 - (b) Cyanobacteria such as Anabaena and Nostoc are important mobilizers of phosphates and for plant nutrition in soil
 - (c) At present it is not possible to grow maize without chemical fertilizers
 - (d) Extensive use of chemical fertilizers may lead to eutrophication of nearby water bodies.
13. Atmospheric nitrogen is converted to nitrogen oxides
 - (a) when fossil fuels are burned at high combustion temperatures.
 - (b) if the sun is shining brightly and there is a temperature inversion.
 - (c) on cloudy days when carbon and soot levels are unusually high.
 - (d) if radon gas is present when gasoline is burned in automobiles.
14. Having covered millions of acres of grasslands, the toxic weed leafy spurge has caused serious damage to the lower regions of Himachal. Being toxic in nature it is avoid by both cattle and horses alike. Leafy spurge has spread completely across many neighboring states of Uttaranchal. If left uncontrolled in these states, leafy spurge populations will
 - (a) keep growing indefinitely.
 - (b) reach their carrying capacity.
 - (c) shift from a S-curve to a constant rate of growth.
 - (d) shift from an S-shaped curve to a *J-shaped* growth curve.
15. Some beetles escape from a ship and fly to a small island covered with grass but with no trees or beetle predators. As the beetles feed, they destroy all the grasses. But with abundant food, the beetle population soars, doubling in size every month. After about a year, the population crashes as thousands of beetles have destroyed almost

- all of the plants and there is little left to feed the large population. This scenario best illustrates
- constant growth followed by equilibrium.
 - a population that has stabilized near its carrying capacity.
 - exponential growth followed by a population crash.
 - logistic growth ending at its carrying capacity.
16. Which of the following is NOT true regarding biotic potential?
- Environmental resistance factors such as disease and available space can prevent a population from maximizing its biotic potential
 - The biotic potential is the average rate of growth of a population over time
 - The ratio of males to females in a population is a characteristic that is used to determine biotic potential
 - The number of individuals of re-productive age in a population is a parameter used to determine biotic potential.
17. Which one of the following statements about the carbon, phosphorus, and nitrogen cycles is true ?
- The major source of carbon used by plants is the soil
 - The major source of nitrogen used by plants is the air.
 - Phosphorus has no atmospheric component.
 - Bacteria drive the phosphorus cycle.
18. Energy
- and nutrients flow through ecosystems.
 - and nutrients ecosystem within ecosystems.
 - cycles within ecosystem nutrients flow through ecosystems.
 - flows through and nutrients cycle within ecosystems
19. Flea beetles alone are unlikely to eliminate all of the leafy spurge in a region.
Instead, the number of leafy spurge plants and the number of flea beetles in a particular community may stabilize. At this point, the leafy spurge and flea beetle populations:
- have exhibited exponential growth followed by a crash.
 - have experienced constant growth.
 - are still experiencing exponential growth
 - are experiencing environmental resistance
20. Humans often manipulate the environment in ways that decrease the overall ecosystem capital because
- short term gains are often local while long-term losses in regulating and cultural services are experienced regionally.
 - short term gains are often regional while long-term losses in regulating and cultural services are experienced locally.
 - long term gains are often local while short term losses in regulating and cultural services are experienced regionally.
 - long term gains are often regional while short-term losses in regulating and cultural services are experienced locally.
21. Primary productivity of the open oceans is very limited because of
- the shortage of water.
 - the shortage of light.
 - the shortage of nutrients.
 - low temperature.
22. The economic gap between developing and industrialized countries may best be narrowed by
- the adoption of democratic forms of government in developing countries.
 - industrialized countries increasing shipments of food supplies to developing countries.
 - industrialized countries harvesting more natural resources in developing countries.
 - stabilizing population growth in developing countries.
23. Certain communities frequently have worms, clams, shrimp, and many other organisms clustered together. These communities nestle around hydrothermal vents where super heated water, springs from the bottom of the ocean. The bacteria present in these communities have special enzyme that allows them to form organic matter through chemosynthesis. They are found off the shore in Japan, deep in the ocean where no sunlight can penetrate. In this unusual ecosystem :
- there is no photosynthesis
 - no energy is transferred between trophic levels
 - temperature is the most important biotic factor
 - producers eat consumers
24. Biomes with permafrost are most likely:
- covered in coniferous forests at high latitudes.
 - in temperate zones with deciduous trees.
 - located near the poles and without any trees.
 - located at high altitudes nearest the equator.
25. Consumers that eat plants rely upon
- chemical energy stored in organic molecules produced by photosynthesis.
 - kinetic energy stored in organic molecules produced by photosynthesis.
 - photosynthesis to convert potential energy to kinetic energy.
 - entropy to generate heat to drive kinetic processes in their bodies.
26. Creation of sustainable ecosystems can be achieved through
- a reduction in world population growth.
 - decreasing poverty so that people are not consumed of immediate needs (food, shelter) and can focus more on human's long-term impact on the environment.
 - decreased human consumption of everything, and thus a lower production of waste.
 - all of the above
27. Deciding to use a natural enemies approach to control the mites that infect her crops, a farmer purchases 10,000 ladybugs in the spring and spreads them over her 100 acre fields. This represents the use of natural:
- predators
 - parasitoids
 - pathogens
 - plant-eaters
28. Deep in the ocean off the shore of Japan, are communities nestled around hydro thermal vents where super heated water, springs from the bottom of the ocean. No sunlight ever penetrates to these deep regions. In these communities, bacteria have special enzymes that allow them to form organic matter by chemosynthesis. These communities frequently have worms, clams, shrimp,

- and many other organisms clustered together. In such a system, fish feed on shrimp that feed on the bacteria. This food chain represents a:
- tertiary consumer eating a secondary consumer eating a primary consumer.
 - primary consumer eating a secondary consumer eating a tertiary consumer.
 - consumer eating a producer which then consumes chemoautotrophic bacteria.
 - secondary consumer eating a primary consumer which then eats a producer.
29. Density-independent factors such as earthquakes and hurricanes are:
- abiotic factors that maintain a population near equilibrium.
 - biotic factors that maintain a population near equilibrium.
 - abiotic factors that are not involved in maintaining a population near its equilibrium.
 - biotic factors that are not involved in maintaining a population near its equilibrium.
30. Developing a new form of ecological pest control, researchers engineer crops to produce the pheromones of the pest. The crops now produce the pest pheromone, overwhelming the fields and causing the male pests to fail to find a mate. This new form of ecological pest control combines:
- natural enemies and cultural control.
 - cultural and natural enemies control.
 - genetic and cultural control.
 - genetic and natural chemical control.
31. Dung beetles feeding on the waste of cattle, grazing on hay in a field, represent
- a decomposer feeding on the wastes of a consumer eating a producer.
 - a producer feeding on the wastes of a producer eating a consumer.
 - a producer feeding on the wastes of a consumer eating a producer.
 - a consumer feeding on the wastes of a decomposer eating a producer.
32. Ecosystem sustainability primarily results from the
- relationships between the organisms in an ecosystem.
 - number of predators found in the ecosystem.
 - frequency of fires or other natural disasters in an ecosystem.
 - total amount of biomass that exists in an ecosystem.
33. Ecotones
- contain only species found in the bordering ecosystems.
 - have the same abiotic characteristics as the bordering ecosystems.
 - consist of two or more landscapes.
 - are transitional regions between ecosystems.
34. Energy transfer between trophic levels in aquatic systems is generally
- less efficient than terrestrial food pyramids.
 - less efficient than a detritus food web because aquatic systems lack fungi.
 - inverted, in which more energy is transferred from one trophic level up to the next.
 - more efficient than terrestrial food pyramids.
35. For more than 20 years, scientists have been analyzing expected climate change and the impact on crops grown in particular regions. For example, states that typically plant corn and soybeans may need to switch to growing cotton. Such a change is an example of :
- mitigation
 - a cap-and-trade policy
 - adaptation
 - stabilization wedge
36. In a forest, deer, raccoons, squirrels, and other animals eat and find shelter. A detritus food web occurs as their wastes accumulate on the forest floor. In this detritus web
- deer and raccoons function as the producers.
 - fungi and earthworms function as producers.
 - decomposers function as consumers.
 - the deer and raccoons represent decomposers.
37. In an ecosystem with many similar species, we typically find
- intense interspecific competition for food.
 - competitors using different resources.
 - intense interspecific competition for nesting sites.
 - adaptations for battles and interspecific competition.
38. In an ecosystem, the replacement of one new species for another because of direct competition for the same resources defines
- intraspecific competition.
 - the competitive exclusion principle.
 - character displacement.
 - resource partitioning.
39. Maintaining sustainable human exploitation of ecosystem capital will be increasingly difficult because of
- the over reliance on grains and other plants as a significant portion of the human diet.
 - the expanding number of viral and bacterial human diseases.
 - the growing human population on Earth.
 - decreases in worldwide ocean levels.
40. Persistent organic pollutants (POPs) reach toxic levels in organisms in natural ecosystems in large part because of biomagnification, in which the highest concentrations of POPs are found in
- primary producers
 - secondary producers
 - primary consumers
 - secondary consumers
41. Some birds have been seen to consume certain soils in what is called geophagy. In some cases, the soils help the birds digest toxins that occur in their diets. These birds eating soil represent a member of the
- biosphere consuming a component of the lithosphere.
 - hydrosphere consuming a component of the atmosphere.
 - lithosphere consuming a component of the biosphere.
 - atmosphere consuming a component of the hydrosphere.
42. The cattle were removed from a field and shrubs and bushes were seen dotting the grass-covered hillsides. A few years later, small pine trees and then larger deciduous trees appeared. Now, there are so many trees and shaded regions it is difficult to tell that this was once a field. What once was an open pasture on a long abandoned

- farm has changed over time, without the direct influence of human. In this wooded region that was once pasture, a mature forest of towering trees has taken over. However, change is still apparent, as a large patch of young shrubs and small trees are growing where several large trees were toppled by a storm several years ago. Collectively, this wooded regions represents
- (a) a climax ecosystem (b) primary succession
(c) a sustained biome (d) intermediate succession
43. What once was an open pasture on a long abandoned farm has changed over time, without the direct influence of humans. After the cattle had been removed, shrubs and bushes could be seen dotting the grass-covered hillsides. A few years later, small pine trees and then larger deciduous trees appeared. Now, there are so many trees and shaded regions it is difficult to tell that this was once a field. The changes in this field, from grasses to shrubs to trees, represent
- (a) ecological succession
(b) climax production
(c) biological evolution
(d) ecological adaptation
44. Which of the following represents a type of mutualism?
- (a) A red-tailed hawk pounces on a field mouse for dinner.
(b) A large herd of zebra graze lazily across the broad savannah
(c) A mosquito draws a blood meal from the back of an elk.
(d) A nectar-feeding bat swoops in to drink nectar from some flowers.
45. Which one of the following illustrates interspecific competition?
- (a) The largest wolves in a pack are the first to feed on a freshly killed deer.
(b) Two honeybees from the same colony converge on a flower to collect pollen & nectar.
(c) Advertising its nectar with red colors, a columbine flower attracts a hummingbird for a meal.
(d) Standing near the dead antelope, vultures wait for the lions to finish their meal.
46. Why are there so few ecosystems with more than four levels of consumers?
- (a) because biomass decreases by about 90% at each level moving up.
(b) because top consumers compete with and kill each other with increasing population size.
(c) because consumers at these highest levels typically form social groups that stop reproducing at high densities.
(d) because predators at the highest levels simply are not intelligent enough to hunt other top predators.
47. Regulating and cultural services provided by natural ecosystems
- (a) are typically the most economically valued components of ecosystems.
(b) are public goods usually provided by markets.
(c) include goods such as fresh water, wild foods, and livestock.
- (d) are essential but difficult to value in monetary terms.
48. The phrase “ecosystem capital” is better than the phrase “natural resources” because ecosystem capital
- (a) includes the ecological value of natural ecosystems.
(b) includes the economic value of an ecosystem’s goods and services.
(c) does not include the value of natural ecosystems.
(d) does not include the economic value of an ecosystem’s goods and services.
49. A sandy and saline area is the natural habitat of an Indian animal species. The animal has no predators in that area but its existence is threatened due to the destruction of its habitat. Which one of the following could be that animal?
- (a) Indian wild buffalo (b) Indian wild ass
(c) Indian wild boar (d) Indian gazelle
50. In addition to global climate change, humans are negatively impacting coral reefs by
- (a) destructive harvesting of fish for food or pets.
(b) using large amounts of coral rock to pave roads.
(c) destroying large regions of coral reefs for commercial aquaculture.
(d) introducing alien species that are thought to be more productive.
51. The concept of sustainable development relates to
- (a) consumption levels
(b) exhaustible resources
(c) social equity
(d) Intergenerational equity
52. The variability among living organisms from all sources including terrestrial, marine and other ecosystems and the ecological complexes of which they are part which includes diversity within species, between species of ecosystems refers to
- (a) geographical diversity
(b) zoological diversity
(c) ecological diversity
(d) biological diversity
53. ‘Population dividend’ refers to
- (a) total number of population
(b) youthful age structure of a population
(c) relatively high proportion of experienced aged people
(d) migration from richer region to poorer region
54. Inclusion strategy does not focus on
- (a) reduction of inequality
(b) reduction of poverty
(c) diversifying livelihood for tribal population
(d) getting poorer countries close
55. Which one of the following is the best description of the term ‘ecosystem’? (CSAT 2015-1)
- (a) A community of organisms interacting with one another.
(b) That part of the Earth which is inhabited by living organisms.
(c) A community of organisms together with the environment in which they live
(d) The flora and fauna of a geographical area

Exercise -2

Statement Based MCQ

1. Which of the following groups contains only easily biodegradable items?

1. Grass, flower and leather
2. Grass, wood and plastic
3. Fruits, peel cake and lime juice
4. Cake, wood and grass.

Which of the above is/are correct?

- (a) 1 and 2 (b) 2 and 3
- (c) 1, 3 and 4 (d) All of these

2. Consider the following statements :

1. The amount of usable energy remains constant as it is passed from one trophic level to another.
2. The energy within an ecosystem is fixed and never changes.

Which of these statement(s) is/are correct ?

- (a) 1 only (b) 2 only
- (c) Both 1 and 2 (d) Neither 1 nor 2

3. Select the correct combination of statements (I - IV) regarding the characteristics of productivity.

1. The rate of biomass production is called productivity and is expressed in terms of a kcal m^{-2} .
2. Gross primary productivity is rate of production of biomass during photosynthesis.
3. Gross primary productivity minus respiration loss is called net primary productivity.
4. Primary productivity depends only on the plant species inhabiting a particular area.

Which of the following statement(s) is/are correct?

- (a) 1, 2 and 3 (b) 2 and 3
- (c) 2, 3 and 4 (d) 2 and 4

4. Food chains differ from food webs in that

1. food chains are single sequence of who eats whom in a community.
2. food chains better represent the entire community.
3. food webs represent the complex interaction among food chains.
4. food chain is the flow of energy in a population.

Which of the following statement(s) is/are correct?

- (a) 1 and 2 (b) 1 and 3
- (c) 1, 2 and 3 (d) 1, 2, 3 and 4

5. Which of following is/are tend (s) in ecological succession ?

1. An increase in complexity of species
2. An increase in productivity
3. An increase in community stability and species diversity
4. A decrease in nonliving organic materials.

Which of the above is/are correct?

- (a) 1 and 2 (b) 1 and 4
- (c) 1, 3 and 3 (d) 1, 2, 3 and 4

6. Which of the following include(s) ecosystem services ?

1. Purification of air and water by forests
2. Forests mitigate droughts and flood
3. Forests act as store house of carbon
4. Forests influence hydrological cycle

Which of the above is/are correct?

- (a) 1 and 3 (b) 1 and 4
- (c) 1, 2 and 3 (d) 1, 2, 3 and 4

7. Which one is not the dynamic aspect of an ecosystem?

1. Producers and mineral cycles
2. Consumers and mineral cycles
3. Producers and energy flow
4. Energy flow and mineral cycles

Which of the above is/are correct?

- (a) 1 and 2 (b) 2 and 4
- (c) 1, 2 and 3 (d) All of these

8. Which of the following statements are true about ecosystem? Select the correct answer from the codes:

1. Ecosystem comprises both biotic and abiotic components.
2. Solar radiation is the main driving force of the ecosystem.
3. Ecosystem is a closed system.
4. Ecosystem does not have its own productivity

Codes:

- (a) 1 and 2 (b) 2 and 3
- (c) 1 and 3 (d) 3 and 4

9. Which of the following are the incorrect statements about 'Keystone species'.

1. Keystone species are the small-sized plants and organisms which have large effect on the environment.
2. Keystone species play critical role in maintaining the structure of an ecological community.
3. Keystone species do not generally affect other organisms.

- (a) 1 and 2 (b) 2 and 3
- (c) 1 and 3 (d) All are correct

10. Which of the following are correct statements about light in aquatic environments?

1. Water selectively reflects and absorbs certain wavelengths of light.
2. Photosynthetic organisms that live in deep water probably utilize red light.
3. Light intensity is an important abiotic factor in limiting the distribution of photosynthetic organisms.

- (a) 1 only (b) 2 only
- (c) 1 and 3 only (d) 2 and 3 only

11. The producers in ecosystems include which of the following?

1. prokaryotes 2. algae
3. plants

- (a) 1 only (b) 2 only
- (c) 3 only (d) 1, 2, and 3

12. Aquatic primary productivity is often limited by which of the following?

1. light 2. nutrients
3. pressure

- (a) 2 only (b) 3 only
- (c) 1 and 2 only (d) 1, 2, and 3

13. Consider the following statements:

1. Interactions between the two organism in which one organism kills and feed on the second organisms, is called Parasitism.
2. Mutualism is the way two organisms of different species biologically interact in a relationship in which each individual derives a fitness benefit.

- Which of the statements given above is/are true?
 (a) 1 only (b) 2 only
 (c) 1 and 2 both (d) None
14. Consider the following:
 1. Bioaccumulation is the increases in concentration of a pollutant from the environment to the first organism in a food chain.
 2. Biomagnification is then increases in concentration of pollutant from one link in a food chain to another.
 Which of the statements given above is/are true?
 (a) 1 only (b) 2 only
 (c) 1 and 2 both (d) None
15. Consider the following statements:
 1. In ecology, an ecosystem is a naturally occurring assemblage of organism (plant, animal and other living organism - also referred to as a biotic community of biocoenosis) living together with their environment (or biotope), function as a unit of sorts.
 2. The term "ecosystem" first coined by Arthur Tansely.
 Which of the statements given above is/are correct?
 (a) 1 only (b) 2 only
 (c) 1 and 2 both (d) None
16. Which of the following statements is true?
 1. Circulation of energy in the biosphere ecosystem is cyclical.
 2. Circulation of matter in the biosphere ecosystem is unidirectional.
 Select the correct answer using the codes given below:
 (a) 1 only (b) 2 only
 (c) 1 and 2 only (d) None
17. Consider the following statements :
 1. Waste are of two types, biodegradable and non-biodegradable.
 2. Blue-green algae are producers.
 3. Biodegradable wastes should be separated and kept in blue colour bins for garbage collectors.
 Which of these statement(s) is/are correct ?
 (a) 1 and 2 (b) 2 and 3
 (c) 1, 2 and 3 (d) None
18. Not all parasitism involves feeding on the body of the host. The exception is
 1. Ectoparasitism 2. Endoparasitism
 3. Parasitoids 4. Brood Parasitism
 Choose the option from the codes given below:
 (a) 1 only (b) 2 only
 (c) 3 only (d) 4 only
19. What factor does not contribute to the rapid loss of nutrients from terrestrial ecosystems?
 1. Clear cutting native forests
 2. Early seral stages
 3. Climax communities
 4. Low diversity
 Choose the correct option from the codes given below:
 (a) 1 and 2 (b) 2 and 3
 (c) 3 only (d) 3 and 4
20. When two organisms attempt to utilize the same resource, the result is:
 1. A fundamental niche
 2. Competition
 3. Commensalism
 4. Mutualism
 Choose the correct option from the codes given below:
 (a) 1 and 2 (b) 2 only
 (c) 3 and 4 (d) 1, 2, 3 and 4
21. With reference to food chains in ecosystems, consider the following statements:
 1. A food chain illustrates the order in which a chain of organisms feed upon each other.
 2. Food chains are found within the populations of a species.
 3. A food chain illustrates the numbers of each organism which are eaten by others.
 Which of the statements given above is/are correct?
 (a) 1 only (b) 1 and 2 only
 (c) 1, 2 and 3 (d) None
22. With reference to the food chains in ecosystems, which of the following kinds of organism is/are known as decomposer organism/organisms?
 1. Virus 2. Fungi
 3. Bacteria
 Select the correct answer using the codes given below.
 (a) 1 only (b) 2 and 3 only
 (c) 1 and 3 only (d) 1, 2 and 3
23. Consider the following :
 1. Bats 2. Bears
 3. Rodents
 The phenomenon of hibernation can be observed in which of the above kinds of animals?
 (a) 1 and 2 only
 (b) 2 only
 (c) 1, 2 and 3
 (d) Hibernation cannot be observed in any of the above
24. Which of the following adds/add carbon dioxide to the carbon cycle on the planet Earth?
 1. Volcanic action
 2. Respiration
 3. Photosynthesis
 4. Decay of organic matter
 Select the correct answer using the code given below.
 (a) 1 and 3 only (b) 2 only
 (c) 1, 2 and 4 only (d) 1, 2, 3 and 4
25. With reference to two non-conventional energy sources called 'coalbed methane' and 'shale gas', consider the following statements :
 1. Coalbed methane is the pure methane gas extracted from coal seams, while shale gas is a mixture of propane and butane only that can be extracted from fine-grained sedimentary rocks.
 2. In India, abundant coalbed methane sources exist, but so far no shale gas sources have been found.
 Which of the statements given above is/are correct?
 (a) 1 only (b) 2 only
 (c) Both 1 and 2 (d) Neither 1 nor 2
26. The study of relationship of living organism with each other and with their environment is called _____.
 1. Economy
 2. Ecology
 3. Geography
 4. Environment
 Which among the following defines the statement
 (a) 1 only (b) 2 only
 (c) 3 only (d) 4 only
27. Consider the following statement related to Abiotic
 1. Abiotic components of ecosystem are the living features of ecosystem on which the living organism depends.
 2. Abiotic component is referred to the physical environment and its numerous interacting variables.

- Which among the following statement is correct
 (a) 1 only (b) 2 only
 (c) Both 1 and 2 (d) None of the Above
28. What is the name of processes of Nitrogen cycle?
 1. Nitrogen Fixation
 2. Nitrification
 3. Assimilation
 4. Denitrification
 Choose the correct option
 (a) 1 and 2 (b) 3 and 4
 (c) 1, 2, 3 (d) 1, 2, 3, 4
29. Which equation explains Nitrification?
 1. $\text{NH}_3 \rightarrow \text{NO}_2 \rightarrow \text{NO}_3^-$
 2. $\text{NH}_3 \rightarrow \text{NO}_3 \rightarrow \text{NO}_2^-$
 3. $\text{NH}_3 \rightarrow \text{NO}_4 \rightarrow \text{NO}_2^-$
 Select the correct answer from the following codes
 (a) 1 only (b) 2 only
 (c) 3 only (d) All of the Above
30. Atmosphere contains 21% of which gas?
 1. Nitrogen
 2. Oxygen
 3. Phosphorous
 (a) 1 only (b) 2 only
 (c) 3 only (d) None of the Above
31. What is the purpose of Ozone layer?
 1. Protects UV rays to reach on earth
 2. Helps UV rays to reach earth
 3. The source of ozone is the oxygen in the atmosphere
 Choose the correct code
 (a) 1 and 2 (b) 2 and 3
 (c) 1 and 3 (d) 1, 2, 3
32. Consider the following statements
 1. Burning fossil fuel decreases oxygen in atmosphere
 2. Burning fossil fuel increases carbon dioxide
 3. The main source of oxygen is atmosphere
 Which among the following code is incorrect?
 (a) 1 only (b) 2 only
 (c) 3 only (d) None of the Above
33. How is the oxygen cycle effected?
 1. Human activities
 2. Running of automobiles
 3. Consumption of fossil fuels
 Choose the correct code
 (a) 1 and 2 (b) 2 and 3
 (c) 1 and 3 (d) 1, 2, 3
34. With reference to the process of photosynthesis for carbon dioxide, which of the following statement is correct?
 1. Through the process of photosynthesis carbon enters into non-living world in the form of carbon dioxide.
 2. Recycling of carbon is done by the burning of fossil fuels.
 Select the correct answer from the following codes
 (a) 1 Only (b) 2 only
 (c) 1 and 2 (d) Neither of the Above
35. Which of the following is not involved in the continuous water exchange?
 1. Stones 2. Air
 3. Land 4. Sea
 Choose the correct code
 (a) 1 only (b) 2 only
 (c) 1 and 2 (d) 3 and 4
36. Consider the following statement
 1. Evaporation takes water into atmosphere in the form of vapours.
2. Clouds are formed when the vaporized water is cooled and condensed.
 Which of the following statement is correct?
 (a) 1 only (b) 2 only
 (c) 1 and 2 (d) None of the Above
37. Name the elements that help in creating phosphorous cycle?
 1. Rocks 2. Earth crust
 3. Air 4. Water
 Choose the correct code
 (a) 1 and 2 (b) 3 and 4
 (c) 1 and 3 (d) 1 and 4
38. Why phosphorous is not found in atmosphere?
 1. It doesn't combine well with other elements
 2. It just cannot be found
 3. The sun evaporates it
 4. At normal temperatures and pressure, phosphorous is at a liquid state
 Choose the correct code from the following statement regarding phosphorous
 (a) 1 only (b) 2 only
 (c) 3 only (d) 4 only
39. How does phosphorous enter plants in the soil?
 1. Photosynthesis 2. Water in the soil
 3. Rocks on the sand
 Choose the correct answer
 (a) 1 only (b) 2 only
 (c) 3 only (d) 1, 2, 3
40. What is the impact of human on phosphorous cycle?
 1. Killing plants
 2. No impact
 3. Pollution
 4. Use fertilizers that are not natural
 Choose the correct code
 (a) 1 only (b) 2 only
 (c) 3 only (d) 4 only
41. Megatherms is defines as
 1. Organisms living in tropical regions
 2. Organisms living in subtropical regions
 3. Organisms living in temperate regions
 Choose the correct definition of Megatherms
 (a) 1 only (b) 2 only
 (c) 3 only (d) None of the Above
42. Consider the following related to humidity
 1. Humidity is the water vapor present in the atmosphere.
 2. Humidity is measured by barometer
 3. Humidity is defined as molecules of water/unit volume
 Choose the correct code
 (a) 1 and 2 (b) 2 and 3
 (c) 1 and 3 (d) 1, 2, 3
43. _____ are the living organism that shapes the ecosystem and comes regularly in contact with each other.
 (a) Abiotic components
 (b) Biotic componenets
 (c) Antibiotic components
 (d) None of the Above
44. Consider the following statement in regards to ecosystem
 1. Components of an ecosystem include Biotic
 2. Components of an ecosystem include Abiotic
 Which of the following statement is correct?
 (a) 1 only (b) 2 only
 (c) 1 and 2 (d) None of the Above

45. The process of cycling of minerals and components through the ecosystem is called _____.
 1. Biological cycle
 2. Biogeochemical cycle
 3. Biochemical cycle
 Choose the correct code
 (a) 1 only (b) 2 only
 (c) 3 only (d) None of the Above
46. Name the forest that is found in Asia, Europe.
 1. Boreal Forest
 2. Temperate Deciduous Forests
 3. Tropical or Rainforests
 Choose the correct name
 (a) 1 only (b) 2 only
 (c) 3 only (d) All of the Above
47. Consider the following statement
 1. The soil of Temperate Deciduous forests is richer than boreal forests.
 2. Temperate Deciduous forests are mostly found in the equatorial belt of the plant.
 Which statement is incorrect?
 (a) 1 only (b) 2 only
 (c) 1 and 2 (d) None of the Above
48. Which forest does not see winter season?
 1. Boreal Forest
 2. Temperate Deciduous Forests
 3. Tropical Forest
 Choose the correct code
 (a) 1 only (b) 1 and 2
 (c) 1, 2, 3 (d) 3 only
49. How much desert area exists on the planet?
 (a) One third (b) Half
 (c) One fifth (d) One fourth
50. Marine lives are found in which place?
 1. Ocean
 2. Lake
 3. Ponds
 4. Sand
 Choose the correct code
 (a) 1, 2 (b) 3, 4
 (c) 1, 2, 3 (d) 1, 2, 3, 4
51. Consider the following statement
 1. Most of the oxygen in the atmosphere is generated by the algae.
 2. Evaporation of salt and water turns into rain
 3. Large amount of carbon dioxide is absorbed by the algae in the atmosphere.
 Which statement is incorrect?
 (a) 1 only (b) 2 only
 (c) 3 only (d) None of the Above
52. The water that has less content of salt is called _____.
 1. Sea
 2. Fresh water
 3. Oil
 Choose the correct code
 (a) 1 only (b) 2 only
 (c) 3 only (d) All of the Above
53. Consider the following statement
 1. Atmospheric temperature of the place depends upon the slope, altitude, latitude, topography.
 2. Temperature increases as we go from equator to poles.
 3. Temperature lowers as we go from equator to poles

Choose the incorrect statement

- (a) 1 only (b) 2 only
 (c) 3 only (d) None of the Above
54. Consider the following statements
 1. Three-fourth of the earth surface is covered by water but less than 3% is fresh water used for human consumption.
 2. Of the total fresh water available, Ice-cap has highest share of 2% followed by ground water 0.68%.
 Select the correct answer using the code given below.
 (a) 1 only (b) 2 only
 (c) 1 and 2 (d) Neither of the Above
55. With reference to phosphorus cycling, consider the following statements
 1. The natural reservoir of phosphorus is atmosphere, which contains phosphorus in the form of phosphates.
 2. Herbivores and other animals obtain phosphorus from plants.
 3. Unlike carbon cycle, there is no respiratory release of phosphorus into atmosphere.
 Which of the statements given above is/are correct?
 (a) 1 and 2 (b) 2 and 3
 (c) 1 and 3 (d) 1, 2, 3
56. What is the name of the cycle where the continuous and balanced process of evaporation, precipitation, transpiration and runoff of water takes place.
 1. Hydrological
 2. Tropological
 3. Hyper logical
 Choose the correct option
 (a) 1 only (b) 2 only
 (c) 3 only (d) None of the Above
57. With reference to ecosystem stability, consider the following statements
 1. A diverse and complex ecosystem is more stable.
 2. Ecosystem stability increases with decrease in number of links in food web.
 Which of the statements given above is/are correct
 (a) 1 only (b) 2 only
 (c) and 2 (d) None of the Above
58. Water in a wetland can be
 1. Fresh
 2. Brackish
 3. Static
 Select the correct answer using the codes given below.
 (a) 1 and 2 (b) 1 and 3
 (c) 2 and 3 (d) 1, 2 and 3
59. Consider the following statement related to humidity
 1. Specific humidity
 2. Absolute humidity
 3. Relative humidity
 4. Non relative humidity
 Which among the following is not the type of humidity?
 (a) 1 only (b) 2 only
 (c) 3 only (d) 4 only
60. Consider the following statement
 1. Steppe is found in low latitudes and middle latitudes.
 2. Prairies are humid and densely covered tall grasslands
 3. Savanna are the areas of thick high grasses
 Choose the correct code
 (a) 1 only (b) 2 only
 (c) 3 only (d) 1, 2, 3

Hints and Explanations

EXERCISE-1

1. (c) An ecosystem may be defined as a structural and functional unit of the biosphere comprising living organisms and their non-living environment that interact by means of food chains and chemical cycles resulting in energy flow, biotic diversity and material cycling to form a stable, self supporting system.
The organisms in an ecosystem are classified into 3 main categories-producers, consumers and decomposers. The consumers utilize materials and energy stored by the producers. Decomposers obtain their food molecules from the organic materials of dead producers and consumers. In a true ecosystem, producers are more than consumers (herbivores and carnivores).
2. (b) The ability to maintain a steady state within constantly changing environment is essential for the survival of living systems. The maintenance of a constant internal environment is called homeostasis.
3. (b) The dead organic matter of plant or animal is called as detritus. While a part of it remains on the soil surface as litter, the other part enters the soil. Many animals such as protozoan's, nematodes, insects etc. depend on detritus and hence they are called as detritivores. Even the human beings are detritivores when they eat cooked food. From detritus, the chain proceeds to detritivores, then to carnivores and finally to top carnivores.
4. (a) Decomposers are saprotrophs which decompose the organic remains by secreting extracellular digestive enzymes. They are also known as mineralisers as they release minerals trapped in organic remains. So in the absence of microorganisms the flow of mineral will stop.
5. (d) Niche is specific part of habitat occupied by individuals of a species which is circumscribed by its range of tolerance, range of movement, microclimate, type of food and its availability, shelter, type of predator, and timing of activity. A habitat has several ecological niches and supports a number of species. An ecological niche is used by a single species. Two or more species cannot use the same niche despite having a mutualistic association. The abundances of a species population within its habitat is called niche density.
6. (b) When succession begins on an area which has not been previously being occupied by a community e.g. a new exposed rock area, sand dunes, new islands, deltas, shore or recent lava flow, it is known as primary succession. The first group of organisms (plants or animals) which become established in such an area is termed the pioneer community.
7. (d) Streaming body is a secondary aquatic adaption. It is found in animals that live permanently in water but most of them are amphibious in nature. The stream

lined body consists of compression of head, body and tail into a curved stream lined form.

There is no protruberance over the body so that the animal can move easily through water. Parasitism is a relationship between two organisms of different species in which one organism called parasite obtains its food directly from another living organism called host.

In xeric adaptation perspiration is reduced to conserve water. Uricotelism is characteristic of terrestrial animals which excrete uric acid.

8. (a) Niche is specific part of habitat occupied by individuals of a species which is circumscribed by its range of tolerance, range of movement, microclimate, type of food and its availability, shelter, type of predator, and timing of activity.

A habitat has several ecological niches and supports a number of species. An ecological niche is used by a single species. Two or more species cannot use the same niche despite having a mutualistic association.

9. (a) In developing countries the conditions are becoming better for survival of human beings. So the mortality rate or the number of individuals dying per unit of time is low.

Mortality or the average number of individuals produced by a population in a unit of time is high. So that there is rapid population growth and there are more individuals in the pre-reproductive age group. So there is young age distribution.

10. (a) Keystone species are those species which has significant and disproportionately large influence on the community structure and characteristics. It has often considerably low abundance and biomass as compared to dominant species. Removal of such species causes serious disruption in structure and function of community.

- | | | | | |
|---------|---------|---------|---------|---------|
| 11. (b) | 12. (d) | 13. (a) | 14. (b) | 15. (c) |
| 16. (b) | 17. (c) | 18. (d) | 19. (d) | 20. (a) |
| 21. (c) | 22. (d) | 23. (a) | 24. (c) | 25. (a) |
| 26. (d) | 27. (a) | 28. (d) | 29. (c) | 30. (d) |
| 31. (a) | 32. (a) | 33. (d) | 34. (c) | 35. (c) |
| 36. (a) | 37. (b) | 38. (b) | 39. (c) | 40. (d) |
| 41. (a) | 42. (a) | 43. (a) | 44. (d) | 45. (d) |
| 46. (a) | 47. (d) | 48. (a) | 49. (b) | 50. (a) |

51. (d) Intergenerational equity is a concept that says that humans 'hold the natural and cultural environment of the Earth in common both with other members of the present generation and with other generations, past and future. It means that we inherit the Earth from previous generations and have an obligation to pass it on in reasonable condition to future generations. The goal of sustainable development is Intergenerational equity.

52. (d) Biological diversity is used to refer to the total number of different species on Earth. A collection of this biodiversity would include human beings,

Bengal tigers, sugar maples, oyster mushrooms, bacteria, and the millions of other living organisms found on Earth.

53. (b) Population dividend or Demographic dividend refers to a period - usually 20 to 30 years - when fertility rates fall due to significant reductions in child and infant mortality rates. It occurs when the proportion of working people in the total population is high because this indicates that more people have the potential to be productive and contribute to growth of the economy.

54. (d)

55. (a) An ecosystem includes all of the living things (plants, animals and organisms) in a given area, interacting with each other, and also with their non-living environments (weather, earth, sun, soil, climate, atmosphere).

EXERCISE-2

1. (c) 2. (d) 3. (b) 4. (a) 5. (c)

6. (d) 7. (c)

8. (a) Ecosystem is the dynamic community of living organism with physical environment. Thus, it comprises of both biotic and abiotic components. Solar energy is the ultimate source of energy in it, so is the main driving force. Dynamic energy transfers occur making it an open system. The autotrophs make the ecosystem's biotic components self-sufficient.

9. (c) Keystone species are not necessarily small sized though they put great effect on the environment. They play very critical role in maintaining the structure of an ecological community by affecting many other organisms. An ecosystem may experience a dramatic shift if a keystone species is removed, even though that species may be small part of the ecosystem by measures of biomass or productivity.

10. (c) 11. (d) 12. (c)

13. (b) In Parasitism, the parasite does not kill the host rather it derives its nutrition from the host in which the later is not affected. The interaction in which an organism kills the other for food is called Predation.

14. (c) Bioaccumulation refers to the accumulation of substances, such as pesticides, or other organic chemicals in an organism. Bioaccumulation occurs when an organism absorbs a toxic substance at a rate greater than that at which the substance is lost. Biological magnification often refers to the process whereby certain substances such as pesticides or heavy metals move up the food chain.

15. (c) Sir Arthur Tansley first used the term ecosystem in 1935.

16. (d) Circulation of energy in the biosphere ecosystem is unidirectional, it is derived from the sun and goes through one trophic level to another. The matter however circulated in cyclic manner.

17. (a) Biodegradable wastes should be separated and kept in green colour bins for garbage collectors.

18. (d) Brood parasites are organisms that use the strategy of brood parasitism, a kind of kleptoparasitism

found among birds, fish or insects, involving the manipulation and use of host individuals either of the same (intraspecific brood-parasitism) or different species (interspecific brood-parasitism) to raise the young of the brood-parasite.

19. (c) In ecology, climax community, or climatic climax community, is a historic term that expressed a biological community of plants and animals and fungi which, through the process of ecological succession — the development of vegetation in an area over time — had reached a steady state. This equilibrium was thought to occur because the climax community is composed of species best adapted to average conditions in that area.

20. (b) Since a community comprises all the species that occur at a particular location, one of the most important things about communities is how the species interact with one another.

Four different types of interactions between different species (inter specific interactions) have been identified:

Competition -Two organisms mutually harm one another

Predator-prey or parasite-host- One organism benefits, the other is harmed

Mutualism- Both organisms benefit

Commensalism- One organism benefits, the other is not affected

21. (a) A food chain illustrates the order in which a chain of organisms feed upon each other.

A food chain is the sequence of who eats whom in a biological community to obtain nutrition.

Sample:- Grassland Biome

GRASS > GRASS HOPPER > RAT > SNAKE > HAWK

22. (b) 23. (c) 24. (c) 25. (d)

26. (b) The branch of science which deals with the study of relation between plants and animals to each other along with their environments is called ecology.

27. (b) Abiotic components of ecosystem are the nonliving features of ecosystem on which the living organism depends. It is basically referred to the physical environment and its numerous interacting variables.

28. (d) Nitrogen cycle is necessary because Plants cannot absorb nitrogen directly instead it is absorbed in the form of nitrate. Nitrogen cycle have 5 important processes i.e. fixation, ammonification, nitrification, assimilation and denitrification.

29. (a) Nitrification is the process of conversion of ammonia.

30. (b) Atmosphere contains 21 % of Oxygen.

31. (c) The source of ozone is the oxygen in the atmosphere. Ozone layer protects the living being from the UV radiation which reaches the earth.

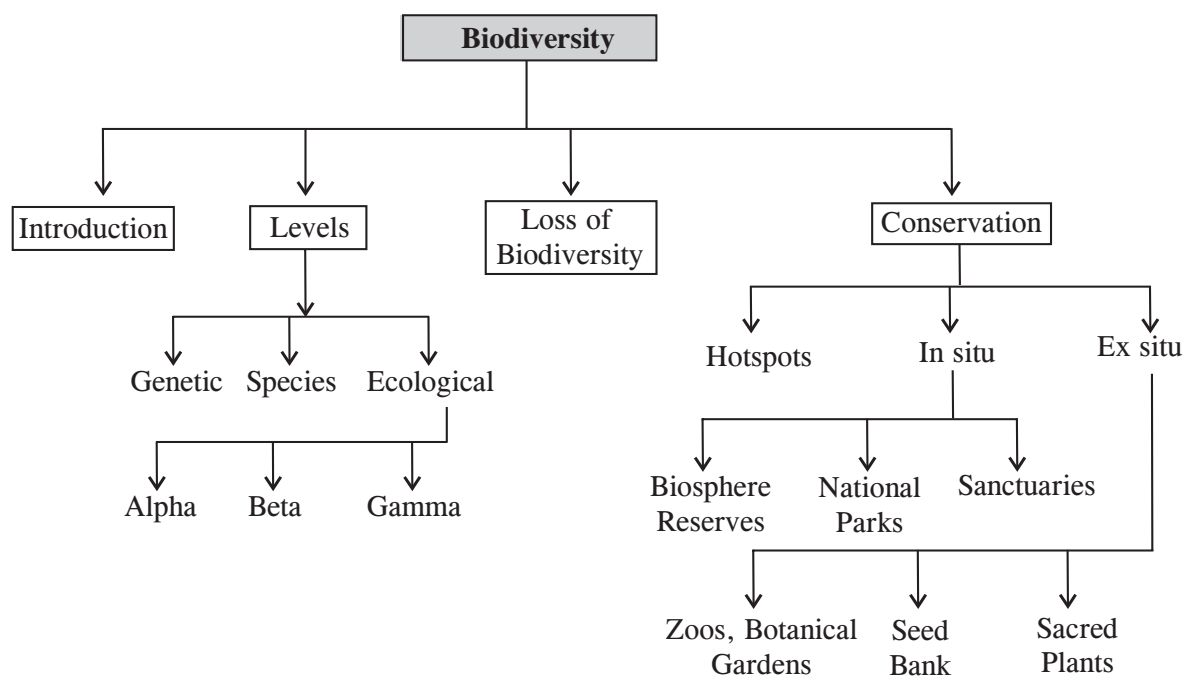
32. (d) Oxygen is a very important element for the existence of all flora and fauna. Atmosphere contains 21 % of oxygen. The main source of oxygen is atmosphere. Plants and animals absorb oxygen through respiration either from water or air and leaves

- through photosynthesis. In respiration process some of the oxygen returns to the atmosphere in the form of carbon dioxide and water vapour. During the process of photosynthesis gaseous oxygen is released completing the oxygen cycle. The source of ozone is the oxygen in the atmosphere. Ozone layer protects the living being from the UV radiation which reaches the earth. By burning fossil fuels man decreases the amount of oxygen in the atmosphere and increases the carbon dioxide content.
33. (d) The oxygen concentration in atmosphere is 21%. The oxygen cycle is effected by human activities such as running automobiles and consumption of fossil fuels which release more carbon dioxide in the atmosphere.
 34. (b) Process of photosynthesis of carbon dioxide is explained in following steps:
 - Through the process of photosynthesis carbon enters into living world in the form of carbon dioxide.
 - This organic compound (food) is then passed from the producers to the consumers (herbivores & carnivores).
 - By the process of respiration or decomposition of dead bodies of plant and animals by decomposers this carbon returns back to the surrounding medium.
 - Recycling of carbon is also done by the burning of fossil fuels.
 35. (a) There is a continuous exchange of water between living organisms, air, land and sea.
 36. (c) Evaporation of water takes place from oceans, rivers and lakes which takes water into atmosphere in the form of vapours. Clouds and water is formed when these vaporized water is cooled and condensed.
 37. (a) Phosphorus cycle is also called as sedimentary cycle because the main reservoir is rocks and the earth crust.
 38. (d) Phosphorous is not found in atmosphere because at normal temperature and pressure phosphorous is at a liquid state.
 39. (b) Phosphorous enters into the soil through water.
 40. (d) Human use fertilizers that are not natural which impact phosphorous cycle.
 41. (a) Organisms living in tropical regions are called regions are called mesotherms, in temperate regions are called regions are called hekisthotherms.
 42. (c) Humidity is the water vapor or water in gaseous form present in the atmosphere. It can also be defined as molecules of water/unit volume. It is measured by Hygrometer.
 43. (b) Biotic components are the living organism that shapes the ecosystem and comes regularly in contact with each other.
 44. (c) Components of an ecosystem include Biotic and Abiotic.
 45. (b) The cycles (carbon and nitrogen cycle) involves phases of weathering of rocks, uptake and storage by organisms and return to the pool of soil, the atmosphere or ocean sediments.
 46. (a) Boreal or taiga forests forests are found in Asia, Europe, Siberia and North America where there are shorter and warmer summers with longer winters.
 47. (a) Tropical or Rainforests are mostly found in the equatorial belt of the plant.
 48. (d) Tropical or Rainforests are found in the equatorial belt of the plant. There is no winter season in these forests, sunlight fall for 12 hours and seasons is mostly rainy or dry with small change in the temperature.
 49. (c) Desert covers one fifth of the planet.
 50. (c) Marine lives are the plants, animals and living organisms which are found oceans, lakes, ponds etc.
 51. (d) Oceans: they are the biggest and the most varied of the ecosystem. Most of the oxygen in the atmosphere is generated by the algae. Here salt water evaporates and turns to rain which in turn falls on land. Large amount of carbon dioxide is absorbed by the algae in the atmosphere. Inter - tidal zone is the zone which connect ocean to the land. Only few species exist in rocky coastal areas as very few tides reach there.
 52. (b) The water that has less content of salt is called fresh water.
 53. (b) Atmospheric temperature of the place depends upon the slope, altitude, latitude, topography, etc. temperature lowers as we go from equator to poles – tropical, subtropical, temperate and arctic.
 54. (c) Three-fourth of the earth surface is covered by water but less than 3% is fresh water used for human consumption. Of the total fresh water available, Ice-cap has highest share of 2% followed by ground water 0.68%.
 55. (b) Herbivores and other animals obtain phosphorus from plants. There is no respiratory release of phosphorus into atmosphere.
 56. (a) Hydrological cycle is the continuous and balanced process of evaporation, precipitation, transpiration and runoff of water.
 57. (a) Diversity increases the resilience of the system to outside invasions of exotic organisms. So statement 1 is correct. A large number of interacting feeding links provide alternative channels for energy flow and this generates a wide variety of adjustments of population to environment changes and stresses within the ecosystem. Therefore, ecosystem stability increases with increase in number of links in food web.
 58. (d) The Ramsar definition of wetlands is fairly wide, including “areas of marine water the depth of which at low tide does not exceed six meters” as well as fish ponds, rice paddies and salt pans. wetlands can be freshwater as well. Freshwater wetlands are not connected to the ocean. They can be found along the boundaries of streams, lakes, ponds or even in large shallow holes that fill up with rainwater. Freshwater wetlands may stay wet all year long, or the water may evaporate during the dry season.
 59. (d) Humidity is of 3 types: Specific, Absolute, Relative
 60. (d) Steppe is found in low latitudes and middle latitudes. Prairies are humid and densely covered tall grasslands. Savanna are the areas of thick high grasses.



Introduction

Biodiversity mean diversity of heterogeneity at all levels of biological organisation, i.e from Micro molecules of the cells to the Biomass. The word Biodiversity was popularized by the sociologist Edward Wilson.



Biodiversity is the existence of a wide variety of plant and animal species in their natural environments, which is the aim of conservationists. Who are mainly concerned about indiscriminate destruction of rainforests and other habitats.

Important Levels of Biodiversity are

1. Genetic diversity

It is the diversity at genetic level, or at subspecies level, i.e. below species level, in a single species.

The genetic diversity helps the population to adapt. If a population has more diversity it can adapt better to the changed environmental conditions. The low diversity leads to uniformity. The genetic variability is therefore, considered to be the raw material for speciation.

2. Species diversity

The measurement of species diversity is its richness, i.e. the number of species per unit area. The greater is the species richness the more will be the species diversity. In nature, the number and kind of species, as well as the number of individual per species, vary, and this leads to greater diversity.

3. Ecological diversity

It is the diversity at community level. It can be of 3-types

- (a) **Alpha (α) diversity** : It is the diversity of organisms within the same community or habitat.
- (b) **Beta (β) diversity** : It is the diversity between communities or different habitats. Higher the heterogeneity in the altitude, Humidity and Temperature of a region, the greater will be the dissimilarity between communities, and higher will be the diversity.
- (c) **Gamma (γ) diversity** : It is the diversity of organisms over the entire geographical area, covering several ecosystems or habitats and various trophic levels and food webs. Such diversity is most stable and productive.

Biodiversity of India:

As per available data, the varieties of species living on the earth are 1753739. Out of the above species, 134781 are residing in India although surface area of India is 2% of the earth's surface. Wild life Institute of India has divided it into ten biogeographical regions and twenty five biotic provinces.

Biogeographical regions are:

- (i) Trans Himalayas,
- (ii) Gangetic plain,
- (iii) Desert,
- (iv) Semiarid zone;
- (v) Western Ghats;
- (vi) Deccan peninsula,
- (vii) North eastern zone,
- (viii) Coastal lands
- (ix) Himalayas,
- (x) Islands.

India is one of the twelve mega diversity nations of the world due to the following reasons:

- (i) It has 7.3% of the global fauna and 10.88% of global flora as per the data collected by Ministry of Environment and forest.
- (ii) It has 350 different mammals, 1200 species of birds-453 different reptiles, 182 amphibians and 45,000 plants species.
- (iii) It has 50,000 known species of insects which include 13,000 butterflies and moths.
- (iv) It has 10 different biogeographical regions and 25 biotic provinces having varieties of lands and species.
- (v) In addition to geographical distribution, geological events in the land mass provide high level of biological diversity.
- (vi) Several crops arose in the country and spread throughout the world.
- (vii) There is wide variety of domestic animals like cows, buffaloes, goats, sheep, pigs, horses etc.
- (viii) The marine biota includes sea weeds, fishes, crustaceans, molluses, corals, reptiles etc.
- (ix) There are a number of hot spots (namely Eastern Ghats, Western Ghats, North Eastern hills etc.).

IUCN at a glance

- Founded in 1948 as the world's first global environmental organisation
- Today the largest professional global conservation network
- A leading authority on the environment and sustainable development
- More than 1,200 member organizations including 200+ government and 900+ non-government organizations
- Almost 11,000 voluntary scientists and experts, grouped in six Commissions in some 160 countries
- IUCN's work is supported by over 1,000 staff in 45 offices and hundreds of partners in public, NGO and private sectors around the world. The Union's headquarters are located in Gland, near Geneva, in Switzerland.
- A neutral forum for governments, NGOs, scientists, business and local communities to find practical solutions to conservation and development challenges
- Thousands of field projects and activities around the world
- Governance by a Council elected by member organizations every four years at the IUCN World Conservation Congress
- Funded by governments, bilateral and multilateral agencies, foundations, member organisations and corporations
- Official Observer Status at the United Nations General Assembly

The **IUCN Red List of Threatened Species™** provides taxonomic, conservation status and distribution information on plants, fungi and animals that have been globally evaluated using the IUCN Red List Categories and Criteria. This system is designed to determine the relative risk of extinction, and the main purpose of the IUCN Red List is to catalogue and highlight those plants and animals that are facing a higher risk of global extinction (i.e. those listed as **Critically Endangered, Endangered and Vulnerable**). The IUCN Red List also includes information on plants, fungi and animals that are categorized as **Extinct** or **Extinct in the Wild**; on taxa that cannot be evaluated because of insufficient information (i.e., are **Data Deficient**); and on plants, fungi and animals that are either close to meeting the threatened thresholds or that would be threatened were it not for an ongoing taxon-specific conservation programme (i.e., are **Near Threatened**).

The IUCN Red List of Threatened Species is widely recognized as the most comprehensive, objective global approach for evaluating the conservation status of plant and animal species. The IUCN Red List was updated three times in 2015. The IUCN Red List now includes 79,837 assessed species, of which 23,250 are threatened with extinction, with habitat loss and degradation identified as the main threat to more than 80% of species assessed.

Red data book

A Red Data Book contains lists of species whose continued existence is threatened. Species are classified into different

categories of perceived risk. Each Red Data Book usually deals with a specific group of animals or plants (e. reptiles, insects, mosses). They are now being published in many different countries and provide useful information on the threat status of the species.

By the end of 2014 India had 988 threatened species on the list, which lists critically endangered, endangered and vulnerable species. In 2013, the number was 973. With 659 species in 2008, the increase over seven years is 50 per cent, in part due to better research identifying more threatened species and deforestation. (The Hindu, 2015)

Indian elephant, Bengal tiger, Indian lion, Indian Rhino, Gaur, lion tailed macaque, Tibetan Antelope, Ganga river dolphin, the Nilgiri Tahr, snow leopard, dhole, black buck, great Indian bustard, forest owlet, white – winged duck and many more are the most endangered animals in India.

ENDANGERED SPECIES IN INDIA

Birds

White-bellied heron
Great Indian bustard (*Ardeotis nigriceps*)
Forest owlet (*Athene blewitti*)
Baer's pochard (*Aythya baeri*)
Spoon-billed sandpiper (*Eurynorhynchus pygmeus*)
Siberian crane (*Grus leucogeranus*)
White-rumped vulture (*Gyps bengalensis*)
Indian vulture (*Gyps indicus*)
Slender-billed vulture (*Gyps tenuirostris*)
Bengal florican (*Houbaropsis bengalensis*)
Himalayan quail (*Ophrysia superciliosa*)
Jerdon's courser (*Rhinoptilus bitorquatus*)
Pink-headed duck (*Rhodonessa caryophyllacea*)
Red-headed vulture (*Sarcogyps calvus*)
Sociable lapwing (*Vanellus gregarius*)
Bugun liocichla (*Liocichla bugunorum*)

Fish

Knifetooth sawfish (*Anoxypristis cuspidata*)
Pondicherry shark (*Carcharhinus hemiodon*)
Ganges shark (*Glyphis gangeticus*)
Deccan labeo (*Labeo potail*)
Largetooth sawfish (*Pristis microdon*)
Longcomb sawfish (*Pristis zijsron*)
Humpback mahseer

Reptiles and Amphibians

Northern river terrapin (*Batagur baska*)
Red-crowned roofed turtle (*Batagur kachuga*)
Hawksbill sea turtle (*Eretmochelys imbricata*)
Gharial (*Gavialis gangeticus*)
Ghats wart frog (*Fejervarya murthii*)
Gundia Indian frog (*Indirana gundia*)
Toad-skinned frog (*Indirana phrynoderma*)
Charles Darwin's frog (*Ingerana charlesdarwini*)
Rao's torrent frog (*Micrixalus kottigeharensis*)
Amboli bush frog (*Pseudophilautus amboli*)
White-spotted bush frog (*Raorchestes chalazodes*)
Griet bush frog (*Raorchestes griet*)
Munnar bush frog (*Raorchestes munnarensis*)
Ponmudi bush frog (*Raorchestes ponmudi*)
Sacred Grove bush frog (*Raorchestes sanctisilvaticus*)
Shillong bubble-nest frog (*Raorchestes shillongensis*)
Resplendent shrubfrog (*Raorchestes resplendens*)
Anaimalai flying frog (*Rhacophorus pseudomalabaricus*)
Patinghe Indian gecko (*Geckoella jeyporensis*)

Mammals

Asiatic cheetah (*Acinonyx jubatus venaticus*)
Namdapha flying squirrel (*Biswamoyopterus biswasi*)
Himalayan wolf (*Canis himalayensis*)
Andaman Shrew (*Crocidura andamanensis*)
Jenkins' shrew (*Crocidura jenkinsi*)
Nicobar shrew (*Crocidura nicobarica*)
Northern Sumatran rhinoceros (*Dicerorhinus sumatrensis lasiotis*)
Kondana soft-furred rat (*Millardia kondana*)
Pygmy hog (*Porcula salvania*)
Indian Javan rhinoceros (*Rhinoceros sondaicus inermis*)
Malabar large-spotted civet (*Viverra civettina*)
Elvira rat (*Cremnomys elvira*)
Chinese pangolin (*Manis pentadactyla*)
Kashmir stag (*Cervus canadensis hanglu*)

Coral

Fire corals (*Millepora boschmai*)
Spiders
Rameshwaram Ornamental or Parachute Spider (*Poecilotheria hanumavilasumica*)
Goody Tarantula, Metallic Tarantula or (*Poecilotheria metallica*)

CITES

CITES (the Convention on International Trade in Endangered Species of Wild Fauna and Flora) is an international agreement between governments. Its aim is to ensure that international trade in specimens of wild animals and plants does not threaten their survival. CITES (the Convention on International Trade in Endangered Species of Wild Fauna and Flora) is an international agreement between governments. Its aim is to ensure that international trade in specimens of wild animals and plants does not threaten their survival.

For many years CITES has been among the conservation agreements with the largest membership, with now 181 Parties. Roughly 5,600 species of animals and 30,000 species of plants are protected by CITES against over-exploitation through international trade. They are listed in the three CITES Appendices. The species are grouped in the Appendices according to how threatened they are by international trade. They include some whole groups, such as primates, cetaceans (whales, dolphins and porpoises), sea turtles, parrots, corals, cacti and orchids. However, in some cases only a subspecies or geographically separate population of a species (for example the population of just one country) is listed.

Ministry of Environment, Forests and Climate Change (INDIA)

The Ministry of Environment & Forests (MoEF) is the nodal agency in the Central Government for overseeing the implementation of India's environment and forest policies and programmes relating to conservation of the country's natural resources including lakes and rivers, its biodiversity, forests and wildlife, ensuring the welfare of animals and prevention and abatement of pollution. While implementing these policies and programmes, the Ministry is guided by the principle of sustainable development. The Ministry is also the nodal agency for the United Nations Environment Programme (UNEP), South Asia Co-operative Environment

Programme (SACEP), International Centre for Integrated Mountain Development (ICIMOD) and the United Nations Conference on Environment and Development (UNCED). The Ministry also coordinates with multilateral bodies such as the Commission on Sustainable Development (CSD), Global Environment Facility (GEF) and regional bodies such as Economic and Social Council for Asia and Pacific (ESCAP) and South Asian Association for Regional Cooperation (SAARC) on matters pertaining to environment.

The broad objectives of the Ministry are:

- Conservation and survey of flora, fauna, forests and wildlife
- Prevention and control of pollution
- Afforestation and regeneration of degraded areas
- Protection of environment, and
- Ensuring the welfare of animals.

National biodiversity authority

The Biological Diversity Act 2002 came into force in 2003. The Act extends to the whole of India. The objectives of the Act are conservation, sustainable utilization and fair and equitable sharing of benefits arising out of the use of biological resources and associated knowledge. The Act is being implemented in a three tiered institutional structures (NBA at National level, State Biodiversity Board at State level and Biodiversity Management Committee at local level).

The NBA is a body corporate established in accordance with the provisions of Sec.8 of the Biological Diversity Act, 2002, at Chennai w.e.f. 1st October 2003. It is an autonomous, statutory and regulatory organization which is intended to implement the provisions of Biological Diversity Act, 2002. The main objectives of NBA are: –

- To regulate access to biological resources of the country to conserve and sustainable use of biological diversity.
- To respect and protect the knowledge of local communities related to biodiversity.
- To secure sharing of benefits with the local people as conservers of biological resources and holders of knowledge and information relating to the use of biological resources.
- Conservation and development of area of importance from the view point of biological diversity by declaring them as biological diversity heritage sites.
- Protection and rehabilitation of threatened species; involvement of institutions of state government in the broad scheme of implementation of the Biological Diversity Act through constitution of committees.

Centres of Excellence (in India)

- Centre for Environment Education (CEE), Ahmedabad .
- CPR Environmental Education Centre (CPREEC), Chennai.
- Centre for Ecological Sciences (CES), Indian Institute of Science (IISc), Bengaluru.
- Centre of Mining Environment (CME), Indian School of Mines, Dhanbad.
- Salim Ali Centre for Ornithology and Natural History (SACON), Coimbatore.
- Centre for Environment Management of Degraded Ecosystem (CEMDE), University of Delhi, Delhi.
- Madras School of Economics (MSE), Chennai.
- Foundation for Revitalization of Local Health Traditions (FRLHT), Bengaluru.

The Tropical Botanic Garden and Research Institute (TBGRI), Thiruvananthapuram .

Centre for Animals and Environment, CARTMAN, Bengaluru.

Loss of Biodiversity

There is continuous loss of the earth's treasure of species. For example, the colonization of tropical Pacific Islands by human has led to extinction of more than 2000 species of native birds.

The Red list of IUCN documented the extinction of 784 species in last 500 years. The last 20 years witnessed the disappearance of 27 species.

Some important examples of recent extinctions are-

Dodo (Mauritius), *Quagga* (Africa), *Thylacine* (Australia), *Steller Sea-cow* (Russia), and subspecies of Tiger, like *bali*, *javan* and *caspian*.

Presently, more than 15,500 species world-wide are facing the threat of extinction. This includes 32% of amphibian species, 23% of mammalian species and 12% of birds' species. About 31% of the gymnosperms species are also facing the extinction. The amphibians are however, more vulnerable in such cases.

Causes of loss of biodiversity

The accelerated rate of species-extinction is largely due to human activities. There are 4-major causes, called '**The Evil Quartet**', for the loss of biodiversity –

1. Habitat loss and fragmentation
2. Over-exploitation
3. Invasion of Alien or exotic species
4. Co-extinctions

1. Habitat loss and fragmentation

The cutting trees and burning of forest destroys the natural habitat of a species. The construction of mines, dams, harbors, industries and buildings for human settlement has also affected the biodiversity. *The Habitat destruction is the primary and major reason for the loss of biodiversity.* The tropical rain forest is the example of the habitat loss where forest covering has been reduced from 14% of land surface to 6%.

The Amazon rain forest, called '*The Lungs of the Planet*', which harbors millions of species, is being cleared for cultivating soybean or developing grasslands for raising cattle. The pollution is also the factor for degradation of habitat.

When large habitats are broken into small fragments due to various human activities, the population of migratory animals, mammals and birds, that require a large territory, are adversely affected.

2. Over-exploitation

When human need turns to human greed, for food and shelter, it leads to overexploitation of natural resources. Many species – extinction, like that of Stellar sea-cow and Passenger pigeon, in last 500 years, are due to over-exploitation by humans. Many marine fishes are also being over harvested. Over fishing from a water body, or over harvesting a product is just like '*killing a goose laying golden eggs*'.

3. Invasion of Alien or exotic species

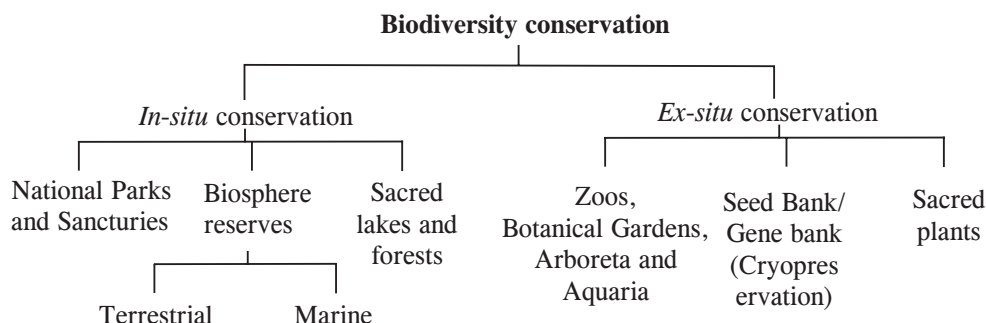
When alien species are introduced into an explored area, some of the species turn invasive and cause decline or extinction of indigenous species. For example –

- Introduction of **Nile perch** into lake Victoria (E. Africa) led to the extinction of more than 200 species of Cichlid fish in the lake
- Introduction of weed species, like **Carrot grass** (*Parthenium*), Lantana and **water hyacinth** (*Eicchornia*) has posed threat to the native species and damage to environment.

- The illegal introduction of **African cat fish** (*Clarias gariepinus*) for aquaculture purposes into the river has threatened indigenous cat fishes.

4. Co-extinctions

Whenever a plant or animal species becomes extinct, its obligatory-associated species also becomes extinct. For example, when a host species becomes extinct, the parasite also meets the same fate. In case of '*plant pollinator mutualism*' the extinction of one species leads to the extinction of the other.



1. In situ conservation

In such conservation the endangered species are protected in their natural habitat with entire ecosystem. The conservationists, on global basis, have identified certain **Biodiversity Hot Spots** (with high level of species richness and high degree of endemism).

(The endemic species are the ones which are confined to a particular region and are not found anywhere else) The hot spots are also the regions of accelerated habitat loss.

Hot spots are the areas with high density of biodiversity or mega diversity which are most threatened at present. The concept was developed by environmental scientist Norman Myers of Oxford University in the United Kingdom in an attempt to identify priority areas for biodiversity conservation. Around the world, 35 areas qualify as hotspots. They represent just 2.3% of Earth's land surface, but they support more than half of the world's plant species as endemics — *i.e.*, species found nowhere else — and nearly 43% of bird, mammal, reptile and amphibian species as endemics.

The number of such hot spots is now 34. These hot spots cover only 1 to 2 percent of earth's land area,

To qualify as a biodiversity hotspot, a region must meet two strict criteria:

- It must have at least 1,500 vascular plants as endemics — which is to say, it must have a high percentage of plant life found nowhere else on the planet. A hotspot, in other words, is irreplaceable.
- It must have 30% or less of its original natural vegetation. In other words, it must be threatened.

The 3-biodiversity hot spots of India, that cover rich-biodiversity regions, are

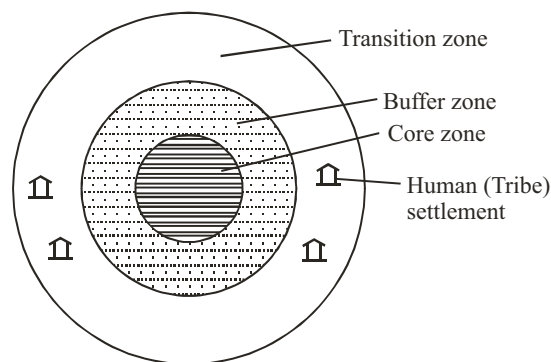
1. Western Ghat
2. Himalaya
3. Indo-Burma

The *in situ* conservation, in India, is done through 15–**Biosphere reserves**, 90–**National Parks**, more than 450 **sanctuaries** and several **Sacred Groves** or the tracts of forests.

1. Biosphere reserves

They represent natural biomes which contain unique biological communities. They include land as well as coastal environment. Biosphere reserves were created under **MAB** (Man and Biosphere) programme of **UNESCO** in 1971. Till May 2000 there were 408 biosphere reserves in 94 countries of the world. In India there are 15 biosphere reserves. There are 3-zones in a biosphere reserve.

- (a) **Core (natural) zone** – It is inner most zone which is legally protected and completely undisturbed from human interference,
- (b) **Buffer zone** – In this zone limited human activity is allowed for research and education purposes.
- (c) **Transition (manipulation) zone** – It is the outermost zone of biosphere reserve in which large number of human activities are permitted, eg. Cultivation, domestication, harvesting of natural product, grazing, forestry, settlement and recreation etc. In this zone the traditional life style of tribals is protected with their live-stock.



Different zones of a Terrestrial Biosphere reserve

Functions of biosphere reserves

1. For conservation of landscape, ecosystem and genetic resources.
2. For economic development.
3. For scientific research, education and for exchange of information at national and global level.

In India the first biosphere reserve, **Nilgiri** was declared in 1986. It includes parts of Karnataka, Kerala and Tamilnadu. The list of biosphere reserves of India is given below :-

- | | |
|-------------------------|-------------------------|
| 1. Nilgiri | 2. Nandadevi |
| 3. Uttrakhand | |
| 4. Nokrek (Meghalaya) | 5. Andamans |
| 6. Simlipal (Orissa) | |
| 7. Kaziranga (Assam) | 8. Gulf of Mannar (T.N) |
| 9. Thar Desert | |
| 10. Sundarbans (W.B.) | 11. Kanha (M.P.) |
| 12. Runn of Kutch (Guj) | |
| 13. Nicobar | 14. Manas (Assam) |
| 15. Namdapha (Ar. P.) | |

2. National Parks

They are reserved for the betterment of wild life, both **fauna and flora**. In national parks private ownership is not allowed. The grazing, cultivation, forestry etc. is also not permitted. The first national park of the world, Yellow stone, in U.S.A., was founded in 1872.

Important state wise national parks of India are –

Jammu and Kashmir	-	Dachigam, Salim Ali
Assam	-	Kaziranga, Manas*
Meghalaya	-	Nokrek
West Bengal	-	Sunderbans
Bihar	-	Hazaribagh, Palamau*
Uttaranchal	-	Corbett* (Hailey), Nanda Devi, Valley of flowers, Rajaji
U. P.	-	Dudhwa*
Gujrat	-	Gir, Marine
Rajasthan	-	Sariska*, Ranthambore*, Desert
Madhya Pradesh	-	Kanha*, Sanjay, Madhav, Panna, Bandhavgarh*, Van Vihar, Fossil
Orissa	-	Simlipal
Karnataka	-	Bandipur*
Kerala	-	Silent Valley, Periyar*

*These national parks are running **Tiger Project** also. (The maximum national parks are present in Madhya Pradesh).

3. Sanctuaries

In sanctuaries the protection is given to **fauna** only. The activity like harvesting of timber, collection of forest products and private ownership rights are permitted so long as they do not interfere with the well being of the animals. The important wild life sanctuaries are Chilka wild life sanctuary (**Orissa**), Bharatpur Bird Sanctuary (**Rajasthan**), Sultanpur Bird sanctuary (**Haryana**) and Jalpara sanctuary (**West Bengal**). Maximum sanctuaries belong to Andaman and Nicobar.

The **Project Tiger** was launched in India in year 1973 with the assistance of **WWF** (World Wild life Fund) after the recommendation of **IBWL** (Indian Board of Wild Life). At present there are more than 20 tiger projects. (**WWF** after its silver jubilee in 1986 has been renamed as **World Fund for Nature (WFN)**). The symbol of **WWF** is **Giant Panda**.

In India National Parks and Sanctuaries were created after formulation of **Wild life (protection) act** in 1972. (This act was amended in 1991).

4. The sacred groves are found in Khasi and Jaintia hills (Meghalaya), Aravalli hills (Rajasthan), Western ghats (Karnataka and Maharashtra) and Sarguja, Chanda and Bastar areas of Madhya Pradesh.

National Parks in India

Name	State	Notability
Bandipur National Park (1974)	Karnataka	Chital, gray langurs, Indian giant squirrel, Gaur, leopard, Sambar deer, indian elephants, honey buzzard, red-headed vulture and other animals.
Bannerghatta National Park (Bannerghatta Biological Park) (1974)	Karnataka	White Tiger, Royal Bengal Tiger, Bear, other animals
Betla National Park (1986)	Jharkhand	Tiger, Sloth Bear, Peacock, Elephant, Sambar deer, mouse deer and other animals.
Bhitarkanika National Park (1988)	Odisha	Mangroves, Saltwater crocodile, white crocodile, Indian python, black ibis, wild pigs, rhesus monkeys, chital and other animals
Buxa Tiger Reserve (1992)	West Bengal	Tiger
Dachigam National Park (1981)	J&K	Only area where Kashmir stag is found
Dudhwa National Park (1977)	U.P	Swamp deer, sambar deer, barking deer, spotted deer, hog deer, tiger, Indian rhinoceros,
Gir Forest National Park (1965)	Gujarat	Asiatic lion

Great Himalayan National Park (1984)	Himachal Pradesh,	UNESCO World Heritage Site
Gulf of Mannar Marine National Park (1980)	Tamil Nadu	Green turtles and Olive Ridley turtles and whales.
Indravati National Park (1981)	Chhattisgarh	Wild Asian Buffalo, Tiger Reserve, Hill Mynas
Jaldapara National Park (2012)	West Bengal	Indian one horned rhinoceros
Jim Corbett National Park (1936)	Uttarakhand	Tiger
Kanha National Park (1955)	M. P.	Swamp Deer, Tigers
Kaziranga National Park (1905)	Assam	Indian rhinoceros, UNESCO World Heritage Site
Keibul Lamjao National Park (1977)	Manipur	only floating park in the world
Keoladeo National Park (1981)	Rajasthan	UNESCO World Heritage Site
Manas National Park (1990)	Assam	UNESCO World Heritage Site
Mandla Plant Fossils National Park (1983)	M. P	Plant Fossils National Park
Marine National Park, Gulf of Kutch (1980)	Gujarat	70 species of sponges, Coral 52 species along with puffer fishes, sea horse and sting ray
Namdapha National Park (1974)	Arunachal Pradesh	Snow Leopards, Clouded Leopards, Common Leopards and Tigers
Nanda Devi National Park (1982)	Uttarakhand	UNESCO World Heritage Site
Neora Valley National Park (1986)	West Bengal	clouded leopard, red panda and musk deer
Nokrek National Park (1986)	Meghalaya	UNESCO World Biosphere Reserve
Periyar National Park (1982)	Kerala	Tigers
Ranthambore National Park (1981)	Rajasthan	Tigers, Leopards, Striped Hyenas, Sambar deer and Chital.
Sariska Tiger Reserve (1955)	Rajasthan	Tiger
Simlipal National Park (1980)	Odisha	Tiger, Leopard, Asian elephant, Sambar, Barking deer, Gaur, Jungle cat, Wild boar, and other animals.
Sultanpur National Park (1989)	Haryana	Siberian crane, greater flamingo, ruff, black-winged stilt, common teal, northern pintail, and yellow wagtail.
Sundarbans National Park (1984)	West Bengal	UNESCO World Heritage Site
Valley of Flowers National Park (1982)	Uttarakhand	Flying squirrel, Himalayan black bear, red fox, Himalayan weasel and Himalayan yellow-throated marten, and Himalayan goral

Wild Life Sanctuaries

India has 515 animal sanctuaries referred to as wildlife sanctuaries category IV protected areas. Among these, the 48 tiger reserves are governed by Project Tiger, and are of special significance in the conservation of the tiger.

WILD LIFE SANCTUARIES IN INDIA

Name of the Sanctuaries	Location	Major Species
Gir Wild Life Sanctuary	Sasan Gir, Junagadh, Amreli	Lion, Leopard, Chausinga, Chital, Hyena, Sambar, Chinkara, Herpetofauna, Crocodiles and birds
Wild Ass Sanctuary	Little Rann of Kachchh	Wild Ass, Chinkara, Blue bull, Houbara bustard, Wolf, Waterfowls, Herpetofauna
Hingolghadh Sanctuary	Hingolghadh, Rajkot	Chinkara, Blue bull, Wolf, Hyena, Fox, Birds, Herpetofauna
Marine Sanctuary	Gulf of Kachchh, Jamnagar	Sponges, Corals, Jellyfish, Sea horse, Octopus, Oyster, Pearloyster, Starfish, Lobster, Dolphin, Dugong, waterfowls
Simlipal Sactuary	Odisha	Elephant, Tiger, Leopard, Gaur, Cheetal
Kutch Desert Sanctuary	Great Rann of Kachchh	Chinkara, Hyena, Fox, Flamingo, Pelicans & other waterfowls, Herpetofauna
Rampara Sanctuary	Rampara, Rajkot	Blue bull, Chinkara, Wolf, Fox, Jackal, Birds, Herpetofauna
Ghana Bird Sanctuary	Rajasthan	Water Bird, Black-buck, Cheetal, Sambar
Panchmarhi	Madhya Pradesh	Tiger, Panther, Sambhar, Nilgai, Baskeng, Deer
Dandeli Sanctuary	Karnataka	Tiger, Panther, Elephant, Cheetal, Sambhar, Wild Boar
Kutch Bustard Sanctuary	Near Naliya, Kachchh	Great Indian Bustard, Lesser Florican, Houbara bustard, Chinkara, Blue bull, Herpetofauna

Biosphere reserves in India Area-wise

Name	State	Key Fauna
Nilgiri Biosphere Reserve	Tamil Nadu, Kerala and Karnataka	Nilgiri tahr, lion-tailed macaque
Nanda Devi National Park & Biosphere Reserve	Uttarakhand	
Gulf of Mannar	Tamil Nadu	Dugong or sea cow
Nokrek	Meghalaya	Red panda
Sundarbans	West Bengal	Royal Bengal tiger
Manas	Assam	Golden langur, red panda
Simlipal	Odisha	Gaur, royal Bengal tiger, elephant
Dihang-Dibang	Arunachal Pradesh	
Pachmarhi Biosphere Reserve	Madhya Pradesh	Giant squirrel, flying squirrel
Achanakmar-Amarkantak Biosphere Reserve	Madhya Pradesh, Chhattisgarh	Four horned antelope (Tetracerus quadricornis), Indian wild dog (Cuon alpinus), Saras crane (Grus antigone), Asian white-backed vulture (Gyps bengalensis), Sacred grove bush frog (Philautus sanctisilvaticus)
Great Rann of Kutch	Gujarat	Indian wild ass

Cold Desert	Himachal Pradesh	Snow leopard
Khangchendzonga	Sikkim	Snow leopard, red panda
Agasthyamali Biosphere Reserve	Kerala, Tamil Nadu	Nilgiri tahr, elephants
Great Nicobar Biosphere Reserve	Andaman and Nicobar Islands	Saltwater crocodile
Dibru-Saikhowa	Assam	Golden langur
Seshachalam Hills	Andhra Pradesh	
Panna	Madhya Pradesh	Tiger, chital, chinkara, sambar and sloth bear

2. Ex situ conservation

In such type of conservation the threatened animals and plants are taken out of their natural habitat and are protected in special parks or areas like, **Zoological parks**, **Wild life safari parks** and **Botanical gardens*** etc. The *ex situ* conservation also includes

- **Cryopreservation** of gametes of threatened species in viable and fertile form.
- Fertilization of eggs *in vitro* and propagation of plants through '**Tissue culture methods**'
- Preservation of seeds through **Seed banks**

The historic conservation on Biodiversity, '**The Earth Summit**' was held in **Rio de Janeiro** (Brazil) in 1992. In a follow-up, in 2002, through '**World Summit on Sustainable development**' in **Johannesburg** (S. Africa), one hundred and ninety countries pledged their commitment for a significant reduction in current rate of biodiversity-loss at global, regional and local level, by 2010. The next summit for the cause of biodiversity is to be held in 2012.

(***Botanical gardens** – There are about 1500 botanical gardens and **arboreta** (a place where specific species of trees or shrubs are cultivated for research or display) in the world. They contain more than 80000 species. Some botanical gardens also have facilities of seed bank and tissue culture.)

Animal Welfare

People for the Ethical Treatment of Animals (PETA)

It is a non-profitable American animal rights organization based in Norfolk, Virginia. Led by Ingrid Newkirk, its international president, founded in 1980 with a slogan of "Animals are not ours to eat, wear, experiment on, use for entertainment, or abuse in any other way." It focuses its attention on the four areas in which the largest numbers of animals suffer the most intensely for the longest periods of time: on factory farms, in the clothing trade, in laboratories, and in the entertainment industry.

World Wide Fund for Nature

The organisation was conceived in Morges, Switzerland (29, April, 1961). It is an international non-governmental organization in nature. Works in the field related to biodiversity conservation, and the reduction of humanity's footprint on the

environment. It is the world's largest conservation organization with the slogan of "For a Living Planet." The method of its working involves Lobbying Research and Consultancy. Basically it's a charitable trust. WWF's giant panda logo originated from a panda named Chi Chi. It has been designed by Sir Peter Scott from preliminary sketches made by Gerald Watterson.

The main missions of WWF are as follows:

- conserving the world's biological diversity
- ensuring that the use of renewable natural resources is sustainable
- Promoting the reduction of pollution and wasteful consumption.

At present WWF's current strategy of achieving its mission which is related to restoring populations of 36 species (species or species groups that are important for their ecosystem or to people, including elephants, tunas, whales, dolphins and porpoises), and ecological footprint in 6 areas (carbon emissions, cropland, grazing land, fishing, forestry and water).

Animal Welfare Board of India

Functions

- To keep the law in force in India for the Prevention of Cruelty to Animals under constant study and to advise the government on the amendments to be undertaken in any such law from time to time.
- To advise the Central Government on the making of rules under the Act with a view to preventing unnecessary pain or suffering to animals and transported.
- To advise in the design of vehicles so as to lessen the burden on draught animals.
- To take all such steps as the Board may think fit for amelioration of animals by encouraging, or providing for the construction of sheds, water troughs and the like and by providing for veterinary assistance to animals.
- To advise in the design of slaughter houses or its maintenance.

India Initiative towards Animal protection

Project Tiger an government of India initiative for conserving its national animal, the tiger. The project was launched in 1973. Since then the no of tiger reserve has been increased from 9 to 47 which accounts for 2.08% the total geographical area of our country. The area of tiger projects have been developed on core/buffer strategy. The core areas are legally termed as National Parks and the buffering areas are a mixture of forest and non-forest land managed as a multiple used area. The project aims at fostering an exclusive tiger agenda in the core areas of tiger reserves, with an inclusive people oriented agenda in the buffer

Project Rhino was joint venture of the Assam Forest Department and Wildlife Trust of India - International Fund for Animal Welfare (WTI-IFAW) and initiated in February 2006 with the trans location of a hand-raised rhino calf to Manas Wildlife Sanctuary. The projects aims at repopulating the one horn rhino by displacing them to Manas wild life sanctuary from Kaziranga National Park. The whole project is supported by Bodoland Territorial Council and the Assam Forest Department.

Project Crocodile Conservation was launched in 1975 in different States for protecting the endangered crocodile species like

Gharial, *Gavialis gangeticus*; Mugger crocodile, *Crocodylus palustris* and Saltwater crocodile, *Crocodylus* were on the verge of extinction by the seventies. The funds and technical support for the project came from UNDP/ FAO through the Government of India.

Project Elephant (PE) is a central government initiative to provide financial and technical support to major elephant bearing states of India. It was launched in February 1992. It aims at protecting the elephants, their habitat and corridor. It also looks after the human elephant issues. It is implemented in 13 States / UTs, viz. Andhra Pradesh, Arunachal Pradesh, Assam, Jharkhand, Karnataka, Kerala, Meghalaya, Nagaland, Orissa, Tamil Nadu, Uttaranchal, Uttar Pradesh and West Bengal.

SAVE (Saving Asia's Vultures from Extinction) is a consortium of regional and international organization to co-ordinate conservation, campaigning and fundraising activities to help the plight of south Asia's vultures. The key strategies of vulture conservation SAVE is involved in a wide range of conservation activities across South Asia including:

- breeding vultures in captivity so that their offspring can be released back in to the wild when the environment is free from diclofenac
- an active advocacy programme targeting the vets and farmers using diclofenac

- legislation controlling the manufacture and sale of veterinary drugs
- in-situ conservation actions focused around the small but key remaining vulture populations in the wild
- an active research programme that underpins these activities and monitors their effectiveness

Project Dolphin Gangetic river dolphins is India's national aquatic animal and is often known as the 'Tiger of the Ganges'. This dolphin species is an indicator animal which represent healthy river ecosystem in a same position as a tiger in a forest. Their population is estimated to be less than 2,000 in the country. Some of the major threats are habitat fragmentation due to construction of dams and barrages, direct killing, indiscriminate fishing and pollution of rivers.

For conservation of dolphins, India's first Dolphin Community Reserve established in West Bengal to protect the endangered mammal, Gangetic river dolphins. The reserve would be set up in the Hooghly River between Malda and Sundarbans as per provisions of Wildlife Protection Act, 1972. State Forest department also has announced that it would also conduct a census to estimate the population of dolphins.

Exercise -1

1. A keystone species
 - (a) has a disproportionately large impact on an ecosystem.
 - (b) typically reduces overall diversity of an ecosystem.
 - (c) is typically an herbivore.
 - (d) is an example of amensalism.
2. Extinction of a weaker species by an aggressive alien species is the results of
 - (a) Endemism of weaker species
 - (b) Habitat loss
 - (c) The Domino Effect
 - (d) All of the above.
3. In the Lower Himalayan Mountains, several species of salamander, an amphibian, live in or near a stream. The largest species lives in the stream and along its edges, a smaller species lives on land within a meter or two of the stream, and a smaller species lives about 3-5 meters away from the stream. In this region, these three salamander species are using
 - (a) different niches within the same habitat
 - (b) the same niche and microhabitat
 - (c) the same landscape but different ecosystems
 - (d) the same habitat but different niches
4. Invasive species are dangerous because
 - (a) they are almost all predators, disturbing ecological relationships by eating other species.
 - (b) they carry viruses that spread disease in new ecosystems.
 - (c) the native species have not evolved with these organisms.
 - (d) they tend to be secretive, going unnoticed in their new ecosystems.
5. Protection and preservation of endangered species away from their natural habitat under human care in zoos, nurseries and laboratories is known as
 - (a) In-situ conservation
 - (b) Ex-situ conservation
 - (c) Biodiversity conservation
 - (d) None of the above
6. Protection of biodiversity around the world requires:
 - (a) basic science to produce government policies and laws that then must be enforced.
 - (b) changes to social structure and political organizations that drive basic science.
 - (c) new technologies and techniques that are still being developed.
 - (d) the introduction of new species into new regions to spread a species range.
7. Within biological communities, some species are important in determining the ability of a large number of other species to persist in the community. Such species are called
 - (a) Keystone species
 - (b) Allopatric species
 - (c) Sympatric species
 - (d) Threatened species
8. The diversity and productivity of coral reefs is most similar to that of
 - (a) desert environments
 - (b) a natural prairie.
 - (c) tropical rain forests
 - (d) a river system.
9. The risk of introducing a natural enemy to control an invasive species is that
 - (a) it might drive the invasive species to extinction.
 - (b) the natural enemy might also become a pest.
 - (c) the natural enemy might evolve into a new species
 - (d) the natural enemy may introduce genetic diversity into the invasive species.
10. Biodiversity is important because:
 - (a) it is necessary to maintain ecosystems.
 - (b) humans can use new sources of food.
 - (c) without certain species, photosynthesis may not be possible.
 - (d) certain species are necessary to provide oxygen in the atmosphere.
11. Compared to forests using sustainable forest management, commercial forests managed for maximum sustainable yield of commercially valuable species will
 - (a) support more biological diversity.
 - (b) be more resistant to pests.
 - (c) produce a greater variety of wood.
 - (d) have greater erosion problems.
12. Endemic species are:
 - (a) secure groups that show the least risk of extinction.
 - (b) limited to just one habitat.
 - (c) widely distributed, found especially on large continents.
 - (d) usually the dominant species within an ecosystem.
12. The greatest loss of biodiversity in the last two centuries has resulted from:
 - (a) the introduction of alien species to new ecosystems.
 - (b) the use of fossil fuels to power transportation and electrical production.
 - (c) the physical alteration of habitats.
 - (d) the use of rivers, lakes, and oceans for transportation.
14. Which of the following represents the greatest conservation of the genetic bank?
 - (a) the human genome center, analyzing the components of the human genome.
 - (b) seed banks that store seeds of thousands of plants from around the world.

- (c) the field of proteomics, investigating the many ways the proteins function in organisms.
 (d) all of the varieties of corn wheat, and rice currently serving as crops.
15. Which one of the following groups of animals belongs to the category of endangered species ?
 (a) Great Indian Bustard, Musk Deer, Red Panda and Asiatic Wild Ass
 (b) Kashmir Stag, Cheetal, Blue Bull and Great Indian Bustard
 (c) Snow Leopard, Swamp Deer, Rhesus Monkey and Saras (Crane)
 (d) Lion-tailed Macaque, Blue Bull, Hanuman Langur and Cheetal
16. Which one of the following national parks is located near Chamoli?
 (a) Dudhwa National Park
 (b) Great Himalayan Park
 (c) Jim Corbett National Park
 (d) Nanda Devi National Park
17. Which one of the following is included in the world list of biosphere reserves by UNESCO?
 (a) Kinnaur Region (b) Spiti Valley
 (c) Nallamalai Hills (d) Sunderbans
18. Which one of the following is a global biodiversity hotspot in India?
 (a) Western Ghats
 (b) Western Himalayas
 (c) Eastern Ghats
 (d) Northern Himalayas
19. In which one of the following states is Ranganathittu Bird Sanctuary located?
 (a) Tamil Nadu (b) Kerala
 (c) Karnataka (d) Andhra Pradesh
20. For which one of the following is Sualkuchi famous?
 (a) Bird sanctuary (b) Temple city
 (c) Silk centre (d) Hill station
21. Which one of the following is the correct sequence of the given tiger reserves of India from North to South?
 (a) Dudwa-Kanha-Indravati-Bandipur
 (b) Kanha-Bandipur-Dudwa-Indravati
 (c) Indravati-Kanha-Dudwa-Bandipur
 (d) Dudwa-Kanha-Bandipur-Indravati
22. Kanha National Park belongs to which one among the following biogeographical areas in the world?
 (a) Tropical Sub-humid Forests
 (b) Tropical Humid Forests
 (c) Tropical Dry Forests
 (d) Tropical Moist Forests
23. In wildlife conservation which one among the following best defines an 'endemic species'?
 (a) When the critical number of a species declines in a forest due to parasitic attack
 (b) A species which is cosmopolitan and can be commonly found in biosphere
 (c) An endangered species which is found in a few restricted areas on the Earth
 (d) A species confined to a particular region and not found anywhere else
24. Veliconda hills, which is a part of Eastern Ghats, is situated in
 (a) Odisha (b) Tamil Nadu
 (c) Karnataka (d) Andhra Pradesh
25. Biodiversity is richer in
 (a) tropical regions (b) polar regions
 (c) temperate regions (d) oceans
26. Which one of the following National Parks has a climate that varies from tropical to subtropical, temperate and arctic? [CSAT 2015-1]
 (a) Khangchendzonga National Park
 (b) Nandadevi National Park
 (c) Neora Valley National Park
 (d) Namdapha National Park
27. In India, in which one of the following types of forests is teak a dominant tree species? [CSAT 2015-1]
 (a) Tropical moist deciduous forest
 (b) Tropical rain forest
 (c) Tropical thorn scrub forest
 (d) Temperate forest with grasslands
28. Which of the following National Parks is unique in being a swamp with floating vegetation that supports a rich biodiversity? [CSAT 2015-1]
 (a) Bhitarkanika National Park
 (b) Keibul Lamjao National Park
 (c) Keoladeo Ghana National Park
 (d) Sultanpur National Park
29. Which one of the following is the national aquatic animal of India? [CSAT 2015-1]
 (a) Saltwater crocodile (b) Olive ridley turtle
 (c) Gangetic dolphin (d) Gharial

Exercise -2

Statement Based MCQ

- Consider the following pairs

Protected area	Well-known for
1. Bhitarkanika, Odisha	— Salt Water Crocodile
2. Desert National Park,	— Great Indian Bustard
	Rajasthan
3. Eravikulam, Kerala	— Hoolak Gibbon

Which of the pairs given above is / are correctly matched ?

(a) 1 only (b) 1 and 2
(c) 2 only (d) 1, 2 and 3
- Three of the following criteria have contributed to the recognition of Western Ghats, Sri Lanka and Indo Burma regions as hotspots of biodiversity
 - Species richness
 - Vegetation density
 - Endemism
 - Ethno-botanical importance
 - Threat perception
 - Adaption of flora and fauna to warm and humid conditions

Which three of the above are correct criteria in this context?

(a) 1, 2 and 6 (b) 2, 4 and 6
(c) 1, 3 and 5 (d) 3, 4 and 6
- Consider the following statements
 - Biodiversity hotspots are located only in tropical regions.
 - India has four biodiversity hotspots i.e., Eastern Himalayas, Western Himalayas, Western Ghats and Andaman and Nicobar Islands.

Which of the statements given above is / are correct?

(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2
- The "Red Data Books" published by the International Union for Conservation of Nature and Natural resources (IUCN) contain lists of ?
 - Endemic plant and animal species present in the biodiversity hotspots.
 - Threatened plant and animal species.
 - Protected sites for conservation of nature and natural resources in various countries.

Select the correct answer using the codes given below:

(a) 1 and 3 (b) 2 only
(c) 2 and 3 (d) 3 only
- How does National Biodiversity Authority (NBA) help in protecting the Indian agriculture?
 - NBA checks the biopiracy and protects the indigenous and traditional genetic resources.
 - National Biodiversity Authority (NBA) directly monitors and supervises the scientific research on genetic modification of crop plants.
 - Application for Intellectual Property Rights related to genetic/biological resources can not be made without the approval of NBA.

Which of the statements given above is/are correct?

(a) 1 only (b) 1 and 2 only
(c) 1 and 3 only (d) 1, 2 and 3
- Consider the following protected areas:
 - Bandipur
 - Bhitarkanika
 - Manas
 - Sunderbans

Which of the above are declared Tiger Reserves?

(a) 1 and 2 only (b) 1, 3 and 4 only
(c) 2, 3 and 4 only (d) 1, 2, 3 and 4
- Which of the following can be threats to the biodiversity of a geographical area?
 - Global warming
 - Fragmentation of habitat
 - Invasion of alien species
 - Promotion of vegetarianism

Select the correct answer using the codes given below:

(a) 1 and 2 only (b) 2 and 3 only
(c) 1, 2 and 3 only (d) 1, 2, 3 and 4 only
- Biodiversity forms the basis for human existence in the following ways:
 - Soil formation
 - Prevention of soil erosion
 - Recycling of waste
 - Pollination of crops

Select the correct answer using the codes given below:

(a) 1, 2 and 3 only (b) 2, 3 and 4 only
(c) 1 and 4 only (d) 1, 2, 3 and 4
- Which of the following regions of India have been designated as biodiversity hotspots?

Select the correct answer from the codes given below:

 - Eastern Himalaya
 - Eastern Ghat
 - Western Ghat
 - Western Himalaya

Codes:

(a) 1 and 2 only (b) 1 and 3 only
(c) 2 and 4 only (d) 3 and 4 only
- The steps taken by the Government of India for conservation of endangered species are
 - The Central Government has enacted the Wild Life (Protection) Act, 1972 for protection of wildlife including birds.
 - Wetland (Conservation and Management) Rules 2010 have been framed for protection of wetlands, in the States, which are habitats of birds.
 - Wildlife Crime Control Bureau has been established for control of illegal trade in wildlife, including endangered species of birds and their parts and products.

4. The Centrally Sponsored Scheme of National Plan for Conservation of Aquatic Eco-System also provides assistance to the States for management of wetlands including Ramsar sites in the country.

Select the answer from the codes given below-

- (a) 1, 2, and 3 (b) 2, 3, and 4
(c) 1, 3, and 4 (d) All of the above

11. Consider the following statements

1. Tree Foundation, an NGO engaged in conservation of the sea turtle found more than 100 dead Olive Ridley Turtles in the shores of Nagapattinam.
2. The Olive Ridley turtles find the coastline of Nagapattinam as a favourable nesting habitat and that's why they reach to the shore from December to March every year.
3. The Olive Ridley looks very similar to the Kemp's Riddle, but has a deeper body and slightly up-turned edges to its carapace (shell).
4. Olive Ridley weighs around 45 kilograms and are 70cm in size and this makes them the smallest of the sea turtles along with Kemp riddles.

Which of the following statements are correct?

- (a) 1, 2 and 3 (b) 2, 3 and 4
(c) 1, 3 and 4 (d) All of the above

12. Which of the following statement is/are correct?

1. First Climate Change theatre in India (second theatre in the world) was opened at Pusa, New Delhi in January 2014.
2. The Inter-governmental Panel on Climate Change (IPCC) UN report on 17 January 2014 reported that during 2000 to 2010, the CO₂ has grown by 2.2 percent per year and this rise is almost twice higher from the growth of the period of 1970 to 2000.

Answer from the codes given below:

- (a) 1 only (b) 2 only
(c) 1 and 2 only (d) None of these

13. Which of the following two criteria have to be met in order to qualify as a 'biodiversity hotspot' on the world hotspots map?

1. The region must contain at least 0.5% or 1500 species of vascular plants as endemic species.
2. The region has to have lost at least 70% of its primary vegetation.

Which of the statements given above is/are correct?

- (a) 1 only (b) 2 only
(c) 1 and 2 both (d) None

14. Sumatran rhino populations have declined steadily to a point near extinction. Because of its population decline, this unusual forest dwelling rhino is near its :

1. carrying capacity
2. officially listed as threatened
3. critical number
4. officially listed as endangered

Which of the above is/are correct?

- (a) 1 and 2 (b) 3 and 4
(c) 1 and 3 (d) 2 and 3

15. Which of the following can be threats to the biodiversity of a geographical area?

1. Global warming
2. Fragmentation of habitat
3. Invasion of alien species
4. Promotion of vegetarianism

Select the correct answer using the codes given below:

- (a) 1, 2 and 3 (b) 2 and 3
(c) 1 and 4 (d) 1, 2, 3 and 4

16. The "Red Data Books" published by the International Union for Conservation of Nature and Natural resources (IUCN) contain lists of ?

1. Endemic plant and animal species present in the biodiversity hotspots.
2. Threatened plant and animal species.
3. Protected sites for conservation of nature and natural resources in various countries.

Select the correct answer using the codes given below:

- (a) 1 and 3 (b) 2 only
(c) 2 and 3 (d) 3 only

17. Consider the following statements:

1. The boundaries of a National Park are defined by legislation.
2. A Biosphere Reserve is declared to conserve a few specific species of flora and fauna.
3. In a Wildlife Sanctuary, limited biotic interference is permitted.

Which of the statements given above is / correct ?

- (a) 1 only (b) 2 and 3
(c) 1 and 3 (d) 1, 2 and 3

18. One Carbon Credit is defined as ____ .

1. Credit permit to release one ton of carbon dioxide.
2. providing loans to establish a unit which produces carbon dioxide for industrial use.
3. Finding out one new business which can use and recycle greenhouse gases.

Which of the following statements(s) is/are correct?

- (a) 3 only (b) 2 only
(c) 1 only (d) 1, 2 and 3

19. How does National Biodiversity Authority (NBA) help in protecting the Indian agriculture?

1. NBA checks the biopiracy and protects the indigenous and traditional genetic resources.
2. NBA directly monitors and supervises the scientific research on genetic modification of crop plants.
3. Application for Intellectual Property Rights related to genetic/biological resources cannot be made without the approval of NBA.

Which of the statements given above is /are correct?

- (a) 1 only (b) 2 and 3
(c) 1 and 3 (d) 1, 2 and 3

20. Three of the following criteria have contributed to the recognition of Western Ghats, Sri Lanka and Indo Burma regions as hotspots of biodiversity
1. Species richness
 2. Vegetation density
 3. Endemism
 4. Ethno-botanical importance
 5. Threat perception
 6. Adaption of flora and fauna to warm and humid conditions
- Which three of the above are correct criteria in this context?
- (a) 1, 2 and 6
 - (b) 2, 4 and 6
 - (c) 1, 3 and 5
 - (d) 3, 4 and 6
21. Biodiversity forms the basis for human existence in the following ways
1. Soil formation
 2. Prevention of soil erosion
 3. Recycling of waste
 4. Pollination of crops
- Select the correct answer using the codes given below:
- (a) 1, 2 and 3
 - (b) 2, 3 and 4
 - (c) 1 and 4
 - (d) 1, 2, 3 and 4
22. Which of the following can be threats to the biodiversity of a geographical area?
1. Global warming
 2. Fragmentation of habitat
 3. Invasion of alien species
 4. Promotion of vegetarianism
- Select the correct answer using the codes given below :
- (a) 1, 2 and 3
 - (b) 2 and 3
 - (c) 1 and 4
 - (d) 1, 2, 3 and 4
23. Consider the following statements :
1. The boundaries of a National Park are defined by legislation.
 2. A Biosphere Reserve is declared to conserve a few specific species of flora and fauna.
 3. In a Wildlife Sanctuary, limited biotic interference is permitted.
- Which of the statements given above is / correct ?
- (a) 1 only
 - (b) 2 and 3
 - (c) 1 and 3
 - (d) 1, 2 and 3
24. In which of the following States is lion-tailed macaque found in its natural habitat?
1. Tamil Nadu
 2. Kerala
 3. Karnataka
 4. Andhra Pradesh
- Select the correct answer using the codes given below.
- (a) 1, 2 and 3 only
 - (b) 2 only
 - (c) 1, 3 and 4 only
 - (d) 1, 2, 3 and 4
25. Consider the following:
1. Star tortoise
 2. Monitor lizard
 3. Pygmy hog
 4. Spider monkey
- Which of the above are naturally found in India?
- (a) 1, 2 and 3 only
 - (b) 2 and 3 only
 - (c) 1 and 4 only
 - (d) 1, 2, 3 and 4
26. Consider the following fauna of India:
1. Gharial
 2. Leatherback turtle
 3. Swamp deer
- Which of the above is/are endangered?
- (a) 1 and 2 only
 - (b) 3 only
 - (c) 1, 2 and 3
 - (d) None
27. If you walk through countryside, you are likely to see some birds stalking alongside the cattle to seize the insects disturbed by their movement through grasses. Which of the following is/are such bird/birds?
1. Painted Stork
 2. Common Myna
 3. Black-necked Crane
- Select the correct answer using the code given below.
- (a) 1 and 2
 - (b) 2 only
 - (c) 2 and 3
 - (d) 3 only
28. With reference to the International Union for Conservation of Nature and Natural Resources (IUCN) and the Convention on International Trade in Endangered Species of Wild Fauna and Flora (CITES), which of the following statements is/are correct? [CSAT 2015-1]
1. IUCN is an organ of the United Nations and CITES is an international agreement between governments.
 2. IUCN runs thousands of field projects around the world to better manage natural environments.
 3. CITES is legally binding on the States that have joined it, but this Convention does not take the place of national laws.
- Select the correct answer using the code given below.
- (a) 1 only
 - (b) 2 and 3 only
 - (c) 1 and 3 only
 - (d) 1, 2 and 3
29. With reference to an organization known as 'Birdlife International', which of the following statements is/are correct? [CSAT 2015-1]
1. It is a Global Partnership of Conservation Organizations.
 2. The concept of 'biodiversity hotspots' originated from this organization.
 3. It identifies the sites known/referred to as 'Important Bird and Biodiversity Areas'.
- Select the correct answer using the code given below.
- (a) 1 only
 - (b) 2 and 3 only
 - (c) 1 and 3 only
 - (d) 1, 2 and 3
30. Consider the following sanctuaries of India
1. Periyar
 2. Dachigam
 3. Sariska
 4. Kanha
- Which one among the following is the correct sequence of location of the above sanctuaries from South to North?
- (a) 1, 4, 2, 3
 - (b) 4, 1, 3, 2
 - (c) 1, 4, 3, 2
 - (d) 3, 1, 4, 2
31. Which of the following are UNESCO recognised world heritage sites?
1. Caves of Ajanta.
 2. Temple and Caves at Ellora.
 3. Mandapas of Mahabalipuram.
 4. Caves of Kanheri.

Select the correct answer using the codes given below

- (a) 1 and 4 (b) 1, 2 and 3
(c) 1, 3 and 4 (d) 2, 3 and 4

32. Consider the following statements

1. Jim Corbett National Park is the oldest national park of India.
2. It was one of the nine tiger reserves created at the launch of the Project Tiger in 1973.
3. Initially it was named as 'Hailey National Park'.

Which of the statements given above are correct?

- (a) 1 and 2 (b) All of these
(c) 2 and 3 (d) 1 and 3

Matching Based MCQ

Directions (Q. 33 to 38): Match List-I with List-II and select the correct answer using the codes given below the lists.

33. List-I

- A. Biodiversity
- B. Wildlife
- C. Ecosystem
- D. Ecology

List-II

1. G. Tansley
2. E.O. Wilson
3. E. Haeckel
4. W.T. Hornaday

Codes:

- | | A | B | C | D |
|-----|---|---|---|---|
| (a) | 2 | 4 | 3 | 1 |
| (b) | 2 | 4 | 1 | 3 |
| (c) | 4 | 2 | 3 | 1 |
| (d) | 4 | 2 | 1 | 3 |

34. Consider the following pairs:

1. Nokrek Bio-sphere Reserve : Garo Hills
2. Logtak (Loktak) Lake : Barail Range
3. Nandapha National Park : Daffa Hills

Which of the above pairs is/are correctly matched?

- (a) 1 only (b) 2 and 3 only
(c) 1, 2 and 3 (d) None

35. Consider the following pairs :

1. Dampa Tiger Reserve : Mizoram
2. Gumti Wildlife Sanctuary : Sikkim
3. Saramati Peak : Nagaland

Which of the above pairs is/are correctly matched?

- (a) 1 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

36.

List-I

(Malor Biome)

- A. The Northern most of the Temperate Formations
- B. Arctic Tundra Vegetation
- C. Marine
- D. The Terrestrial Biomes of the Tropics

List-II

(Physical characteristics)

1. Foristically poor (i.e., a continuous belt across North America and Northern Eurasia)
2. Boreal Forest
3. Pelagic division
4. Savanna woodland
5. Soviet Steppe and North American Prairie

Select the correct option from the codes given below:

- | | A | B | C | D |
|-----|---|---|---|---|
| (a) | 2 | 4 | 3 | 1 |
| (b) | 2 | 1 | 3 | 4 |
| (c) | 4 | 3 | 5 | 2 |
| (d) | 4 | 1 | 3 | 2 |

37. Match the following

List I	List II
(Biosphere Reserve)	(Places)
A. Manas	1. Meghalaya
B. Pachmarhi	2. Asom
C. Nokrek	3. Madhya Pradesh
D. Achanakmar Amarkantak	4. Chhattisgarh

Codes

- | | A | B | C | D | | A | B | C | D |
|-----|---|---|---|---|-----|---|---|---|---|
| (a) | 4 | 3 | 1 | 2 | (b) | 2 | 1 | 3 | 4 |
| (c) | 4 | 1 | 3 | 2 | (d) | 2 | 3 | 1 | 4 |

38. Match the following

List I	List II
(Biosphere Reserve)	(State)
A. Nilgiri	1. Odisha
B. Manas	2. Madhya Pradesh
C. Panchmarhi	3. Tamil Nadu
D. Simes lipal	4. Asom

Codes

- | | A | B | C | D | | A | B | C | D |
|-----|---|---|---|---|-----|---|---|---|---|
| (a) | 3 | 2 | 4 | 1 | (b) | 1 | 4 | 2 | 3 |
| (c) | 3 | 4 | 2 | 1 | (d) | 1 | 2 | 4 | 3 |

Hints and Explanations

EXERCISE-2

1. (a) 2. (a) 3. (a) 4. (c) 5. (d)
6. (a) 7. (a) 8. (c) 9. (b) 10. (a)
11. (d) 12. (b) 13. (c) 14. (b)
15. (a) Red Panda and Asiatic Wild Ass, are endangered species.
16. (d) The Nanda Devi National Park is a national park situated around the peak of Nanda Devi (7,816 m) in Uttarakhand.
17. (d) The Sundarbans is the largest single block of tidal halophytic mangrove forest in the world. The Sundarban forest lies in the vast delta on the Bay of Bengal formed by the super confluence of the Ganges, Padma, Brahmaputra and Meghna rivers across southern Bangladesh. It is a UNESCO World Heritage Site.
18. (a) Western Ghats are UNESCO World Heritage Site and is one of the eight “hottest hotspots” of biological diversity in the world.
19. (c) Ranganathittu Bird Sanctuary is located in Karnataka.
20. (c) Sualkuchi is located in Assam. This is the textile center of Assam. Muga silk and Pat silk along with Eri silk and Endi cloth from this region is famous for its quality.
21. (a) 1. Dudwa National park - Uttar Pradesh
2. Kanha National Park- Madhya Pradesh
3. Indravati National Park - Chattisgarh
4. Bandipur National Park- Karnataka
22. (c) Kanha National Park belongs to tropical moist dry deciduous forest. It is a tiger reserve of India and the largest national park of Madhya Pradesh.
23. (d) An endemic species is one whose habitat is restricted to a particular area. The term could refer to an animal, a plant, a fungus, or even a microorganism. Some of the endemic species in India are Grey-headed Bulbul, Malabar Lark, Nilgiri Flycatcher and Grey Jungle fowl etc.
24. (d) Veliconda Hills are situated in southeastern Andhra Pradesh state. They form the eastern flank of the Eastern Ghats.
25. (a) Biodiversity is richer in tropical regions. Biodiversity is a measure of the health of ecosystems. Greater biodiversity implies greater health. Biodiversity is in part a function of climate. In terrestrial habitats, tropical regions are typically rich whereas Polar Regions support fewer species.
26. (d) Namdapha National Park is located in Arunachal Pradesh. The climate of this area varies from tropical to subtropical, temperate and arctic. It is tropical and subtropical in southern regions and arctic type found in northern part of the park.

27. (a) The tropical moist deciduous forests are found in Sahyadris, the north-eastern parts of the peninsula and along the foothills of the Himalayas. Teak and sal are found in these forests.
28. (b) The Keibul Lamjao National Park is a national park located in Manipur. It is 40 km in area and the only floating park in the world which is located in North East India, and an integral part of Loktak Lake.
29. (c) The Gangetic dolphins have been declared as the National Aquatic Animal of India. River Dolphin is the National Aquatic Animal of India. The Ministry of Environment and Forests notified the Ganges River Dolphin as the National Aquatic Animal on 18th May 2010. This mammal is also said to represent the purity of the holy Ganga as it can only survive in pure and fresh water.

EXERCISE-2

1. (b) • Bhiterkanika, Odisha is a protected area for salt water crocodile, where breeding is the main purpose of that protected area.
• Great Indian Bustard is protected in desert area of Rajasthan.
• The Eravikulam National Park was established to protect the Nilgiri tahr (wild goat) species.
2. (c) To qualify as a hotspot, a region must meet two strict criteria: it must contain at least 1,500 species of vascular plants (> 0.5 percent of the world's total) as endemics, and it has to have lost at least 70 percent of its original habitat. So we choose Species richness as well as Endemism. Along with this Threat perception is necessary to take, because it makes the base of this concept. Adaptation of flora is an arbitrary option, Ethno-botanical importance does nothing with the Biodiversity Hotspot selection criteria, vegetation Density is also discarded.
3. (d) Biodiversity hot spots are located in temperate regions and hotspots are present in eastern Himalayas, Western Ghats and Andaman Islands.
4. (b) The red data book is contain only 8 lists of threatened plant and animal species.
5. (c) The National Biodiversity Authority (NBA) an autonomous body was established in 2003 to implement India's Biological Diversity Act (2002).
6. (b) Bandipur National Park, a tiger reserve is located in the south Indian state of Karnataka.
Manas National Park or Manas Wildlife Sanctuary is a National Park, UNESCO Natural World Heritage site, a Project Tiger Reserve, an Elephant Reserve and a Biosphere Reserve in Assam. The Sundarban National Park is a National Park, Tiger Reserve,

and a Biosphere Reserve in India. It is a part of the Sundarbans on the Ganges Delta of India and Bangladesh.

Bhitarkanika National Park is a national park located in the Kendrapara District Odisha, which is not specifically for Tiger reserve.

7. (a) Global Warming, fragmentation of habitat and invasion of alien species can be threats to the biodiversity of a geographical area.
8. (d) Biological diversity helps in the formation and maintenance of soil structure and the retention of moisture and nutrient levels. Trees on the other hand, lower the water table and remove deposited salt from the upper soil horizons.
9. (b) A biodiversity hotspot is a biogeographic region with a significant reservoir of biodiversity that is under threat from humans. Around the world, as many as 25 areas qualify to be the hotspots. Out of which India has 2 hotspots: Eastern Himalayas and Western Ghats.
10. (d) For conversion endangered species Wildlife Crime Control Bureau has been established for control of illegal trade in wildlife, including endangered species of birds and their parts and products. Research and monitoring activities on birds are promoted by the Government through reputed research organizations. Wildlife Institute of India, Bombay Natural History society and Salim Ali Centre for Ornithology and Natural History are some of the research organizations undertaking research on conservation of birds. The Indian government has banned the veterinary use of diclofenac drug that has caused rapid decline in vulture population across the Indian Subcontinent. Conservation Breeding Programmes to conserve these vulture species have been initiated at Pinjore (Haryana), Buxa (West Bengal) and Rani, Guwahati (Assam) by the Bombay Natural History Society.
11. (d) The Olive Ridley turtles are rusty coloured carapace and have slightly smaller head and shell than the Kemp turtles. These Olive Ridley turtles generally occur through the Antilles, around the north coast of South America, in West Africa, the Indian Ocean, Australia and Southeast Asia. As per the reports the populations of Olive Riddles have declined in Pakistan, Myanmar, Malaysia and Thailand, and possibly on the east coast of India, south of Orissa and in the Andaman and Nicobar islands.
12. (b) First Climate Change theatre was opened at Pushpa Gujral Science City in Kapurthala, Punjab on 16 January 2014. The theatre will educate people on climate change. This is the second theatre in the world to be opened after Canada. The theatre is 18 metre in diameter and it is set up in a dome-shaped building with a seating capacity of 125 persons. The 25-minute film shows what worst can happen if humans do not take action on current or impending problems which could threaten civilization. The film starts by giving a glimpse of future - floods, droughts, earthquakes and other natural disasters.
13. (c) To qualify as a biodiversity hotspot on Myers 2000 edition of the hotspot-map, a region must meet two strict criteria: it must contain at least 0.5% or 1,500 species of vascular plants as endemics, and it has to have lost at least 70% of its primary vegetation.
14. (b) The Sumatran Rhino Crisis Summit opened with the shocking news that rather than 130-190 Sumatran rhinos as previously estimated, there are in fact fewer than 100 individual animals.
15. (a) Except promotion of vegetarianism all other acts are threats to the biodiversity of a geographical area.
16. (b) The red data book contains only 8 lists of threatened plant and animal species.
17. (c) The fix boundary of national park is described in Wild Life Protection Act, 1972 and the actual area of the national park is notified by state government. A biosphere reserve conserves an ecosystem and not just few specific species of plants and animals.
18. (c) A carbon credit is a generic term for any tradable certificate or permit representing the right to emit one tonne of carbon dioxide or the mass of another greenhouse gas with a carbon dioxide equivalent (tCO₂e) equivalent to one tonne of carbon dioxide.
19. (d) National Biodiversity Authority (NBA) checks the biopiracy and protects the indigeneous and traditional genetic resources. It directly monitors and supervises the scientific research on genetic modification of crop plants application for intellectual property Rights related to genetic biological resources cannot be made without the approval of NBA.
20. (c) To qualify as a hotspot, a region must meet two strict criteria: it must contain at least 1,500 species of vascular plants (> 0.5 percent of the world's total) as endemics, and it has to have lost at least 70 percent of its original habitat. So we choose Species richness as well as Endemism. Along with this Threat perception is necessary to take, because it makes the base of this concept. Adaptation of flora is an arbitrary option, Ethno-botanical importance does nothing with the Biodiversity Hotspot selection criteria, vegetation Density is also discarded.
21. (d) Biological diversity helps in the formation and maintenance of soil structure and the retention of moisture and nutrient levels. Biodiversity supports ecosystem services including air quality, climate water purification, pollination, and prevention of erosion.

22. (a) Except promotion of vegetarianism all other acts are threats to the biodiversity of a geographical area.
23. (c) The fix boundary of national park is described in Wild Life Protection Act, 1972 and the actual area of the national park is notified by state government. A biosphere reserve conserves an ecosystem and not just few specific species of plants and animals.
24. (a) Lion-tailed Macaques are found in the mountain forests scattered across three Indian states stated above. The lion-tailed Macaques are endangered as per IUCN.
25. (a) Star tortoise is found in India in the dry and scrub forests. Pygmy Hog is an endangered species found in Assam. Only 150 animals are left. Monitor Lizard is found in India, Sri Lanka and Pakistan. Spider Monkey is the inhabitant of tropical forests of Central and South America.
26. (c) Gharial is critically endangered according to IUCN. Overhunting for skin and trophies, habitat loss due to construction of dams and barrages has been the reason for their decline. Leather back turtles are endangered due to human carelessness. Swamp deer occupies a place in the list of the endangered species of the world. Deforestation, draining of swamps and marshes for farming has led to the destruction of their natural habitat.
27. (b) Common Myna are birds stalking alongside the cattle to seize the insects disturbed by their movement through grasses. The common myna is readily identified by the brown body, black hooded head and the bare yellow patch behind the eye.
28. (b) IUCN is not an organ of UN. It has observer and consultative status at the United Nations.
29. (c) BirdLife international is a global partnership of conservation organisations that strives to conserve birds, their habitats and global biodiversity. It is working with people towards sustainability in the use of natural resources. It is the World's largest partnership of conservation organisations, with over 120 partner organizations. An Important Bird and Biodiversity Area (IBA) is an area recognized as being globally important habitat for the conservation of bird populations. The program was developed and sites are identified by BirdLife International.
30. (c) Periyar National Park and Wildlife Sanctuary is a protected area in the districts of Idukki and Pathanamthitta in Kerala. Kanha National Park is one of the tiger reserves of India and the largest national park of Madhya Pradesh. The Sariska Tiger Reserve is an Indian national park located in the Alwar district of Rajasthan. Dachigam National Park is located in Jammu and Kashmir.
31. (b)
32. (b) All statements are correct.
33. (b) The concept of biodiversity was propounded by E.O. Wilson. The concept of wildlife was propounded by W. Hornaday. The concept of Ecosystem propounded by G. Tansley. The concept of Ecology was propounded by E. Haeckel.
34. (a) Nokrek Biosphere Reserve is situated in Garo Hills in Meghalaya. Logtak Lake is in Manipur. Barail Range is in Assam. Though Namdapha National Park and Dafla Hill both are in Arunachal Pradesh, the two are separate entities.
35. (c) Dampa Tiger Reserve, the largest wildlife sanctuary in Mizoram. Saramati peak is in Nagaland. It is located near Tuensang town with a height of 3,826 m. Gumti Wildlife Sanctuary is famous wildlife reserve in Tripura.
36. (b)
37. (d) 1. Manas- Asom
2. Pachmarhi - Madhya Pradesh
3. Nokrek - Meghalaya
4. Achanakmar-Amarkantak- Chhattisgarh,
38. (c) The Nilgiri Biosphere Reserve is an International Biosphere Reserve in the Western Ghats and Nilgiri Hills ranges of South India. Manas Wildlife Sanctuary is situated in Assam and UNESCO Natural World Heritage site which is a Project Tiger Reserve. The Panchmarhi Biosphere Reserve is situated in the Satpura Range of Madhya Pradesh state. Simlipal National Park is a national park and a tiger reserve situated in the Indian state of Odisha.



ENVIRONMENTAL ISSUES

3

Chapter

Introduction

Environmental issues are harmful effects of human activity on the biophysical environment. Environmentalism, a social and environmental movement, addresses environmental issues through advocacy, education and activism.

Our environment is constantly changing, which no one can deny. With these great environment changes, it becomes highly important for us to become increasingly aware of the environmental problems as well. With a monumental inundation of natural disasters, warming and cooling periods, different types of weather forms and much more, people should be aware of what types of environmental problems our Earth is facing.

Our planet is on the verge of a severe environmental crisis. Current environmental problems make us susceptible to disasters and tragedies, now as well as in the future. We are in a phase of planetary emergency, with environmental problems blooming around us. Unless we address the various issues proactively and sincerely we are surely going to be wrecked with these disasters. All the current environmental problems need urgent attention.

DIFFERENT ENVIRONMENTAL ISSUES AND ITS EFFECT ON CLIMATE

Environmental issues are increasing day by day and it has a very adverse effect on climate. Some of the Environmental issues are discussed below:

Global Warming

Climate changes like global warming are the result of human practices like emission of Greenhouse gases. Global warming leads to rising temperatures of the oceans and the earth's surface causing melting of polar ice caps, rise in sea levels and also unnatural patterns of precipitation such as flash *floods*, excessive snow or desertification.

Effects of Global Warming

The Intergovernmental Panel on Climate Change (IPCC) was established in 1988 by the World Meteorological Organization (WMO) and the United Nations Environment Programme (UNEP) in recognition of the problem of global warming. IPCC has estimated the following effects of global warming:

- Earth's temperature will rise by 1-30°C in next few decades, leading to extreme weather changes (heat waves, hurricanes and severe winters), changes in ocean currents and marine life. The largest glacier chain in the tropics is melting fast because of rising temperatures and peaks are turning brown. This trend is endangering future water supplies. Glaciers serve agriculture, hydel plants and feed rivers that supply water to the sprawling cities and shanty towns on Peru's bone-dry Pacific coast. **Quelccaya**, in southern Peru, the world's largest tropical ice-cap, is retreating at about 200 feet per year, up from 20 feet per year in the 1960s. **Lonnie Thompson**, a leading glacier expert of Ohio State University, monitoring glacier retreat on the Andes, Himalayas and Kilimanjaro, said that the rate of ice loss in glaciers all over the world is actually accelerating.
- If CO₂ concentration doubles, Earth's temperature may rise by 50°C. Coastal areas will see a rise in water levels by 0.5 - 5.0 feet due to melting of mountain glaciers, polar ice-caps, etc.

- Islands like **Maldives** would get submerged. In 1999, two uninhabited islands in the South Pacific (**Tebua Tarawa and Abanuea**) were submerged by rising sea levels, and two neighboring inhabited islands (**Kiribati and Tuvalu**) are on the brink of submersion as well.
- The biggest glacier in the **Peruvian Andes** was retreating by 5 meters per year some 20 years ago; today it is shrinking by 33 meters per year. The second largest glacier on Earth, the Greenland ice sheet, is thinning at an unprecedented rate of one meter each year.
- The Arctic Sea ice has thinned by 40% in the last two decades, while Mount Everest is losing height at the rate of 1.5 meters per year.

As global warming's, terrifying threat increases, our planet's nations come together at the paries climate conference (2015) to fight for our future.

In the wake of the paries conference America must lead the fight against global warming. We need to embrace clean energy and leave our dirty fossil fuels in the ground.

Here are five key steps president Obama and other U.S. leaders should take to protect our planet.

(i) Suppost a just, ambitions and binding international climate treaty:

Under the paris framework and beyond, the U.S. should back efforts to end fossil fuel use in developed nations by 2050.

(ii) Strengthen the clean power plan:

The U.S. needs to move rapidly away from all fossil fuels and toward wildlife-friendly sources of clean energy.

(iii) Cut pollution from airoplanes and other unregulated sources:

The Environmental protection Agency acknowledges that airoplane pollution endangers our climate.

(iv) Halt new fossil fuel development in America's oceans and on our public lands:

Ending new fossil fuel leasing on public lands and offshore areas controlled by the U.S. would keep upto 450 billion tons of green house gases from polluting the atmosphere, according, to a recent analysis prepared for the centre by scientists at *Ecoshift*.

(v) Crack down on fossil fuel exports and transport

We must halt the dangerous push to send America's dirty fossil fuels abroad.

Green House Effect

The Earth gets energy from the sun in the form of sunlight. The Earth's surface absorbs some of this energy and heats up. That's why the surface of a road can feel hot even after the sun has gone down because it has absorbed a lot of energy from the sun. The Earth cools down by giving off a different form of energy, called infrared radiation. But before all this radiation can escape to outer space, greenhouse gases in the atmosphere absorb some of it, which makes the atmosphere warmer. As the atmosphere gets warmer, it makes the Earth's surface warmer, too. Without this Greenhouse effect the Earth would be at least 30 degrees cooler, in which life would not exist.

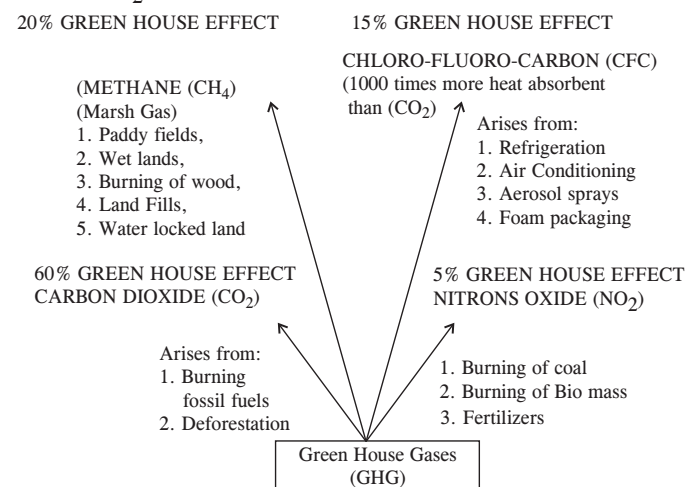
Greenhouse Gases

A greenhouse gas (GHG) is a gas in an atmosphere that absorbs and emits radiation within the thermal infrared range. This process is the fundamental cause of the greenhouse effect. The primary greenhouse gases in Earth's atmosphere are *water vapour, carbon dioxide, methane, nitrous oxide, ozone, and chlorofluorocarbons*.

- Water vapour** contributes to 36-72% of Greenhouse effect.
- Carbon Dioxide (CO₂)** arises from burning fossil fuels and as a result of deforestation. It contributes to 9-26% of Greenhouse effect. It is the primary greenhouse gas emitted through human activities.
- Methane (CH₄)**, also called "*Marsh gas*", arises from rice paddies, wetlands, enteric fermentation in cattle, burning of wood, and landfills. It is responsible for about 4-9% of Greenhouse effect.
- Nitrous Oxide (NO₂)** contributes (5%) which arises from coal burning, biomass burning, and breakdown of chemical fertilizers.
- Chlorofluorocarbons (CFCs)** and their replacements (15%) are 1000 times more heat absorbent than carbon dioxide. They reach the atmosphere from *refrigeration and air conditioning, aerosol sprays, and foam packaging industry*.
- Ozone** contributes to 3-7% of Greenhouse effect. The largest net source of tropospheric ozone is influx from the stratosphere. Large amounts of ozone are also produced in the troposphere by photochemical reactions, the amounts increasing with high levels of air pollution.

Unfortunately, recent human activities such as burning fossil fuels to run automobiles, heat homes and businesses, and power factories are causing increased concentrations of greenhouse gases, thereby resulting in more heat being trapped. The planet is losing less heat and, as a result we are beginning to experience Global Warming.

Estimates indicate there has been a 25% increase in CO₂, concentration in the last 100 years and this is expected to double in the next 50 years, e.g. **Brazil** alone contributes billions of tons of CO₂, every year due to *deforestation*.



Ozone depletion

Ozone (O_3) is a gas found throughout the atmosphere, but most highly concentrated in the stratosphere, between 10 and 50 km above the sea level, where it is known as the “Ozone layer”.

- This Ozone layer forms a *protective shield* for the earth from the harmful *ultra-violet radiation* from outer space, particularly *UV-B* rays which affects *DNA molecules*, causing damage to the outer surface of plants and animals and also marine life. In humans it causes skin cancer, eye cataracts and is a general immuno-suppressant.
- “**Ozone Holes**” were first discovered over *Antarctica* by the British Antarctica Survey in 1983. Levels of ozone are dropping very fast, resulting in parts of the layer becoming thin and ‘holes’ developing because only a small percentage of O_3 gets naturally replenished every year.
- In 1974, **Mario Molina** and Sherwood Roland of the **University of California** discovered that a group of synthetic chemical substances known as *CFCs* and *HCFCs* destroy ozone in the stratosphere. These chemicals are inert, non-flammable, non-toxic, and lighter than air and can remain intact for years. They contain Chlorine and Fluorine, common being CFC-II, CFC-12, CFC-22 and CFC-13.
- ‘**Halons**’ containing ‘**Bromine**’ and used in the fire-fighting industry, are 100 times more potent than CFCs. CFCs are commonly used in Air-conditioners and the Refrigeration industry (Freon gas), aerosol propellants (in perfumes and deodorants), in the foam packaging industry (Styropor, Thermocol) and as solvents for greases and glues.
- Compounds like Carbon Tetrachloride and Methyl Chloroform are also found to release Chlorine (Halogens) which ultimately destroy the stratospheric Ozone.
- Du Pont (USA) and ICI (UK) have developed certain substitutes like HFC (Hydrofluorocarbon) and HCFC (Hydro-Chloro-Fluoro Carbon), e.g. HCFC-123 which contain less Chlorine than CFC, but these are not effective, permanent solutions.
- Cheaper alternatives for refrigeration being developed are Propane and Ammonia as coolants, which are completely environment friendly.

Deforestation

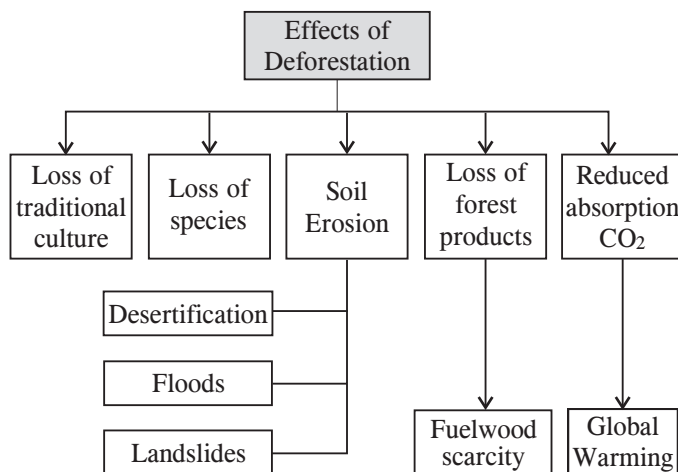
The process of clearance of forest by burning or logging is called deforestation. The main reasons for deforestation are trees or derived charcoal are used as, or sold, for fuel or as a commodity, while cleared land is used as grassland for livestock, plantations of commodities, and settlements. Deforested areas usually sustain extensive adverse soil erosion and regularly damage into wasteland.

Causes of deforestation

There are numerous causes of deforestation such as

- Expansion of farming land
- Logging and fuel wood
- Overgrazing

- Fires
- Mining
- urbanization/Industrialization and Infra-structure.
- Air Pollution
- Wars and role of military
- Tourism
- Over population and poverty
- land rights, land tenure and inequitable land distribution resources
- Economic, i.e. development/land conversion value, fiscal policies, etc.
- Under valuing the forest
- Corruption and political cause



Strategies to reduce deforestation

Strategies to reduce deforestation should be a combination of pro-active role of national, state, municipalities, civil societies and private sector in the following ways-

- Reducing Population growth
- Reducing emissions from deforestation and forest degradation
- Increase the area and standard of management of protected areas
- Increase the area of forest reserved for timber production
- launch the mass awareness programme regarding value of forest.
- Encouraging substitutes
- Increase area of forest plantation
- Government initiatives through policies and action-plan
- Participatory forest management and rights
- Increase investment in research, education and extension
- Improve the information base and monitoring

Air Pollution

The main issue among environmentalists and researchers especially in developed countries is *air pollution*.

- The main pollutants of air pollution are *particulate matter*, *PAHs*, *lead*, *ground-level ozone*, *heavy metals*, *Sulphur dioxide*, *benzene*, *carbon monoxide* and *nitrogen dioxide*.

- Air pollution is also responsible for climate change due to the higher greenhouse effect, acid rain, and the depletion of the ozone layer that constitute important global environmental problems.
- Air pollution is the main reason of ill health and death by natural and man-made sources.
- Tobacco smoke, house cleaning items, insecticides industries, automobiles, power generation, combustion of solid fuels for cooking, poor maintenance of cars and other automobile, etc. are the main cause of air pollution.
- Air pollution can be of two types: *indoor* and *outdoor*.
- *Indoor air pollution* is restricted to buildings only. It is the amount of chemical, biological and physical contaminants in the air inside a building. Building materials, central heating and cooling devices, painting colours, stoves, gas heater, and tobacco smoke, etc. are the examples of indoor air pollution.
- Release of several air pollutants into the atmosphere which causes severe threat to living organisms or upsetting the functioning of environment is called outdoor air pollution.

Contributors of Pollutants

- There are natural pollutants of air like animal decay.
- Volcanic eruptions release more sulphur fumes than all power plants and all industries in the world.
- Lightning bolts create nitrogen oxides just as automobiles and industrial furnaces do.
- Trees emit hydrocarbons called **terpenes** causing bluish haze.

Effects

- There is mounting evidence that our whole planet is affected by pollution. The South Pole seems fairly clean because 90 % of earth's population lives in northern hemisphere. Yet in 1985 scientists detected a major hole in protective ozone screen over Antarctica.
- The North Pole, on the other hand, resembles a cool town. In winter when the Arctic is tilted into long nights and the sun cannot generate cleansing winds and precipitation, the largest single mass of pollution sits atop the globe like a dirty cap composed of mixture of gases and particles, sulphates and soot.
- Tracing pollutants has become a vital necessity as air currents do not respect political boundaries. Many countries are blaming their neighbors for polluting their air space. Air pollutants weaken the trees and then they are killed by drought and pathogens. Trees weakened by climatic variations are finished off by air pollution. Perhaps the most controversial issue of the decade is *acid rain*. We are trying to understand the full effects on an atmosphere acidified by burning fossil fuel.

For millions of years the ingredients of such substances have been cycling through the ecosystem, constantly changing their form. They pass in animal and plant tissues, sink in sea, return to the earth and are vaulted aloft in some geologic event to begin the cycle again. An atom of oxygen completes this cycle generally once in 2000 years. We are in the midst of a chemical revolution in which some 65,000 commercial compounds enter our environment each year. Some are proven carcinogens. A modern city suffers from many air pollutants which form a complex mixture of smog.

Pollutants and their Effects

Sr. No.	Pollutant	Origin	Effect
1.	Arsenic (As)	Coal, oil furnaces, glass factories	Lung and skin cancer
2.	Benzene (C ₆ H ₆)	Refineries, motor vehicles	Leukemia
3.	Cadmium (Cd)	Smelters, coals, oil furnaces	Damage to lung, kidney, bones
4.	Chlorine (Cl)	Chemical Industries, volcanic activities	Causes irritation
5.	Carbon monoxide (CO)	Motor vehicles, smelters, coal steel plants	Starves body of oxygen, damages heart
6.	Fluoride (F)	Smelters, steel plants	Mottles teeth in children
7.	Hydrocarbons	Unburnt gasoline fumes, motor vehicles	Combines with nitrogen oxides in sunlight to form smog, causes irritation in eyes and nose
8.	Formaldehyde (HCHO)	Chemical plants	Allergenic, carcinogenic, headaches, burning sensation in the throat, and can aggravate <i>asthma symptoms</i>
9.	HCl (Hydrogen chloride)	Incinerators	Irritates eyes and lungs
10.	Hydrogen fluoride (HF)	Fertilizer plants Smelters	Irritates skin, eyes, mucous membrane
11.	Mercury (Hg)	Coal, smelters oil furnaces	Tremors, nerve troubles
12.	Nitric acid (HNO ₃)	Formed from NO ₂ causes acid rain	Respiratory diseases
13.	Nitrous acid (HNO ₂)	Formed from NO ₂ and water vapour	Respiratory disease
14.	Hydrogen sulphide (H ₂ S)	Refineries, Pulp mills	Nausea, irritates eyes
15.	Sulphuric acid (H ₂ SO ₄)	Formed from SO ₂ in sunlight with	Respiratory diseases hydroxyl ions
16.	Manganese (Mn)	Steel and sulphur dioxide power plants	Parkinson's diseases
17.	Nickel (Ni)	Smelters, coal, oil	Lung Cancer furnaces
18.	Nitric Oxide (NO)	Motor Vehicles, coal, oil furnaces	Oxidizes to NO ₂

19.	Nitrogen dioxide (NO ₂)	Formed in Sunlight from NO	Bronchitis Loss of resistance to influenza forms Ozone
20.	Ozone (O ₃)	Ground level ozone formed from nitrogen oxides (NO _x) and volatile organic compounds (VOCs)	Asthma, irritates eyes sunlight from nitrogen oxides and hydrocarbons
21.	Lead (Pb)	Motor vehicles, high smelters	Brain damage
22.	Silicon Tetra fluoride (SiF ₄)	Chemical plants	Lung diseases
23.	Sulfur dioxide (SO ₂)	Smelters Coal, Oil furnaces	Irritates eyes, breathing problems

Prevention of air pollution

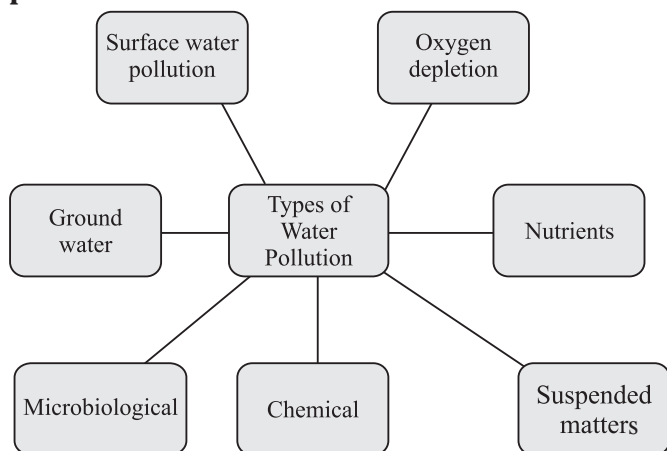
Air pollution can be reduced with the help of-

- using smokeless sources of energy like smokeless stoves, biogas, solar and wind energy
- using devices for filtering smoke in chimneys of factories and power houses
- Planting more trees
- Locating industries away from residential areas
- Strictly checking pollution levels in automobiles exhaust emission.

Water Pollution

- Water pollution is a kind of pollution which involves the contamination of water sources or bodies on which several aquatic animals depends on for their life support.
- Polluted water comprises of Industrial discharged wastes, sewage water, and rain water pollution.
- Quality of soil and vegetation is affected by the polluted water. Pollutants in water comprise a extensive kind of chemicals, pathogens, and physical chemistry or sensory changes. Many of the chemical substances are toxic or even dangerous.
- Pathogens can produce water borne disease in humans and animals.
- Polluted water is discharged in water polluting the aquatic flora and fauna.
- Washing clothes near lakes and rivers is one of the reason of water pollution since, detergents cause a condition called **“Eutrophication”** which blocks sunlight from entering inside that water body thus reducing oxygen standards in the water and causing an inhabitable environment.

Types of Water Pollution based on causes & pollutants

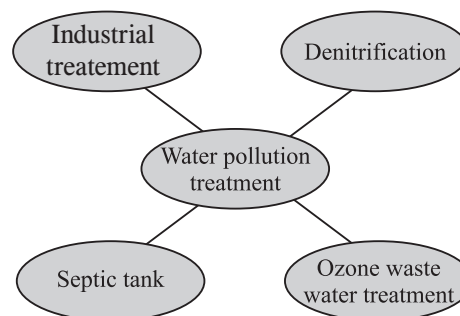


Diseases caused by Water Pollution

The most common water pollution diseases involve digestive system and infections diseases, it may cause many others like-

- Infections diseases caused by pathogens* from animal fecal origins involving :- Typhoid, Giardiasis, Amoebiasis, Ascariasis, Hookworm
- Diseases caused by polluted beach water* :- Gastroenteritis, Diarrhea, Encephalitis stomach craps and aches, vomiting, Hepatitis, Respiratory infections
- Liver damage and cancer caused by chlorinated solvents, MTBE.
- Kidney damage caused by a series of chemicals found in contaminated water.
- Neurological problems due to pesticides (eg. DDT)
- Reproductive and endocrine damage
- Bathing in polluted water causes rashes, ear aches and pink eyes.

Water Pollution treatment



Water can be treated by many ways, i.e. denitrification, industrial treatment, Septic tank and Ozone waste water treatment. Raw sewage should be treated in water treatment plant before releasing it in environment. In water treatment plant sewage goes through many chambers and chemical processes which reduce its toxicity. Denitrification is an ecological method to prevent the discharge of nitrates in soil, and stops ground water pollution with nutrients. Septic tanks treat sewage at the place where it is located and used to treat sewage from an individual building. Untreated sewage from a property streams into the septic tank and the solids are separated from the liquid. Breaking of pollutants into water sources is done by ozone generator. By using Ultraviolet radiation and Electric discharge field oxygen is converted into ozone by the generators.

BOD (Biochemical Oxygen Demand)

BOD (Biochemical Oxygen Demand), also often referred to as biological oxygen demand, is a test performed to measure the potential of wastewater and other waters to deplete the oxygen level of receiving waters. In other words, the BOD test is performed to determine what effect dirty water, containing bacteria and organic materials, will have on animal and plant life when released into a stream or lake. When there is an abundance of bacteria and organic materials, the bacteria will take in oxygen in order to breakdown these molecules. If bacteria are taking in large amounts of oxygen, this will have a detrimental effect on the surrounding ecosystem. On the contrary, when there are low levels of organic waste in the water, there are fewer bacteria present, the BOD will be lower and the dissolved oxygen levels higher. In wastewater treatment plants, they often calculate the percentage removal of BOD to determine the efficiency of the treatment process. For this reason, BOD is sometimes referred to as a water contaminant.

A BOD level of 1-2 ppm is considered very good. There will not be much organic waste present in the water supply. A water supply with a BOD level of 3-5 ppm is considered moderately clean. In water with a BOD level of 6-9 ppm, the water is considered somewhat polluted because there is usually organic matter present and bacteria are decomposing this waste. At BOD levels of 100 ppm or greater, the water supply is considered very polluted with organic waste.

A pH of 6.5 to 8.2 is optimal for most organisms. Rapidly growing algae or submerged aquatic vegetation remove CO₂ from the water during photosynthesis, significantly increasing pH levels. pH levels > 9.0 begin to be harmful to salmonids (trout) and perch. Rainwater naturally has a pH of 5.5; pH < 5.5 is harmful to freshwater shrimp, snails, and clams; metals normally trapped in sediments may be released into the acidified water.

Acidic < 6.5 pH

Basic > 7.5 pH

Sound Pollution

Unwanted sounds created by humans, animals and machines which disturbs the environment and humans is called as sound pollution. The word noise comes from the Latin word *nausea* meaning *seasickness*.

Sources of Sound

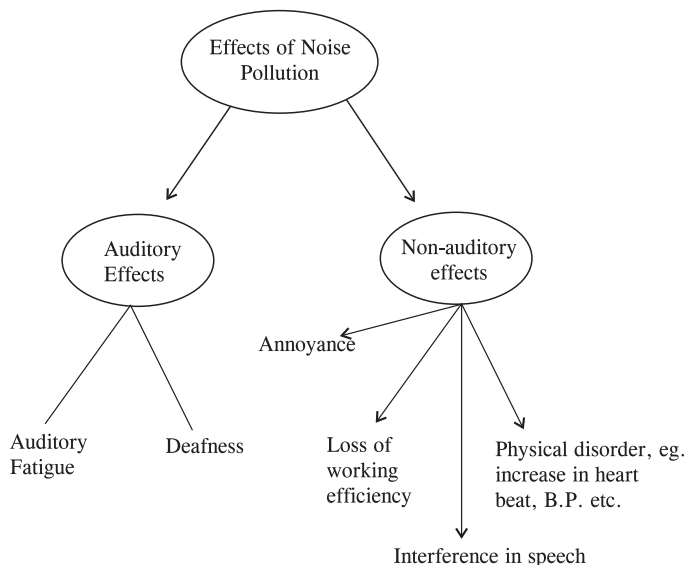
The main source of noise is transportation system including rail noise, aircraft noise and vehicle noise. People living near factories experience sound pollution because of the unwanted sounds coming from factories. Other sources of sound pollution are car alarms, emergency service sirens, office equipment, factory machinery, construction work, grounds keeping equipment, barking of dogs, appliances, power tools, lighting hum, audio entertainment systems, loudspeakers, and noisy people. Use of loudspeakers for political purposes and other purposes is also the cause of sound pollution.

Measurement of sound

Sound pollution is measured in *decibels*. Humans can't sleep at 45 decibels; hearing begins to damage in 85 decibels and pain in ears start at 120 decibels.

Effects of sound pollution

Health and behaviour of humans are disturbed by the noise pollution. Unwanted sounds can damage physiological and psychological health. Sound pollution can cause annoyance and aggression, hypertension, high stress levels, tinnitus, hearing loss, sleep disturbances, and other harmful effects. Due to increase in sound level there can be lack of concentration at work which can lead to low productivity and performance. High sound levels can increase in cardiovascular effects in humans which is very dangerous for health.



Central Government's Regulation & Control of Noise Pollution

These regulations are meant for the following :

- (i) Implementation of noise standards in different zones/ areas.
- (ii) Restrict the use of loud-speakers.
- (iii) Restrict the over-usage of horns , sound creating equipments for construction and fire-crackers.
- (iv) Allotting responsibility to state pollution control boards and central pollution control board to provide data about the noise pollution, so that measures may be taken to control noise pollution.

Agro-chemicals & Its effect

Agro chemicals are developed by the use of modern technology that depends on inorganic fertilizers and pesticides. Excess use of these fertilizers can lead to immediate harmful effect or can also be long lasting. Although many benefits are there by the use of agro chemicals which are related to increase yield of plants and animal crops and less wastage during storing. These profits are substantial. In combination with genetically enhanced varieties of crop species, agrochemicals have made significant contributions to the accomplishments of the "green revolution." However there are certain environmental and ecological damages also related to the use of agro chemicals. For example excess use of fertilizers can contaminate the ground water with nitrate, making it unfit for the consumption of humans

and livestock. If there is large concentration of nitrogen in water it can poison animals by immobilizing some hemoglobin in blood and hence reducing the ability to transport oxygen. However if the fertilizers are in the water such as streams, lakes, etc. can cause an increased productivity of those aquatic ecosystems, a problem known as eutrophication. Due to eutrophication there can be excessive growth of algae, wide mortality of fish and other aquatic animals and a bad taste of water.

There can be many environmental problems due to the use of pesticides. As we all know that pesticides are used to reduce the abundance of species of pests. But while using pesticides humans also come in contact with them which are harmful to them. When the entire fields are sprayed with pesticides by the use of tractor or airplane or helicopter many non-targeted organisms come in contact with the pesticides. This happens on the treated site, and also on nearby off-sites as a result of “drift” of the sprayed agrochemical. These non-target contacts cause several unnecessary poisonings and deaths of organisms that are not agricultural pests.

There is a global contamination of environment with pesticides such as *DDT*, *Dieldrin*, and *Aldrin*. This contamination includes the extensive presence of pesticide residues in almost all wildlife, well water, food, and even in humans. Residues of some of the chemicals used in animal husbandry are also thought by some people to be a problem, for example, when traces of antibiotics and bovine growth hormones arise in consumer products such as meat or milk.

The worst examples of the use of pesticides are the use of DDT. Modern use of pesticides includes the use of pesticides that are less persistent than DDT and related chlorinated hydrocarbon. However humans are also at the risk of the use of some pesticides. There are about almost one million pesticides poisoning all over the world with nearly 20,000 fatalities. About one-half of the human poisonings happen in poorer, less-developed countries, even though these places account for only 20% of the world's use of pesticides. This is due to the illiteracy in these countries and to negligent enforcement of regulations about the use of pesticides.

Example of the damage cause by the pesticides to humans includes *Bhopal tragedy* in India which occurred in 1984, in the area of a factory that was manufacturing an *agricultural insecticide*. In that case, there was an accidental discharge of about 45 tons (40 tonnes) of deadly *methyl isocyanate* vapor to the atmosphere. This agrochemical-related emission caused the deaths of about 3,000 people, and more than 20,000 others were seriously injured.

Researchers are continuously searching for non-chemical means of dealing with several of these agricultural requirements. Organic methods are been invented in enhancing the soil fertility and dealing with pests. Therefore, modern agricultural industries will continue to depend on heavily on the use of agrochemicals to achieve their problems of fertility, soil quality, and pests.

Acid Rain

Acid rain is a rain or any other form of precipitation that is unusually acidic, meaning that it possesses elevated levels of hydrogen ions (low pH). The term “**acid rain**” was coined

in 1872 by **Robert Angus Smith**, after a link was established between *sulfur dioxide* (SO_2) emissions from the burning of coal in Manchester and acidification of nearby rainfall. *Rainfall with pH less than 5.6 is called Acid rain.*

Sources of Acid Rain

- Acid rain is caused by a chemical reaction that starts when sulfur dioxide and nitrogen oxides are released into the air.
- These compounds can rise very high into the atmosphere, where they mix and react with water, oxygen, and other chemicals to form more acidic pollutants, called as acid rain. Sulfur dioxide and nitrogen oxides are highly soluble in water and can be carried very far by the wind.
- Consequently, the two compounds can travel long distances where they become part of the rain, sleet, snow, and fog that humans experience on specific days.
- Though there are few natural causes also for acid rain, but still human activities are considered to be the main cause of acid rain.
- During last few decades, human beings have released various chemicals into the air, which has altogether changed the composition of gases in the atmosphere. Power plants release huge amount of sulfur dioxide and nitrogen oxides when they burn fossil fuels, like coal, to produce electricity. Additionally, the exhaust from light and heavy vehicles releases nitrogen oxides and sulfur dioxide into the air. These pollutants are the main cause of acid rain.

Effects of Acid Rain

- Acid rain leads to acidification of water bodies such as lakes and streams. It contributes to the damage of trees at high elevations (e.g. red spruce trees at the height of 2,000 feet and above) and many sensitive forest soils.
- Additionally, acid rain stimulates the decay of building materials and paints. Before reaching the earth, sulfur dioxide and nitrogen oxide gases and their particulate matter derivatives (**sulfates** and **nitrates**) lead to visibility degradation and harm people health.
- Acid rain has been held responsible for ruining the marble walls and pillars of one of the Seven Wonders of the World, *Taj Mahal* in India.
- Acid rain reacts with calcium to form calcium bicarbonate, which can be easily washed away.
- **St. Paul's Cathedral in London** and the Statue of Liberty in New York are known to be few victims of acid rain. As the concentration of Sulphur dioxide is increasing in the air of Delhi, there may be a danger of corrosion of the Red Fort and similar other historical buildings and monuments made up of stones.
- In Calcutta also, architectures such as the marble-built Victoria Memorial Hall may be in similar danger in the near future.

Measures to control Acid Rain

Some of the major procedures that must be followed to control acid rain are as follows:

- Reduce amount of sulphur dioxide and oxides of nitrogen released into the atmosphere.
- Use cleaner fuels
- Flue (waste) gas desulphurisation (FGD)
- Use other sources of electricity (i.e. nuclear power, hydro-electricity, wind energy, geothermal energy, and solar energy).
- Reduce the effects of Acid Rain by liming the soil and water.

Solid Waste Management

Solid waste management is one of the major challenges faced by many countries around the globe. Inadequate collection, recycling or treatment and uncontrolled disposal of waste in dumps can lead to severe hazards, such as *health risks* and *environmental pollution*.

No other pollutant is discussed about more vociferously among environmentalists, politicians and the people at large than the garbage which is the *bulky plastic* and refuse heap variety trash that accumulates in different corners of the cities. These piles are quite often taller than the city halls. What we throw away is the closest we come to the pollution problem as we rarely see the acid rain or spot those CFCs.

The growing accumulation of garbage reduces the land values, increases truck traffic, and ruins health, aesthetics and the necessities of life - the air we breathe and the water we drink.

Sources

Solid waste refers to the non-liquid waste materials arising from domestic (*garbage debris* and night soil) activities, trade and commercial activities (hazardous and non-hazardous) industrial activities, agricultural activities, mining and public services (office and hospital wastes).

Effects

It may be emphasized that unsanitary disposal and utilization of wastes result in high incidence of illness and death from *faecal borne diseases*. The faecal borne diseases are *bacillary dysentery*, *typhoid fever* and *enteritis*. Therefore, it is necessary to provide adequate and sanitary measures of disposal of wastes.

Types of Wastes

Urban wastes are discarded as organic and inorganic substances in the form of solid, semi-solid, liquid and gases, which are residues or derivatives of human, vegetable material and industry. These wastes are broadly classified into the following categories:

- **HOUSEHOLD WASTES:** Waste generated in the preparation and consumption of food, human excreta, generally termed as garbage.
- **COMMERCIAL WASTES:** These wastes include a high proportion of paper, cardboards and plastics. They result from activities in office buildings, stores, markets, theatres, hospitals, and restaurants.
- **INDUSTRIAL WASTES:** Wastes due to different types of production activities. This category of waste, contains, hazardous wastes that are harmful to human beings and hence should be stored and treated separately.
- **EFFLUENT WASTES:** Domestic as well as industrial effluents that contaminate river water if allowed to flow unchecked.
- **CITY WASTES:** All the waste resulting from the maintenance of streets, roads, parks, and schools, paper, dry leaves, animal wastes, sludge, carcasses of small animals and slaughter house wastes. The responsibility of the solid waste management department is only to look after the disposal of domestic wastes, city wastes and the domestic sewage.

Method of Collection

The 90% of the refuse is collected from refuse bins and 10% of the refuse is collected by house to house collection method, as described below.

- *Refuse storage:* which may sometimes require delivery of refuse by the householder over a considerable distance.
- Where the householder delivers the refuse to the vehicle at the time of collection.
- Door-to-door collection, where the collector enters the premises and collects the refuse and the householder is not involved in the collection process.

Radioactive Waste Management

Radioactive waste which arises from civil nuclear activities as well as from defense related nuclear weapon activities, poses a terrible problem for handling and keeping the environment to be safe to the present and future generations. The techniques used emphasizes on waste minimization and volume reduction. Nuclear waste is categorized into high, intermediate and low levels depending on the level of radioactivity in it.

Spent fuel is stored for long time to reduce the level of radioactivity in it and then reprocessed at reprocessing plants for gathering fissile elements. The generation of high level waste is at reprocessing plants. The amount of this waste in our country is much lesser due to our adoption of the closed fuel cycle. High level waste produced from the reprocessing plant is vitrified into a glassy form, enclosed in multiple barrier vessels and stored for a temporary period of three to four decades in engineered vaults with essential observation services. After cooling down in these storage facilities, waste vessels will be stored for long term in deep geological repositories.

Reprocessing and Waste Management plants are currently being operated by Bhabha Atomic Research Centre (BARC).

Harmful Effects

Radioactive wastes have the harmful effects in the following ways:

- (i) Pollutes the earth to a dangerous level of toxicity.
- (ii) Are absorbed in water and then enter in living beings through food chains.
- (iii) Emit harmful radiations which damage cells, tissues and Red blood corpuscles (RBC).
- (iv) Can cause cancer, leukemia, etc.
- (v) Are threats for aquatic life.

Control Measures for Radioactive wastes

With respect to control of the materials composition radioactive waste can be grouped into:

- (i) The treatment and packaging has to be performed according to a qualified process.
- (ii) Conditioned waste products being qualified with respect to the radiological requirements.
- (iii) Legacy waste products need to be qualified by spot checking according to composition requirements.

Plastic Waste Management

Plastics have become an indispensable part of our daily lives. Invented in 1935, they are wonderful products of polymer chemistry produced from the by-products of petroleum refining. They are classified as into two main categories:

- **Thermoplastics:** They are substances that become plastic on heating. A plastic material can be repeatedly melted or softened by heat without change of properties. This property makes it possible to recycle the used plastic articles.
- **Thermosetting plastics:** Plastics that have once been subjected to heat and pressure, lose their plasticity.

Environmental aspects of Plastic Manufacturing

All the varieties of plastics are manufactured from *petrochemical* based *hydrocarbons*. These hydrocarbons, and the plastic manufacturing processes involved possess environmentally critical characteristics.

The raw materials and intermediate products used in the manufacture of *Polyvinyl chloride* (PVC) - Ethylene, Chlorine, Hydrogen chloride, Vinyl Chloride Monomer (VCM), and Ethylene Dichloride (EDC) — are known hazardous materials. Additives, fillers, and coloring pigments used in plastic goods can also exhibit hazardous properties.

Environmental issues

- Escape of gaseous hydrocarbons, chlorine, and hydrogen chloride gas into the atmosphere.
- Waste-water from the processes and wash-waters can carry pollutants.
- Dioxins can be liberated due to mishaps in the process.
- Most significant health and safety issue in the manufacture of PVC is the exposure of plant operators to Vinyl Chloride Monomer (VCM). In earlier days over exposure of workers to high concentrations of VCM was found to increase the risk of angiosarcoma of the liver — a rare cancer of the blood vessels of the liver. Exposure levels are now monitored and controlled; fugitive emissions are avoided by good house-keeping in the plastic industry, greatly reducing the VCM-related health problems.
- Fire hazards in plastic industry and in godowns storing plastic goods release toxic dioxins. PVC being a very widely used plastic in electrical cable insulation, and building construction has a high fire risk, one particular advantage of PVC is that it does not itself burn, but is charred by the heat of a fire. If there are no other fuels present; it will self-extinguish. This is one of the strengths of PVC in the electrical cabling and building construction industry.

Heavy-Metal Pollution from Plastics

Lead and *cadmium* compounds are added as stabilizers in PVC. Manufacturing these chemicals are used in the manufacture of soft plastic items such as vinyl flooring sheets, soft toys, etc. to increase their durability. Lead and cadmium can leach out during human contact, or when disposed in land-fills. Incineration of such rejected plastic items produces ash with high heavy metal content. Use of lead compounds in the manufacturing process can be a potential hazard to workers in the PVC industry. *Lead and cadmium are known neurotoxins and nephrotoxins respectively*. Neurotoxins damage the *nervous system*, whereas nephrotoxins affect the kidneys. Strict adherence to process controls in industry, and quality control of products can greatly reduce the risk of heavy-metal pollution.

Nature of Plastics

Articles made of plastics are environment friendly if properly used and handled. Plastics are *non-biodegradable* because of their chemical structure. They cannot be bio-chemically decomposed by the microbes and as such, there is no threat of pollution. Being non-biodegradable they become virtually inert materials and remain in the environment for very long periods. They obstruct the natural and man-made activities in a physical way and do not easily participate in any reactions. These activities can be avoided by little care and common sense while discarding used plastics. That proves the fact that plastic, an 'environment friendly' product has been made an enemy by the callousness of humans. It is the misuse or abuse of plastic that is creating problems and not the plastics that are derived from the depths of Mother Earth.

The importance of plastic should be seen in their utility some of their uses and contributions to the environment are presented below.

Use Negative effects of not using plastics, and using other materials

(i) Carry bags and packing amount environment

If paper is used in place of plastic, a great amount of biomass has to be extracted either from Natural forests or commercial wood plantations for paper-making. In any case either environment loses greenery. Instead, the nutrients circulated these plants and trees could be used for producing food. Further, manufacturing of paper in pulp and paper mills produces large quantities of highly polluting waste-waters, which are difficult and expensive to treat and clean of their pollution effects. If discharged without treatment they pose an equally big threat as that of unscientific discarding of plastics to the environment.

(ii) Food packaging using plastics

Milk Supply: In earlier days milk supplied in glass bottles which involved Jobs, like supply the milk and then next day, to collect and transport back empty bottles to the dairy plant. Thus the glass bottles were adding to transportation cost and transport related air pollution problems. Now more milk can be transported per truck in plastic sachets whereas glass bottles used to occupy large space as they were bulky and heavy. Loading and unloading of milk bottles and empty bottles was

also difficult, and required more labour. Elaborate washing of milk bottles for refilling was another big job. Bottle washing was consuming large quantity of scarce fresh water releasing highly polluted wastewater adding to water treatment costs. If discharged without treatment wastes create severe pollution problems including odour. Use of plastic sachets for milk transportation has eliminated these problems. Earlier dairies used to receive complaints of milk getting spoiled due to improper washing of bottles, which does not happen with plastic sachets. Food packaging: food ingredients, precooked and ready to cook and use foods are widely available in the market and are popular in cities. Plastics are the mainstay of the packaged food industry.

(iii) Drinking water supplied in plastic bottle

Drinking water outside the home was a big menace earlier. It is a recognized fact that drinking contaminated or unsafe water is a major cause for spread of contagious diseases. Though expensive, packaged water bottles have reduced the risk of contamination.

(iv) In the field of medicine and health services

Before the advent of plastic injection syringes and needles, glucose bottles and many other glass appliances had to be sterilized and reused in hospitals. There was always a risk of improper sterilization, which had the potential for spreading infection. This could have happened even in blood collected in bottled. Imagine the use of improperly sterilized glass articles in a scenario where HIV-AIDS infected patients are handled. Disposable syringes, blood sachets and many other hospital items made of plastic have greatly reduced the risks and hazards of contamination.

(v) Water supply pipes and Industrial piping

Transportation of water in PVC pipes is very common as it has several advantages over conventional GI Pipes.

- They are light and plumbing work is easy
- Corrosion problem is eliminated and hence corrosion related contamination of water is avoided and life of the pipes increases.
- Inner surface of pipes may be made smooth to reduce friction losses, thus saving on electricity bills and conserving energy. In industries whenever aggressive chemicals are to be transported PVC pipes of different varieties are used. PVC containers are commonly used for storage of water and many industrial chemicals.

(vi) Construction material

Plastics are extensively used as construction materials such as doors and windows house hold. Articles such as chairs, table holding racks, bucket and more. Use of plastics is reducing the burden on conventional materials such as wood and metals. If these are all the advantages of using plastics, why so much of a hue and cry is being raised about the ill effects of plastics on the environment? In this regard we have to recognize the fact that plastic is not the culprit; it is humans and society who, after use of plastics, discard them indiscriminately into the environment creating problems that are easily avoidable.

Problems Arising from Indiscriminate Discarding of used Plastics, particularly Carry Bags

Thin plastic carry bags are found strewn everywhere on land, water bodies and in drains. They have become a main component of municipal solid waste. Being non-biodegradable, they stay in the environment for very long periods. Several ill effects of these plastic items have been recognized:

- They choke storm water drains, often causing overflow of storm water on roads.
- Some people have a callous attitude of dumping solid wastes including plastic carry bags into the sewer system with utter disregard to the consequences. This happens when manhole covers are missing, extending an invitation for people to dump the garbage into them. The obvious effect is the *choking of sewers*.
- It is a common scene to sight floating plastic bags of various colours in vicinity of towns, preventing the entry of sunlight into the water, thus hindering, photosynthesis process, which is a source of oxygen supply for the fish life. Direct transfer of molecular oxygen into water is also affected. Another aspect of inadequate presence of dissolved oxygen in such water bodies is the development of anaerobic conditions leading to foul odours.
- Plastic carry bags often end up in agricultural fields directly or indirectly when domestic solid wastes stored in plastic bags are used as manure along with cattle dung. During tilling they get embedded in the soil and cannot be removed easily. Labour costs hinders removal of their plastics, thereby leaving them in the soil. They obstruct the root zone activities reducing the agricultural productivity. Being non-biodegradable they remain in the soil for a very long time, thus affecting the farm economy.
- Attracted by food left-over in plastic carry bag in marriage halls, religious places, tourist places large catering houses, cows and other domestic animals eat them. The plastic accumulated in their digestive system, often resulting in their death.

Prevention of Plastic-Disposal — Related Problems

Do not burn the plastics. They are valuable resource. Also gaseous emissions from combustion of plastics pollute the air and some of them are considered to be toxic. Combustion of plastics, particularly in high temperature incinerators produces (**Dioxins**) and (**Furan**) as by-products. (**Dioxins**) are a family of more than 75 different chlorinated hydrocarbons while Plastic or PVC – polyvinyl chloride is a chlorinated hydrocarbon. Some of them are highly toxic and are persistent chemicals that stay in the environment for a very long period. (**Furans**) (e.g. Furfuran) a colorless liquid, with a low boiling point of 320°C- used as an intermediate chemical in manufacture of synthetic resins are also toxic in nature.

Recycling

The best way to tackle the plastic-disposal problem is to adopt suitable methods for collection and conveyance of plastic

articles and practice recycling. With proper quality control, reprocessed plastics can be made as good as first-generation products are cheaper than good reprocessed plastic goods. Methods should be devised to make up this extra cost through taxes or subsidy. The extra monetary expenses involved in purifying the used plastic may become small compared to the environmental burden imposed by indiscriminate discarding of plastics.

Should Plastics be banned?

Some environmental activists are arguing in favour of completely banning low-end plastic items and recycled plastics. This is not very sound thinking particularly in India. Thousands of families are dependent on the collection of plastic wastes and small-scale reprocessing. Besides, more area would be deforested to supply raw materials to paper manufacturing industries for producing packing paper and carry bags.

Government Regulations on Manufacture and Use of Recycled Plastics

The Ministry of Environment and Forests issued the Recycled Plastics Manufacture and Usage Rules 1999, and amended it in 2003 under the Environment (Protection) Act, 1986 for regulating and managing plastic carry bags and containers. Salient features of the Rules are:-

- No vendor shall use carry bags and containers of recycled plastics for storing carrying and / or packaging of foodstuffs.
- Carry bags and Container used for packaging of foodstuff shall be made of virgin plastic and of natural shade or white.
- Carry bags and Container made from recycled plastics must be manufactured using pigments and colorants as per IS: 9833/ 1981 notified by the Bureau of Indian Standards (BIS);
- Minimum thickness of Carry bags made of virgin or recycled plastics must not be less than 20 microns.
- Manufactures of recycled carry bag shall code/ mark carry bags and containers as per IS: 14534: 1998 and mark them as “Recycled “along with percentage of recycled material.
- Manufactures shall print on words packet containing carry bags the words “Recycled marital” or “Virgin Plastics’ as the case may be.
- No vendor shall use carry bags made of virgin or recycled plastic below 8×12 inches $\{20 \times 30 \text{ cm}\}$ in size; and 50 bags of such will have minimum weight of 105 grams; and proportionate increase in weight to the increased size of the carry bags; for selling any commodity.
- Every occupier manufacturing carry bags or containers shall apply in prescribed form to the State Pollution Control Board/Pollution Control Committee for grant of Registration and renewal of Registration.
- The State Pollution Control Board / Pollution Committee shall issue and renew the Registration after ascertaining that the unite meets the norms prescribed under these rules and also possesses a valid cosset under Air and Water Act as per requirements of the State Pollution Control Board / Pollution Control Committee.

- The prescribed authority of enforcement of provisions relating to use, collection, segregation, transpiration and disposal is with District collector, Deputy Commissioner of the concerned district, where no such Authority has been constituted by the State Government/Union Territory administration under any law regarding non-biodegradable garbage.
- SPCB's / PCC's are the prescribed authority for enforcement of provisions relating to manufacture and recycling.

Difficulties in Recycling

Almost the entire content of the discarded plastics is made up of thin carry bags and small food-packaging pouches. Laminations made on books, electronic goods, compact discs, cassettes, and many consumer goods add to this problem. The tendency of people particularly house wives is to preserve the thick large bags and discard the thin carry bags with the domestic garbage. Once they get mixed up with the garbage these carry bags are difficult to separate. Cost of retrieval becomes more. Even the rag pickers do not find it attractive to collect the carry bags. If housewives, shopkeepers, and other users discard the carry bags in a container for periodic collection by some designated agency for recycling, the problem of plastics in environment can be greatly reduced.

Bio-degradable Plastics

An alternative to the plastic disposal problem has evolved in the form of ‘Bio-degradable plastics’. It is possible to collect these items along with the municipal solid waste (MSW) for suitable disposal. MSW collection and disposal is not satisfactory in most of the towns in India. The bio-degradable plastics will add to the already piling up municipal garbage. The immediate benefits of recovery and recycle of normal plastics is also lost if bio-degradable plastics are introduced. As of now, compared to normal plastic, the bio-degradable plastics are expensive and the technology for manufacture is not easily available. It may be possible to treat bio-degradable plastics in countries where solid waste management systems are working satisfactorily and extensively.

Reuse of Plastics

Mixing shredded PVC bags with asphalt for *road making* has been experimented with reasonable success. *Rope-making* using fiber removed from knitted cement bags is done in rural areas. These ropes are used in farming activities.

Mercury Pollution

Mercury pollution arises from a variety of sources. There is mercury mining although this is overwhelmingly now only in China and some Central Asian countries such as Kyrgyzstan. It is also very heavily used in artisanal and small-scale gold mining to separate gold from the ore. Mercury is used in the chemical and petrochemical industries and also in household products like compact fluorescent lamps (CFLs) and thermometers. Mercury emissions to the atmosphere also take place from coal-fired power plants. Mercury is present in industrial effluents that are let into water bodies and the sea and enters the human food chain through the consumption of fish. This is, in fact, what caused the disaster at Minamata in the 1950s. Contaminated

sites including old mines, landfills and waste disposal locations are also important sources of mercury pollution.

India dragged its feet a bit, but in the end signed up to the Minamata Convention on Mercury on 30 September, a year after it was adopted. The Minamata Convention gives India five years to control and, where feasible, to reduce emissions from new power plants and 10 years to do so for existing power plants. The Minamata Convention on Mercury is a global treaty to protect human health and the environment from the adverse effects of mercury. It was agreed at the fifth session of the Intergovernmental Negotiating Committee in Geneva, Switzerland at 7 a.m. on the morning of Saturday, 19 January 2013.

The major highlights of the Minamata Convention on Mercury include a ban on new mercury mines, the phase-out of existing ones, control measures on air emissions, and the international regulation of the informal sector for artisanal and small-scale gold mining.

EARTH HOUR

WWF's Earth Hour is an annual global celebration where people switch off their lights for one hour to show they care about the future of our planet. This year's celebrations will be on Saturday 19 March from 8.30 pm to 9.30 pm.

Since it first began in Sydney Australia in 2007, the number of countries taking part in Earth Hour has grown to an incredible 172 countries and territories.

India joined the Earth Hour campaign in 2009 and over the years has seen an exponential growth in participation across cities, towns and even far-flung villages of rural India. Iconic monuments of the country like the Rashtrapati Bhawan, Gateway of India, India Gate, Howrah Bridge and the Victoria Memorial, among others, switch off their non-essential lights in support of this global campaign.

ECO SENSITIVE ZONES

The concept of ecologically sensitive areas is very much an Indian invention, rooted in attempts by civil society to use the EPA to promote sustainable development alongside protection of the natural heritage. The term 'Ecologically Fragile Area' was first used in 1991 for Dahanu Taluka in Maharashtra, followed by the declaration of other ESAs like Mahabaleshwar-Panchgani and Matheran. These are all initiatives of civil society organisations or are a consequence of a resolution of the Indian Board for Wildlife in 2002 to protect areas up to 10 kilometres from the boundaries of wildlife sanctuaries and national parks.

Initially, there were no guidelines available on what areas may be considered ecologically sensitive, nor on working out an appropriate management regime. These issues were addressed in 2000 by the Pronab Sen Committee. The Sen Committee's

foremost criterion for identification of ESA is endemism. Western Ghats harbours well over two thousand endemic species of flowering plants, fish, frogs, birds and mammals amongst the better known groups of organisms, and thousands more amongst less studied groups. Amongst themselves these span the entire Western Ghats and all conceivable habitats, including highly disturbed ones. The Western Ghats region also qualifies as an ESA under several other Sen Committee criteria.

Under ESZ, commercial mining, polluting industries and large hydro-power projects are prohibited as per the ministry guidelines. At present the 29 ESZ notified by the Central government are spread across Haryana, Gujarat, Jharkhand, Karnataka, Sikkim, Goa, Maharashtra, Rajasthan, Gujarat and Uttarakhand.

INTERNATIONAL CONVENTIONAL FOR CLIMATE CHANGE

Convention	Place	Crucial documents
United nation Conference on environment and Development or Earth Summit (1992)	Rio de Janeiro, Brazil	Draft Earth Charter Convention on climate change and Biological Diversity Convention on Forest Agenda 21
The United Nations Framework Convention on Climate Change (1994)	Kyoto, Japan	Gather and share information on greenhouse gas emissions, national policies and best practices Launch national strategies for addressing greenhouse gas emissions and adapting to expected impacts, including the provision of financial and technological support to developing countries Cooperate in preparing for adaptation to the impacts of climate change
Convention on Biological Diversity (1993)		The conservation of biological diversity The sustainable use of the components of biological diversity The fair and equitable sharing of the benefits arising out of the utilization of genetic resources
Ramsar Convention on Wetland (1971)	Ramsar, Iran	Halt the worldwide loss of wetlands and To conserve, use and management, those that remain. This requires international cooperation, policy making, capacity building and technology transfer.

Convention on International Trade in Endangered Species of Wild Fauna and Flora (1963)	Washington, U.S.A.	Help in conservation of species
International Tropical Timber Organization (1983)	Geneva	Provide an effective framework for cooperation between tropical timber producers and consumers To encourage the development of national policies aimed at sustainable utilization Conservation of tropical forests and their genetic resources
United Nations Forum on Forests (2000)		Implementation of agreements and foster a common understanding on sustainable forest management; To provide policy development and dialogue among Governments and international organizations, To enhance cooperation To foster international cooperation and To monitor, assess and report on progress of the above functions and objectives To strengthen political commitment to the management, conservation and sustainable development.
Global tiger forum (1994)	New Delhi, India	Set up to embark on a worldwide campaign to save the wild tiger
Stockholm Convention on Persistent Organic Pollutants (2001)	Stockholm, Sweden	It develops a risk management evaluation Determines whether the substance fulfills POP screening. Manage and dispose of POPs wastes in an environmentally sound manner
Basel Convention on the Control of Trans boundary Movements of Hazardous Wastes and Their Disposal (1989)	Basel, Switzerland	To reduce hazardous waste generation and promote environmental sound management system for their disposal. Restrict trans boundary movement of such wastes and Provide regulatory system applying to cases where such movement is allowable.
United Nations Convention to Combat Desertification (1994)		promotes a global response to desertification, land degradation and drought
International Whaling Commission (1946)	Washington, D.C., United States,	to keep under review and revise as necessary the measures laid down in the Schedule to the Convention which govern the conduct of whaling throughout the world
Montreal Protocol on Substances that Deplete the Ozone Layer (1987)	Helsinki, Finland	Play role in controlling the ozone depletion

COP

The UNFCCC entered into force on 21 March 1994. Today, it has near-universal membership. The 195 countries that have ratified the Convention are called Parties to the Convention. The COP is the supreme decision-making body of the Convention. All States that are Parties to the Convention are represented at the COP, at which they review the implementation of the Convention and any other legal instruments that the COP adopts and take decisions necessary to promote the effective implementation of the Convention, including institutional and administrative arrangements.

The COP meets every year, unless the Parties decide otherwise. The first COP meeting was held in Berlin, Germany in March, 1995.

COP (from 2009-2015)

15 th session	7 th - 18 th Sec. 2009	Copenhagen (Denmark)
16 th session	29 th Nov. -10 th Dec. 2010	Cancun (Mexico)

17 th session	28 th Nov.-09 th Dec. 2011	Durban (South Africa)
18 th session	26 th Nov.- 7 th Dec. 2012	Doha (Qatar)
19 th session	11 th Nov. - 22 nd Nov. 2013	Warsaw (Poland)
20 th session	1 st – 12 th Sec. 2014	Lima (Peru)
21 st session	30 th Nov. - 11 th Dec. 2015	Paris (France)

Cop 21 Agreement - In Brief

- Reaffirm the goal of limiting global temperature increase well below 2 degrees Celsius, while urging efforts to limit the increase to 1.5 degrees;
- Establish binding commitments by all parties to make “nationally determined contributions” (NDCs), and to pursue domestic measures aimed at achieving them;
- Commit all countries to report regularly on their emissions and “progress made in implementing and achieving” their NDCs, and to undergo international review;

- Commit all countries to submit new NDCs every five years, with the clear expectation that they will “represent a progression” beyond previous ones;
- Reaffirm the binding obligations of developed countries under the UNFCCC to support the efforts of developing countries, while for the first time encouraging voluntary contributions by developing countries too;
- Extend the current goal of mobilizing \$100 billion a year in support by 2020 through 2025, with a new, higher goal to be set for the period after 2025;
- Extend a mechanism to address “loss and damage” resulting from climate change, which explicitly will not “involve or provide a basis for any liability or compensation”
- Require parties engaging in international emissions trading to avoid “double counting;”
- Call for a new mechanism, similar to the Clean Development Mechanism under the Kyoto Protocol, enabling emission reductions in one country to be counted toward another country’s NDC.

Environmental Acts

- The Water (Prevention and Control of Pollution) Act, 1974
- The Water (Prevention and Control of Pollution) Rules, 1975
- The Water (Prevention and Control of Pollution) Cess Act, 1977
- The Water (Prevention and Control of Pollution) Cess Rules, 1978
- The Air (Prevention and Control of Pollution) Act, 1981
- The Air (Prevention and Control of Pollution) Rules, 1982
- The Environment (Protection) Act, 1986
- The Environment (Protection) Rules, 1986
- Hazardous Wastes (Management and Handling) Rules, 1989
- Manufacture, Storage and Import of Hazardous Chemical Rules, 1989
- The Forest (Conservation) Act, 1980
- The Forest (Conservation) Rules, 1981
- The Wildlife Protection Act, 1972
- The Wildlife (Transactions and Taxidermy) Rules, 1973
- The Wildlife (Stock Declaration) Central Rules, 1973
- The Wildlife (Protection) Licensing (Additional Matters for Consideration) Rules, 1983
- The Wildlife (Protection) Rules, 1995
- The Wildlife (Specified Plants - Conditions for Possession by Licensee) Rules, 1995
- The Public Liability Insurance Act, 1991
- The Public Liability Insurance Rules, 1991
- The National Environment Tribunal Act, 199
- The National Environment Appellate Authority Act, 1997

Exercise -1

1. Green house effect is warming due to
 - (a) infra-red rays reaching earth
 - (b) moisture layer in atmosphere
 - (c) increase in temperature due to increase in carbon dioxide concentration of atmosphere
 - (d) ozone layer of atmosphere.
2. Acid rain is due to increase in atmospheric concentration of
 - (a) ozone and dust
 - (b) CO_2 and CO
 - (c) SO_2 and CO
 - (d) SO_2 and NO_2 .
3. The two great industrial tragedies namely, MIC and Chernobyl tragedies respectively occurred where and at which time?
 - (a) Bhopal 1984, Ukrain 1986
 - (b) Bhopal 1986, Russia 1988
 - (c) Bhopal 1984, Ukrain 1990
 - (d) Bhopal 1984, Ukrain 1988
4. If there was no CO_2 in the earth's atmosphere, the temperature of earth's surface would be
 - (a) higher than the present
 - (b) dependent on the amount of oxygen in the atmosphere
 - (c) same as present
 - (d) less than the present.
5. How carbon monoxide, emitted by automobiles, prevents transport of oxygen in the body tissues?
 - (a) by forming a stable compound with haemoglobin
 - (b) by obstructing the reaction of oxygen with haemoglobin
 - (c) by changing oxygen into carbon dioxide
 - (d) by destroying the haemoglobin.
6. What is B.O.D.?
 - (a) The amount of O_2 utilized by organisms in water
 - (b) The amount of O_2 utilized by micro-organisms for decomposition
 - (c) The total amount of P_2 present in water
 - (d) All of the above.
7. Which one of the following pairs is mismatched?
 - (a) fossil fuel burning – release of CO_2
 - (b) nuclear power – radioactive wastes
 - (c) solar energy – greenhouse effect
 - (d) biomass burning – release of CO_2 .
8. Acid rain concentrated in the eastern portions of the United States is primarily the result of
 - (a) nuclear power plants in the region
 - (b) coal-burning power plants in the Midwest
 - (c) hydro electric power plants in northeastern Canada
 - (d) off-shore oil drilling rigs along the east coast of the United States
9. Alien species that cause the most harm are those that
 - (a) struggle to fit into new ecosystems and eventually die out.
 - (b) eventually become naturalized
 - (c) become invasive
 - (d) become agricultural products
10. Environmental pollutants such as PCB's contaminate oceans and other aquatic systems. Plankton in the ocean become contaminated by PCB's and pass this along through the food chain. A pregnant woman has been cautioned to limit her consumption of food that may be high in PCBs. If she consumes food from a nearby contaminated ocean, we would expect that the most contaminated foods would be:
 - (a) shrimp
 - (b) clams
 - (c) fish that eat shrimp
 - (d) kelp and other ocean plants
11. In clean water the concentration of
 - (a) BOD is low but DO is high
 - (b) Both BOD and DO are high
 - (c) BOD is high but DO is low
 - (d) Both BOD and DO are low
12. Examining an old abandoned home several months after a flood, a relief worker suddenly experiences difficulty breathing. At the hospital she learns that she experienced:
 - (a) a chronic respiratory illness more common in the developed world.
 - (b) a chronic respiratory illness more common in the developing world.
 - (c) an acute respiratory illness more common in the developed world.
 - (d) an acute respiratory illness more common in the developing world.
13. Harmful algal blooms appear to be linked to unusually high levels of nutrient pollution. Which of the following is the most likely source of nutrient pollution in a river drainage system associated with an algal bloom?
 - (a) a coal-fired power plant
 - (b) a nuclear power plant
 - (c) chicken and hog farms
 - (d) a large shopping mall
14. Health problems associated with indoor air pollution in developing countries is most commonly associated with:
 - (a) chlorine gas released from tap water.
 - (b) the use of biofuels for cooking and heating.
 - (c) poor hygiene and sanitation inside the home.
 - (d) the widespread use of pesticides to control disease vectors.
15. The formation of ozone hole in the Antarctic region has been a cause of concern. What could be the reason for the formation of this hole ?
 - (a) Presence of prominent tropo-spheric turbulence; and inflow of chlorofluorocarbons.
 - (b) Presence of prominent polar front and stratospheric clouds; and inflow of chlorofluorocarbons.
 - (c) Absence of polar front and stratospheric clouds; and inflow of methane and chlorofluorocarbons.
 - (d) Increased temperature at polar region due to global warming.

16. Recent studies indicate that two of the most dangerous components of air pollution around major cities in the developed nations are:
 - (a) fine particles and sulfur pollution.
 - (b) carbon monoxide and ozone.
 - (c) lead and volatile organic compounds.
 - (d) *radon* and carbon monoxide.
17. The decline of the polar ice caps because of increasing temperatures at the poles will
 - (a) increase the amount of fresh water available for human use.
 - (b) decrease the largest reserve of fresh water on Earth.
 - (c) increase the amount of fresh water available in aquifers.
 - (d) decrease global sea levels.
18. The developed countries of the world have contributed the most to global climate change. By applying the polluter pays and equity principles, we would expect that the:
 - (a) developed countries will provide funds for adaptations in the developing countries.
 - (b) developing countries will provide funds for adaptations in the developed countries.
 - (c) precautionary principle will guide the payment of compensation to developed countries.
 - (d) stabilization wedge approach to global climate change will help to equalize the funds for adaptations.
19. The evolution of pesticide resistance resurgence, and secondary-pest outbreaks are only some of the problems that result from reliance on
 - (a) crop rotation and biological controls, which disrupt the natural dynamics of ecosystems.
 - (b) pesticides, creating the need to alternate between a pesticide and an herbicide every other year.
 - (c) rodenticides to kill weeds and insect pests and prevent the spread of viral diseases.
 - (d) pesticides, creating a never-ending pesticide treadmill requiring new pest-fighting strategies.
20. The most dramatic temperature shifts in the past few decades have been
 - (a) on land near the equator.
 - (b) in the oceans nearest the equator.
 - (c) in the north and south polar regions.
 - (d) in the innermost regions of the North American and African continents.
21. The most effective way to reduce GHG emissions is to increase
 - (a) the production of electric cars.
 - (b) energy efficiency and renewable energy.
 - (c) our reliance on widely available natural gas
 - (d) the use of coal gasification plants and scrubber technologies to reduce sulfur emissions.
22. The most likely sustainable solutions of ecological problems are
 - (a) Incorporate the concerns of economists, ecologists, and sociologists.
 - (b) emphasize ecology over all other fields.
 - (c) emphasize economics over all other fields.
 - (d) emphasize ecological and social is - "Use over economic concerns.
23. The negative impacts of ozone pollution on forests are expected to increase as
 - (a) ocean levels rise and wind patterns shift.
 - (b) organisms spread northward because of warming climate conditions.
 - (c) the increasing demand for timber further stresses the growth of trees,
 - (d) temperatures increase and precipitation becomes more unpredictable.
24. The pesticide that directly attacks the nervous system is
 - (a) Aldrin
 - (b) DDT
 - (c) Organic Phosphates
 - (d) None of the above
25. The pesticides that also function as endocrine disruptors cause disease by
 - (a) causing excessive secretion of stomach acids.
 - (b) mimicking the effects of estrogenic hormones.
 - (c) causing muscle spasms and cramping in major muscle groups.
 - (d) greatly reducing the ability of the intestines to absorb nutrients.
26. The quality of the final treated wastewater effluent from a modern treatment plant is typically:
 - (a) lower in organic and nutrient content than the body of water into which it is discharged.
 - (b) lower in organic content but higher in nutrient content than the body of water into which it is discharged.
 - (c) higher in organic and nutrient content than the body of water into which it is discharged.
 - (d) higher in organic content but lower in nutrient content than the body of water into which it is discharged.
27. Brightly colored antique children's toys from before 1970 may be colored with paints that are contaminated with:
 - (a) heavy metals
 - (b) toxic plastic compounds.
 - (c) synthetic organic compounds.
 - (d) synthetic inorganic compounds.
28. By design, the molecules that resist biodegradation and include some of the most problematic persistent organic pollutants are the :
 - (a) synthetic organic compounds
 - (b) synthetic inorganic compounds
 - (c) recycled heavy metals
 - (d) chlorinated heavy metals
29. Carcinogens are dangerous because they affect
 - (a) oxygen-carrying red blood cells.
 - (b) the ability of the lining of the lungs to absorb oxygen.
 - (c) DNA molecules inside cells.
 - (d) the ability to absorb nutrients in the wall of the intestines.
30. CFCs primarily contribute to the destruction of the ozone by:
 - (a) producing chlorinated gases that reflect back a significant amount of ultraviolet light.
 - (b) releasing carbon monoxide into the stratosphere, which reacts with the oxygen in ozone.

- (c) releasing gases into the stratosphere that block the enzymes that create ozone.
- (d) contributing chlorine, which acts as a catalyst in the breakdown of ozone.
31. At an international conference on global climate change, a representative of a developing country admits that each year, his country contributes the same amount of greenhouse gases to the atmosphere as a particular developed nation. However, he argues that because his country has only 10% of the wealth of that developed nation, the developed nation should pay much more of the costs of adaptation. This representative's arguments illustrate the:
- (a) polluter pays principle.
- (b) precautionary principle.
- (c) equity principle.
- (d) conservation of the commons principle.
32. Concerned with the expensive disposal of their hazardous wastes, a company learns that it can purchase another chemical that will neutralize the company's hazardous wastes into a nontoxic form. The strategy used by this company is most consistent with which of the following hazardous-waste disposal system methods?
- (a) secure landfill
- (b) deep-well injection
- (c) on-site surface impoundment
- (d) best-demonstrated available technology
33. Freshwater becomes polluted:
- (a) by oil spills in ocean water moving inland.
- (b) primarily by contaminants from aquifers moving to surface waters.
- (c) as a result of eutrophication.
- (d) from runoff associated with urban areas chemicals used in farming in rural areas.
34. In developing countries, contaminated water is responsible for the deaths of more than 1.6 million people. Contributing to this problem is the use of
- (a) groundwater for consumption and the disposal of human sewage.
- (b) groundwater for consumption and the disposal of human sewage in surface waters.
- (c) surface waters for consumption and the disposal of human sewage.
- (d) surface waters for consumption and the disposal of human sewage in groundwater.
35. In general, temperatures along an ocean coastline vary less than temperatures 100 miles inland. This moderation of temperatures along coastlines is because
- (a) as the oceans evaporate it cools off the coastlines.
- (b) the sun shines more intensely away from the ocean coastlines
- (c) ocean temperatures change more quickly than air temperatures.
- (d) ocean temperatures do not change as quickly as air temperatures.
36. Ozone levels increase in the atmosphere when volatile organic compounds (VOCs) are present because:
- (a) less nitric oxide is available to react with ozone.
- (b) VOCs react with atmospheric nitrogen to form ozone.
- (c) VOCs release ozone when they are broken apart by solar energy.
- (d) more carbon dioxide is available to contribute additional oxygen for ozone formation.
37. The greatest progress in reducing atmospheric levels of lead pollution resulted from
- (a) the elimination of leaded gas.
- (b) the switch from lead to graphite in pencils.
- (c) the development of new types of batteries that use lithium instead of lead.
- (d) new types of lead scrubbers on smokestacks that removed lead from the air.
38. The inside of a car or greenhouse would not heat up as much in the presence of sunshine if
- (a) air was circulated within the car or within the greenhouse.
- (b) infrared radiation passed through glass as easily as sunlight.
- (c) infrared radiation could not pass through glass as easily as sunlight.
- (d) sunlight could pass through glass more easily than through air.
39. Villagers living in a heavily forested region surrounding a remote district in Punjab decided to reduce air pollution in their village. In the autumn season, after the leaves had fallen from the trees, the villagers blew all of the dead leaves into the village pond. About 8 months later, they noticed a large number of dead fish in the pond. What is the most likely cause of the fish kill?
- (a) the dead leaves released poisons that killed the fish.
- (b) the decomposing leaves depleted the levels of oxygen.
- (c) bacteria fed on the leaves and then the bacteria infected the fish.
- (d) carbon dioxide from the decaying leaves reached toxic levels and killed the fish.
40. Which of the following represents an alarming positive feedback loop of global warming?
- (a) increasing temperature raises humidity, which further increases temperatures.
- (b) decreased pH of the ocean increases the rate at which carbon dioxide is absorbed by the oceans from the atmosphere.
- (c) increased use of fossil fuels adds sulfate aerosols into the atmosphere, which traps more heat.
- (d) increased levels of atmospheric carbon dioxide increase photosynthesis, which further increases carbon dioxide atmospheric levels.
41. Which one of the following generally increases the pollution of the air?
- (a) bright sunlight.
- (b) generation of hydroxyl radicals.
- (c) sea salt aerosols entering the air over an ocean.
- (d) gases released by a volcanic eruption.
42. Wood pellets are produced from the waste sawdust of lumber and paper mills. Home-heating stoves burning these pellets can heat homes directly, instead of relying on other energy sources. Heating your home with wood pellets is:

- (a) sustainable, less polluting, and about 3 times as efficient as heating a home using electricity from a coal-fired power plant.
- (b) sustainable, slightly more polluting, and is about 30% more efficient than using electricity from a coal-fired power plant.
- (c) not sustainable but is less polluting and is about as efficient as using electricity from a coal-fired power plant.
- (d) not sustainable and actually pollutes more than using electricity from a coal-fired power plant.
43. The Clean Development Mechanism (CDM), a mechanism to reduce greenhouse gas emission as per Kyoto Protocol implies that
- (a) industrial countries receive carbon credits by funding carbon saving projects in another relatively affluent nation
- (b) industrial countries reduce their carbon emission by using environment friendly technology in production
- (c) developed countries invest in carbon reduction in developing countries and receive carbon credit in return
- (d) developed nations purchase carbon credit from other nations
44. A gardener applied heavy doses of the same insecticide to his garden for two consecutive years to kill squash bugs. During the third year, the man called in an expert to explain why he had an abundance of new pests that were destroying her garden. The expert explained that the abundant new pests were largely due to his previous use of a insecticide in a phenomenon known as
- (a) pesticide resistance
- (b) secondary-pest outbreak
- (c) triennial pest emergence
- (d) bounce back resurgence
45. A gardener has a large garden and decides this year he will not let the pests get beyond control. At the earliest sign of insect pests, he applies an organic insecticide and continues to apply it every month throughout the growing season. The next year he decides not to use any insecticides, thinking that he must have eliminated the pests with the prior year's treatments. Unfortunately, the pests reappear in numbers greater than he has ever seen before, and his plants are destroyed. Investigating this phenomenon, he learns that he has just experienced a phenomenon known as
- (a) resurgence (b) pesticide resistance
- (c) natural selection (d) emergence
46. Moss invades and establishes itself on bare rock, accumulating the beginnings of soil. After several years, enough soil has become established that grasses begin to grow where there was once bare rock.
- Without the moss building up soil, the grasses would have had no chance. The mosses changed the environment enough to permit grasses to grow in a process called:
- (a) sublimation. (b) facilitation.
- (c) regeneration. (d) improvisation.
47. Most of the wheat, rice and corn raised in the world has resulted from genetic engineering of one sort or another, either by crossing certain varieties or deliberately transferring genes using transgenic techniques. These methods select for plants that produce their own defenses against pests with chemicals or physical barriers. Helping to feed the world, this represents an example of :
- (a) cultural control
- (b) natural enemies control
- (c) genetic control
- (d) natural chemical control
48. The eagle predators, the amount of acorns produced annually, nesting sites in the trees, and cold winter temperatures limits the squirrel population in the Punjab region. The many factors listed above that can affect the squirrel population represent:
- (a) environmental resistance.
- (b) the carrying capacity of the squirrel population.
- (c) the squirrel's life history.
- (d) the biotic potential of the squirrel population.
49. The most widespread negative health impact of air pollution is the
- (a) destruction of the cellular component of the immune system.
- (b) loss of the ability to absorb vital nutrients by the digestive system.
- (c) disruption of the signaling processes of the endocrine system.
- (d) chronic stress that weakens many systems of the body.
50. The population of a particular type of fish, called Kubani found and in the Chilka lake only, is under heavy fishing pressure. If too many Kubians are caught, its population will crash and future years of fishing Kubani will suffer. Kubani can exhibit logistic growth under certain circumstances. Assuming logistic growth, it would be best to manage Kubani population by permitting the harvesting of just enough fish to keep the Kubani population.
- (a) at 1/10 of its carrying capacity.
- (b) at half its carrying capacity.
- (c) at its full carrying capacity.
- (d) above its carrying capacity.
51. In the lower regions of Uttaranchal, a toxic weed called leafy spurge was accidentally introduced and has grown and spread rapidly, covering millions of acres of grasslands. Leafy spurge is generally avoided by cattle and horses and may be toxic to them. Thus, grasslands where leafy spurge has spread has been damaged by the invasion of this plant. Plants such as leafy spurge can double their population size every year in part because of their efficient production of large amounts of seeds. Populations that can double every year, such as leafy spurge.
- (a) can do so endlessly, eventually covering all of the land on Earth.
- (b) exhibit constant growth increasing by the same amount every year.
- (c) exhibit a state of equilibrium when they are spreading.
- (d) exhibit exponential growth as they spread to new regions.

52. Which one of the following statements best reflects the overall position of current science on the role of biodiversity in ecosystems?
- The more species in an ecosystem, the greater the biomass production.
 - The more species in an ecosystem, the greater the drought resistance.
 - The effects of biodiversity on the functioning of an ecosystem are not consistent.
 - Almost every species in an ecosystem is essential to maintain the overall ecosystem.
53. Widely applying pesticides may lead to resurgence and secondary-pest outbreak because:
- the insecticide also killed the natural predators of the pests.
 - the plants have now lost their ability to fight the pests.
 - pesticides typically harm plants in ways that take several years to appear.
 - new species that are more resistant to insecticides have evolved.
54. Protection of endangered species by preserving the entire ecosystem is known as:
- In-situ conservation
 - Ex-situ conservation
 - Biodiversity conservation
 - None of the above
55. Ramsar Convention 1971 aimed at the conservation of
- Wasteland
 - Wetland
 - Desert
 - All of the above
56. Building on scientific research and careful measurements, the 1987 Montreal Protocol represented :
- global stewardship to limit the destruction of the ozone.
 - agreements to maintain sustainable levels of agricultural productivity.
 - sound science to better understand the impact of acid precipitation.
 - stewardship by the Canadian government to limit the production of greenhouse gases.
57. Over the past 20 years, vultures in India and Pakistan have declined by more than 95% due to :
- increased hunting and fear from villagers that the vultures will kill their domestic cattle.
 - the destruction of their nesting habitat in cliffs bordering the Indus River.
 - the spread of respiratory viruses common in domestic chickens.
 - the widespread use of an anti-inflammatory drug in cattle that were eaten by vultures.
58. 'El Nino' associated with the formation of the South West Monsoon of India is
- an abnormally warm ocean current
 - a periodic warm air-mass
 - a periodic warm wind
 - a periodic low pressure centre
59. Ozone holes are more pronounced at the
- Equator
 - Tropic of Cancer
 - Tropic of Capricorn
 - Poles
60. Acid rains are produced by
- excess NO_2 and SO_2 from burning fossil fuels
 - excess production of NH_3 by industry and coal gas
 - excess release of carbon monoxide by incomplete combustion
 - excess formation of CO_2 by combustion and animal respiration. (1988, 89)
61. 'BioCarbon Fund Initiative for Sustainable Forest Landscapes' is managed by the **(CSAT 2015-I)**
- Asian Development Bank
 - International Monetary Fund
 - United Nations Environment Programme
 - World Bank
62. Which one of the following is associated with the issue of control and phasing out of the use of ozone-depleting substances? **(CSAT 2015-I)**
- Bretton Woods Conference
 - Montreal Protocol
 - Kyoto Protocol
 - Nagoya Protocol
63. What is Rio+20 Conference, of ten mentioned in the news? **(CSAT 2015-I)**
- It is the United Nations Conference on Sustainable Development
 - It is a Ministerial Meeting of the World Trade Organization
 - It is a Conference of the Inter-governmental Panel on Climate Change
 - It is a Conference of the Member Countries of the Convention on Biological Diversity
64. Which of the following statements regarding 'Green Climate Fund' is/are correct? **(CSAT 2015-I)**
- It is intended to assist the developing countries in adaptation and mitigation practices to counter climate change.
 - It is founded under the aegis of UNEP, OECD, Asian Development Bank and World Bank.
- Select the correct answer using the code given below.
- 1 only
 - 2 only
 - Both 1 and 2
 - Neither 1 nor 2

Exercise -2

Statement Based MCQ

- Biomass gasification is considered to be one of the sustainable solutions to the power crisis in India. In this context, which of the following statements is/are correct?
 - Coconut shells, groundnut shells and rice husk can be used in biomass gasification.
 - The combustible gases generated from biomass gasification consist of hydrogen and carbon dioxide only.
 - The combustible gases generated from biomass gasification can be used for direct heat generation but not in internal combustion engines.
 Select the correct answer using the codes given below :

(a) 1 only	(b) 2 and 3
(c) 1 and 3	(d) 1, 2 and 3
- Eutrophication in the Chesapeake Bay along the eastern edge of Maryland has resulted in low oxygen levels in the water and alteration of food webs. The cause of this eutrophication appears to be pollution that contains high levels of:

1. nitrogen	2. carbon
3. phosphorus	4. sulphur

 Which of the above is/are correct?

(a) 1 and 3	(b) 2 and 4
(c) 1, 2 and 3	(d) All of these
- Consider the following statements:
 - Kyoto protocol came into force in the year 2005.
 - Kyoto protocol deals primarily with the depletion of the ozone layer.
 - Methane as a green house gas is more harmful than carbon dioxide.
 Which of the statements given above is/are correct?

(a) 1 and 2	(b) 1 and 3
(c) 1 only	(d) 3 only
- Compared to 50 years ago, the thinning ozone layer has produced dramatic increases in cases of :

1. asthma	2. color blindness
3. cataracts	4. skin cancer

 Which of the above is/are correct?

(a) 1 and 2	(b) 3 and 4
(c) 1, 2 and 3	(d) All of these
- If atmospheric carbon dioxide was eliminated from our atmosphere, we would expect that :
 - the Earth would cool considerably
 - photosynthesis would dramatically increase
 - the Earth would heat up considerably
 - photosynthesis would dramatically decrease
 Which of the above is/are correct?

(a) 1 and 2	(b) 2 and 3
(c) 1 and 4	(d) 2 and 4
- Due to their extensive rice cultivation, some regions may be contributing to global warming. To what possible reason/reasons is this attributable ?
 - The anaerobic conditions associated with rice cultivation cause the emission of methane.
 - When nitrogen based fertilizers are used, nitrous oxide is emitted from the cultivated soil.
 Which of the statements given above is / are correct ?

(a) 1 only	(b) 2 only
(c) Both 1 and 2	(d) Neither 1 nor 2
- Gases commonly referred as green house gases are :

1. CH_4	2. CO_2
3. CFC	4. NH_3

 Which of the above is/are correct?

(a) 1 and 4	(b) 2 and 3
(c) 1, 2 and 3	(d) All of these
- Fertilizers cause
 - eutrophication of water bodies
 - survival of most microorganisms
 - destruction of crumb structure of soil
 - all the above
 Which of the above is/are correct?

(a) 1 and 3	(b) 2 and 4
(c) 1, 2 and 3	(d) All of these
- Chipko movement is
 - A movement of political strength held in Assam under the supervision of Rajeev Gandhi
 - A environment movement held in uttarakhand under the guidance of Sunderlal Bahuguna.
 - A movement for independence under the guidance of Nehruji
 - A movement of independence under the guidance of Gandhi ji.
 Which of the following statements(s) is/are correct?

(a) 1 only	(b) 2 only
(c) 3 only	(d) Both 1 and 4
- Eutrophication of a lake most likely is the result of
 - Elevated nitrogen gas level in water.
 - Elevated phosphorus levels in water.
 - Excessive concentration of CO_2 .
 - A decrease in oxygen content of the water.
 Which of the above is/are correct?

(a) 1 only	(b) 1 and 3
(c) 1 and 2	(d) 1 and 4
- Ozone layer can be destroyed by pollutants such as

1. Hydro carbons	2. Carbon mono oxide
3. Sulphur dioxide	4. Nitrogen oxides

(a) 1 and 2	(b) 2 and 3
(c) 1 and 3	(d) 1 and 4
- Climatologist warn of a tipping point when global temperature trigger catastrophic events and rise levels rise more than 50 feet. What would cause the sea level rise so greatly and how much warmer does the world need to get for this tripping point to happen
 - 1°C
 - 3°C
 - Melting of Greenland ice sheet
 - thermal expansion of oceans

Which of the above is/are correct?

- (a) 1 and 3 (b) 1 and 3
(c) 2 and 3 (d) 2 and 4

13. One Carbon Credit is defined as ____ .

1. Credit permit to release one ton of carbon dioxide.
2. providing loans to establish a unit which produces carbon dioxide for industrial use.
3. Finding out one new business which can use and recycle greenhouse gases.

Which of the following statements(s) is/are correct?

- (a) 3 only (b) 2 only
(c) 1 only (d) All 1, 2 and 3

14. Humans have contributed to habitat destruction by

1. clearing land for farming
2. excessive use of chemicals
3. producing green-house gases through use of fossil fuels
4. exploitation of land and water for mining of scarce resources

Which of the above is/are correct?

- (a) 1 and 2 (b) 2 and 3
(c) 1, 2 and 3 (d) All of these

15. Global climate change threatens coral reefs by

1. increasing the temperature
2. decreasing the temperature
3. increasing the pH of the oceans
4. decreasing the pH of the oceans

Which of the above is/are correct?

- (a) 1 and 2 (b) 2 and 3
(c) 1 and 4 (d) 2 and 4

16. The acidification of oceans is increasing. Why is this phenomenon a cause of concern?

1. The growth and survival of calcareous phytoplankton will be adversely affected.
2. The growth and survival of coral reefs will be adversely affected.
3. The survival of some animals that have phytoplanktonic larvae will be adversely affected.
4. The cloud seeding and formation of clouds will be adversely affected.

Which of the statements given above is /are correct?

- (a) 1, 2 and 3 (b) 2 only
(c) 1 and 3 (d) 1, 2, 3 and 4

17. Which of the following countries suffer from the acid rains?

1. Canada 2. France
3. Norway 4. Germany

Select the correct answers from the codes given below:

Codes:

- (a) 1 and 2 (b) 1 and 3
(c) 2 and 3 (d) 3 and 4

18. Which of the following statements about Radioactive pollution are correct?

1. It causes genetic changes in the animals.
2. It causes disbalance among different minerals in the soil.
3. It hinders blood circulation.
4. It causes cancers.

Select the correct answer from the codes given below:

Codes:

- (a) 1 and 2 (b) 1 and 4
(c) 1, 3 and 4 (d) 2, 3 and 4

19. Which of the following conditions indicate the impact of global warming?

1. Melting of glaciers
2. Rise in sea level
3. Changes in weather conditions
4. Rise in global temperature

Select the correct answer from the codes given below:

Codes:

- (a) 1 and 2 (b) 1, 2 and 3
(c) 2, 3 and 4 (d) 1, 2, 3 and 4

20. As per National Aeronautics and Space Administration (NASA) research scientists found that concentrations of mercury near the ground level had increased in the Arctic Sea by mercury-pumping reaction which takes place because -

1. of open water in a lead is much warmer than the air above it.
2. of the temperature difference, the air above the lead churns like the air above a boiling pot.
3. the mixing is so strong that it actually pulls down mercury from a higher layer of the atmosphere to near the surface.

Select the answer from the codes given below-

- (a) 1 only (b) 1 and 2 only
(c) 2 and 3 only (d) All of the above

21. A new nuclear waste disposal strategy announced by United States include-

1. a "pilot interim store" will become operational in 2021
2. a larger "full-scale interim store" will open by 2025
3. an underground disposal facility to be established by 2048 to permanently dispose of the material.
4. a new organisation will be established to manage the siting, development and operation of the future waste stores.

Select the answer from the codes given below-

- (a) 1, 2 and 3 (b) 2, 3 and 4
(c) 1, 3 and 4 (d) All of the above

22. Which of the following statements are correct ?

1. A new study has found that changes in solar activity contributed no more than 10 per cent to global warming in the 20th century published in the journal Environmental Research Letters.
2. It has been proposed that cosmic rays may have a role in cooling the earth by encouraging clouds to form, which subsequently reflect the sun's rays back into space.
3. Researchers found high correlation between cosmic rays and global temperatures occurring every 22 years.

Select the answer from the codes given below-

- (a) 1 only (b) 1 and 2 only
(c) 2 and 3 only (d) All of the above

23. There is a concern, over the increases in harmful algal blooms in the seawaters of India. What could be the causative factors for this phenomenon?
1. Discharge of nutrients from the estuaries.
 2. Run-off from the land during the monsoon.
 3. Upwelling in the seas.
- Select the correct answer from the codes given below:
- (a) 1 only
 - (b) 1 and 2
 - (c) 2 and 3
 - (d) 1, 2 and 3
24. With reference to India, consider the following Central Acts
1. Import and Export (Control) Act, 1947
 2. Mining and Mineral Development (Regulation) Act, 1957
 3. Customs Act, 1962
 4. Indian Forest Act, 1927
- Which of the above Acts have relevance to/bearing on the biodiversity conservation in the country ?
- (a) 1 and 3
 - (b) 2,3 and 4
 - (c) 1,2,3 and 4
 - (d) None
25. Biomass gasification is considered to be one of the sustainable solutions to the power crisis in India. In this context, which of the following statements is/are correct?
1. Coconut shells, groundnut shells and rice husk can be used in biomass gasification.
 2. The combustible gases generated from biomass gasification consist of hydrogen and carbon dioxide only.
 3. The combustible gases generated from biomass gasification can be used for direct heat generation but not in internal combustion engines.
- Select the correct answer using the codes given below :
- (a) 1 only
 - (b) 2 and 3
 - (c) 1 and 3
 - (d) 1, 2 and 3
26. The safest method for biomedical waste disposal is:
1. Incineration
 2. Autoclaving
 3. Sharp pit encapsulation
 4. Precipitation
- Which of the above is/are correct?
- (a) 1 and 2
 - (b) 1 and 3
 - (c) 1, 2 and 3
 - (d) 1, 2, 3 and 4
27. Forests that experience high levels of acid precipitation expose trees to soil that has :
1. more dissolved calcium
 2. more dissolved aluminium
 3. less dissolved calcium
 4. less dissolved aluminium
- Which of the above is/are correct?
- (a) 1 and 2
 - (b) 2 and 3
 - (c) 1 and 4
 - (d) 2 and 4
28. Excessive exposure of humans to UV-rays results in
1. damage to immune system
 2. damage to lungs
 3. skin cancer
 4. peptic ulcers
- Which of the above is/are correct?
- (a) 1 and 2
 - (b) 2 and 3
 - (c) 1 and 3
 - (d) 2 and 4
29. Gases commonly referred as green house gases are :
1. CH_4
 2. CO_2
 3. CFC
 4. NH_3
- Which of the above is/are correct?
- (a) 1 and 4
 - (b) 2 and 3
 - (c) 1, 2 and 3
 - (d) 1, 2, 3 and 4
30. Climatologist warn of a tipping point when global temperature trigger catastrophic events and rise levels rise more than 50 feet. What would cause the sea level rise so greatly and how much warmer does the world need to get for this tripping point to happen
1. 1°C
 2. 3°C
 3. Melting of Greenland ice sheet
 4. thermal expansion of oceans
- Which of the above is/are correct?
- (a) 1 and 3
 - (b) 1 and 3
 - (c) 2 and 3
 - (d) 2 and 4
31. Due to improper/ indiscriminate disposal of old and used computers or their parts, which of the following are re-released into the environment as e-waste?
1. Beryllium
 2. Cadmium
 3. Chromium
 4. Heptachlor
 5. Mercury
 6. Lead
 7. Plutonium
- Select the correct answer using the codes given below.
- (a) 1, 3, 4, 6 and 7 only
 - (b) 1, 2, 3, 5 and 6 only
 - (c) 2, 4, 5 and 7 only
 - (d) 1, 2, 3, 4, 5, 6 and 7
32. Which of the following can be found as pollutants in the drinking water in some parts of India?
1. Arsenic
 2. Sorbitol
 3. Fluoride
 4. Formaldehyde
 5. Uranium
- Select the correct answer using the codes given below.
- (a) 1 and 3 only
 - (b) 2, 4 and 5 only
 - (c) 1, 3 and 5 only
 - (d) 1, 2, 3, 4 and 5
33. Consider the following international agreements :
1. The International Treaty on Plant Genetic Resources for Food and Agriculture
 2. The United Nations Convention to Combat Desertification
 3. The World Heritage Convention
- Which of the above has/have a bearing on the biodiversity?
- (a) 1 and 2 only
 - (b) 3 only
 - (c) 1 and 3 only
 - (d) 1, 2 and 3
34. Consider the following statements regarding 'Earth Hour':
1. It is an initiative of UNEP and UNESCO.
 2. It is a movement in which the participants switch off the lights for one hour on a certain day every year.
 3. It is a movement to raise the awareness about the climate change and the need to save the planet.
- Which of the statements given above is/are correct?
- (a) 1 and 3 only
 - (b) 2 only
 - (c) 2 and 3 only
 - (d) 1, 2 and 3
35. With reference to a conservation organization called "Wetlands International", which of the following statements is/ are correct?

1. It is an intergovernmental organization formed by the countries which are signatories to Ramsar Convention.
 2. It works at the field level to develop and mobilize knowledge, and use the practical experience to advocate for better policies.
- Select the correct answer using the code given below.
- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2
36. Other than poaching, what are the possible reasons for the decline in the population of Ganges River Dolphins?
1. Construction of dams and barrages on rivers
 2. Increase in the population of crocodiles in rivers
 3. Getting trapped in fishing nets accidentally
 4. Use of synthetic fertilizers and other agricultural chemicals in crop-fields in the vicinity of rivers
- Select the correct answer using the code given below.
- (a) 1 and 2 only (b) 2 and 3 only
(c) 1, 3 and 4 only (d) 1, 2, 3 and 4
37. Brominated flame retardants are used in many household products like mattresses and upholstery. Why is there some concern about their use?
1. They are highly resistant to degradation in the environment.
 2. They are able to accumulate in humans and animals.
- Select the correct answer using the code given below.
- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2
38. With reference to 'Eco-Sensitive Zones', which of the following statements is/are correct?
1. Eco-Sensitive Zones are the areas that are declared under the Wildlife (Protection) Act, 1972.
 2. The purpose of the declaration of Eco-Sensitive Zones is to prohibit all kinds of human activities in those zones except agriculture.
- Select the correct answer using the code given below.
- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2
39. Consider the following statements :
1. Animal Welfare Board of India is established under the Environment (Protection) Act, 1986.
 2. National Tiger Conservation Authority is a statutory body.
 3. National Ganga River Basin Authority is chaired by the Prime Minister.
- Which of the statements given above is/are correct?
- (a) 1 only (b) 2 and 3 only
(c) 2 only (d) 1, 2 and 3
40. Which of the following have coral reefs?
1. Andaman and Nicobar Islands
 2. Gulf of Kachchh
 3. Gulf of Mannar
 4. Sunderbans
- Select the correct answer using the code given below.
- (a) 1, 2 and 3 only (b) 2 and 4 only
(c) 1 and 3 only (d) 1, 2, 3 and 4
41. In India, the problem of soil erosion is associated with which of the following?
1. Terrace cultivation
 2. Deforestation
 3. Tropical climate
- Select the correct answer using the code given below.
- (a) 1 and 2 only (b) 2 only
(c) 1 and 3 only (d) 1, 2 and 3
42. The scientific view is that the increase in global temperature should not exceed 2 °C above pre-industrial level. If the global temperature increases beyond 3 °C above the pre-industrial level, what can be its possible impact/impacts on the world?
1. Terrestrial biosphere tends toward a net carbon source.
 2. Widespread coral mortality will occur.
 3. All the global wetlands will permanently disappear.
 4. Cultivation of cereals will not be possible anywhere in the world.
- Select the correct answer using the code given below.
- (a) 1 only (b) 1 and 2 only
(c) 2, 3 and 4 only (d) 1, 2, 3 and 4
43. Which of the following are some important pollutants released by steel industry in India?
1. Oxides of sulphur
 2. Oxides of nitrogen
 3. Carbon monoxide
 4. Carbon dioxide
- Select the correct answer using the code given below.
- (a) 1, 3 and 4 only (b) 1 and 3 only
(c) 1 and 4 only (d) 1, 2, 3 and 4
44. Global climate change threatens coral reefs by
1. increasing the temperature
 2. decreasing the temperature
 3. increasing the pH of the oceans
 4. decreasing the pH of the oceans
- Which of the above is/are correct?
- (a) 1 and 2 (b) 2 and 3
(c) 1 and 4 (d) 2 and 4
45. With reference to 'Forest Carbon Partnership Facility', which of the following statements is/ are correct?
1. It is a global partnership of governments, businesses, civil society and indigenous peoples.
 2. It provides financial aid to universities, individual scientists and institutions involved in scientific forestry research to develop eco-friendly and climate adaptation technologies for sustainable forest management.
 3. It assists the countries in their 'REDD+ (Reducing Emissions from Deforestation and Forest Degradation+)' efforts by providing them with financial and technical assistance.
- Select the correct answer using the code given below.
- (a) 1 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3
46. With reference to 'dugong', a mammal found in India, which of the following statements is/an; correct?
1. It is a herbivorous marine animal.
 2. It is found along the entire coast of India.
 3. It is given legal protection under Schedule I of the Wildlife (Protection) Act, 1972.
- Select the correct answer using the code given below.
- (a) 1 and 2 (b) 2 only
(c) 1 and 3 (d) 3 only

47. What are the effects that IPCC has predicted regarding global warming?
- Earth's temperature will rise by 1-30°C in next few decades, leading to extreme weather changes (heat waves, hurricanes and severe winters), changes in ocean currents and marine life.
 - If CO₂ concentration doubles, Earth's temperature may rise by 50°C.
 - The biggest glacier in the Peruvian Andes was retreating by 5 meters per year some 20 years ago; today it is shrinking by 33 meters per year.
 - The Arctic Sea ice has thinned by 40% in the last two decades, while Mount Everest is losing height at the rate of 1.5 meters per year.
- (a) Only I (b) I & II
(c) Only III (d) All the above
48. Which statement is correct regarding the steps taken to reduce global warming?
- Cleaning up and gasification of coal (for which technology is available) will result in lesser pollution.
 - Increased use of natural gas contains only half of the carbon and no Sulphur.
 - Renewable energy sources, such as wind, solar, photo-voltaic and fuel cells, tidal, etc.
 - Manufacture of fuel-efficient vehicles.
- (a) I & II (b) Only III
(c) Only II (d) All the above
49. Which statement is correct regarding greenhouse gases?
- A greenhouse gas (GHG) is a gas in an atmosphere that absorbs and emits radiation within the thermal infrared range
 - Water vapour contributes to 36-72% of Greenhouse effect
 - Nitrous Oxide contributes to about 10% of Greenhouse effect
 - Ozone contributes to 3-7% of Greenhouse effect
- (a) I & II (b) I, II & III
(c) I, II & IV (d) All the above
50. Ozone layer in the outer atmosphere helps in:
- Reflecting radio waves and makes radio communication possible
 - Absorbing U-V radiations
 - Regulating the temperature of atmosphere
 - Absorbing cosmic ray particles
- (a) I & II (b) Only II
(c) Only III (d) All the above
51. Ozone depletion would impact the plant community in several ways. These include :
- Increase in photosynthesis
 - Decline in water use efficiency
 - Decline in yield of plants
- (a) I & II (b) Only III
(c) II & III (d) All the above
52. Nitrogen Oxide is responsible for the depletion of Ozone layer, which of following is source of Nitrogen Oxide?
- Industrial emission
 - Fertilizers which are used in agricultural activities
 - Thermonuclear weapons
- (a) I & II (b) Only III
(c) Only II (d) All the above
53. Which of the following statement is correct regarding CFC's and HCFC's?
- These chemicals are inert, non-flammable, non-toxic, and lighter than air and can remain intact for years
 - CFCs are commonly used in Air-conditioners and the Refrigeration industry (Freon gas), aerosol propellants (in perfumes and deodorants), in the foam packaging industry (Styropor, Thermocol) and as solvents for greases and glues.
 - They contain Chlorine and Fluorine, common being CFC-II, CFC-12, CFC-22 and CFC- 13.
- (a) I & II (b) II & III
(c) Only III (d) All the above
54. Which statement is correct regarding deforestation?
- The process of clearance of forest by burning or logging is called deforestation
 - The main reasons for deforestation are trees or derived charcoal are used as, or sold, for fuel or as a commodity, while cleared land is used as grassland for livestock, plantations of commodities, and settlements.
 - Deforested areas usually sustain extensive adverse soil erosion and regularly damage into wasteland.
- (a) I & II (b) II & III
(c) Only III (d) All the above
55. Deforestation includes which of the following?
- Felling, and removal of forest litter
 - Browsing
 - Grazing and trampling of seedlings
 - Repeated lopping
- Select the correct answer:
- (a) Only I (b) I and III
(c) III and IV (d) All the above
56. Consider the following statements
- As a greenhouse gas (GHG) methane is even more harmful than carbon dioxide
 - Methane has been included in the list of six GHGs in Kyoto Protocol.
- Which of the statements given above is/are correct?
- (a) Only I (b) Only II
(c) Both I & II
57. Increased level of carbon dioxide in the atmosphere would impact the plants in many ways. These can be:
- Decrease in photosynthetic productivity of plants
 - Proliferation of weeds
 - Increase in number of insects and other pests.
- (a) I & II (b) II only
(c) II & III (d) All the above
58. Relative contributions of CO₂, CH₄, CFCs and N₂O towards global warming are:
- 50 %, 30 %, 10 %, and 10 % respectively
 - 60%, 20%, 14%, and 6% respectively
 - 40 %, 30%, 20% and 10% respectively
 - None of the above
- (a) Only II (b) Only III
(c) Only I (d) None of the above

59. What is denitrification?
- It prevents the discharge of nitrates in soil, and stops ground water pollution with nutrients.
 - In this solids are separated from the liquid
 - In this sewage goes through many chambers and chemical processes which reduce its toxicity.
- I & II
 - Only I
 - II & III
 - All the above
60. Which statement is correct regarding noise pollution?
- The word noise comes from the Latin word nausea meaning seasickness.
 - The main source of noise is transportation system including rail noise, aircraft noise and vehicle noise
 - Sound pollution is measured in decibels
- I & II
 - II & III
 - Only III
 - All the above
61. Chlorofluorocarbons, known as ozone-depleting substances, are used:
- In the production of plastic foams
 - In the production of tubeless tyres
 - In cleaning certain electronic components
 - As pressurizing agents in aerosol cans
- Only I
 - I, III & IV
 - II & III
 - All the above
62. Which of the following gases are responsible for acid rain in environment?
- Carbon dioxide and nitrogen
 - Carbon monoxide and carbon dioxide
 - Ozone and carbon dioxide
 - Nitrous oxide and Sulphur dioxide
- I & II
 - Only II
 - Only IV
 - All the above
63. Spraying of DDT on crops causes pollution of:
- Soil and Water
 - Air and Soil
 - Crops and Air
 - Air and Water
- I & II
 - II & III
 - Only I
 - All the above
64. Which statement is correct regarding Agro-Chemicals?
- Agro chemicals are developed by the use of modern technology that depends on inorganic fertilizers and pesticides
 - Excess use of these fertilizers can lead to immediate harmful effect or can also be long lasting.
 - In combination with genetically enhanced varieties of crop species, agrochemicals have made significant contributions to the accomplishments of the "green revolution."
- I & II
 - Only II
 - Only III
 - All the above
65. Which statement is correct regarding acid rain?
- Acid rain is caused by a chemical reaction that starts when sulfur dioxide (SO_2) and nitrogen oxides (NO_2) are released into the air
 - Sulfur dioxide and nitrogen oxides are highly soluble in water and can be carried very far by the wind
 - Power plants release huge amount of sulfur dioxide and nitrogen oxides when they burn fossil fuels, like coal, to produce electricity which can cause acid rain
- I & II
 - II & III
 - Only III
 - All the above
66. Acid rain reacting with calcium forms:
- Calcium bicarbonate
 - Calcium Nitrate
 - Calcium Sulphate
 - Calcium Carbonate
- I & IV
 - Only I
 - Only II
 - Only III
67. What among the below is/are the cause of ground water contamination?
- Septic tanks
 - Septic tanks
 - Landfills
- Only II
 - I & III
 - II & III
 - All the above
68. Which statement is correct regarding Ozone Hole?
- Ozone destruction rate is equal to the its formation rate
 - Ozone formation and destruction keep on happening
 - Ozone destruction rate is higher than its formation rate
- I & II
 - II & III
 - Only III
 - Only I
69. Which statement is not true regarding Ozone?
- Ozone is covered under Montreal Protocol.
 - Montreal protocol binds countries to adopt measures to curb ozone depleting substances.
 - HFC was used to replace ozone depleting substances.
 - Ozone is also covered under Kyoto Protocol.
- I & II
 - Only III
 - Only IV
 - II & III
70. Which of the following is not the feedback of the example in which human activity is responsible for the global climatic changes in the temperature?
- Global warming causes snow to melt in polar regions
 - Global warming causes increased rainfall, plant growth and photosynthesis
 - Global warming causes increased CO_2 release from biomass decomposition
 - Tropical deforestation causes warming and drying so that remaining forests begin to decline
- I & II
 - Only iii
 - Only II
 - II & IV
71. What is effluent waste?
- This category of waste, contains, hazardous wastes that are harmful to human beings and hence should be stored and treated separately.
 - These wastes include a high proportion of paper, cardboards and plastics.
 - Domestic as well as industrial effluents that contaminate river water if allowed to flow unchecked.
 - All the waste resulting from the maintenance of streets, roads, parks, and schools, paper, dry leaves, animal wastes, sludge, carcasses of small animals and slaughter house wastes.

- (a) I & II (b) Only III
(c) Only IV (d) II & IV
72. What are the methods of collection of solid wastes?
- Refuse storage which may sometimes require delivery of refuse by the householder over a considerable distance
 - Door-to-door collection, where the collector enters the premises and collects the refuse and the householder is not involved in the collection process.
 - Dumping in river or stream
- (a) I & II (b) Only II
(c) Only I (d) All the above
73. Which statement is correct regarding radioactive waste material?
- This arises from civil nuclear activities as well as from defense related nuclear weapon activities
 - The techniques used emphasizes on waste minimization and volume reduction
 - Nuclear waste is categorized into high, intermediate and low levels depending on the level of radioactivity in it
- (a) I & III (b) Only III
(c) Only I (d) All the above
74. Which statement is correct regarding the disposal of High level waste produced from radioactive material?
- High level waste produced from the reprocessing plant is vitrified into a glassy form, enclosed in multiple barrier vessels
 - They are stored for a temporary period of three to four decades in engineered vaults with essential observation services
 - After cooling down in these storage facilities, waste vessels will be stored for long term in deep geological repositories
- (a) I & II (b) Only II
(c) Only III (d) All the above
75. Which statement is correct regarding plastic waste management?
- Plastic was invented in 1960
 - They are the products of polymer chemistry produced from the by- products of petroleum refining
 - They are characterized into thermoplastics and thermosetting plastics
- (a) I & II (b) II & III
(c) Only III (d) All the above
76. Which sentence is correct regarding the manufacturing of plastic?
- All the varieties of plastics are manufactured from petrochemical based hydrocarbons
 - The raw materials and intermediate products used in the manufacture of Polyvinyl chloride (PVC) - Ethylene, Chlorine, Hydrogen chloride, Vinyl Chloride Monomer (VCM), and Ethylene Dichloride (EDC) — are known hazardous materials
 - Additives, fillers, and coloring pigments used in plastic goods can also exhibit non-hazardous properties
- (a) I & II (b) Only ii
(c) Only III (d) All the above
77. What are the environmental issues regarding the manufacturing of plastics?
- Escape of gaseous hydrocarbons, chlorine, and hydrogen chloride gas into the atmosphere.
 - Waste-water from the processes and wash-waters can carry pollutants.
 - Dioxins can be liberated due to mishaps in the process.
 - Health and safety issue in the manufacture of PVC is the exposure of plant operators to Vinyl Chloride Monomer (VCM)
- (a) I & II (b) II & III
(c) Only IV (d) All the above
78. What are the advantages of using PVC in Water supply pipes and Industrial piping over GI pipes?
- They are heavy and plumbing work is difficult
 - Corrosion problem is eliminated and hence corrosion related contamination of water is avoided and life of the pipes increases
 - Inner surface of pipes may be made smooth to reduce friction losses, thus saving on electricity bills and conserving energy
- (a) I & II (b) II & III
(c) Only III (d) All the above
79. What are the problems from Indiscriminate Discarding of used Plastics?
- They choke storm water drains, often causing overflow of storm water on roads.
 - Being non-biodegradable they remain in the soil for a very long time, thus affecting the farm economy
 - Direct transfer of molecular oxygen into water is also affected
- (a) I & II (b) Only II
(c) Only III (d) All the above
80. Which statement is correct regarding the rule of the Forests issued the Recycled Plastics Manufacture and Usage Rules 1999 which was amended in 2003 under the Environment (Protection) Act, 1986?
- No vendor shall use carry bags and containers of recycled plastics for storing carrying and / or packaging of foodstuffs
 - Carry bags and Container used for packaging of foodstuff shall be made of virgin plastic and of natural shade or white
 - Minimum thickness of Carry bags made of virgin or recycled plastics must not be less than 20 microns
- (a) I & II (b) II & III
(c) Only III (d) All the above
81. What is the main health risks associated with greater UV radiation through the atmosphere due to depletion of stratospheric ozone?
- Increased skin cancer
 - Damage to eyes
 - Increased liver cancer
 - Reduced immune system
- (a) I & II (b) III
(c) II & III (d) All the above

82. What are the negative effect regarding bio degradable plastics?
- The bio-degradable plastics will add to the already piling up municipal garbage
 - bio-degradable plastics are expensive and the technology for manufacture is not easily available
- (a) only I (b) only II
(c) Both I & II (d) only III
83. Which of the following reasons does not help regulate global carbon dioxide concentrations?
- Alterations in rainfall patterns
 - Storing carbon in the soil and biomass
 - Absorbing carbon dioxide for photosynthesis
 - Releasing carbon dioxide following decay
- (a) I & II (b) II & III
(c) Only I (d) All the above
84. The main function of ozone layer is:
- Heating the stratosphere
 - Maintaining the temperature of atmosphere
 - Absorbing the ultraviolet solar radiation
- (a) I & II (b) Only II
(c) Only III (d) All the above
85. Which statement is correct regarding methane?
- Methane (CH_4), also called "Marsh gas"
 - It arises from rice paddies, wetlands, enteric fermentation in cattle, burning of wood, and landfills
 - It is responsible for about 4-9% of Greenhouse effect.
- (a) I & II (b) II & III
(c) Only III (d) All the above
86. Which statement is correct regarding CFC's?
- They are 1000 times more heat absorbent than carbon dioxide
 - They reach the atmosphere from refrigeration & air conditioning, aerosol sprays, and foam packaging industry.
 - They are responsible for 30% of greenhouse effect
- (a) I & II (b) II & III
(c) Only III (d) All the above
87. Consider the following statements regarding Ozone:
- Ozone contributes to 3-7% of Greenhouse effect
 - The largest net source of tropospheric ozone is influx from the stratosphere
 - Large amounts of ozone are also produced in the troposphere by photochemical reactions, the amounts increasing with high levels of air pollution.
- (a) I & II (b) II & III
(c) Only III (d) All the above
88. What are the causes of deforestation?
- Population growth and overpopulation and urbanization
 - Globalization
 - Dishonesty of government institutions
- (a) Only I (b) Only II
(c) I & II (d) All the above
89. Which statement is correct regarding acid rain?
- The term "acid rain" was coined in 1972 by Robert Angus Smith
 - Rainfall with pH less than 5.6 is called Acid rain.
 - Acid rain is caused by a chemical reaction that starts when sulfur dioxide (SO_2) and nitrogen oxides (NOX) are released into the air
- (a) I & II (b) Only II
(c) II & III (d) All the above
90. Lead and cadmium compounds are added as stabilizers in PVC. Which statement is correct regarding lead and cadmium?
- Lead and cadmium can leach out during human contact, or when disposed in land-fills
 - Lead and cadmium are known neurotoxins and nephrotoxins respectively
 - These chemicals are used in the manufacture of soft plastic items such as vinyl flooring sheets, soft toys etc. to increase their durability
- (a) I & II (b) II & III
(c) Only III (d) All the above

Hints and Explanations

1. (c) The mean global temperature rise by $2^{\circ} - 6^{\circ}\text{C}$ and the concentration of carbon dioxide increases in the troposphere upto 600 ppm. Hence, the surface of the earth becomes warm which causes global warming. The phenomenon is similar to that of green house in which the glass enclosed atmosphere gets heated up due to its insulation from the rest of the environment. Hence, global warming is also known as green house effect and the gases responsible for it are called green house gases e.g CH_4 , CO_2 etc.
2. (d) SO_2 and NO when present in large quantities dissolved in water vapour form sulphuric acid and nitric acid which dissolve in rain water resulting in acid rain (H_2SO_4) and (HNO_3) which in turn causes great damage to forests and vegetation.
3. (a) The Bhopal gas tragedy occurred on 3rd Dec. 1984 in which methyl isocyanate gas was released from a fertilizer manufacturing plant of Union Carbide causing death of approximately 2500 persons. Chernobyl disaster occurred on April 26, 1986, from an explosion at the chernobyl power station which released a huge radioactive cloud into the atmosphere in Ukraine.
4. (d) Carbon dioxide is one of the important green house gas. It allows the shorter wavelength of infra red radiations to pass through it but does not allow these radiations to leave the earth's atmosphere. This results in warming of the atmosphere. If the amount of CO_2 decreases then there will not be any increase in temperature.
5. (a)
6. (b) Strength of sewage or degree of water pollution is measured in terms of BOD (Biochemical Oxygen Demand) value. BOD may be defined as, 'number of milligrams of O_2 required for decomposition of one litre of waste or water by decomposing microorganisms (bacteria)'.
7. (c) Due to heavy industrilization and transportation (modernization), CO_2 concentration is increasing day by day in the atmosphere. CO_2 has capacity for absorbing heat radiations and thus increases temperature. This increase in global temperature (global warming) is mainly due to CO_2 concentration is called green house effect. Complete combustion of fossil fuels and biomass releases carbon dioxide. Nuclear power plants releases radioactive wastes.
8. (b)
9. (c)
10. (b)
11. (a)
12. (c)
13. (c)
14. (b)
15. (b)
16. (a)
17. (b)
18. (a)
19. (d)
20. (c)
21. (b)
22. (a)
23. (d)
24. (a)
25. (b)
26. (a)
27. (a)
28. (a)
29. (c)
30. (d)
31. (c)
32. (d)
33. (d)
34. (c)
35. (d)
36. (a)
37. (a)
38. (b)
39. (b)
40. (a)
41. (d)
42. (a)
43. (b) The Kyoto protocol is a protocol to the United Nations Framework convention on climate change aimed at fighting global warming. This protokol adopted on 11 December 1997 in Kyoto, Japan. Under the protocol 37 countries commit themselves to a reduction of four greenhouse gases and two groups of gases.
44. (b)
45. (a)
46. (b)
47. (c)
48. (a)
49. (d)
50. (b)
51. (d)
52. (c)
53. (a)
54. (b)
55. (b)
56. (a)
57. (d)
58. (a) 'El Nino' is a warm ocean current. The term El Niño refers to the large-scale oceanatmosphere climate interaction linked to a periodic warming in sea surface temperatures across the central and east-central Equatorial Pacific.
59. (d) Ozone holes are more pronounced at the polar regions especially over Antarctica.
60. (a) Acid rain refers to the precipitation with a pH of less than 5. It is a mixture of H_2SO_4 and HNO_3 , the ratio of the two acids vary depending on the relative quantities of sulphur oxides and nitrogen oxides present in the atmosphere. These oxides are mainly produced by combustion of fossil fuels, smelters, industries, power plants, automobile exhausts etc.
61. (d) Housed within the Carbon Finance Unit of the World Bank, the Bio Carbon Fund is a public-private sector initiative mobilizing financing to help for development of projects that sequester or conserve carbon in forest and agro-ecosystems. It was created in 2004.
62. (b) The Montreal Protocol is an international treaty designed to protect the ozone layer by phasing out the production of numerous substances that are responsible for ozone depletion.
63. (a) The United Nations Conference on Environment and Development (UNCED), also known as the Rio Summit or Earth Summit. It was a major United Nations conference held in Rio de Janeiro from 3 to 14 June 1992. In 2012, the United Nations Conference on Sustainable Development was also held in Rio, and is also commonly called Rio+20 or Rio Earth Summit 2012.
64. (a) The Green Climate Fund (GCF) is a fund within the framework of the UNFCCC founded as a mechanism to redistribute money from the developed to the developing world, in order to assist the developing countries in adaptation and mitigation practices to counter climate change. The Fund is governed by the GCF Board. The assets of the GCF will be administered by a trustee only for the purpose of, and in accordance with, the relevant decisions of the GCF Board. The World Bank was invited by the COP to serve as the interim trustee of the GCF, subject to a review three years after operation of the Fund.

EXERCISE-2

1. (c) Coconut shells, groundnut shells and rice husk can be used in biomass gasification. Combustible gases generated from biomass gasification can be used for direct heat generation but not in internal combustion engines.
2. (a)
3. (b) The protocol was initially adopted on 11th December 1997. But it entered into force on 16th February 2005, after Kyoto, Japan. So, it is called Kyoto protocol.
Kyoto protocol deals with reducing the green house gases emission to a level at least 5% below 1990 level. Methane is more effective or harmful green house gas than carbon dioxide.
4. (b)
5. (c)
6. (c) Methane and nitrous oxide are the two green house gases emitted from rice field. In rice field, the flooding a field cuts off the oxygen supply from atmosphere resulting anaerobic condition which emits methane and nitrogen fertilizer which generally used in rice field for high production, emits nitrous oxide by the reaction with atmospheric oxygen.
7. (c)
8. (a)
9. (b)
10. (c)
11. (d)
12. (c)
13. (c)
14. (d)
15. (c)
16. (a)
17. (b) United States and Canada are the countries which are mostly affected by acid rain because of high number of factories, power plants and large number of automotive plants. Europe, Poland, Germany, Czech Republic, Sweden, Norway and Finland are affected because of British and European factories. In Asia, India and China are mostly affected mainly because of the large number of factories.
18. (b) Radioactive contamination or pollution causes severe life- threatening consequences in organisms. Because of the radioactive decay of the contaminants, which emit harmful ionising radiation such as alpha or beta particles, gamma rays or neutrons, genetic mutations occur which are hereditary. Carcinogenic contaminants cause cancers.
19. (d) Global warming is the increase in the global temperature. It has put many negative impacts on glaciers causing them to start melting, so, rise of sea level. Unpredictable weather conditions prevailing in some geographical areas are some of the other effects of global warming.
20. (d) Almost all of the mercury in the Arctic atmosphere is transported there in gaseous form from sources in areas farther south, from sources such as wildfires, coal burning and gold mining. Scientists have long known that mercury in the air near ground level undergoes complex chemical reactions that deposit the element on the surface. Once the mercury is completely removed from the air, these reactions stop. However, this newly discovered mixing ice forces down additional mercury to restart and sustains the reactions.
21. (d) The schedule is meant to reduce the growth of the US government's liabilities under the 1982 Nuclear Waste Policy Act, under which it was to begin taking spent reactor fuel from power companies in 1998. About 68,000 tonnes of used reactor fuel remains at 72 different power plant sites across the country, with the Department of Energy (DoE) reimbursing power companies the cost. The current production rate of spent fuel is 2,000 tonnes a year. The two interim facilities will accept used reactor fuel at a rate faster than this in order to reduce gradually the inventory at power companies.
22. (b) The Researchers found a small correlation between cosmic rays and global temperatures occurring every 22 years; however, the changing cosmic ray rate lagged behind the change in temperatures by between one and two years, suggesting that the cause might not be down to cosmic rays and cloud formation but might be due to the direct effects of the sun. By comparing the small oscillations in the cosmic ray rate, which were taken from data from two neutron monitors, and temperature with the overall trends in both since 1955, the research team found that less than 14 per cent of the global warming seen during this period could be attributable to solar activity.
23. (d) There is no single factor which cause an algal bloom. A combination of optimum factors such as the presence of good nutrients, warm temperatures, surface runoff, upwelling in the sea can all contribute harmful algal blooms.
24. (c) Imports and Exports (Control) Act, 1947 and the Foreign Trade (Development and Regulation) Ordinance, 1992 (Ord. 11 of 1992) got repealed with the enactment of Foreign Trade (Development And Regulation) Act, 1992. But still the answer remains the same because the new act also has "bearing" on biodiversity conservation.
25. (c) Coconut shells, groundnut shells and rice husk can be used in biomass gasification. Combustible gases generated from biomass gasification can be used for direct heat generation but not in internal combustion engines.
26. (b) Autoclaves use pressurized steam to destroy microorganisms, and are the most dependable systems available for the decontamination of laboratory waste and the sterilization of laboratory glassware, media, and reagents. For efficient heat transfer, steam must flush the air out of the autoclave chamber. Before using the autoclave, check the drain screen at the bottom of the chamber and clean if blocked. If the sieve is blocked with debris, a layer of air may form at the bottom of the autoclave, preventing efficient operation.
27. (b) Forests that experience high levels of acid precipitation expose trees to soil that has more dissolved aluminum and less dissolved calcium.
28. (c) Prolonged human exposure to solar UV radiation may result in acute and chronic health effects on the skin, eye and immune system. Over the longer

term, UV radiation induces degenerative changes in cells of the skin, fibrous tissue and blood vessels leading to premature skin aging, photodermatoses and actinic keratoses.

29. (c) Greenhouse gases are those that can absorb and emit infrared radiation, but not radiation in or near the visible spectrum. In order, the most abundant greenhouse gases in Earth's atmosphere are:
 - Water vapor (H_2O)
 - Carbon dioxide (CO_2)
 - Methane (CH_4)
 - Nitrous oxide (N_2O)
 - Ozone (O_3)
 - CFCs
30. (c) Greenland ice sheet melt water, which moves to the sea under the ice in contact with the land surface, may transport solids or dissolved material such as iron to the ocean. Measurements of the amount of available iron in melt water from the Greenland ice sheet shows that extensive melting of the ice sheet might add an amount of iron to the Atlantic Ocean equivalent to that added by airborne dust. This would increase biological activity in the Atlantic.
31. (b) Electronic waste or E-waste has ferrous and non-ferrous metals both. Non-ferrous metals like copper, aluminium, silver, gold, platinum, palladium etc. The presence of elements like lead, mercury arsenic, cadmium, selenium and hexavalent chromium are classified as hazardous waste.
32. (a) Drinking water in some parts of India has contaminants like Arsenic, Fluoride other than many other contaminants. The sources of Arsenic are run off from orchards. The sources of fluoride are erosion of natural deposits, discharge from fertilizers and aluminum factories.
33. (d) The International Treaty on Plant Genetic Resources for Food and Agriculture aims at guaranteeing food security through the conservation, exchange and sustainable use of the world's plant genetic resources for food and agriculture. The United Nations Convention to Combat Desertification is a Convention to combat desertification and mitigate the effects of drought. The World Heritage Convention is concerned with the protection of the world cultural and natural heritage. All three of them have a bearing on the biodiversity.
34. (c) Earth Hour is a worldwide movement for the planet organized by the World Wide Fund for Nature (WWF). The event is held worldwide annually encouraging individuals, communities, households and businesses to turn off their non-essential lights for one hour, from 8:30 to 9:30 p.m. to raise the awareness about the climate change and the need to save the planet.
35. (b) This organization is in partnership but not formed by Ramsar convention. The organization was formed in 1954 and Ramsar convention was signed in 1971. "Wetlands International", It is an intergovernmental organization formed by the countries which are signatories to Ramsar Convention. It works at the field level to develop and mobilize knowledge, and use the practical experience to advocate for better policies.
36. (c) Statement 1, 3 and 4 are correct as per wwf and environment ministry.
37. (c) Brominated flame retardants used in many household products are highly resistant to degradation in the environment and they are able to accumulate in humans and animals.
38. (d) Eco-sensitive zones are the ecologically important areas designated to be protected from industrial pollution and unregulated development under the Environment Protection Act of 1986. Only environmentally hazardous human activities are prohibited in those areas. Therefore, both the statements are incorrect.
39. (b) The Animal Welfare Board of India was established in 1962 under Section 4 of The Prevention of Cruelty to Animals Act, 1960. The National Tiger Conservation Authority is set up under the Chairmanship of the Minister for Environment and Forests and is a statutory body. National Ganga River Basin Authority was established by the Central Government of India, on 20 February 2009. The Prime Minister is the chair of the Authority.
40. (a) Coral reefs are diverse underwater ecosystems held together by calcium carbonate structures secreted by corals. Andaman & Nicobar, Gulf of Kachchh and Gulf of Mannar have coral reefs. However Sunderbans do not have coral reef.
41. (b) In India, the problem of soil erosion is associated with deforestation. Terrace cultivation helps in less erosion of soil.
42. (b) If the global temperature increases beyond $3^{\circ}C$ above the pre-industrial level then Terrestrial biosphere tends toward a net carbon source and Widespread coral mortality will occur.
43. (d) In steel furnace coke reacts with iron to release pollutants like Oxides of sulphur, Oxides of nitrogen, Carbon monoxide & Carbon dioxide.
44. (c) Global climate change threatens coral reef by increasing the temperature and decreasing the pH level of the ocean. Much of the carbon dioxide that enters the atmosphere dissolves into the ocean. In fact, the oceans have absorbed about 1/3 of the carbon dioxide produced from human activities since 1800 and about 1/2 of the carbon dioxide produced by burning fossil fuels (Sabine et al. 2004). As carbon dioxide in the ocean increases, ocean pH decreases or becomes more acidic.
45. (c) The Forest Carbon Partnership Facility is a global partnership of governments, businesses, civil society, and Indigenous Peoples focused on reducing emissions from deforestation and forest degradation, forest carbon stock conservation, the sustainable management of forests, and the enhancement of forest carbon stocks in developing countries (activities commonly referred to as REDD+).

46. (c) The dugong is a medium-sized marine mammal. Dugong is listed under schedule 1 of India Wildlife Protection Act, 1972. In 2008, a MoU was signed between the Ministry of Environment and Forests and the Government of India, in order to conserve dugongs. In fact the highest level of legal protection is accorded to dugongs in India.
47. (d) 48. (d)
49. (c) Nitrous Oxide (NO_2) (5%) arises from coal burning, biomass burning, and breakdown of chemical fertilizers.
50. (b)
51. (c) Photosynthesis would decrease as increased UV radiations (due to ozone depletion) would make it difficult for leaves of green plants to exchange gases with the atmosphere. Due to increased temperature consequent upon increased UV radiation, the evaporation rate of surface water will increase and soil moisture would decrease leading to drying of agricultural crops and hence reduced yield.
52. (d)
53. (d) CFCs and HCFCs destroy ozone in the stratosphere. These chemicals are inert, non-flammable, non-toxic, and lighter than air and can remain intact for years. They contain Chlorine and Fluorine, common being CFC-II, CFC-12, CFC-22 and CFC- 13.
54. (d) The process of clearance of forest by burning or logging is called deforestation. The main reasons for deforestation are trees or derived charcoal are used as, or sold, for fuel or as (a) commodity, while cleared land is used as grassland for livestock, plantations of commodities, and settlements. Deforested areas usually sustain extensive adverse soil erosion and regularly damage into wasteland.
55. (d) Deforestation is the removal of vegetation in (a) forest to the extent that it no longer supports its natural flora and fauna. Deforestation is (a) very broad term, which consists of cutting of trees including repeated lopping, felling, and removal of forest litter, browsing, grazing and trampling of seedlings.
56. (c) The six main greenhouse gases are – • Carbon dioxide (CO_2); • Methane (CH_4); • Nitrous oxide (N_2O); • Hydrofluorocarbons (HFCs); • Perfluorocarbons (PFCs); and • Sulphur hexafluoride (SF_6). Methane (CH_4) is the second most important greenhouse gas. CH_4 is more potent than CO_2 because the radiative forcing produced per molecule is greater. In addition, the infrared window is less saturated in the range of wavelengths of radiation absorbed by CH_4 , so more molecules may fill in the region.
57. (c)
58. (a) CO_2 is largest contributor towards global warming.
59. (b) 60. (d) 61. (b) 62. (c)
63. (c) spraying of DDT in soil results in the pollution of soil and water.
64. (d) Agro chemicals are developed by the use of modern technology that depends on inorganic fertilizers and pesticides. Excess use of these fertilizers can lead to immediate harmful effect or can also be long lasting. Although many benefits are there by the use of agro chemicals which are related to increase yield of plants and animal crops and less wastage during storing. These profits are substantial.
65. (d)
66. (b) Acid rain reacts with calcium to form calcium bicarbonate, which can be easily washed away.
67. (d) 68. (b) 69. (c) 70. (c)
71. (c) Domestic as well as industrial effluents that contaminate river water if allowed to flow unchecked.
72. (a) Refuse storage: which may sometimes require delivery of refuse by the householder over (a) considerable distance. Where the householder delivers the refuse to the vehicle at the time of collection. Door-to-door collection, where the collector enters the premises and collects the refuse and the householder is not involved in the collection process.
73. (d) Radioactive waste which arises from civil nuclear activities as well as from defense related nuclear weapon activities, poses a terrible problem for handling and keeping the environment to be safe to the present and future generations. The techniques used emphasizes on waste minimization and volume reduction. Nuclear waste is categorized into high, intermediate and low levels depending on the level of radioactivity in it.
74. (d) High level waste produced from the reprocessing plant is vitrified into a glassy form, enclosed in multiple barrier vessels and stored for a temporary period of three to four decades in engineered vaults with essential observation services. After cooling down in these storage facilities, waste vessels will be stored for long term in deep geological repositories.
75. (d) Plastics have become an indispensable part of our daily lives. Invented in 1935, they are wonderful products of polymer chemistry produced from the by- products of petroleum refining. They are classified as into two main categories
76. (a) All the varieties of plastics are manufactured from petrochemical based hydrocarbons. These hydrocarbons, and the plastic manufacturing processes involved possess environmentally critical characteristics. The raw materials and intermediate products used in the manufacture of Polyvinyl chloride (PVC) - Ethylene, Chlorine, Hydrogen chloride, Vinyl Chloride Monomer (VCM), and Ethylene Dichloride (EDC) — are known hazardous materials. Additives, fillers, and coloring pigments used in plastic goods can also exhibit hazardous properties.
77. (d) 78. (b) 79. (d)
80. (d) No vendor shall use carry bags and containers of recycled plastics for storing carrying and / or packaging of foodstuffs, Carry bags and Container used for packaging of foodstuff shall be made of virgin plastic and of natural shade or white, Carry

bags and Container made from recycled plastics must be manufactured using pigments and colorants as per IS: 9833/ 1981 notified by the Bureau of Indian Standards (BIS), Minimum thickness of Carry bags made of virgin or recycled plastics must not be less than 20 microns.

81. (b)
82. (c) The bio-degradable plastics will add to the already piling up municipal garbage. The immediate benefits of recovery and recycle of normal plastics is also lost if bio-degradable plastics are introduced. As of now, compared to normal plastic, the bio-degradable plastics are expensive and the technology for manufacture is not easily available. It may be possible to treat bio-degradable plastics in countries where solid waste management systems are working satisfactorily and extensively.
83. (c)
84. (c) The main function of the ozone gas found in the ozonosphere is to absorb the ultraviolet solar radiation. Ozone absorbs the ultraviolet solar radiation and through this process, the most harmful ultraviolet radiation is effectively filtered, thus safeguarding life at earth.
85. (d) Methane (CH_4), also called "Marsh gas", arises from rice paddies, wetlands, enteric fermentation in cattle, burning of wood, and landfills. It is responsible for about 4-9% of Greenhouse effect.
86. (a) Chlorofluorocarbons (CFCs) and their replacements (15%) are 1000 times more heat absorbent than carbon dioxide. They reach the atmosphere from refrigeration & air conditioning, aerosol sprays, and foam packaging industry
87. (d) Ozone contributes to 3-7% of Greenhouse effect. The largest net source of tropospheric ozone is influx from the stratosphere. Large amounts of ozone are also produced in the troposphere by photochemical reactions, the amounts increasing with high levels of air pollution.
88. (d) There are numerous causes of current deforestation such as dishonesty of government institutions, the imbalanced distribution of wealth and power, population growth and overpopulation, and urbanization. Globalization is also main cause of deforestation, though there are cases in which the effects of globalization have supported localized forest recuperate.
89. (c) Acid rain is a rain or any other form of precipitation that is unusually acidic, meaning that it possesses elevated levels of hydrogen ions (low pH). The term "acid rain" was coined in 1872 by Robert Angus Smith, after a link was established between sulfur dioxide (SO_2) emissions from the burning of coal in Manchester and acidification of nearby rainfall. Rainfall with pH less than 5.6 is called Acid rain.
90. (d) Lead and cadmium compounds are added as stabilizers in PVC. These chemicals are used in the manufacture of soft plastic items such as vinyl flooring sheets, soft toys etc. to increase their durability. Lead and cadmium can leach out during human contact, or when disposed in land-fills. Incineration of such rejected plastic items produces ash with high heavy metal content. Use of lead compounds in the manufacturing process can be (a) potential hazard to workers in the PVC industry. Lead and cadmium are known neurotoxins and nephrotoxins respectively